```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                     2 ************************************************************************
                                                     3 *
                                                     4 *    Zvector E7 instruction tests for VRR-c encoded:
                                                     5 *
                                                     6 *        E7B0 VDL    - VECTOR DIVIDE LOGICAL
                                                     7 *        E7B1 VRL    - VECTOR REMAINDER LOGICAL
                                                     8 *        E7B2 VD     - VECTOR DIVIDE
                                                     9 *        E7B3 VR     - VECTOR REMAINDER
                                                    10 *
                                                    11 *    and partial testing of
                                                    12 *
                                                    13 *        E7AA VMAL    - VECTOR MULTIPLY AND ADD LOW
                                                    14 *
                                                    15 *    during cross check tests.
                                                    16 *
                                                    17 *        James Wekel March 2026
                                                    18 ************************************************************************


                                                    20 ************************************************************************
                                                    21 *
                                                    22 *        basic instruction tests
                                                    23 *
                                                    24 ************************************************************************
                                                    25 *  This program tests proper functioning of the z/arch E7 VRR-c
                                                    26 *  Vector Divide Logical, Vector Remainder Logical, Vector Divide,
                                                    27 *  and Vector Remainder instructions.
                                                    28 *
                                                    29 *  Exceptions are not tested.
                                                    30 *
                                                    31 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                    32 *  obvious coding errors.  None of the tests are thorough.  They are
                                                    33 *  NOT designed to test all aspects of any of the instructions.
                                                    34 *
                                                    35 ************************************************************************
                                                    36 *
                                                    37 *    *Testcase zvector-e7-30-DivideRem
                                                    38 *    *
                                                    39 *    *   Zvector E7 instruction tests for VRR-c encoded:
                                                    40 *    *
                                                    41 *    *   E7B0 VDL    - VECTOR DIVIDE LOGICAL
                                                    42 *    *   E7B1 VRL    - VECTOR REMAINDER LOGICAL
                                                    43 *    *   E7B2 VD     - VECTOR DIVIDE
                                                    44 *    *   E7B3 VR     - VECTOR REMAINDER
                                                    45 *    *
                                                    46 *    *
                                                    47 *    *   # ------------------------------------------------------------
                                                    48 *    *   #  This tests only the basic function of the instructions.
                                                    49 *    *   #  Exceptions are NOT tested.
                                                    50 *    *   # ------------------------------------------------------------
                                                    51 *    *
                                                    52 *    mainsize    2
                                                    53 *    numcpu      1
                                                    54 *    sysclear
                                                    55 *    archlvl     z/Arch
                                                    56 *
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 *    loadcore    "$(testpath)/zvector-e7-30-DivideRem.core" 0x0
                                                 *
                                                 *    diag8cmd    enable    # (needed for messages to Hercules console)
                                                 *    runtest     5
                                                 *    diag8cmd    disable   # (reset back to default)
                                                 *
                                                 *    *Done
                                                 *
                                                 *
                                                 ************************************************************************
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  ***********************************************************************
                                                  *        FCHECK Macro - Is a Facility Bit set?
                                                  *
                                                  *        If the facility bit is NOT set, an message is issued and
                                                  *        the test is skipped.
                                                  *
                                                  *        Fcheck uses R0, R1 and R2
                                                  *
                                                  * eg.    FCHECK 134,'vector-packed-decimal'
                                                  ***********************************************************************
                                                           MACRO
                                                           FCHECK &BITNO,&NOTSETMSG
                                                  .*                       &BITNO : facility bit number to check
                                                  .*                       &NOTSETMSG : 'facility name'
                                                           LCLA  &FBBYTE           Facility bit in Byte
                                                           LCLA  &FBBIT            Facility bit within Byte

                                                           LCLA  &L(8)
                                                  &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte

                                                  &FBBYTE  SETA  &BITNO/8
                                                  &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                  .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'

                                                           B     X&SYSNDX
                                                  *                                      Fcheck data area
                                                  *                                      skip messgae
                                                  SKT&SYSNDX DC  C'    Skipping tests: '
                                                           DC    C&NOTSETMSG
                                                           DC    C' (bit &BITNO) is not installed.'
                                                  SKL&SYSNDX EQU *-SKT&SYSNDX
                                                  *                                      facility bits
                                                           DS    FD                      gap
                                                  FB&SYSNDX DS   4FD
                                                           DS    FD                      gap
                                                  *
                                                  X&SYSNDX EQU *
                                                           LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                           STFLE FB&SYSNDX               get facility bits

                                                           XGR   R0,R0
                                                           IC    R0,FB&SYSNDX+&FBBYTE    get fbit byte
                                                           N     R0,=F'&FBBIT'           is bit set?
                                                           BNZ   XC&SYSNDX
                                                  *
                                                  * facility bit not set, issue message and exit
                                                  *
                                                           LA    R0,SKL&SYSNDX           message length
                                                           LA    R1,SKT&SYSNDX           message address
                                                           BAL   R2,MSG

                                                           B     EOJ
                                                  XC&SYSNDX EQU *
                                                           MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   123 ************************************************************************
                                                   124 *     Instruction Macros
                                                   125 ************************************************************************
                                                   126 ************************************************************************
                                                   127 * (pending E7B0 VDL    - VECTOR DIVIDE LOGICAL
                                                   128 *  inclusion in SATK ASAM)
                                                   129 *
                                                   130 *     VDL Macro to help build VDL instruction
                                                   131 *        VDL   m4, m5
                                                   132 *
                                                   133 *        Note: v1, v2, v3 are fixed vector registers:
                                                   134 *                        v1 = 1; v2 = 2, v3 = 3
                                                   135 *              m4 are specified in hex eg. x'0'
                                                   136 *              m5 are specified in hex eg. x'0'
                                                   137 ************************************************************************
                                                   138          MACRO
                                                   139          VDL   &M4,&M5
                                                   140 .*                                  &M4  - m4 for VDL instruction
                                                   141 .*                                  &M5  - m5 for VDL instruction
                                                   142          LCLA  &M4M5
                                                   143 &M4M5    SETA  +((+&M5*256)+(+&M4*16))
                                                   144                                                        SPACE 1
                                                   145          DS    0H                     E7B0 VDL
                                                   146          DC    X'E7'                  - VECTOR DIVIDE LOGICAL
                                                   147          DC    X'12'                   v1, v2
                                                   148          DC    X'30'                   v3, reserved
                                                   149          DC    HL2'&M4M5'              reserved, m5, m4, RXB
                                                   150          DC    X'B0'
                                                   151                                                        SPACE 1
                                                   152          MEND


                                                   154 ************************************************************************
                                                   155 * (pending E7B1 VRL    - VECTOR REMAINDER LOGICAL
                                                   156 *  inclusion in SATK ASAM)
                                                   157 *
                                                   158 *     VDL Macro to help build VDL instruction
                                                   159 *        VDL   m4, m5
                                                   160 *
                                                   161 *        Note: v1, v2, v3 are fixed vector registers:
                                                   162 *                        v1 = 1; v2 = 2, v3 = 3
                                                   163 *              m4 are specified in hex eg. x'0'
                                                   164 *              m5 are specified in hex eg. x'0'
                                                   165 ************************************************************************
                                                   166          MACRO
                                                   167          VRL   &M4,&M5
                                                   168 .*                                  &M4  - m4 for VRL instruction
                                                   169 .*                                  &M5  - m5 for VRL instruction
                                                   170          LCLA  &M4M5
                                                   171 &M4M5    SETA  +((+&M5*256)+(+&M4*16))
                                                   172                                                        SPACE 1
                                                   173          DS    0H                     E7B1 VRL
                                                   174          DC    X'E7'                  - VECTOR REMAINDER LOGICAL
                                                   175          DC    X'12'                   v1, v2
                                                   176          DC    X'30'                   v3, reserved
                                                   177          DC    HL2'&M4M5'              reserved, m5, m4, RXB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  178          DC    X'B1'
                                                  179                                                  SPACE 1
                                                  180          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 ***********************************************************************
                                                 * (pending E7B2 VD     - VECTOR DIVIDE
                                                 *  inclusion in SATK ASAM)
                                                 *
                                                 *     VDL Macro to help build VD instruction
                                                 *        VDL   m4, m5
                                                 *
                                                 *        Note: v1, v2, v3 are fixed vector registers:
                                                 *                         v1 = 1; v2 = 2, v3 = 3
                                                 *              m4 are specified in hex eg. x'0'
                                                 *              m5 are specified in hex eg. x'0'
                                                 ***********************************************************************
                                                         MACRO
                                                         VD    &M4,&M5
                                                 .*                                  &M4  - m4 for VD instruction
                                                 .*                                  &M5  - m5 for VD instruction
                                                         LCLA  &M4M5
                                                 &M4M5   SETA  +((+&M5*256)+(+&M4*16))
                                                                                                          SPACE 1
                                                         DS    0H                    E7B2 VD
                                                         DC    X'E7'                 - VECTOR DIVIDE
                                                         DC    X'12'                  v1, v2
                                                         DC    X'30'                  v3, reserved
                                                         DC    HL2'&M4M5'             reserved, m5, m4, RXB
                                                         DC    X'B2'
                                                                                                          SPACE 1
                                                         MEND


                                                 ***********************************************************************
                                                 * (pending E7B3 VR     - VECTOR REMAINDER
                                                 *  inclusion in SATK ASAM)
                                                 *
                                                 *     VDL Macro to help build VDL instruction
                                                 *        VDL   m4, m5
                                                 *
                                                 *        Note: v1, v2, v3 are fixed vector registers:
                                                 *                         v1 = 1; v2 = 2, v3 = 3
                                                 *              m4 are specified in hex eg. x'0'
                                                 *              m5 are specified in hex eg. x'0'
                                                 ***********************************************************************
                                                         MACRO
                                                         VR    &M4,&M5
                                                 .*                                  &M4  - m4 for VR instruction
                                                 .*                                  &M5  - m5 for VR instruction
                                                         LCLA  &M4M5
                                                 &M4M5   SETA  +((+&M5*256)+(+&M4*16))
                                                                                                          SPACE 1
                                                         DS    0H                    E7B3 VR
                                                         DC    X'E7'                 - VECTOR REMAINDER
                                                         DC    X'12'                  v1, v2
                                                         DC    X'30'                  v3, reserved
                                                         DC    HL2'&M4M5'             reserved, m5, m4, RXB
                                                         DC    X'B3'
                                                                                                          SPACE 1
                                                         MEND
```

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				237

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 ************************************************************************
                                                 *        Low core PSWs
                                                 ************************************************************************
                             00000000  00006F37  ZVE7TST  START 0
00000000                     00000000            USING ZVE7TST,R0            Low core addressability

                             00000140  00000000  SVOLDPSW EQU   ZVE7TST+X'140'        z/Arch Supervisor call old PSW

00000000                     00000000  000001A0          ORG   ZVE7TST+X'1A0'        z/Architecure RESTART PSW
000001A0  00000001 80000000                      DC    X'0000000180000000'
000001A8  00000000 00000200                      DC    AD(BEGIN)

000001B0                     000001B0  000001D0          ORG   ZVE7TST+X'1D0'        z/Architecure PROGRAM CHECK PSW
000001D0  00020001 80000000                      DC    X'0002000180000000'
000001D8  00000000 0000DEAD                      DC    AD(X'DEAD')

000001E0                     000001E0  00000200          ORG   ZVE7TST+X'200'        Start of actual test program...

                                                 ************************************************************************
                                                 *               The actual "ZVE7TST" program itself...
                                                 ************************************************************************
                                                 *
                                                 *  Architecture Mode: z/Arch
                                                 *  Register Usage:
                                                 *
                                                 *   R0       (work)
                                                 *   R1-4     (work)
                                                 *   R5       Testing control table - current test base
                                                 *   R6-R7    (work)
                                                 *   R8       First base register
                                                 *   R9       Second base register
                                                 *   R10      Third base register
                                                 *   R11      E7TEST call return
                                                 *   R12      E7TESTS register
                                                 *   R13      (work)
                                                 *   R14      Subroutine call
                                                 *   R15      Secondary Subroutine call or work
                                                 *
                                                 ************************************************************************

00000200                     00000200            USING  BEGIN,R8        FIRST  Base Register
00000200                     00001200            USING  BEGIN+4096,R9   SECOND Base Register
00000200                     00002200            USING  BEGIN+8192,R10  THIRD  Base Register

00000200  0580                                   BEGIN    BALR  R8,0             Initalize FIRST base register
00000202  0680                                           BCTR  R8,0             Initalize FIRST base register
00000204  0680                                           BCTR  R8,0             Initalize FIRST base register

00000206  4190 8800                  00000800            LA    R9,2048(,R8)     Initalize SECOND base register
0000020A  4190 9800                  00000800            LA    R9,2048(,R9)     Initalize SECOND base register
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000020E  41A0 9800                    00000800   290          LA    R10,2048(,R9)    Initalize THIRD base register
00000212  41A0 A800                    00000800   291          LA    R10,2048(,R10)   Initalize THIRD base register
                                                  292
00000216  B600 851C                    0000071C   293          STCTL R0,R0,CTLR0      Store CR0 to enable AFP
0000021A  9604 851D                    0000071D   294          OI    CTLR0+1,X'04'    Turn on AFP bit
0000021E  9602 851D                    0000071D   295          OI    CTLR0+1,X'02'    Turn on Vector bit
00000222  B700 851C                    0000071C   296          LCTL  R0,R0,CTLR0      Reload updated CR0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    298 ************************************************************************
                                                    299 * Is z/Architecture Vector-enhancements facility 3 installed  (bit 198)
                                                    300 ************************************************************************
                                                    301
                                                    302          FCHECK 198,'Vector-enhancements facility 3'
00000226  47F0 80A8                    000002A8     303+         B     X0001
                                                    304+*                                     Fcheck data area
                                                    305+*                                     skip messgae
0000022A  40404040 E2928997                         306+SKT0001  DC    C'    Skipping tests: '
0000023E  E58583A3 96996085                         307+         DC    C'Vector-enhancements facility 3'
0000025C  404D8289 A340F1F9                         308+         DC    C' (bit 198) is not installed.'
                             0000004E  00000001     309+SKL0001  EQU   *-SKT0001
                                                    310+*                                     facility bits
00000278  00000000 00000000                         311+         DS    FD                     gap
00000280  00000000 00000000                         312+FB0001   DS    4FD
000002A0  00000000 00000000                         313+         DS    FD                     gap
                                                    314+*
                             000002A8  00000001     315+X0001    EQU   *
000002A8  4100 0004                    00000004     316+         LA    R0,((X0001-FB0001)/8)-1
000002AC  B2B0 8080                    00000280     317+         STFLE FB0001                 get facility bits
000002B0  B982 0000                                 318+         XGR   R0,R0
000002B4  4300 8098                    00000298     319+         IC    R0,FB0001+24           get fbit byte
000002B8  5400 8558                    00000758     320+         N     R0,=F'2'               is bit set?
000002BC  4770 80D0                    000002D0     321+         BNZ   XC0001
                                                    322+*
                                                    323+* facility bit not set, issue message and exit
                                                    324+*
000002C0  4100 004E                    0000004E     325+         LA    R0,SKL0001             message length
000002C4  4110 802A                    0000022A     326+         LA    R1,SKT0001             message address
000002C8  4520 8438                    00000638     327+         BAL   R2,MSG
000002CC  47F0 8500                    00000700     328+         B     EOJ
                             000002D0  00000001     329+XC0001   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 331 ***********************************************************************
                                                 332 *              Do tests in the E7TESTS table
                                                 333 ***********************************************************************
                                                 334
000002D0  58C0 855C                    0000075C   335          L     R12,=A(E7TESTS)       get table of test addresses
                                                 336
                             000002D4  00000001   337 NEXTE7   EQU   *
000002D4  5850 C000                    00000000   338          L     R5,0(0,R12)       get test address
000002D8  1255                                    339          LTR   R5,R5                 have a test?
000002DA  4780 838C                    0000058C   340          BZ    ENDTEST                 done?
                                                 341
000002DE                     00000000             342          USING E7TEST,R5
                                                 343
000002DE  4800 5004                    00000004   344          LH    R0,TNUM           save current test number
000002E2  5000 8E04                    00001004   345          ST    R0,TESTING        for easy reference
                                                 346
000002E6  E710 8EF8 0006               000010F8   347          VL    V1,V1FUDGE
000002EC  58B0 5000                    00000000   348          L     R11,TSUB          get address of test routine
000002F0  05BB                                    349          BALR  R11,R11           do test
                                                 350
                                                 351 * do cross check
000002F2  45F0 810E                    0000030E   352          BAL   R15,XCHECK
                                                 353
                                                 354 *  validate results
                                                 355
000002F6  E310 5020 0014               00000020   356          LGF   R1,READDR         get address of expected result
000002FC  D50F 5030 1000     00000030  00000000   357          CLC   V1OUTPUT,0(R1)    valid?
00000302  4770 8378                    00000578   358          BNE   FAILMSG              no, issue failed message
                                                 359
00000306  41C0 C004                    00000004   360          LA    R12,4(0,R12)      next test address
0000030A  47F0 80D4                    000002D4   361          B     NEXTE7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   363 *-----------------------------------------------------------------------
                                                   364 * cross check that the result can be converted back to the source
                                                   365 * Note: skip xcheck - if skip requested (S or Y)
                                                   366 *                   - if instruction is not VDL, VRL, VD, VR
                                                   367 *-----------------------------------------------------------------------
                                 0000030E  00000001  368 XCHECK   EQU   *
0000030E  D207 5048 8528         00000048  00000728  369          MVC   SKIPXC,=CL8'SKIP XC '
00000314  D500 5007 8568         00000007  00000768  370          CLC   XCSKIP,=CL1'S'         skip xcheck requested
0000031A  478F 0000                        00000000  371          BE    0(R15)                    skip == S, so exit
0000031E  D500 5007 8569         00000007  00000769  372          CLC   XCSKIP,=CL1'Y'         skip xcheck requested
00000324  478F 0000                        00000000  373          BE    0(R15)                    skip == Y, so exit
                                                   374 *
                                                   375 *        cross check depends on the instruction
                                                   376 *
00000328  D207 5048 8530         00000048  00000730  377          MVC   SKIPXC,=CL8'        '
0000032E  D507 500A 8538         0000000A  00000738  378          CLC   OPNAME,=CL8'VDL     '
00000334  4780 8158                        00000358  379          BE    XCVDL
00000338  D507 500A 8540         0000000A  00000740  380          CLC   OPNAME,=CL8'VRL     '
0000033E  4780 81AC                        000003AC  381          BE    XCVRL
00000342  D507 500A 8548         0000000A  00000748  382          CLC   OPNAME,=CL8'VD      '
00000348  4780 8200                        00000400  383          BE    XCVD
0000034C  D507 500A 8550         0000000A  00000750  384          CLC   OPNAME,=CL8'VR      '
00000352  4780 8254                        00000454  385          BE    XCVR
                                                   386 *                                       Not part of this xcheck test
00000356  07FF                                     387          BR    R15                       so return
                                                   388 * ---------------------------------------------------------------------
                                                   389 * cross check result in V20, saved V1 in V21
                                                   390 *
                                                   391 * do cross check: VDL    - VECTOR DIVIDE LOGICAL
                                                   392 *
00000358                                           393 XCVDL    DS    0H
00000358  E751 0000 0856                           394          VLR   V21,V1                 save VDL result
                                                   395
0000035E  D500 5008 856A         00000008  0000076A  396          CLC   M4,=X'2'
00000364  4770 8178                        00000378  397          BNE   XCVDL3
                                                   398          VRL   X'2',x'8'

00000368                                           400+         DS    0H                        E7B1 VRL
00000368  E7                                       401+         DC    X'E7'                     - VECTOR REMAINDER LOGICAL
00000369  12                                       402+         DC    X'12'                      v1, v2
0000036A  30                                       403+         DC    X'30'                      v3, reserved
0000036B  0820                                     404+         DC    HL2'2080'                  reserved, m5, m4, RXB
0000036D  B1                                       405+         DC    X'B1'

0000036E  E745 3200 1CAA                           407          VMAL  V20,V21,V3,V1,x'2'
00000374  47F0 82A8                        000004A8  408          B     XCCONT
                                                   409
00000378                                           410 XCVDL3   DS    0H
00000378  D500 5008 856B         00000008  0000076B  411          CLC   M4,=X'3'
0000037E  4770 8192                        00000392  412          BNE   XCVDL4
                                                   413          VRL   x'3',x'8'

00000382                                           415+         DS    0H                        E7B1 VRL
00000382  E7                                       416+         DC    X'E7'                     - VECTOR REMAINDER LOGICAL
00000383  12                                       417+         DC    X'12'                      v1, v2
00000384  30                                       418+         DC    X'30'                      v3, reserved
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00000385  0830                                     419+         DC    HL2'2096'                reserved, m5, m4, RXB
00000387  B1                                       420+         DC    X'B1'

00000388  E745 3300 1CAA                           422          VMAL  V20,V21,V3,V1,x'3'
0000038E  47F0 82A8                     000004A8   423          B     XCCONT
                                                   424
00000392                                           425 XCVDL4   DS    0H
00000392  D500 5008 856C      00000008  0000076C   426          CLC   M4,=X'4'
00000398  477F 0000                     00000000   427          BNE   0(R15)
                                                   428          VRL   X'4',x'8'

0000039C                                           430+         DS    0H                      E7B1 VRL
0000039C  E7                                       431+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
0000039D  12                                       432+         DC    X'12'                    v1, v2
0000039E  30                                       433+         DC    X'30'                    v3, reserved
0000039F  0840                                     434+         DC    HL2'2112'                reserved, m5, m4, RXB
000003A1  B1                                       435+         DC    X'B1'

000003A2  E745 3400 1CAA                           437          VMAL  V20,V21,V3,V1,x'4'
000003A8  47F0 82A8                     000004A8   438          B     XCCONT
                                                   439 *
                                                   440 * do cross check: VRL    - VECTOR REMAINDER LOGICAL
                                                   441 *
000003AC                                           442 XCVRL    DS    0H
000003AC  E751 0000 0856                           443          VLR   V21,V1                save VRL result
                                                   444
000003B2  D500 5008 856A      00000008  0000076A   445          CLC   M4,=X'2'
000003B8  4770 81CC                     000003CC   446          BNE   XCVRL3
                                                   447          VDL   X'2',x'8'

000003BC                                           449+         DS    0H                      E7B0 VDL
000003BC  E7                                       450+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
000003BD  12                                       451+         DC    X'12'                    v1, v2
000003BE  30                                       452+         DC    X'30'                    v3, reserved
000003BF  0820                                     453+         DC    HL2'2080'                reserved, m5, m4, RXB
000003C1  B0                                       454+         DC    X'B0'

000003C2  E741 3200 59AA                           456          VMAL  V20,V1,V3,V21,x'2'
000003C8  47F0 82A8                     000004A8   457          B     XCCONT
                                                   458
000003CC                                           459 XCVRL3   DS    0H
000003CC  D500 5008 856B      00000008  0000076B   460          CLC   M4,=X'3'
000003D2  4770 81E6                     000003E6   461          BNE   XCVRL4
                                                   462          VDL   x'3',x'8'

000003D6                                           464+         DS    0H                      E7B0 VDL
000003D6  E7                                       465+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
000003D7  12                                       466+         DC    X'12'                    v1, v2
000003D8  30                                       467+         DC    X'30'                    v3, reserved
000003D9  0830                                     468+         DC    HL2'2096'                reserved, m5, m4, RXB
000003DB  B0                                       469+         DC    X'B0'

000003DC  E741 3300 59AA                           471          VMAL  V20,V1,V3,V21,x'3'
000003E2  47F0 82A8                     000004A8   472          B     XCCONT
                                                   473
000003E6                                           474 XCVRL4   DS    0H
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000003E6  D500 5008 856C      00000008  0000076C   475          CLC   M4,=X'4'
000003EC  477F 0000                     00000000   476          BNE   0(R15)
                                                   477          VDL   X'4',x'8'

000003F0                                           479+         DS    0H                      E7B0 VDL
000003F0  E7                                       480+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
000003F1  12                                       481+         DC    X'12'                    v1, v2
000003F2  30                                       482+         DC    X'30'                    v3, reserved
000003F3  0840                                     483+         DC    HL2'2112'                reserved, m5, m4, RXB
000003F5  B0                                       484+         DC    X'B0'

000003F6  E741 3400 59AA                           486          VMAL  V20,V1,V3,V21,x'4'
000003FC  47F0 82A8                     000004A8   487          B     XCCONT
                                                   488 *
                                                   489 * do cross check: VD    - VECTOR DIVIDE
                                                   490 *
00000400                                           491 XCVD     DS    0H
00000400  E751 0000 0856                           492          VLR   V21,V1               save VDL result
                                                   493
00000406  D500 5008 856A      00000008  0000076A   494          CLC   M4,=X'2'
0000040C  4770 8220                     00000420   495          BNE   XCVD3
                                                   496          VR    X'2',x'8'

00000410                                           498+         DS    0H                      E7B3 VR
00000410  E7                                       499+         DC    X'E7'                   - VECTOR REMAINDER
00000411  12                                       500+         DC    X'12'                    v1, v2
00000412  30                                       501+         DC    X'30'                    v3, reserved
00000413  0820                                     502+         DC    HL2'2080'                reserved, m5, m4, RXB
00000415  B3                                       503+         DC    X'B3'

00000416  E745 3200 1CAA                           505          VMAL  V20,V21,V3,V1,x'2'
0000041C  47F0 82A8                     000004A8   506          B     XCCONT
                                                   507
00000420                                           508 XCVD3    DS    0H
00000420  D500 5008 856B      00000008  0000076B   509          CLC   M4,=X'3'
00000426  4770 823A                     0000043A   510          BNE   XCVD4
                                                   511          VR    x'3',x'8'

0000042A                                           513+         DS    0H                      E7B3 VR
0000042A  E7                                       514+         DC    X'E7'                   - VECTOR REMAINDER
0000042B  12                                       515+         DC    X'12'                    v1, v2
0000042C  30                                       516+         DC    X'30'                    v3, reserved
0000042D  0830                                     517+         DC    HL2'2096'                reserved, m5, m4, RXB
0000042F  B3                                       518+         DC    X'B3'

00000430  E745 3300 1CAA                           520          VMAL  V20,V21,V3,V1,x'3'
00000436  47F0 82A8                     000004A8   521          B     XCCONT
                                                   522
0000043A                                           523 XCVD4    DS    0H
0000043A  D500 5008 856C      00000008  0000076C   524          CLC   M4,=X'4'
00000440  477F 0000                     00000000   525          BNE   0(R15)
                                                   526          VR    X'4',x'8'

00000444                                           528+         DS    0H                      E7B3 VR
00000444  E7                                       529+         DC    X'E7'                   - VECTOR REMAINDER
00000445  12                                       530+         DC    X'12'                    v1, v2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00000446  30                                      531+         DC    X'30'                    v3, reserved
00000447  0840                                    532+         DC    HL2'2112'                reserved, m5, m4, RXB
00000449  B3                                      533+         DC    X'B3'

0000044A  E745 3400 1CAA                          535          VMAL  V20,V21,V3,V1,x'4'
00000450  47F0 82A8                    000004A8   536          B     XCCONT
                                                  537 *
                                                  538 * do cross check: VR     - VECTOR REMAINDER
                                                  539 *
00000454                                          540 XCVR     DS    0H
00000454  E751 0000 0856                          541          VLR   V21,V1               save VR  result
                                                  542
0000045A  D500 5008 856A      00000008  0000076A  543          CLC   M4,=X'2'
00000460  4770 8274                    00000474   544          BNE   XCVR3
                                                  545          VD    X'2',x'8'

00000464                                          547+         DS    0H                     E7B2 VD
00000464  E7                                      548+         DC    X'E7'                  - VECTOR DIVIDE
00000465  12                                      549+         DC    X'12'                   v1, v2
00000466  30                                      550+         DC    X'30'                   v3, reserved
00000467  0820                                    551+         DC    HL2'2080'               reserved, m5, m4, RXB
00000469  B2                                      552+         DC    X'B2'

0000046A  E741 3200 59AA                          554          VMAL  V20,V1,V3,V21,x'2'
00000470  47F0 82A8                    000004A8   555          B     XCCONT
                                                  556
00000474                                          557 XCVR3    DS    0H
00000474  D500 5008 856B      00000008  0000076B  558          CLC   M4,=X'3'
0000047A  4770 828E                    0000048E   559          BNE   XCVR4
                                                  560          VD    x'3',x'8'

0000047E                                          562+         DS    0H                     E7B2 VD
0000047E  E7                                      563+         DC    X'E7'                  - VECTOR DIVIDE
0000047F  12                                      564+         DC    X'12'                   v1, v2
00000480  30                                      565+         DC    X'30'                   v3, reserved
00000481  0830                                    566+         DC    HL2'2096'               reserved, m5, m4, RXB
00000483  B2                                      567+         DC    X'B2'

00000484  E741 3300 59AA                          569          VMAL  V20,V1,V3,V21,x'3'
0000048A  47F0 82A8                    000004A8   570          B     XCCONT
                                                  571
0000048E                                          572 XCVR4    DS    0H
0000048E  D500 5008 856C      00000008  0000076C  573          CLC   M4,=X'4'
00000494  477F 0000                    00000000   574          BNE   0(R15)
                                                  575          VD    X'4',x'8'

00000498                                          577+         DS    0H                     E7B2 VD
00000498  E7                                      578+         DC    X'E7'                  - VECTOR DIVIDE
00000499  12                                      579+         DC    X'12'                   v1, v2
0000049A  30                                      580+         DC    X'30'                   v3, reserved
0000049B  0840                                    581+         DC    HL2'2112'               reserved, m5, m4, RXB
0000049D  B2                                      582+         DC    X'B2'

0000049E  E741 3400 59AA                          584          VMAL  V20,V1,V3,V21,x'4'
000004A4  47F0 82A8                    000004A8   585          B     XCCONT
                                                  586
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  * ---------------------------------------------------------------------
                                                  * check result of cross check
                                                  *
000004A8                                          XCCONT DS   0H
000004A8  E740 5058 080E                00000058          VST   V20,XCOUTPUT
000004AE  E715 0000 0456                                  VLR   V1,V21                restore V1 for expected result test

000004B4  E310 5014 0014                00000014          LGF   R1,V2ADDR             expected source address
000004BA  D50F 5058 1000                00000000          CLC   XCOUTPUT(16),0(R1)
000004C0  4770 82C6                     000004C6          BNE   XCFAILMSG
000004C4  07FF                                            BR    R15                   return from xcheck


                                                  * xcheck failed message
000004C6                                          XCFAILMSG DS   0H
000004C6  4820 5004                     00000004          LH    R2,TNUM               get test number and convert
000004CA  4E20 8ED6                     000010D6          CVD   R2,DECNUM
000004CE  D211 8EC0 8EAA      000010C0  000010AA          MVC   PRT3,EDIT
000004D4  DE11 8EC0 8ED6      000010C0  000010D6          ED    PRT3,DECNUM
000004DA  D202 8E66 8ECD      00001066  000010CD          MVC   XCPTNUM(3),PRT3+13    fill in message with test #

000004E0  D207 8E8F 500A      0000108F  0000000A          MVC   XCPNAME,OPNAME        fill in message with instruction

000004E6  B982 0022                                       XGR   R2,R2
000004EA  4320 5008                     00000008          IC    R2,M4                 get i3 and convert
000004EE  4E20 8ED6                     000010D6          CVD   R2,DECNUM
000004F2  D211 8EC0 8EAA      000010C0  000010AA          MVC   PRT3,EDIT
000004F8  DE11 8EC0 8ED6      000010C0  000010D6          ED    PRT3,DECNUM
000004FE  D201 8EA0 8ECE      000010A0  000010CE          MVC   XCPM4(2),PRT3+14      fill in message with m4 field

00000504  B982 0022                                       XGR   R2,R2
00000508  4320 5009                     00000009          IC    R2,M5                 get m4 and convert
0000050C  4E20 8ED6                     000010D6          CVD   R2,DECNUM
00000510  D211 8EC0 8EAA      000010C0  000010AA          MVC   PRT3,EDIT
00000516  DE11 8EC0 8ED6      000010C0  000010D6          ED    PRT3,DECNUM
0000051C  D201 8EA7 8ECE      000010A7  000010CE          MVC   XCPM5(2),PRT3+14      fill in message with m5 field
                                                  *
00000522  50F0 8370                     00000570          ST    R15,XCR15             save r15
00000526  4100 0051                     00000051          LA    R0,XCPLNG             message length
0000052A  4110 8E59                     00001059          LA    R1,XCPLINE            messagfe address
0000052E  45F0 839A                     0000059A          BAL   R15,RPTERROR
00000532  58F0 8370                     00000570          L     R15,XCR15

00000536  5800 8560                     00000760          L     R0,=F'1'              set failed test indicator
0000053A  5000 8E00                     00001000          ST    R0,FAILED
0000053E  07FF                                            BR    R15                   return from xcheck

00000540                                          DS    0FD
00000540  00000000 00000000                       XCRESULT DS   XL16
00000550  00000000 00000000                       XCV1     DS   XL16
00000560  00000000 00000000                       XCV2     DS   XL16
00000570  00000000 00000000                       XCR15    DS   FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                           640 ***********************************************************************
                                           641 * result not as expected:
                                           642 *        issue message with test number, instruction under test
                                           643 *              and instruction m4
                                           644 ***********************************************************************
                             00000578  00000001   645 FAILMSG  EQU   *
00000578  45F0 839A                    0000059A   646          BAL   R15,RPTERROR


                                           648 ***********************************************************************
                                           649 * continue after a failed test
                                           650 ***********************************************************************
                             0000057C  00000001   651 FAILCONT EQU   *
0000057C  5800 8560                    00000760   652          L     R0,=F'1'          set failed test indicator
00000580  5000 8E00                    00001000   653          ST    R0,FAILED
                                                  654
00000584  41C0 C004                    00000004   655          LA    R12,4(0,R12)      next test address
00000588  47F0 80D4                    000002D4   656          B     NEXTE7


                                           658 ***********************************************************************
                                           659 * end of testing; set ending psw
                                           660 ***********************************************************************
                             0000058C  00000001   661 ENDTEST  EQU   *
0000058C  5810 8E00                    00001000   662          L     R1,FAILED         did a test fail?
00000590  1211                                    663          LTR   R1,R1
00000592  4780 8500                    00000700   664          BZ    EOJ                     No, exit
00000596  47F0 8518                    00000718   665          B     FAILTEST                Yes, exit with BAD PSW
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  667 ***********************************************************************
                                                  668 *        RPTERROR                 Report instruction test in error
                                                  669 ***********************************************************************

0000059A  50F0 841C                    0000061C   671 RPTERROR ST    R15,RPTSAVE          Save return address
0000059E  5050 8420                    00000620   672          ST    R5,RPTSVR5           Save R5
                                                  673 *
000005A2  4820 5004                    00000004   674          LH    R2,TNUM              get test number and convert
000005A6  4E20 8ED6                    000010D6   675          CVD   R2,DECNUM
000005AA  D211 8EC0 8EAA      000010C0 000010AA   676          MVC   PRT3,EDIT
000005B0  DE11 8EC0 8ED6      000010C0 000010D6   677          ED    PRT3,DECNUM
000005B6  D202 8E15 8ECD      00001015 000010CD   678          MVC   PRTNUM(3),PRT3+13    fill in message with test #
                                                  679
000005BC  D207 8E3E 500A      0000103E 0000000A   680          MVC   PRTNAME,OPNAME       fill in message with instruction
                                                  681 *
000005C2  E320 5008 0076               00000008   682          LB    R2,m4                get m4 and convert
000005C8  4E20 8ED6                    000010D6   683          CVD   R2,DECNUM
000005CC  D211 8EC0 8EAA      000010C0 000010AA   684          MVC   PRT3,EDIT
000005D2  DE11 8EC0 8ED6      000010C0 000010D6   685          ED    PRT3,DECNUM
000005D8  D201 8E4F 8ECE      0000104F 000010CE   686          MVC   PRTM4(2),PRT3+14     fill in message with m4 field
                                                  687 *
000005DE  B982 0022                               688          XGR   R2,R2
000005E2  4320 5009                    00000009   689          IC    R2,M5                get m5 and convert
000005E6  4E20 8ED6                    000010D6   690          CVD   R2,DECNUM
000005EA  D211 8EC0 8EAA      000010C0 000010AA   691          MVC   PRT3,EDIT
000005F0  DE11 8EC0 8ED6      000010C0 000010D6   692          ED    PRT3,DECNUM
000005F6  D201 8E56 8ECE      00001056 000010CE   693          MVC   PRTM5(2),PRT3+14     fill in message with m5 field
                                                  694 *
                                                  695 *        Use Hercules Diagnose for Message to console
                                                  696 *
000005FC  9002 8428                    00000628   697          STM   R0,R2,RPTDWSAV       save regs used by MSG
00000600  4100 0051                    00000051   698          LA    R0,PRTLNG            message length
00000604  4110 8E08                    00001008   699          LA    R1,PRTLINE           messagfe address
00000608  4520 8438                    00000638   700          BAL   R2,MSG               call Hercules console MSG display
0000060C  9802 8428                    00000628   701          LM    R0,R2,RPTDWSAV       restore regs


00000610  5850 8420                    00000620   703          L     R5,RPTSVR5         Restore R5
00000614  58F0 841C                    0000061C   704          L     R15,RPTSAVE        Restore return address
00000618  07FF                                    705          BR    R15                Return to caller

0000061C  00000000                                707 RPTSAVE  DC    F'0'               R15 save area
00000620  00000000                                708 RPTSVR5  DC    F'0'               R5 save area

00000628  00000000 00000000                       710 RPTDWSAV DC    2D'0'              R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 712 ***********************************************************************
                                                 713 *        Issue HERCULES MESSAGE pointed to by R1, length in R0
                                                 714 *              R2 = return address
                                                 715 ***********************************************************************

00000638  4900 8564                    00000764   717 MSG      CH    R0,=H'0'                Do we even HAVE a message?
0000063C  07D2                                    718          BNHR  R2                      No, ignore

0000063E  9002 8474                    00000674   720          STM   R0,R2,MSGSAVE           Save registers

00000642  4900 8566                    00000766   722          CH    R0,=AL2(L'MSGMSG)       Message length within limits?
00000646  47D0 844E                    0000064E   723          BNH   MSGOK                   Yes, continue
0000064A  4100 005F                    0000005F   724          LA    R0,L'MSGMSG             No, set to maximum

0000064E  1820                                    726 MSGOK    LR    R2,R0                   Copy length to work register
00000650  0620                                    727          BCTR  R2,0                    Minus-1 for execute
00000652  4420 8480                    00000680   728          EX    R2,MSGMVC               Copy message to O/P buffer

00000656  4120 200A                    0000000A   730          LA    R2,1+L'MSGCMD(,R2)      Calculate true command length
0000065A  4110 8486                    00000686   731          LA    R1,MSGCMD               Point to true command

0000065E  83120008                                733          DC    X'83',X'12',X'0008'     Issue Hercules Diagnose X'008'
00000662  4780 846E                    0000066E   734          BZ    MSGRET                  Return if successful
                                                  735
00000666  1222                                    736          LTR   R2,R2                   Is Diag8 Ry (R2) 0?
00000668  4780 846E                    0000066E   737          BZ    MSGRET                    an error occurred but coninue
                                                  738
0000066C  0000                                    739          DC    H'0'                    CRASH for debugging purposes

0000066E  9802 8474                    00000674   741 MSGRET   LM    R0,R2,MSGSAVE           Restore registers
00000672  07F2                                    742          BR    R2                      Return to caller



00000674  00000000 00000000                       744 MSGSAVE  DC    3F'0'                   Registers save area
00000680  D200 848F 1000      0000068F  00000000  745 MSGMVC   MVC   MSGMSG(0),0(R1)         Executed instruction


00000686  D4E2C7D5 D6C8405C                       747 MSGCMD   DC    C'MSGNOH * '            *** HERCULES MESSAGE COMMAND ***
0000068F  40404040 40404040                       748 MSGMSG   DC    CL95' '                 The message text to be displayed
                                                  749
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    751 ************************************************************************
                                                    752 *        Normal completion or Abnormal termination PSWs
                                                    753 ************************************************************************


000006F0  00020001 80000000                         755 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)

00000700  B2B2 84F0                       000006F0  757 EOJ      LPSWE EOJPSW               Normal completion


00000708  00020001 80000000                         759 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')

00000718  B2B2 8508                       00000708  761 FAILTEST LPSWE FAILPSW              Abnormal termination


                                                    763 ************************************************************************
                                                    764 *        Working Storage
                                                    765 ************************************************************************


0000071C  00000000                                  767 CTLR0    DS    F                    CR0
00000720  00000000                                  768          DS    F

00000728                                            770          LTORG ,                    Literals pool
00000728  E2D2C9D7 40E7C340                         771                =CL8'SKIP XC '
00000730  40404040 40404040                         772                =CL8'        '
00000738  E5C4D340 40404040                         773                =CL8'VDL     '
00000740  E5D9D340 40404040                         774                =CL8'VRL     '
00000748  E5C44040 40404040                         775                =CL8'VD      '
00000750  E5D94040 40404040                         776                =CL8'VR      '
00000758  00000002                                  777                =F'2'
0000075C  00006D50                                  778                =A(E7TESTS)
00000760  00000001                                  779                =F'1'
00000764  0000                                      780                =H'0'
00000766  005F                                      781                =AL2(L'MSGMSG)
00000768  E2                                        782                =CL1'S'
00000769  E8                                        783                =CL1'Y'
0000076A  02                                        784                =X'2'
0000076B  03                                        785                =X'3'
0000076C  04                                        786                =X'4'
                                                    787
                                                    788 *        some constants
                                                    789
                                  00000400  00000001 790 K        EQU   1024                 One KB
                                  00001000  00000001 791 PAGE     EQU   (4*K)                Size of one page
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                              00010000  00000001   792 K64      EQU   (64*K)           64 KB
                              00100000  00000001   793 MB       EQU   (K*K)             1 MB
                                                   794
                              AABBCCDD  00000001   795 REG2PATT EQU   X'AABBCCDD'    Polluted Register pattern
                              000000DD  00000001   796 REG2LOW  EQU         X'DD'    (last byte above)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    798 *=====================================================================
                                                    799 *
                                                    800 *  NOTE: start data on an address that is easy to display
                                                    801 *        within Hercules
                                                    802 *
                                                    803 *=====================================================================
                                                    804
0000076D                      0000076D  00001000    805          ORG   ZVE7TST+X'1000'
00001000  00000000                                  806 FAILED   DC    F'0'                      some test failed?
00001004  00000000                                  807 TESTING  DC    F'0'                      current test number


                                                    809 *
                                                    810 *        failed message and associated editting
                                                    811 *
00001008  40404040 4040E385                         812 PRTLINE  DC    C'      Test # '
00001015  A7A7A7                                    813 PRTNUM   DC    C'xxx'
00001018  40868189 9385847A                         814          DC    c' failed: wrong result for instruction '
0000103E  A7A7A7A7 A7A7A7A7                         815 PRTNAME  DC    CL8'xxxxxxxx'
00001046  40A689A3 884094F4                         816          DC    C' with m4='
0000104F  A7A7                                      817 PRTM4    DC    C'xx'
00001051  6B                                        818          DC    C','
00001052  4094F57E                                  819          DC    C' m5='
00001056  A7A7                                      820 PRTM5    DC    C'xx'
00001058  4B                                        821          DC    C'.'
                              00000051  00000001    822 PRTLNG   EQU   *-PRTLINE


                                                    824 ***********************************************************************
                                                    825 *        TEST failed : XCHECK
                                                    826 ***********************************************************************
                                                    827 *
                                                    828 *        XCHECK failed message
                                                    829 *
00001059  40404040 4040E385                         830 XCPLINE  DC    C'      Test # '
00001066  A7A7A7                                    831 XCPTNUM  DC    C'xxx'
00001069  40868189 9385847A                         832          DC    c' failed: cross check  for instruction '
0000108F  A7A7A7A7 A7A7A7A7                         833 XCPNAME  DC    CL8'xxxxxxxx'
00001097  40A689A3 884094F4                         834          DC    C' with m4='
000010A0  A7A7                                      835 XCPM4    DC    C'xx'
000010A2  6B                                        836          DC    C','
000010A3  4094F57E                                  837          DC    C' m5='
000010A7  A7A7                                      838 XCPM5    DC    C'xx'
000010A9  4B                                        839          DC    C'.'
                              00000051  00000001    840 XCPLNG   EQU   *-XCPLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    842 ************************************************************************
                                                    843 *        TEST failed : message working storge
                                                    844 ************************************************************************
000010AA  40212020 20202020                         845 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                    846
000010BC  7E7E7E6E                                  847          DC    C'===>'
000010C0  40404040 40404040                         848 PRT3     DC    CL18' '
000010D2  4C7E7E7E                                  849          DC    C'<===='
000010D6  00000000 00000000                         850 DECNUM   DS    CL16


                                                    852 ************************************************************************
                                                    853 *        Vector instruction results, pollution and input
                                                    854 ************************************************************************
000010E8                                            855          DS    0F
000010E8  00000000 00000000                         856          DS    XL16                                        gap
000010F8  FFFFFFFF FFFFFFFF                         857 V1FUDGE  DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'    V1 FUDGE
00001108  00000000 00000000                         858          DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    860 ************************************************************************
                                                    861 *        E7TEST DSECT
                                                    862 ************************************************************************


                                                    864 E7TEST   DSECT ,
00000000  00000000                                  865 TSUB     DC    A(0)           pointer  to test
00000004  0000                                      866 TNUM     DC    H'00'          Test Number
00000006  00                                        867          DC    X'00'
00000007  40                                        868 XCSKIP   DC    CL1' '            Y or S = skip cross check
00000008  00                                        869 M4       DC    HL1'00'        m4 used
00000009  00                                        870 M5       DC    HL1'00'        m5 used
                                                    871
0000000A  40404040 40404040                         872 OPNAME   DC    CL8' '         E7 name
00000014  00000000                                  873 V2ADDR   DC    A(0)           address of v2 source
00000018  00000000                                  874 V3ADDR   DC    A(0)           address of v3 source
0000001C  00000000                                  875 RELEN    DC    A(0)           RESULT LENGTH
00000020  00000000                                  876 READDR   DC    A(0)           result (expected) address
00000028  00000000 00000000                         877          DS    FD                gap
00000030  00000000 00000000                         878 V1OUTPUT DS    XL16           V1 Output
00000040  00000000 00000000                         879          DS    FD                gap
00000048  40404040 40404040                         880 SKIPXC   DC    CL8' '         was cross check skipped?
00000050  00000000 00000000                         881          DS    FD                gap
00000058  00000000 00000000                         882 XCOUTPUT DS    XL16           Cross check Output
00000068  00000000 00000000                         883          DS    FD                gap
                                                    884
                                                    885 *        test routine will be here (from VRR-c macro)
                                                    886 *
                                                    887 *        followed by
                                                    888 *              EXPECTED RESULT


                          00000000  00006F37        890 ZVE7TST  CSECT ,
00001118                                            891          DS    0F


                                                    893 ************************************************************************
                                                    894 *     Macros to help build test tables
                                                    895 ************************************************************************


                                                    897 *
                                                    898 * macro to generate individual test
                                                    899 *
                                                    900          MACRO
                                                    901          VRR_C &INST,&M4,&M5,&SKIP
                                                    902 .*                              &INST   - VRR-c instruction under test
                                                    903 .*                              &m4     - m4 field
                                                    904
                                                    905          GBLA  &TNUM
                                                    906 &TNUM    SETA  &TNUM+1
                                                    907
                                                    908          DS    0FD
                                                    909          USING *,R5             base for test data and test routine
                                                    910
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                  911 T&TNUM   DC    A(X&TNUM)         address of test routine
                                                  912          DC    H'&TNUM'          test number
                                                  913          DC    X'00'
                                                  914          DC    CL1'&SKIP'        S or Y = skip cross check
                                                  915
                                                  916          DC    &M4                m4
                                                  917          DC    &M5                m5
                                                  918
                                                  919          DC    CL8'&INST'        instruction name
                                                  920          DC    A(RE&TNUM+16)     address of v2 source
                                                  921          DC    A(RE&TNUM+32)     address of v3 source
                                                  922          DC    A(16)             result length
                                                  923 REA&TNUM DC    A(RE&TNUM)        result address
                                                  924          DS    FD                   gap
                                                  925 V1O&TNUM DS    XL16              V1 output
                                                  926          DS    FD                   gap
                                                  927          DC    CL8' '            was cross check skipped?
                                                  928          DS    FD                   gap
                                                  929 XCO&TNUM DS    XL16              Cross check Output
                                                  930          DS    FD                   gap
                                                  931 .*
                                                  932 *
                                                  933 X&TNUM   DS    0F
                                                  934          LGF   R1,V2ADDR         load v2 source
                                                  935          VL    v2,0(R1)
                                                  936
                                                  937          LGF   R1,V3ADDR         load v3 source
                                                  938          VL    v3,0(R1)
                                                  939
                                                  940          &INST &M4,&m5           test instruction
                                                  941          VST   V1,V1O&TNUM       save v1 output
                                                  942
                                                  943          BR    R11               return
                                                  944
                                                  945 RE&TNUM  DC    0F                xl16 expected result
                                                  946
                                                  947          DROP  R5
                                                  948          MEND


                                                  950 *
                                                  951 * macro to generate table of pointers to individual tests
                                                  952 *
                                                  953          MACRO
                                                  954          PTTABLE
                                                  955          GBLA  &TNUM
                                                  956          LCLA  &CUR
                                                  957 &CUR     SETA  1
                                                  958 .*
                                                  959 TTABLE   DS    0F
                                                  960 .LOOP    ANOP
                                                  961 .*
                                                  962          DC    A(T&CUR)
                                                  963 .*
                                                  964 &CUR     SETA  &CUR+1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   965          AIF   (&CUR LE &TNUM).LOOP
                                                   966 *
                                                   967          DC    A(0)              END OF TABLE
                                                   968          DC    A(0)
                                                   969 .*
                                                   970          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    ***********************************************************************
                                                    *        E7 VRR-c tests
                                                    ***********************************************************************
                                                            PRINT DATA
                                                    *
                                                    *   E7B0 VDL    - VECTOR DIVIDE LOGICAL
                                                    *   E7B1 VRL    - VECTOR REMAINDER LOGICAL
                                                    *   E7B2 VD     - VECTOR DIVIDE
                                                    *   E7B3 VR     - VECTOR REMAINDER
                                                    *
                                                    *        VRR-c instruction, m4, m5
                                                    *              followed by
                                                    *                 16 byte expected result (V1)
                                                    *                 16 byte V2 source
                                                    *                 16 byte V3 source
                                                    *
                                                    *-----------------------------------------------------------------
                                                    *   VDL    - VECTOR DIVIDE LOGICAL
                                                    *-----------------------------------------------------------------
                                                    *Word    M4:  2, m5= 8 (IDC=1)

                                                            VRR_C VDL,X'2',X'8',N
00001118                                                    DS    0FD
00001118                        00001118                    USING *,R5              base for test data and test routine
00001118  00001188                                  T1      DC    A(X1)             address of test routine
0000111C  0001                                              DC    H'1'              test number
0000111E  00                                                DC    X'00'
0000111F  D5                                                DC    CL1'N'            S or Y = skip cross check
00001120  02                                                DC    X'2'               m4
00001121  08                                                DC    X'8'               m5
00001122  E5C4D340 40404040                                 DC    CL8'VDL'          instruction name
0000112C  000011C0                                          DC    A(RE1+16)         address of v2 source
00001130  000011D0                                          DC    A(RE1+32)         address of v3 source
00001134  00000010                                          DC    A(16)             result length
00001138  000011B0                                  REA1    DC    A(RE1)            result address
00001140  00000000 00000000                                 DS    FD                   gap
00001148  00000000 00000000                         V1O1    DS    XL16              V1 output
00001150  00000000 00000000
00001158  00000000 00000000                                 DS    FD                   gap
00001160  40404040 40404040                                 DC    CL8' '            was cross check skipped?
00001168  00000000 00000000                                 DS    FD                   gap
00001170  00000000 00000000                         XCO1    DS    XL16              Cross check Output
00001178  00000000 00000000
00001180  00000000 00000000                                 DS    FD                   gap
                                                    *
00001188                                            X1      DS    0F
00001188  E310 5014 0014                00000014            LGF   R1,V2ADDR         load v2 source
0000118E  E721 0000 0006                00000000            VL    v2,0(R1)
00001194  E310 5018 0014                00000018            LGF   R1,V3ADDR         load v3 source
0000119A  E731 0000 0006                00000000            VL    v3,0(R1)

000011A0                                                    DS    0H                     E7B0 VDL
000011A0  E7                                                DC    X'E7'                  - VECTOR DIVIDE LOGICAL
000011A1  12                                                DC    X'12'                   v1, v2
000011A2  30                                                DC    X'30'                   v3, reserved
000011A3  0820                                              DC    HL2'2080'               reserved, m5, m4, RXB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000011A5  B0                                       1027+         DC    X'B0'

000011A6  E710 5030 000E                00001148   1029+         VST   V1,V1O1           save v1 output
000011AC  07FB                                     1030+         BR    R11               return
000011B0                                           1031+RE1      DC    0F                xl16 expected result
000011B0                                           1032+         DROP  R5
000011B0  00000000 00000000                        1033          DC    XL16'00000000 00000000 00000000 00000000'   result
000011B8  00000000 00000000
000011C0  00000000 00000000                        1034          DC    XL16'00000000 00000000 00000000 00000000'   v2
000011C8  00000000 00000000
000011D0  01020304 05060708                        1035          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000011D8  090A0B0C 0D0E0F10
                                                   1036
                                                   1037          VRR_C VDL,X'2',X'8',N
000011E0                                           1038+         DS    0FD
000011E0                      000011E0             1039+         USING *,R5              base for test data and test routine
000011E0  00001250                                 1040+T2       DC    A(X2)             address of test routine
000011E4  0002                                     1041+         DC    H'2'              test number
000011E6  00                                       1042+         DC    X'00'
000011E7  D5                                       1043+         DC    CL1'N'            S or Y = skip cross check
000011E8  02                                       1044+         DC    X'2'               m4
000011E9  08                                       1045+         DC    X'8'               m5
000011EA  E5C4D340 40404040                        1046+         DC    CL8'VDL'          instruction name
000011F4  00001288                                 1047+         DC    A(RE2+16)         address of v2 source
000011F8  00001298                                 1048+         DC    A(RE2+32)         address of v3 source
000011FC  00000010                                 1049+         DC    A(16)             result length
00001200  00001278                                 1050+REA2     DC    A(RE2)            result address
00001208  00000000 00000000                        1051+         DS    FD                   gap
00001210  00000000 00000000                        1052+V1O2     DS    XL16              V1 output
00001218  00000000 00000000
00001220  00000000 00000000                        1053+         DS    FD                   gap
00001228  40404040 40404040                        1054+         DC    CL8' '            was cross check skipped?
00001230  00000000 00000000                        1055+         DS    FD                   gap
00001238  00000000 00000000                        1056+XCO2     DS    XL16              Cross check Output
00001240  00000000 00000000
00001248  00000000 00000000                        1057+         DS    FD                   gap
                                                   1058+*
00001250                                           1059+X2       DS    0F
00001250  E310 5014 0014                00000014   1060+         LGF   R1,V2ADDR         load v2 source
00001256  E721 0000 0006                00000000   1061+         VL    v2,0(R1)
0000125C  E310 5018 0014                00000018   1062+         LGF   R1,V3ADDR         load v3 source
00001262  E731 0000 0006                00000000   1063+         VL    v3,0(R1)

00001268                                           1066+         DS    0H                     E7B0 VDL
00001268  E7                                       1067+         DC    X'E7'                  - VECTOR DIVIDE LOGICAL
00001269  12                                       1068+         DC    X'12'                   v1, v2
0000126A  30                                       1069+         DC    X'30'                   v3, reserved
0000126B  0820                                     1070+         DC    HL2'2080'               reserved, m5, m4, RXB
0000126D  B0                                       1071+         DC    X'B0'

0000126E  E710 9010 000E                00001210   1073+         VST   V1,V1O2           save v1 output
00001274  07FB                                     1074+         BR    R11               return
00001278                                           1075+RE2      DC    0F                xl16 expected result
00001278                                           1076+         DROP  R5
00001278  00000000 00000000                        1077          DC    XL16'00000000 00000000 00000000 00000000'   result
00001280  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001288  01020303 05060707                        1078          DC    XL16'01020303 05060707 090A0B0B 0D0E0F0F'   v2
00001290  090A0B0B 0D0E0F0F
00001298  01020304 05060708                        1079          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000012A0  090A0B0C 0D0E0F10
                                                   1080
                                                   1081          VRR_C VDL,X'2',X'8',N
000012A8                                           1082+         DS    0FD
000012A8                       000012A8            1083+         USING *,R5              base for test data and test routine
000012A8  00001318                                 1084+T3       DC    A(X3)             address of test routine
000012AC  0003                                     1085+         DC    H'3'              test number
000012AE  00                                       1086+         DC    X'00'
000012AF  D5                                       1087+         DC    CL1'N'            S or Y = skip cross check
000012B0  02                                       1088+         DC    X'2'               m4
000012B1  08                                       1089+         DC    X'8'               m5
000012B2  E5C4D340 40404040                        1090+         DC    CL8'VDL'          instruction name
000012BC  00001350                                 1091+         DC    A(RE3+16)         address of v2 source
000012C0  00001360                                 1092+         DC    A(RE3+32)         address of v3 source
000012C4  00000010                                 1093+         DC    A(16)             result length
000012C8  00001340                                 1094+REA3     DC    A(RE3)            result address
000012D0  00000000 00000000                        1095+         DS    FD                   gap
000012D8  00000000 00000000                        1096+V1O3     DS    XL16              V1 output
000012E0  00000000 00000000
000012E8  00000000 00000000                        1097+         DS    FD                   gap
000012F0  40404040 40404040                        1098+         DC    CL8' '            was cross check skipped?
000012F8  00000000 00000000                        1099+         DS    FD                   gap
00001300  00000000 00000000                        1100+XCO3     DS    XL16              Cross check Output
00001308  00000000 00000000
00001310  00000000 00000000                        1101+         DS    FD                   gap
                                                   1102+*
00001318                                           1103+X3       DS    0F
00001318  E310 5014 0014                00000014   1104+         LGF   R1,V2ADDR         load v2 source
0000131E  E721 0000 0006                00000000   1105+         VL    v2,0(R1)
00001324  E310 5018 0014                00000018   1106+         LGF   R1,V3ADDR         load v3 source
0000132A  E731 0000 0006                00000000   1107+         VL    v3,0(R1)

00001330                                           1110+         DS    0H                     E7B0 VDL
00001330  E7                                       1111+         DC    X'E7'                  - VECTOR DIVIDE LOGICAL
00001331  12                                       1112+         DC    X'12'                   v1, v2
00001332  30                                       1113+         DC    X'30'                   v3, reserved
00001333  0820                                     1114+         DC    HL2'2080'               reserved, m5, m4, RXB
00001335  B0                                       1115+         DC    X'B0'

00001336  E710 5030 000E                000012D8   1117+         VST   V1,V1O3           save v1 output
0000133C  07FB                                     1118+         BR    R11               return
00001340                                           1119+RE3      DC    0F                xl16 expected result
00001340                                           1120+         DROP  R5
00001340  00000001 00000001                        1121          DC    XL16'00000001 00000001 00000001 00000001'   result
00001348  00000001 00000001
00001350  01020304 05060708                        1122          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v2
00001358  090A0B0C 0D0E0F10
00001360  01020304 05060708                        1123          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00001368  090A0B0C 0D0E0F10
                                                   1124
                                                   1125          VRR_C VDL,X'2',X'8',N
00001370                                           1126+         DS    0FD
00001370                       00001370            1127+         USING *,R5              base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001370  000013E0                                 1128+T4       DC    A(X4)              address of test routine
00001374  0004                                     1129+         DC    H'4'               test number
00001376  00                                       1130+         DC    X'00'
00001377  D5                                       1131+         DC    CL1'N'             S or Y = skip cross check
00001378  02                                       1132+         DC    X'2'                m4
00001379  08                                       1133+         DC    X'8'                m5
0000137A  E5C4D340 40404040                        1134+         DC    CL8'VDL'           instruction name
00001384  00001418                                 1135+         DC    A(RE4+16)          address of v2 source
00001388  00001428                                 1136+         DC    A(RE4+32)          address of v3 source
0000138C  00000010                                 1137+         DC    A(16)              result length
00001390  00001408                                 1138+REA4     DC    A(RE4)             result address
00001398  00000000 00000000                        1139+         DS    FD                    gap
000013A0  00000000 00000000                        1140+V1O4     DS    XL16               V1 output
000013A8  00000000 00000000
000013B0  00000000 00000000                        1141+         DS    FD                    gap
000013B8  40404040 40404040                        1142+         DC    CL8' '             was cross check skipped?
000013C0  00000000 00000000                        1143+         DS    FD                    gap
000013C8  00000000 00000000                        1144+XCO4     DS    XL16               Cross check Output
000013D0  00000000 00000000
000013D8  00000000 00000000                        1145+         DS    FD                    gap
                                                   1146+*
000013E0                                           1147+X4       DS    0F
000013E0  E310 5014 0014                00000014   1148+         LGF   R1,V2ADDR          load v2 source
000013E6  E721 0000 0006                00000000   1149+         VL    v2,0(R1)
000013EC  E310 5018 0014                00000018   1150+         LGF   R1,V3ADDR          load v3 source
000013F2  E731 0000 0006                00000000   1151+         VL    v3,0(R1)

000013F8                                           1154+         DS    0H                      E7B0 VDL
000013F8  E7                                       1155+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
000013F9  12                                       1156+         DC    X'12'                    v1, v2
000013FA  30                                       1157+         DC    X'30'                    v3, reserved
000013FB  0820                                     1158+         DC    HL2'2080'                reserved, m5, m4, RXB
000013FD  B0                                       1159+         DC    X'B0'

000013FE  E710 5030 000E                000013A0   1161+         VST   V1,V1O4            save v1 output
00001404  07FB                                     1162+         BR    R11                return
00001408                                           1163+RE4      DC    0F                 xl16 expected result
00001408                                           1164+         DROP  R5
00001408  FFFFFFFF 7FFFFFFF                        1165          DC    XL16'FFFFFFFF 7FFFFFFF 19999999 11111111'   result
00001410  19999999 11111111
00001418  FFFFFFFF FFFFFFFF                        1166          DC    XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'   v2
00001420  FFFFFFFF FFFFFFFF
00001428  00000001 00000002                        1167          DC    XL16'00000001 00000002 0000000A 0000000F'   v3
00001430  0000000A 0000000F
                                                   1168
                                                   1169          VRR_C VDL,X'2',X'8',N
00001438                                           1170+         DS    0FD
00001438                        00001438           1171+         USING *,R5               base for test data and test routine
00001438  000014A8                                 1172+T5       DC    A(X5)              address of test routine
0000143C  0005                                     1173+         DC    H'5'               test number
0000143E  00                                       1174+         DC    X'00'
0000143F  D5                                       1175+         DC    CL1'N'             S or Y = skip cross check
00001440  02                                       1176+         DC    X'2'                m4
00001441  08                                       1177+         DC    X'8'                m5
00001442  E5C4D340 40404040                        1178+         DC    CL8'VDL'           instruction name
0000144C  000014E0                                 1179+         DC    A(RE5+16)          address of v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001450  000014F0                                 1180+         DC    A(RE5+32)         address of v3 source
00001454  00000010                                 1181+         DC    A(16)             result length
00001458  000014D0                                 1182+REA5     DC    A(RE5)            result address
00001460  00000000 00000000                        1183+         DS    FD                   gap
00001468  00000000 00000000                        1184+V1O5     DS    XL16              V1 output
00001470  00000000 00000000
00001478  00000000 00000000                        1185+         DS    FD                   gap
00001480  40404040 40404040                        1186+         DC    CL8' '            was cross check skipped?
00001488  00000000 00000000                        1187+         DS    FD                   gap
00001490  00000000 00000000                        1188+XCO5     DS    XL16              Cross check Output
00001498  00000000 00000000
000014A0  00000000 00000000                        1189+         DS    FD                   gap
                                                   1190+*
000014A8                                           1191+X5       DS    0F
000014A8  E310 5014 0014                 00000014  1192+         LGF   R1,V2ADDR         load v2 source
000014AE  E721 0000 0006                 00000000  1193+         VL    v2,0(R1)
000014B4  E310 5018 0014                 00000018  1194+         LGF   R1,V3ADDR         load v3 source
000014BA  E731 0000 0006                 00000000  1195+         VL    v3,0(R1)

000014C0                                           1198+         DS    0H                      E7B0 VDL
000014C0  E7                                       1199+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
000014C1  12                                       1200+         DC    X'12'                    v1, v2
000014C2  30                                       1201+         DC    X'30'                    v3, reserved
000014C3  0820                                     1202+         DC    HL2'2080'                reserved, m5, m4, RXB
000014C5  B0                                       1203+         DC    X'B0'

000014C6  E710 5030 000E                 00001468  1205+         VST   V1,V1O5           save v1 output
000014CC  07FB                                     1206+         BR    R11               return
000014D0                                           1207+RE5      DC    0F                xl16 expected result
000014D0                                           1208+         DROP  R5
000014D0  000000FE 00000032                        1209          DC    XL16'000000FE 00000032 0000001C 00000013'   result
000014D8  0000001C 00000013
000014E0  FFFFFFFF FFFFFFFF                        1210          DC    XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'   v2
000014E8  FFFFFFFF FFFFFFFF
000014F0  01020304 05060708                        1211          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000014F8  090A0B0C 0D0E0F10
                                                   1212
                                                   1213 *Doubleword    M4:  3, m5= 8 (IDC=1)
                                                   1214
                                                   1215          VRR_C VDL,X'3',X'8',N
00001500                                           1216+         DS    0FD
00001500                      00001500             1217+         USING *,R5              base for test data and test routine
00001500  00001570                                 1218+T6       DC    A(X6)             address of test routine
00001504  0006                                     1219+         DC    H'6'              test number
00001506  00                                       1220+         DC    X'00'
00001507  D5                                       1221+         DC    CL1'N'            S or Y = skip cross check
00001508  03                                       1222+         DC    X'3'               m4
00001509  08                                       1223+         DC    X'8'               m5
0000150A  E5C4D340 40404040                        1224+         DC    CL8'VDL'          instruction name
00001514  000015A8                                 1225+         DC    A(RE6+16)         address of v2 source
00001518  000015B8                                 1226+         DC    A(RE6+32)         address of v3 source
0000151C  00000010                                 1227+         DC    A(16)             result length
00001520  00001598                                 1228+REA6     DC    A(RE6)            result address
00001528  00000000 00000000                        1229+         DS    FD                   gap
00001530  00000000 00000000                        1230+V1O6     DS    XL16              V1 output
00001538  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001540  00000000 00000000                        1231+         DS    FD                   gap
00001548  40404040 40404040                        1232+         DC    CL8' '             was cross check skipped?
00001550  00000000 00000000                        1233+         DS    FD                   gap
00001558  00000000 00000000                        1234+XCO6     DS    XL16               Cross check Output
00001560  00000000 00000000
00001568  00000000 00000000                        1235+         DS    FD                   gap
                                                   1236+*
00001570                                           1237+X6       DS    0F
00001570  E310 5014 0014                 00000014  1238+         LGF   R1,V2ADDR          load v2 source
00001576  E721 0000 0006                 00000000  1239+         VL    v2,0(R1)
0000157C  E310 5018 0014                 00000018  1240+         LGF   R1,V3ADDR          load v3 source
00001582  E731 0000 0006                 00000000  1241+         VL    v3,0(R1)

00001588                                           1244+         DS    0H                      E7B0 VDL
00001588  E7                                       1245+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001589  12                                       1246+         DC    X'12'                    v1, v2
0000158A  30                                       1247+         DC    X'30'                    v3, reserved
0000158B  0830                                     1248+         DC    HL2'2096'                reserved, m5, m4, RXB
0000158D  B0                                       1249+         DC    X'B0'

0000158E  E710 5030 000E                 00001530  1251+         VST   V1,V1O6            save v1 output
00001594  07FB                                     1252+         BR    R11                return
00001598                                           1253+RE6      DC    0F                 xl16 expected result
00001598                                           1254+         DROP  R5
00001598  00000000 00000000                        1255          DC    XL16'0000000000000000 0000000000000000'   result
000015A0  00000000 00000000
000015A8  00000000 00000000                        1256          DC    XL16'0000000000000000 0000000000000000'   v2
000015B0  00000000 00000000
000015B8  01020304 05060708                        1257          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000015C0  090A0B0C 0D0E0F10
                                                   1258
                                                   1259          VRR_C VDL,X'3',X'8',N
000015C8                                           1260+         DS    0FD
000015C8                         000015C8          1261+         USING *,R5               base for test data and test routine
000015C8  00001638                                 1262+T7       DC    A(X7)              address of test routine
000015CC  0007                                     1263+         DC    H'7'               test number
000015CE  00                                       1264+         DC    X'00'
000015CF  D5                                       1265+         DC    CL1'N'             S or Y = skip cross check
000015D0  03                                       1266+         DC    X'3'                m4
000015D1  08                                       1267+         DC    X'8'                m5
000015D2  E5C4D340 40404040                        1268+         DC    CL8'VDL'           instruction name
000015DC  00001670                                 1269+         DC    A(RE7+16)          address of v2 source
000015E0  00001680                                 1270+         DC    A(RE7+32)          address of v3 source
000015E4  00000010                                 1271+         DC    A(16)              result length
000015E8  00001660                                 1272+REA7     DC    A(RE7)             result address
000015F0  00000000 00000000                        1273+         DS    FD                   gap
000015F8  00000000 00000000                        1274+V1O7     DS    XL16               V1 output
00001600  00000000 00000000
00001608  00000000 00000000                        1275+         DS    FD                   gap
00001610  40404040 40404040                        1276+         DC    CL8' '             was cross check skipped?
00001618  00000000 00000000                        1277+         DS    FD                   gap
00001620  00000000 00000000                        1278+XCO7     DS    XL16               Cross check Output
00001628  00000000 00000000
00001630  00000000 00000000                        1279+         DS    FD                   gap
                                                   1280+*
00001638                                           1281+X7       DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001638  E310 5014 0014                00000014  1282+         LGF   R1,V2ADDR         load v2 source
0000163E  E721 0000 0006                00000000  1283+         VL    v2,0(R1)
00001644  E310 5018 0014                00000018  1284+         LGF   R1,V3ADDR         load v3 source
0000164A  E731 0000 0006                00000000  1285+         VL    v3,0(R1)

00001650                                          1288+         DS    0H                      E7B0 VDL
00001650  E7                                      1289+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001651  12                                      1290+         DC    X'12'                    v1, v2
00001652  30                                      1291+         DC    X'30'                    v3, reserved
00001653  0830                                    1292+         DC    HL2'2096'                reserved, m5, m4, RXB
00001655  B0                                      1293+         DC    X'B0'

00001656  E710 5030 000E                000015F8  1295+         VST   V1,V1O7           save v1 output
0000165C  07FB                                    1296+         BR    R11               return
00001660                                          1297+RE7      DC    0F                xl16 expected result
00001660                                          1298+         DROP  R5
00001660  00000000 00000000                       1299          DC    XL16'0000000000000000 0000000000000000'   result
00001668  00000000 00000000
00001670  01020303 05060707                       1300          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00001678  090A0B0B 0D0E0F0F
00001680  01020304 05060708                       1301          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00001688  090A0B0C 0D0E0F10
                                                  1302
                                                  1303          VRR_C VDL,X'3',X'8',N
00001690                                          1304+         DS    0FD
00001690                        00001690          1305+         USING *,R5              base for test data and test routine
00001690  00001700                                1306+T8       DC    A(X8)             address of test routine
00001694  0008                                    1307+         DC    H'8'              test number
00001696  00                                      1308+         DC    X'00'
00001697  D5                                      1309+         DC    CL1'N'            S or Y = skip cross check
00001698  03                                      1310+         DC    X'3'               m4
00001699  08                                      1311+         DC    X'8'               m5
0000169A  E5C4D340 40404040                       1312+         DC    CL8'VDL'          instruction name
000016A4  00001738                                1313+         DC    A(RE8+16)         address of v2 source
000016A8  00001748                                1314+         DC    A(RE8+32)         address of v3 source
000016AC  00000010                                1315+         DC    A(16)             result length
000016B0  00001728                                1316+REA8     DC    A(RE8)            result address
000016B8  00000000 00000000                       1317+         DS    FD                   gap
000016C0  00000000 00000000                       1318+V1O8     DS    XL16              V1 output
000016C8  00000000 00000000
000016D0  00000000 00000000                       1319+         DS    FD                   gap
000016D8  40404040 40404040                       1320+         DC    CL8' '            was cross check skipped?
000016E0  00000000 00000000                       1321+         DS    FD                   gap
000016E8  00000000 00000000                       1322+XCO8     DS    XL16              Cross check Output
000016F0  00000000 00000000
000016F8  00000000 00000000                       1323+         DS    FD                   gap
                                                  1324+*
00001700                                          1325+X8       DS    0F
00001700  E310 5014 0014                00000014  1326+         LGF   R1,V2ADDR         load v2 source
00001706  E721 0000 0006                00000000  1327+         VL    v2,0(R1)
0000170C  E310 5018 0014                00000018  1328+         LGF   R1,V3ADDR         load v3 source
00001712  E731 0000 0006                00000000  1329+         VL    v3,0(R1)

00001718                                          1332+         DS    0H                      E7B0 VDL
00001718  E7                                      1333+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001719  12                                      1334+         DC    X'12'                    v1, v2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000171A  30                                      1335+         DC    X'30'                    v3, reserved
0000171B  0830                                    1336+         DC    HL2'2096'                reserved, m5, m4, RXB
0000171D  B0                                      1337+         DC    X'B0'

0000171E  E710 5030 000E                000016C0  1339+         VST   V1,V1O8           save v1 output
00001724  07FB                                    1340+         BR    R11               return
00001728                                          1341+RE8      DC    0F                xl16 expected result
00001728                                          1342+         DROP  R5
00001728  00000000 00000001                       1343          DC    XL16'0000000000000001 0000000000000001'   result
00001730  00000000 00000001
00001738  01020304 05060708                       1344          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
00001740  090A0B0C 0D0E0F10
00001748  01020304 05060708                       1345          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00001750  090A0B0C 0D0E0F10
                                                  1346
                                                  1347          VRR_C VDL,X'3',X'8',N
00001758                                          1348+         DS    0FD
00001758                      00001758            1349+         USING *,R5              base for test data and test routine
00001758  000017C8                                1350+T9       DC    A(X9)             address of test routine
0000175C  0009                                    1351+         DC    H'9'              test number
0000175E  00                                      1352+         DC    X'00'
0000175F  D5                                      1353+         DC    CL1'N'            S or Y = skip cross check
00001760  03                                      1354+         DC    X'3'               m4
00001761  08                                      1355+         DC    X'8'               m5
00001762  E5C4D340 40404040                       1356+         DC    CL8'VDL'          instruction name
0000176C  00001800                                1357+         DC    A(RE9+16)         address of v2 source
00001770  00001810                                1358+         DC    A(RE9+32)         address of v3 source
00001774  00000010                                1359+         DC    A(16)             result length
00001778  000017F0                                1360+REA9     DC    A(RE9)            result address
00001780  00000000 00000000                       1361+         DS    FD                   gap
00001788  00000000 00000000                       1362+V1O9     DS    XL16              V1 output
00001790  00000000 00000000
00001798  00000000 00000000                       1363+         DS    FD                   gap
000017A0  40404040 40404040                       1364+         DC    CL8' '            was cross check skipped?
000017A8  00000000 00000000                       1365+         DS    FD                   gap
000017B0  00000000 00000000                       1366+XCO9     DS    XL16              Cross check Output
000017B8  00000000 00000000
000017C0  00000000 00000000                       1367+         DS    FD                   gap
                                                  1368+*
000017C8                                          1369+X9       DS    0F
000017C8  E310 5014 0014                00000014  1370+         LGF   R1,V2ADDR         load v2 source
000017CE  E721 0000 0006                00000000  1371+         VL    v2,0(R1)
000017D4  E310 5018 0014                00000018  1372+         LGF   R1,V3ADDR         load v3 source
000017DA  E731 0000 0006                00000000  1373+         VL    v3,0(R1)

000017E0                                          1376+         DS    0H                     E7B0 VDL
000017E0  E7                                      1377+         DC    X'E7'                  - VECTOR DIVIDE LOGICAL
000017E1  12                                      1378+         DC    X'12'                   v1, v2
000017E2  30                                      1379+         DC    X'30'                   v3, reserved
000017E3  0830                                    1380+         DC    HL2'2096'               reserved, m5, m4, RXB
000017E5  B0                                      1381+         DC    X'B0'

000017E6  E710 5030 000E                00001788  1383+         VST   V1,V1O9           save v1 output
000017EC  07FB                                    1384+         BR    R11               return
000017F0                                          1385+RE9      DC    0F                xl16 expected result
000017F0                                          1386+         DROP  R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000017F0  FFFFFFFF FFFFFFFF                         1387          DC    XL16'FFFFFFFFFFFFFFFF 7FFFFFFFFFFFFFFF'   result
000017F8  7FFFFFFF FFFFFFFF
00001800  FFFFFFFF FFFFFFFF                         1388          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00001808  FFFFFFFF FFFFFFFF
00001810  00000000 00000001                         1389          DC    XL16'0000000000000001 0000000000000002'   v3
00001818  00000000 00000002
                                                    1390
                                                    1391          VRR_C VDL,X'3',X'8',N
00001820                                            1392+         DS    0FD
00001820                      00001820              1393+         USING *,R5              base for test data and test routine
00001820  00001890                                  1394+T10      DC    A(X10)            address of test routine
00001824  000A                                      1395+         DC    H'10'             test number
00001826  00                                        1396+         DC    X'00'
00001827  D5                                        1397+         DC    CL1'N'            S or Y = skip cross check
00001828  03                                        1398+         DC    X'3'               m4
00001829  08                                        1399+         DC    X'8'               m5
0000182A  E5C4D340 40404040                         1400+         DC    CL8'VDL'          instruction name
00001834  000018C8                                  1401+         DC    A(RE10+16)        address of v2 source
00001838  000018D8                                  1402+         DC    A(RE10+32)        address of v3 source
0000183C  00000010                                  1403+         DC    A(16)             result length
00001840  000018B8                                  1404+REA10    DC    A(RE10)           result address
00001848  00000000 00000000                         1405+         DS    FD                   gap
00001850  00000000 00000000                         1406+V1O10    DS    XL16              V1 output
00001858  00000000 00000000
00001860  00000000 00000000                         1407+         DS    FD                   gap
00001868  40404040 40404040                         1408+         DC    CL8' '            was cross check skipped?
00001870  00000000 00000000                         1409+         DS    FD                   gap
00001878  00000000 00000000                         1410+XCO10    DS    XL16              Cross check Output
00001880  00000000 00000000
00001888  00000000 00000000                         1411+         DS    FD                   gap
                                                    1412+*
00001890                                            1413+X10      DS    0F
00001890  E310 5014 0014                 00000014   1414+         LGF   R1,V2ADDR         load v2 source
00001896  E721 0000 0006                 00000000   1415+         VL    v2,0(R1)
0000189C  E310 5018 0014                 00000018   1416+         LGF   R1,V3ADDR         load v3 source
000018A2  E731 0000 0006                 00000000   1417+         VL    v3,0(R1)

000018A8                                            1420+         DS    0H                      E7B0 VDL
000018A8  E7                                        1421+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
000018A9  12                                        1422+         DC    X'12'                    v1, v2
000018AA  30                                        1423+         DC    X'30'                    v3, reserved
000018AB  0830                                      1424+         DC    HL2'2096'                reserved, m5, m4, RXB
000018AD  B0                                        1425+         DC    X'B0'

000018AE  E710 5030 000E                 00001850   1427+         VST   V1,V1O10          save v1 output
000018B4  07FB                                      1428+         BR    R11               return
000018B8                                            1429+RE10     DC    0F                xl16 expected result
000018B8                                            1430+         DROP  R5
000018B8  19999999 99999999                         1431          DC    XL16'1999999999999999 1111111111111111'   result
000018C0  11111111 11111111
000018C8  FFFFFFFF FFFFFFFF                         1432          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
000018D0  FFFFFFFF FFFFFFFF
000018D8  00000000 0000000A                         1433          DC    XL16'000000000000000A 000000000000000F'   v3
000018E0  00000000 0000000F
                                                    1434
                                                    1435          VRR_C VDL,X'3',X'8',N
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000018E8                                           1436+         DS    0FD
000018E8                     000018E8              1437+         USING *,R5              base for test data and test routine
000018E8  00001958                                 1438+T11      DC    A(X11)            address of test routine
000018EC  000B                                     1439+         DC    H'11'             test number
000018EE  00                                       1440+         DC    X'00'
000018EF  D5                                       1441+         DC    CL1'N'            S or Y = skip cross check
000018F0  03                                       1442+         DC    X'3'               m4
000018F1  08                                       1443+         DC    X'8'               m5
000018F2  E5C4D340 40404040                        1444+         DC    CL8'VDL'          instruction name
000018FC  00001990                                 1445+         DC    A(RE11+16)        address of v2 source
00001900  000019A0                                 1446+         DC    A(RE11+32)        address of v3 source
00001904  00000010                                 1447+         DC    A(16)             result length
00001908  00001980                                 1448+REA11    DC    A(RE11)           result address
00001910  00000000 00000000                        1449+         DS    FD                   gap
00001918  00000000 00000000                        1450+V1O11    DS    XL16              V1 output
00001920  00000000 00000000
00001928  00000000 00000000                        1451+         DS    FD                   gap
00001930  40404040 40404040                        1452+         DC    CL8' '            was cross check skipped?
00001938  00000000 00000000                        1453+         DS    FD                   gap
00001940  00000000 00000000                        1454+XCO11    DS    XL16              Cross check Output
00001948  00000000 00000000
00001950  00000000 00000000                        1455+         DS    FD                   gap
                                                   1456+*
00001958                                           1457+X11      DS    0F
00001958  E310 5014 0014                00000014   1458+         LGF   R1,V2ADDR         load v2 source
0000195E  E721 0000 0006                00000000   1459+         VL    v2,0(R1)
00001964  E310 5018 0014                00000018   1460+         LGF   R1,V3ADDR         load v3 source
0000196A  E731 0000 0006                00000000   1461+         VL    v3,0(R1)

00001970                                           1464+         DS    0H                      E7B0 VDL
00001970  E7                                       1465+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001971  12                                       1466+         DC    X'12'                    v1, v2
00001972  30                                       1467+         DC    X'30'                    v3, reserved
00001973  0830                                     1468+         DC    HL2'2096'                reserved, m5, m4, RXB
00001975  B0                                       1469+         DC    X'B0'

00001976  E710 5030 000E                00001918   1471+         VST   V1,V1O11          save v1 output
0000197C  07FB                                     1472+         BR    R11               return
00001980                                           1473+RE11     DC    0F                xl16 expected result
00001980                                           1474+         DROP  R5
00001980  00000000 0000007F                        1475          DC    XL16'000000000000007F 000000000000000E'   result
00001988  00000000 0000000E
00001990  7FFFFFFF FFFFFFFF                        1476          DC    XL16'7FFFFFFFFFFFFFFF 7FFFFFFFFFFFFFFF'   v2
00001998  7FFFFFFF FFFFFFFF
000019A0  01020304 05060708                        1477          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000019A8  090A0B0C 0D0E0F10
                                                   1478
                                                   1479 *Quadword      M4:  4, m5= 8 (IDC=1)
                                                   1480
                                                   1481          VRR_C VDL,X'4',X'8',N
000019B0                                           1482+         DS    0FD
000019B0                     000019B0              1483+         USING *,R5              base for test data and test routine
000019B0  00001A20                                 1484+T12      DC    A(X12)            address of test routine
000019B4  000C                                     1485+         DC    H'12'             test number
000019B6  00                                       1486+         DC    X'00'
000019B7  D5                                       1487+         DC    CL1'N'            S or Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000019B8  04                                       1488+         DC    X'4'                m4
000019B9  08                                       1489+         DC    X'8'                m5
000019BA  E5C4D340 40404040                        1490+         DC    CL8'VDL'           instruction name
000019C4  00001A58                                 1491+         DC    A(RE12+16)         address of v2 source
000019C8  00001A68                                 1492+         DC    A(RE12+32)         address of v3 source
000019CC  00000010                                 1493+         DC    A(16)              result length
000019D0  00001A48                                 1494+REA12    DC    A(RE12)            result address
000019D8  00000000 00000000                        1495+         DS    FD                    gap
000019E0  00000000 00000000                        1496+V1O12    DS    XL16               V1 output
000019E8  00000000 00000000
000019F0  00000000 00000000                        1497+         DS    FD                    gap
000019F8  40404040 40404040                        1498+         DC    CL8' '             was cross check skipped?
00001A00  00000000 00000000                        1499+         DS    FD                    gap
00001A08  00000000 00000000                        1500+XCO12    DS    XL16               Cross check Output
00001A10  00000000 00000000
00001A18  00000000 00000000                        1501+         DS    FD                    gap
                                                   1502+*
00001A20                                           1503+X12      DS    0F
00001A20  E310 5014 0014                00000014   1504+         LGF   R1,V2ADDR          load v2 source
00001A26  E721 0000 0006                00000000   1505+         VL    v2,0(R1)
00001A2C  E310 5018 0014                00000018   1506+         LGF   R1,V3ADDR          load v3 source
00001A32  E731 0000 0006                00000000   1507+         VL    v3,0(R1)

00001A38                                           1510+         DS    0H                      E7B0 VDL
00001A38  E7                                       1511+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001A39  12                                       1512+         DC    X'12'                    v1, v2
00001A3A  30                                       1513+         DC    X'30'                    v3, reserved
00001A3B  0840                                     1514+         DC    HL2'2112'                reserved, m5, m4, RXB
00001A3D  B0                                       1515+         DC    X'B0'

00001A3E  E710 5030 000E                000019E0   1517+         VST   V1,V1O12           save v1 output
00001A44  07FB                                     1518+         BR    R11                return
00001A48                                           1519+RE12     DC    0F                 xl16 expected result
00001A48                                           1520+         DROP  R5
00001A48  00000000 00000000                        1521          DC    XL16'0000000000000000 0000000000000000'   result
00001A50  00000000 00000000
00001A58  00000000 00000000                        1522          DC    XL16'0000000000000000 0000000000000000'   v2
00001A60  00000000 00000000
00001A68  01020304 05060708                        1523          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00001A70  090A0B0C 0D0E0F10
                                                   1524
                                                   1525          VRR_C VDL,X'4',X'8',N
00001A78                                           1526+         DS    0FD
00001A78                        00001A78           1527+         USING *,R5               base for test data and test routine
00001A78  00001AE8                                 1528+T13      DC    A(X13)             address of test routine
00001A7C  000D                                     1529+         DC    H'13'              test number
00001A7E  00                                       1530+         DC    X'00'
00001A7F  D5                                       1531+         DC    CL1'N'             S or Y = skip cross check
00001A80  04                                       1532+         DC    X'4'                m4
00001A81  08                                       1533+         DC    X'8'                m5
00001A82  E5C4D340 40404040                        1534+         DC    CL8'VDL'           instruction name
00001A8C  00001B20                                 1535+         DC    A(RE13+16)         address of v2 source
00001A90  00001B30                                 1536+         DC    A(RE13+32)         address of v3 source
00001A94  00000010                                 1537+         DC    A(16)              result length
00001A98  00001B10                                 1538+REA13    DC    A(RE13)            result address
00001AA0  00000000 00000000                        1539+         DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001AA8  00000000 00000000                        1540+V1O13    DS    XL16              V1 output
00001AB0  00000000 00000000
00001AB8  00000000 00000000                        1541+         DS    FD                   gap
00001AC0  40404040 40404040                        1542+         DC    CL8' '            was cross check skipped?
00001AC8  00000000 00000000                        1543+         DS    FD                   gap
00001AD0  00000000 00000000                        1544+XCO13    DS    XL16              Cross check Output
00001AD8  00000000 00000000
00001AE0  00000000 00000000                        1545+         DS    FD                   gap
                                                   1546+*
00001AE8                                           1547+X13      DS    0F
00001AE8  E310 5014 0014                 00000014  1548+         LGF   R1,V2ADDR         load v2 source
00001AEE  E721 0000 0006                 00000000  1549+         VL    v2,0(R1)
00001AF4  E310 5018 0014                 00000018  1550+         LGF   R1,V3ADDR         load v3 source
00001AFA  E731 0000 0006                 00000000  1551+         VL    v3,0(R1)

00001B00                                           1554+         DS    0H                      E7B0 VDL
00001B00  E7                                       1555+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001B01  12                                       1556+         DC    X'12'                    v1, v2
00001B02  30                                       1557+         DC    X'30'                    v3, reserved
00001B03  0840                                     1558+         DC    HL2'2112'                reserved, m5, m4, RXB
00001B05  B0                                       1559+         DC    X'B0'

00001B06  E710 5030 000E                 00001AA8  1561+         VST   V1,V1O13          save v1 output
00001B0C  07FB                                     1562+         BR    R11               return
00001B10                                           1563+RE13     DC    0F                xl16 expected result
00001B10                                           1564+         DROP  R5
00001B10  00000000 00000000                        1565          DC    XL16'0000000000000000 0000000000000000'   result
00001B18  00000000 00000000
00001B20  01020303 05060707                        1566          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00001B28  090A0B0B 0D0E0F0F
00001B30  01020304 05060708                        1567          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00001B38  090A0B0C 0D0E0F10
                                                   1568
                                                   1569          VRR_C VDL,X'4',X'8',N
00001B40                                           1570+         DS    0FD
00001B40                         00001B40          1571+         USING *,R5              base for test data and test routine
00001B40  00001BB0                                 1572+T14      DC    A(X14)            address of test routine
00001B44  000E                                     1573+         DC    H'14'             test number
00001B46  00                                       1574+         DC    X'00'
00001B47  D5                                       1575+         DC    CL1'N'            S or Y = skip cross check
00001B48  04                                       1576+         DC    X'4'               m4
00001B49  08                                       1577+         DC    X'8'               m5
00001B4A  E5C4D340 40404040                        1578+         DC    CL8'VDL'          instruction name
00001B54  00001BE8                                 1579+         DC    A(RE14+16)        address of v2 source
00001B58  00001BF8                                 1580+         DC    A(RE14+32)        address of v3 source
00001B5C  00000010                                 1581+         DC    A(16)             result length
00001B60  00001BD8                                 1582+REA14    DC    A(RE14)           result address
00001B68  00000000 00000000                        1583+         DS    FD                   gap
00001B70  00000000 00000000                        1584+V1O14    DS    XL16              V1 output
00001B78  00000000 00000000
00001B80  00000000 00000000                        1585+         DS    FD                   gap
00001B88  40404040 40404040                        1586+         DC    CL8' '            was cross check skipped?
00001B90  00000000 00000000                        1587+         DS    FD                   gap
00001B98  00000000 00000000                        1588+XCO14    DS    XL16              Cross check Output
00001BA0  00000000 00000000
00001BA8  00000000 00000000                        1589+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1590+*
00001BB0                                           1591+X14      DS    0F
00001BB0  E310 5014 0014                 00000014  1592+         LGF   R1,V2ADDR         load v2 source
00001BB6  E721 0000 0006                 00000000  1593+         VL    v2,0(R1)
00001BBC  E310 5018 0014                 00000018  1594+         LGF   R1,V3ADDR         load v3 source
00001BC2  E731 0000 0006                 00000000  1595+         VL    v3,0(R1)

00001BC8                                           1598+         DS    0H                      E7B0 VDL
00001BC8  E7                                       1599+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001BC9  12                                       1600+         DC    X'12'                    v1, v2
00001BCA  30                                       1601+         DC    X'30'                    v3, reserved
00001BCB  0840                                     1602+         DC    HL2'2112'                reserved, m5, m4, RXB
00001BCD  B0                                       1603+         DC    X'B0'

00001BCE  E710 5030 000E                 00001B70  1605+         VST   V1,V1O14          save v1 output
00001BD4  07FB                                     1606+         BR    R11               return
00001BD8                                           1607+RE14     DC    0F                xl16 expected result
00001BD8                                           1608+         DROP  R5
00001BD8  00000000 00000000                        1609          DC    XL16'0000000000000000 0000000000000001'   result
00001BE0  00000000 00000001
00001BE8  01020304 05060708                        1610          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
00001BF0  090A0B0C 0D0E0F10
00001BF8  01020304 05060708                        1611          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00001C00  090A0B0C 0D0E0F10
                                                   1612
                                                   1613          VRR_C VDL,X'4',X'8',N
00001C08                                           1614+         DS    0FD
00001C08                      00001C08             1615+         USING *,R5              base for test data and test routine
00001C08  00001C78                                 1616+T15      DC    A(X15)            address of test routine
00001C0C  000F                                     1617+         DC    H'15'             test number
00001C0E  00                                       1618+         DC    X'00'
00001C0F  D5                                       1619+         DC    CL1'N'            S or Y = skip cross check
00001C10  04                                       1620+         DC    X'4'               m4
00001C11  08                                       1621+         DC    X'8'               m5
00001C12  E5C4D340 40404040                        1622+         DC    CL8'VDL'          instruction name
00001C1C  00001CB0                                 1623+         DC    A(RE15+16)        address of v2 source
00001C20  00001CC0                                 1624+         DC    A(RE15+32)        address of v3 source
00001C24  00000010                                 1625+         DC    A(16)             result length
00001C28  00001CA0                                 1626+REA15    DC    A(RE15)           result address
00001C30  00000000 00000000                        1627+         DS    FD                   gap
00001C38  00000000 00000000                        1628+V1O15    DS    XL16              V1 output
00001C40  00000000 00000000
00001C48  00000000 00000000                        1629+         DS    FD                   gap
00001C50  40404040 40404040                        1630+         DC    CL8' '            was cross check skipped?
00001C58  00000000 00000000                        1631+         DS    FD                   gap
00001C60  00000000 00000000                        1632+XCO15    DS    XL16              Cross check Output
00001C68  00000000 00000000
00001C70  00000000 00000000                        1633+         DS    FD                   gap
                                                   1634+*
00001C78                                           1635+X15      DS    0F
00001C78  E310 5014 0014                 00000014  1636+         LGF   R1,V2ADDR         load v2 source
00001C7E  E721 0000 0006                 00000000  1637+         VL    v2,0(R1)
00001C84  E310 5018 0014                 00000018  1638+         LGF   R1,V3ADDR         load v3 source
00001C8A  E731 0000 0006                 00000000  1639+         VL    v3,0(R1)

00001C90                                           1642+         DS    0H                      E7B0 VDL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001C90  E7                                      1643+         DC    X'E7'                    - VECTOR DIVIDE LOGICAL
00001C91  12                                      1644+         DC    X'12'                     v1, v2
00001C92  30                                      1645+         DC    X'30'                     v3, reserved
00001C93  0840                                    1646+         DC    HL2'2112'                 reserved, m5, m4, RXB
00001C95  B0                                      1647+         DC    X'B0'

00001C96  E710 5030 000E                00001C38  1649+         VST   V1,V1O15          save v1 output
00001C9C  07FB                                    1650+         BR    R11               return
00001CA0                                          1651+RE15     DC    0F                xl16 expected result
00001CA0                                          1652+         DROP  R5
00001CA0  FFFFFFFF FFFFFFFF                       1653          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   result
00001CA8  FFFFFFFF FFFFFFFF
00001CB0  FFFFFFFF FFFFFFFF                       1654          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00001CB8  FFFFFFFF FFFFFFFF
00001CC0  00000000 00000000                       1655          DC    XL16'0000000000000000 0000000000000001'   v3
00001CC8  00000000 00000001
                                                  1656
                                                  1657          VRR_C VDL,X'4',X'8',N
00001CD0                                          1658+         DS    0FD
00001CD0                        00001CD0          1659+         USING *,R5              base for test data and test routine
00001CD0  00001D40                                1660+T16      DC    A(X16)            address of test routine
00001CD4  0010                                    1661+         DC    H'16'             test number
00001CD6  00                                      1662+         DC    X'00'
00001CD7  D5                                      1663+         DC    CL1'N'            S or Y = skip cross check
00001CD8  04                                      1664+         DC    X'4'               m4
00001CD9  08                                      1665+         DC    X'8'               m5
00001CDA  E5C4D340 40404040                       1666+         DC    CL8'VDL'          instruction name
00001CE4  00001D78                                1667+         DC    A(RE16+16)        address of v2 source
00001CE8  00001D88                                1668+         DC    A(RE16+32)        address of v3 source
00001CEC  00000010                                1669+         DC    A(16)             result length
00001CF0  00001D68                                1670+REA16    DC    A(RE16)           result address
00001CF8  00000000 00000000                       1671+         DS    FD                   gap
00001D00  00000000 00000000                       1672+V1O16    DS    XL16              V1 output
00001D08  00000000 00000000
00001D10  00000000 00000000                       1673+         DS    FD                   gap
00001D18  40404040 40404040                       1674+         DC    CL8' '            was cross check skipped?
00001D20  00000000 00000000                       1675+         DS    FD                   gap
00001D28  00000000 00000000                       1676+XCO16    DS    XL16              Cross check Output
00001D30  00000000 00000000
00001D38  00000000 00000000                       1677+         DS    FD                   gap
                                                  1678+*
00001D40                                          1679+X16      DS    0F
00001D40  E310 5014 0014                00000014  1680+         LGF   R1,V2ADDR         load v2 source
00001D46  E721 0000 0006                00000000  1681+         VL    v2,0(R1)
00001D4C  E310 5018 0014                00000018  1682+         LGF   R1,V3ADDR         load v3 source
00001D52  E731 0000 0006                00000000  1683+         VL    v3,0(R1)

00001D58                                          1686+         DS    0H                       E7B0 VDL
00001D58  E7                                      1687+         DC    X'E7'                    - VECTOR DIVIDE LOGICAL
00001D59  12                                      1688+         DC    X'12'                     v1, v2
00001D5A  30                                      1689+         DC    X'30'                     v3, reserved
00001D5B  0840                                    1690+         DC    HL2'2112'                 reserved, m5, m4, RXB
00001D5D  B0                                      1691+         DC    X'B0'

00001D5E  E710 5030 000E                00001D00  1693+         VST   V1,V1O16          save v1 output
00001D64  07FB                                    1694+         BR    R11               return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001D68                                           1695+RE16     DC    0F                xl16 expected result
00001D68                                           1696+         DROP  R5
00001D68  7FFFFFFF FFFFFFFF                        1697          DC    XL16'7FFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   result
00001D70  FFFFFFFF FFFFFFFF
00001D78  FFFFFFFF FFFFFFFF                        1698          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00001D80  FFFFFFFF FFFFFFFF
00001D88  00000000 00000000                        1699          DC    XL16'0000000000000000 0000000000000002'   v3
00001D90  00000000 00000002
                                                   1700
                                                   1701          VRR_C VDL,X'4',X'8',N
00001D98                                           1702+         DS    0FD
00001D98                      00001D98             1703+         USING *,R5              base for test data and test routine
00001D98  00001E08                                 1704+T17      DC    A(X17)            address of test routine
00001D9C  0011                                     1705+         DC    H'17'             test number
00001D9E  00                                       1706+         DC    X'00'
00001D9F  D5                                       1707+         DC    CL1'N'            S or Y = skip cross check
00001DA0  04                                       1708+         DC    X'4'               m4
00001DA1  08                                       1709+         DC    X'8'               m5
00001DA2  E5C4D340 40404040                        1710+         DC    CL8'VDL'          instruction name
00001DAC  00001E40                                 1711+         DC    A(RE17+16)        address of v2 source
00001DB0  00001E50                                 1712+         DC    A(RE17+32)        address of v3 source
00001DB4  00000010                                 1713+         DC    A(16)             result length
00001DB8  00001E30                                 1714+REA17    DC    A(RE17)           result address
00001DC0  00000000 00000000                        1715+         DS    FD                   gap
00001DC8  00000000 00000000                        1716+V1O17    DS    XL16              V1 output
00001DD0  00000000 00000000
00001DD8  00000000 00000000                        1717+         DS    FD                   gap
00001DE0  40404040 40404040                        1718+         DC    CL8' '            was cross check skipped?
00001DE8  00000000 00000000                        1719+         DS    FD                   gap
00001DF0  00000000 00000000                        1720+XCO17    DS    XL16              Cross check Output
00001DF8  00000000 00000000
00001E00  00000000 00000000                        1721+         DS    FD                   gap
                                                   1722+*
00001E08                                           1723+X17      DS    0F
00001E08  E310 5014 0014                00000014   1724+         LGF   R1,V2ADDR         load v2 source
00001E0E  E721 0000 0006                00000000   1725+         VL    v2,0(R1)
00001E14  E310 5018 0014                00000018   1726+         LGF   R1,V3ADDR         load v3 source
00001E1A  E731 0000 0006                00000000   1727+         VL    v3,0(R1)

00001E20                                           1730+         DS    0H                      E7B0 VDL
00001E20  E7                                       1731+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001E21  12                                       1732+         DC    X'12'                    v1, v2
00001E22  30                                       1733+         DC    X'30'                    v3, reserved
00001E23  0840                                     1734+         DC    HL2'2112'                reserved, m5, m4, RXB
00001E25  B0                                       1735+         DC    X'B0'

00001E26  E710 5030 000E                00001DC8   1737+         VST   V1,V1O17          save v1 output
00001E2C  07FB                                     1738+         BR    R11               return
00001E30                                           1739+RE17     DC    0F                xl16 expected result
00001E30                                           1740+         DROP  R5
00001E30  19999999 99999999                        1741          DC    XL16'1999999999999999 9999999999999999'   result
00001E38  99999999 99999999
00001E40  FFFFFFFF FFFFFFFF                        1742          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00001E48  FFFFFFFF FFFFFFFF
00001E50  00000000 00000000                        1743          DC    XL16'0000000000000000 000000000000000A'   v3
00001E58  00000000 0000000A
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1744
                                                   1745          VRR_C VDL,X'4',X'8',N
00001E60                                           1746+         DS    0FD
00001E60                     00001E60              1747+         USING *,R5              base for test data and test routine
00001E60  00001ED0                                 1748+T18      DC    A(X18)            address of test routine
00001E64  0012                                     1749+         DC    H'18'             test number
00001E66  00                                       1750+         DC    X'00'
00001E67  D5                                       1751+         DC    CL1'N'            S or Y = skip cross check
00001E68  04                                       1752+         DC    X'4'               m4
00001E69  08                                       1753+         DC    X'8'               m5
00001E6A  E5C4D340 40404040                        1754+         DC    CL8'VDL'          instruction name
00001E74  00001F08                                 1755+         DC    A(RE18+16)        address of v2 source
00001E78  00001F18                                 1756+         DC    A(RE18+32)        address of v3 source
00001E7C  00000010                                 1757+         DC    A(16)             result length
00001E80  00001EF8                                 1758+REA18    DC    A(RE18)           result address
00001E88  00000000 00000000                        1759+         DS    FD                   gap
00001E90  00000000 00000000                        1760+V1O18    DS    XL16              V1 output
00001E98  00000000 00000000
00001EA0  00000000 00000000                        1761+         DS    FD                   gap
00001EA8  40404040 40404040                        1762+         DC    CL8' '            was cross check skipped?
00001EB0  00000000 00000000                        1763+         DS    FD                   gap
00001EB8  00000000 00000000                        1764+XCO18    DS    XL16              Cross check Output
00001EC0  00000000 00000000
00001EC8  00000000 00000000                        1765+         DS    FD                   gap
                                                   1766+*
00001ED0                                           1767+X18      DS    0F
00001ED0  E310 5014 0014                 00000014  1768+         LGF   R1,V2ADDR         load v2 source
00001ED6  E721 0000 0006                 00000000  1769+         VL    v2,0(R1)
00001EDC  E310 5018 0014                 00000018  1770+         LGF   R1,V3ADDR         load v3 source
00001EE2  E731 0000 0006                 00000000  1771+         VL    v3,0(R1)

00001EE8                                           1774+         DS    0H                      E7B0 VDL
00001EE8  E7                                       1775+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001EE9  12                                       1776+         DC    X'12'                    v1, v2
00001EEA  30                                       1777+         DC    X'30'                    v3, reserved
00001EEB  0840                                     1778+         DC    HL2'2112'                reserved, m5, m4, RXB
00001EED  B0                                       1779+         DC    X'B0'

00001EEE  E710 5030 000E                 00001E90  1781+         VST   V1,V1O18          save v1 output
00001EF4  07FB                                     1782+         BR    R11               return
00001EF8                                           1783+RE18     DC    0F                xl16 expected result
00001EF8                                           1784+         DROP  R5
00001EF8  11111111 11111111                        1785          DC    XL16'1111111111111111 1111111111111111'   result
00001F00  11111111 11111111
00001F08  FFFFFFFF FFFFFFFF                        1786          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00001F10  FFFFFFFF FFFFFFFF
00001F18  00000000 00000000                        1787          DC    XL16'0000000000000000 000000000000000F'   v3
00001F20  00000000 0000000F
                                                   1788
                                                   1789          VRR_C VDL,X'4',X'8',N
00001F28                                           1790+         DS    0FD
00001F28                     00001F28              1791+         USING *,R5              base for test data and test routine
00001F28  00001F98                                 1792+T19      DC    A(X19)            address of test routine
00001F2C  0013                                     1793+         DC    H'19'             test number
00001F2E  00                                       1794+         DC    X'00'
00001F2F  D5                                       1795+         DC    CL1'N'            S or Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001F30  04                                       1796+         DC    X'4'                m4
00001F31  08                                       1797+         DC    X'8'                m5
00001F32  E5C4D340 40404040                        1798+         DC    CL8'VDL'           instruction name
00001F3C  00001FD0                                 1799+         DC    A(RE19+16)         address of v2 source
00001F40  00001FE0                                 1800+         DC    A(RE19+32)         address of v3 source
00001F44  00000010                                 1801+         DC    A(16)              result length
00001F48  00001FC0                                 1802+REA19    DC    A(RE19)            result address
00001F50  00000000 00000000                        1803+         DS    FD                    gap
00001F58  00000000 00000000                        1804+V1O19    DS    XL16               V1 output
00001F60  00000000 00000000
00001F68  00000000 00000000                        1805+         DS    FD                    gap
00001F70  40404040 40404040                        1806+         DC    CL8' '             was cross check skipped?
00001F78  00000000 00000000                        1807+         DS    FD                    gap
00001F80  00000000 00000000                        1808+XCO19    DS    XL16               Cross check Output
00001F88  00000000 00000000
00001F90  00000000 00000000                        1809+         DS    FD                    gap
                                                   1810+*
00001F98                                           1811+X19      DS    0F
00001F98  E310 5014 0014                00000014   1812+         LGF   R1,V2ADDR          load v2 source
00001F9E  E721 0000 0006                00000000   1813+         VL    v2,0(R1)
00001FA4  E310 5018 0014                00000018   1814+         LGF   R1,V3ADDR          load v3 source
00001FAA  E731 0000 0006                00000000   1815+         VL    v3,0(R1)

00001FB0                                           1818+         DS    0H                      E7B0 VDL
00001FB0  E7                                       1819+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00001FB1  12                                       1820+         DC    X'12'                    v1, v2
00001FB2  30                                       1821+         DC    X'30'                   v3, reserved
00001FB3  0840                                     1822+         DC    HL2'2112'               reserved, m5, m4, RXB
00001FB5  B0                                       1823+         DC    X'B0'

00001FB6  E710 5030 000E                00001F58   1825+         VST   V1,V1O19           save v1 output
00001FBC  07FB                                     1826+         BR    R11                return
00001FC0                                           1827+RE19     DC    0F                 xl16 expected result
00001FC0                                           1828+         DROP  R5
00001FC0  00000000 00000000                        1829          DC    XL16'0000000000000000 000000000000007F'   result
00001FC8  00000000 0000007F
00001FD0  7FFFFFFF FFFFFFFF                        1830          DC    XL16'7FFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00001FD8  FFFFFFFF FFFFFFFF
00001FE0  01020304 05060708                        1831          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00001FE8  090A0B0C 0D0E0F10
                                                   1832
                                                   1833          VRR_C VDL,X'4',X'8',N
00001FF0                                           1834+         DS    0FD
00001FF0                        00001FF0           1835+         USING *,R5               base for test data and test routine
00001FF0  00002060                                 1836+T20      DC    A(X20)             address of test routine
00001FF4  0014                                     1837+         DC    H'20'              test number
00001FF6  00                                       1838+         DC    X'00'
00001FF7  D5                                       1839+         DC    CL1'N'             S or Y = skip cross check
00001FF8  04                                       1840+         DC    X'4'                m4
00001FF9  08                                       1841+         DC    X'8'                m5
00001FFA  E5C4D340 40404040                        1842+         DC    CL8'VDL'           instruction name
00002004  00002098                                 1843+         DC    A(RE20+16)         address of v2 source
00002008  000020A8                                 1844+         DC    A(RE20+32)         address of v3 source
0000200C  00000010                                 1845+         DC    A(16)              result length
00002010  00002088                                 1846+REA20    DC    A(RE20)            result address
00002018  00000000 00000000                        1847+         DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002020  00000000 00000000                        1848+V1O20    DS    XL16              V1 output
00002028  00000000 00000000
00002030  00000000 00000000                        1849+         DS    FD                   gap
00002038  40404040 40404040                        1850+         DC    CL8' '            was cross check skipped?
00002040  00000000 00000000                        1851+         DS    FD                   gap
00002048  00000000 00000000                        1852+XCO20    DS    XL16              Cross check Output
00002050  00000000 00000000
00002058  00000000 00000000                        1853+         DS    FD                   gap
                                                   1854+*
00002060                                           1855+X20      DS    0F
00002060  E310 5014 0014                 00000014  1856+         LGF   R1,V2ADDR         load v2 source
00002066  E721 0000 0006                 00000000  1857+         VL    v2,0(R1)
0000206C  E310 5018 0014                 00000018  1858+         LGF   R1,V3ADDR         load v3 source
00002072  E731 0000 0006                 00000000  1859+         VL    v3,0(R1)

00002078                                           1862+         DS    0H                      E7B0 VDL
00002078  E7                                       1863+         DC    X'E7'                   - VECTOR DIVIDE LOGICAL
00002079  12                                       1864+         DC    X'12'                    v1, v2
0000207A  30                                       1865+         DC    X'30'                    v3, reserved
0000207B  0840                                     1866+         DC    HL2'2112'                reserved, m5, m4, RXB
0000207D  B0                                       1867+         DC    X'B0'

0000207E  E710 5030 000E                 00002020  1869+         VST   V1,V1O20          save v1 output
00002084  07FB                                     1870+         BR    R11               return
00002088                                           1871+RE20     DC    0F                xl16 expected result
00002088                                           1872+         DROP  R5
00002088  00000000 0000000E                        1873          DC    XL16'000000000000000E 29164CB5F714841E'   result
00002090  29164CB5 F714841E
00002098  7FFFFFFF FFFFFFFF                        1874          DC    XL16'7FFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
000020A0  FFFFFFFF FFFFFFFF
000020A8  00000000 00000000                        1875          DC    XL16'0000000000000000 090A0B0C0D0E0F10'   v3
000020B0  090A0B0C 0D0E0F10
                                                   1876
                                                   1877 *-----------------------------------------------------------------
                                                   1878 *   VRL    - VECTOR REMAINDER LOGICAL
                                                   1879 *-----------------------------------------------------------------
                                                   1880 *Word    M4:  2, m5= 8 (IDC=1)
                                                   1881
                                                   1882          VRR_C VRL,X'2',X'8',N
000020B8                                           1883+         DS    0FD
000020B8                         000020B8          1884+         USING *,R5              base for test data and test routine
000020B8  00002128                                 1885+T21      DC    A(X21)            address of test routine
000020BC  0015                                     1886+         DC    H'21'             test number
000020BE  00                                       1887+         DC    X'00'
000020BF  D5                                       1888+         DC    CL1'N'            S or Y = skip cross check
000020C0  02                                       1889+         DC    X'2'               m4
000020C1  08                                       1890+         DC    X'8'               m5
000020C2  E5D9D340 40404040                        1891+         DC    CL8'VRL'          instruction name
000020CC  00002160                                 1892+         DC    A(RE21+16)        address of v2 source
000020D0  00002170                                 1893+         DC    A(RE21+32)        address of v3 source
000020D4  00000010                                 1894+         DC    A(16)             result length
000020D8  00002150                                 1895+REA21    DC    A(RE21)           result address
000020E0  00000000 00000000                        1896+         DS    FD                   gap
000020E8  00000000 00000000                        1897+V1O21    DS    XL16              V1 output
000020F0  00000000 00000000
000020F8  00000000 00000000                        1898+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002100  40404040 40404040                        1899+         DC    CL8' '            was cross check skipped?
00002108  00000000 00000000                        1900+         DS    FD                   gap
00002110  00000000 00000000                        1901+XCO21    DS    XL16              Cross check Output
00002118  00000000 00000000
00002120  00000000 00000000                        1902+         DS    FD                   gap
                                                   1903+*
00002128                                           1904+X21      DS    0F
00002128  E310 5014 0014                 00000014  1905+         LGF   R1,V2ADDR         load v2 source
0000212E  E721 0000 0006                 00000000  1906+         VL    v2,0(R1)
00002134  E310 5018 0014                 00000018  1907+         LGF   R1,V3ADDR         load v3 source
0000213A  E731 0000 0006                 00000000  1908+         VL    v3,0(R1)

00002140                                           1911+         DS    0H                      E7B1 VRL
00002140  E7                                       1912+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002141  12                                       1913+         DC    X'12'                    v1, v2
00002142  30                                       1914+         DC    X'30'                    v3, reserved
00002143  0820                                     1915+         DC    HL2'2080'                reserved, m5, m4, RXB
00002145  B1                                       1916+         DC    X'B1'

00002146  E710 5030 000E                 000020E8  1918+         VST   V1,V1O21          save v1 output
0000214C  07FB                                     1919+         BR    R11               return
00002150                                           1920+RE21     DC    0F                xl16 expected result
00002150                                           1921+         DROP  R5
00002150  00000000 00000000                        1922          DC    XL16'00000000 00000000 00000000 00000000'   result
00002158  00000000 00000000
00002160  00000000 00000000                        1923          DC    XL16'00000000 00000000 00000000 00000000'   v2
00002168  00000000 00000000
00002170  01020304 05060708                        1924          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00002178  090A0B0C 0D0E0F10
                                                   1925
                                                   1926          VRR_C VRL,X'2',X'8',N
00002180                                           1927+         DS    0FD
00002180                         00002180          1928+         USING *,R5              base for test data and test routine
00002180  000021F0                                 1929+T22      DC    A(X22)            address of test routine
00002184  0016                                     1930+         DC    H'22'             test number
00002186  00                                       1931+         DC    X'00'
00002187  D5                                       1932+         DC    CL1'N'            S or Y = skip cross check
00002188  02                                       1933+         DC    X'2'               m4
00002189  08                                       1934+         DC    X'8'               m5
0000218A  E5D9D340 40404040                        1935+         DC    CL8'VRL'          instruction name
00002194  00002228                                 1936+         DC    A(RE22+16)        address of v2 source
00002198  00002238                                 1937+         DC    A(RE22+32)        address of v3 source
0000219C  00000010                                 1938+         DC    A(16)             result length
000021A0  00002218                                 1939+REA22    DC    A(RE22)           result address
000021A8  00000000 00000000                        1940+         DS    FD                   gap
000021B0  00000000 00000000                        1941+V1O22    DS    XL16              V1 output
000021B8  00000000 00000000
000021C0  00000000 00000000                        1942+         DS    FD                   gap
000021C8  40404040 40404040                        1943+         DC    CL8' '            was cross check skipped?
000021D0  00000000 00000000                        1944+         DS    FD                   gap
000021D8  00000000 00000000                        1945+XCO22    DS    XL16              Cross check Output
000021E0  00000000 00000000
000021E8  00000000 00000000                        1946+         DS    FD                   gap
                                                   1947+*
000021F0                                           1948+X22      DS    0F
000021F0  E310 5014 0014                 00000014  1949+         LGF   R1,V2ADDR         load v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000021F6  E721 0000 0006                00000000  1950+         VL    v2,0(R1)
000021FC  E310 5018 0014                00000018  1951+         LGF   R1,V3ADDR         load v3 source
00002202  E731 0000 0006                00000000  1952+         VL    v3,0(R1)

00002208                                          1955+         DS    0H                      E7B1 VRL
00002208  E7                                      1956+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002209  12                                      1957+         DC    X'12'                    v1, v2
0000220A  30                                      1958+         DC    X'30'                    v3, reserved
0000220B  0820                                    1959+         DC    HL2'2080'                reserved, m5, m4, RXB
0000220D  B1                                      1960+         DC    X'B1'

0000220E  E710 5030 000E                000021B0  1962+         VST   V1,V1O22          save v1 output
00002214  07FB                                    1963+         BR    R11               return
00002218                                          1964+RE22     DC    0F                xl16 expected result
00002218                                          1965+         DROP  R5
00002218  01020303 05060707                       1966          DC    XL16'01020303 05060707 090A0B0B 0D0E0F0F'   result
00002220  090A0B0B 0D0E0F0F
00002228  01020303 05060707                       1967          DC    XL16'01020303 05060707 090A0B0B 0D0E0F0F'   v2
00002230  090A0B0B 0D0E0F0F
00002238  01020304 05060708                       1968          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00002240  090A0B0C 0D0E0F10
                                                  1969
                                                  1970          VRR_C VRL,X'2',X'8',N
00002248                                          1971+         DS    0FD
00002248                        00002248          1972+         USING *,R5              base for test data and test routine
00002248  000022B8                                1973+T23      DC    A(X23)            address of test routine
0000224C  0017                                    1974+         DC    H'23'             test number
0000224E  00                                      1975+         DC    X'00'
0000224F  D5                                      1976+         DC    CL1'N'            S or Y = skip cross check
00002250  02                                      1977+         DC    X'2'               m4
00002251  08                                      1978+         DC    X'8'               m5
00002252  E5D9D340 40404040                       1979+         DC    CL8'VRL'          instruction name
0000225C  000022F0                                1980+         DC    A(RE23+16)        address of v2 source
00002260  00002300                                1981+         DC    A(RE23+32)        address of v3 source
00002264  00000010                                1982+         DC    A(16)             result length
00002268  000022E0                                1983+REA23    DC    A(RE23)           result address
00002270  00000000 00000000                       1984+         DS    FD                   gap
00002278  00000000 00000000                       1985+V1O23    DS    XL16              V1 output
00002280  00000000 00000000
00002288  00000000 00000000                       1986+         DS    FD                   gap
00002290  40404040 40404040                       1987+         DC    CL8' '            was cross check skipped?
00002298  00000000 00000000                       1988+         DS    FD                   gap
000022A0  00000000 00000000                       1989+XCO23    DS    XL16              Cross check Output
000022A8  00000000 00000000
000022B0  00000000 00000000                       1990+         DS    FD                   gap
                                                  1991+*
000022B8                                          1992+X23      DS    0F
000022B8  E310 5014 0014                00000014  1993+         LGF   R1,V2ADDR         load v2 source
000022BE  E721 0000 0006                00000000  1994+         VL    v2,0(R1)
000022C4  E310 5018 0014                00000018  1995+         LGF   R1,V3ADDR         load v3 source
000022CA  E731 0000 0006                00000000  1996+         VL    v3,0(R1)

000022D0                                          1999+         DS    0H                      E7B1 VRL
000022D0  E7                                      2000+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
000022D1  12                                      2001+         DC    X'12'                    v1, v2
000022D2  30                                      2002+         DC    X'30'                    v3, reserved
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000022D3  0820                                     2003+         DC    HL2'2080'                reserved, m5, m4, RXB
000022D5  B1                                       2004+         DC    X'B1'

000022D6  E710 5030 000E                 00002278  2006+         VST   V1,V1O23          save v1 output
000022DC  07FB                                     2007+         BR    R11               return
000022E0                                           2008+RE23     DC    0F                xl16 expected result
000022E0                                           2009+         DROP  R5
000022E0  00000000 00000000                        2010          DC    XL16'00000000 00000000 00000000 00000000'   result
000022E8  00000000 00000000
000022F0  01020304 05060708                        2011          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v2
000022F8  090A0B0C 0D0E0F10
00002300  01020304 05060708                        2012          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00002308  090A0B0C 0D0E0F10
                                                   2013
                                                   2014          VRR_C VRL,X'2',X'8',N
00002310                                           2015+         DS    0FD
00002310                         00002310          2016+         USING *,R5              base for test data and test routine
00002310  00002380                                 2017+T24      DC    A(X24)            address of test routine
00002314  0018                                     2018+         DC    H'24'             test number
00002316  00                                       2019+         DC    X'00'
00002317  D5                                       2020+         DC    CL1'N'            S or Y = skip cross check
00002318  02                                       2021+         DC    X'2'               m4
00002319  08                                       2022+         DC    X'8'               m5
0000231A  E5D9D340 40404040                        2023+         DC    CL8'VRL'          instruction name
00002324  000023B8                                 2024+         DC    A(RE24+16)        address of v2 source
00002328  000023C8                                 2025+         DC    A(RE24+32)        address of v3 source
0000232C  00000010                                 2026+         DC    A(16)             result length
00002330  000023A8                                 2027+REA24    DC    A(RE24)           result address
00002338  00000000 00000000                        2028+         DS    FD                   gap
00002340  00000000 00000000                        2029+V1O24    DS    XL16              V1 output
00002348  00000000 00000000
00002350  00000000 00000000                        2030+         DS    FD                   gap
00002358  40404040 40404040                        2031+         DC    CL8' '            was cross check skipped?
00002360  00000000 00000000                        2032+         DS    FD                   gap
00002368  00000000 00000000                        2033+XCO24    DS    XL16              Cross check Output
00002370  00000000 00000000
00002378  00000000 00000000                        2034+         DS    FD                   gap
                                                   2035+*
00002380                                           2036+X24      DS    0F
00002380  E310 5014 0014                 00000014  2037+         LGF   R1,V2ADDR         load v2 source
00002386  E721 0000 0006                 00000000  2038+         VL    v2,0(R1)
0000238C  E310 5018 0014                 00000018  2039+         LGF   R1,V3ADDR         load v3 source
00002392  E731 0000 0006                 00000000  2040+         VL    v3,0(R1)

00002398                                           2043+         DS    0H                      E7B1 VRL
00002398  E7                                       2044+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002399  12                                       2045+         DC    X'12'                    v1, v2
0000239A  30                                       2046+         DC    X'30'                    v3, reserved
0000239B  0820                                     2047+         DC    HL2'2080'                reserved, m5, m4, RXB
0000239D  B1                                       2048+         DC    X'B1'

0000239E  E710 5030 000E                 00002340  2050+         VST   V1,V1O24          save v1 output
000023A4  07FB                                     2051+         BR    R11               return
000023A8                                           2052+RE24     DC    0F                xl16 expected result
000023A8                                           2053+         DROP  R5
000023A8  00000000 00000001                        2054          DC    XL16'00000000 00000001 00000005 00000000'   result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000023B0  00000005 00000000
000023B8  FFFFFFFF FFFFFFFF                         2055          DC    XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'   v2
000023C0  FFFFFFFF FFFFFFFF
000023C8  00000001 00000002                         2056          DC    XL16'00000001 00000002 0000000A 0000000F'   v3
000023D0  0000000A 0000000F
                                                    2057
                                                    2058          VRR_C VRL,X'2',X'8',N
000023D8                                            2059+         DS    0FD
000023D8                      000023D8              2060+         USING *,R5              base for test data and test routine
000023D8  00002448                                  2061+T25      DC    A(X25)            address of test routine
000023DC  0019                                      2062+         DC    H'25'             test number
000023DE  00                                        2063+         DC    X'00'
000023DF  D5                                        2064+         DC    CL1'N'            S or Y = skip cross check
000023E0  02                                        2065+         DC    X'2'               m4
000023E1  08                                        2066+         DC    X'8'               m5
000023E2  E5D9D340 40404040                         2067+         DC    CL8'VRL'          instruction name
000023EC  00002480                                  2068+         DC    A(RE25+16)        address of v2 source
000023F0  00002490                                  2069+         DC    A(RE25+32)        address of v3 source
000023F4  00000010                                  2070+         DC    A(16)             result length
000023F8  00002470                                  2071+REA25    DC    A(RE25)           result address
00002400  00000000 00000000                         2072+         DS    FD                   gap
00002408  00000000 00000000                         2073+V1O25    DS    XL16              V1 output
00002410  00000000 00000000
00002418  00000000 00000000                         2074+         DS    FD                   gap
00002420  40404040 40404040                         2075+         DC    CL8' '            was cross check skipped?
00002428  00000000 00000000                         2076+         DS    FD                   gap
00002430  00000000 00000000                         2077+XCO25    DS    XL16              Cross check Output
00002438  00000000 00000000
00002440  00000000 00000000                         2078+         DS    FD                   gap
                                                    2079+*
00002448                                            2080+X25      DS    0F
00002448  E310 5014 0014                 00000014   2081+         LGF   R1,V2ADDR         load v2 source
0000244E  E721 0000 0006                 00000000   2082+         VL    v2,0(R1)
00002454  E310 5018 0014                 00000018   2083+         LGF   R1,V3ADDR         load v3 source
0000245A  E731 0000 0006                 00000000   2084+         VL    v3,0(R1)

00002460                                            2087+         DS    0H                     E7B1 VRL
00002460  E7                                        2088+         DC    X'E7'                  - VECTOR REMAINDER LOGICAL
00002461  12                                        2089+         DC    X'12'                   v1, v2
00002462  30                                        2090+         DC    X'30'                   v3, reserved
00002463  0820                                      2091+         DC    HL2'2080'               reserved, m5, m4, RXB
00002465  B1                                        2092+         DC    X'B1'

00002466  E710 5030 000E                 00002408   2094+         VST   V1,V1O25          save v1 output
0000246C  07FB                                      2095+         BR    R11               return
00002470                                            2096+RE25     DC    0F                xl16 expected result
00002470                                            2097+         DROP  R5
00002470  00010207 04D2A06F                         2098          DC    XL16'00010207 04D2A06F 02E6CAAF 07F4E1CF'   result
00002478  02E6CAAF 07F4E1CF
00002480  FFFFFFFF FFFFFFFF                         2099          DC    XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'   v2
00002488  FFFFFFFF FFFFFFFF
00002490  01020304 05060708                         2100          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00002498  090A0B0C 0D0E0F10
                                                    2101
                                                    2102 *Doubleword    M4:  3, m5= 8 (IDC=1)
                                                    2103
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   2104          VRR_C VRL,X'3',X'8',N
000024A0                                           2105+         DS    0FD
000024A0                     000024A0              2106+         USING *,R5              base for test data and test routine
000024A0  00002510                                 2107+T26      DC    A(X26)            address of test routine
000024A4  001A                                     2108+         DC    H'26'             test number
000024A6  00                                       2109+         DC    X'00'
000024A7  D5                                       2110+         DC    CL1'N'            S or Y = skip cross check
000024A8  03                                       2111+         DC    X'3'               m4
000024A9  08                                       2112+         DC    X'8'               m5
000024AA  E5D9D340 40404040                        2113+         DC    CL8'VRL'          instruction name
000024B4  00002548                                 2114+         DC    A(RE26+16)        address of v2 source
000024B8  00002558                                 2115+         DC    A(RE26+32)        address of v3 source
000024BC  00000010                                 2116+         DC    A(16)             result length
000024C0  00002538                                 2117+REA26    DC    A(RE26)           result address
000024C8  00000000 00000000                        2118+         DS    FD                   gap
000024D0  00000000 00000000                        2119+V1O26    DS    XL16              V1 output
000024D8  00000000 00000000
000024E0  00000000 00000000                        2120+         DS    FD                   gap
000024E8  40404040 40404040                        2121+         DC    CL8' '            was cross check skipped?
000024F0  00000000 00000000                        2122+         DS    FD                   gap
000024F8  00000000 00000000                        2123+XCO26    DS    XL16              Cross check Output
00002500  00000000 00000000
00002508  00000000 00000000                        2124+         DS    FD                   gap
                                                   2125+*
00002510                                           2126+X26      DS    0F
00002510  E310 5014 0014                 00000014  2127+         LGF   R1,V2ADDR         load v2 source
00002516  E721 0000 0006                 00000000  2128+         VL    v2,0(R1)
0000251C  E310 5018 0014                 00000018  2129+         LGF   R1,V3ADDR         load v3 source
00002522  E731 0000 0006                 00000000  2130+         VL    v3,0(R1)

00002528                                           2133+         DS    0H                      E7B1 VRL
00002528  E7                                       2134+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002529  12                                       2135+         DC    X'12'                    v1, v2
0000252A  30                                       2136+         DC    X'30'                    v3, reserved
0000252B  0830                                     2137+         DC    HL2'2096'                reserved, m5, m4, RXB
0000252D  B1                                       2138+         DC    X'B1'

0000252E  E710 5030 000E                 000024D0  2140+         VST   V1,V1O26          save v1 output
00002534  07FB                                     2141+         BR    R11               return
00002538                                           2142+RE26     DC    0F                xl16 expected result
00002538                                           2143+         DROP  R5
00002538  00000000 00000000                        2144          DC    XL16'0000000000000000 0000000000000000'   result
00002540  00000000 00000000
00002548  00000000 00000000                        2145          DC    XL16'0000000000000000 0000000000000000'   v2
00002550  00000000 00000000
00002558  01020304 05060708                        2146          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00002560  090A0B0C 0D0E0F10
                                                   2147
                                                   2148          VRR_C VRL,X'3',X'8',N
00002568                                           2149+         DS    0FD
00002568                     00002568              2150+         USING *,R5              base for test data and test routine
00002568  000025D8                                 2151+T27      DC    A(X27)            address of test routine
0000256C  001B                                     2152+         DC    H'27'             test number
0000256E  00                                       2153+         DC    X'00'
0000256F  D5                                       2154+         DC    CL1'N'            S or Y = skip cross check
00002570  03                                       2155+         DC    X'3'               m4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002571  08                                       2156+         DC    X'8'                m5
00002572  E5D9D340 40404040                        2157+         DC    CL8'VRL'           instruction name
0000257C  00002610                                 2158+         DC    A(RE27+16)         address of v2 source
00002580  00002620                                 2159+         DC    A(RE27+32)         address of v3 source
00002584  00000010                                 2160+         DC    A(16)              result length
00002588  00002600                                 2161+REA27    DC    A(RE27)            result address
00002590  00000000 00000000                        2162+         DS    FD                    gap
00002598  00000000 00000000                        2163+V1O27    DS    XL16               V1 output
000025A0  00000000 00000000
000025A8  00000000 00000000                        2164+         DS    FD                    gap
000025B0  40404040 40404040                        2165+         DC    CL8' '             was cross check skipped?
000025B8  00000000 00000000                        2166+         DS    FD                    gap
000025C0  00000000 00000000                        2167+XCO27    DS    XL16               Cross check Output
000025C8  00000000 00000000
000025D0  00000000 00000000                        2168+         DS    FD                    gap
                                                   2169+*
000025D8                                           2170+X27      DS    0F
000025D8  E310 5014 0014                 00000014  2171+         LGF   R1,V2ADDR          load v2 source
000025DE  E721 0000 0006                 00000000  2172+         VL    v2,0(R1)
000025E4  E310 5018 0014                 00000018  2173+         LGF   R1,V3ADDR          load v3 source
000025EA  E731 0000 0006                 00000000  2174+         VL    v3,0(R1)

000025F0                                           2177+         DS    0H                       E7B1 VRL
000025F0  E7                                       2178+         DC    X'E7'                    - VECTOR REMAINDER LOGICAL
000025F1  12                                       2179+         DC    X'12'                     v1, v2
000025F2  30                                       2180+         DC    X'30'                     v3, reserved
000025F3  0830                                     2181+         DC    HL2'2096'                    reserved, m5, m4, RXB
000025F5  B1                                       2182+         DC    X'B1'

000025F6  E710 5030 000E                 00002598  2184+         VST   V1,V1O27           save v1 output
000025FC  07FB                                     2185+         BR    R11                return
00002600                                           2186+RE27     DC    0F                 xl16 expected result
00002600                                           2187+         DROP  R5
00002600  01020303 05060707                        2188          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   result
00002608  090A0B0B 0D0E0F0F
00002610  01020303 05060707                        2189          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00002618  090A0B0B 0D0E0F0F
00002620  01020304 05060708                        2190          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00002628  090A0B0C 0D0E0F10
                                                   2191
                                                   2192          VRR_C VRL,X'3',X'8',N
00002630                                           2193+         DS    0FD
00002630                      00002630             2194+         USING *,R5               base for test data and test routine
00002630  000026A0                                 2195+T28      DC    A(X28)             address of test routine
00002634  001C                                     2196+         DC    H'28'              test number
00002636  00                                       2197+         DC    X'00'
00002637  D5                                       2198+         DC    CL1'N'             S or Y = skip cross check
00002638  03                                       2199+         DC    X'3'                m4
00002639  08                                       2200+         DC    X'8'                m5
0000263A  E5D9D340 40404040                        2201+         DC    CL8'VRL'           instruction name
00002644  000026D8                                 2202+         DC    A(RE28+16)         address of v2 source
00002648  000026E8                                 2203+         DC    A(RE28+32)         address of v3 source
0000264C  00000010                                 2204+         DC    A(16)              result length
00002650  000026C8                                 2205+REA28    DC    A(RE28)            result address
00002658  00000000 00000000                        2206+         DS    FD                    gap
00002660  00000000 00000000                        2207+V1O28    DS    XL16               V1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002668  00000000 00000000
00002670  00000000 00000000                         2208+         DS    FD                   gap
00002678  40404040 40404040                         2209+         DC    CL8' '             was cross check skipped?
00002680  00000000 00000000                         2210+         DS    FD                   gap
00002688  00000000 00000000                         2211+XCO28    DS    XL16               Cross check Output
00002690  00000000 00000000
00002698  00000000 00000000                         2212+         DS    FD                   gap
                                                    2213+*
000026A0                                            2214+X28      DS    0F
000026A0  E310 5014 0014                 00000014   2215+         LGF   R1,V2ADDR          load v2 source
000026A6  E721 0000 0006                 00000000   2216+         VL    v2,0(R1)
000026AC  E310 5018 0014                 00000018   2217+         LGF   R1,V3ADDR          load v3 source
000026B2  E731 0000 0006                 00000000   2218+         VL    v3,0(R1)

000026B8                                            2221+         DS    0H                     E7B1 VRL
000026B8  E7                                        2222+         DC    X'E7'                  - VECTOR REMAINDER LOGICAL
000026B9  12                                        2223+         DC    X'12'                   v1, v2
000026BA  30                                        2224+         DC    X'30'                   v3, reserved
000026BB  0830                                      2225+         DC    HL2'2096'               reserved, m5, m4, RXB
000026BD  B1                                        2226+         DC    X'B1'

000026BE  E710 5030 000E                 00002660   2228+         VST   V1,V1O28           save v1 output
000026C4  07FB                                      2229+         BR    R11                return
000026C8                                            2230+RE28     DC    0F                 xl16 expected result
000026C8                                            2231+         DROP  R5
000026C8  00000000 00000000                         2232          DC    XL16'0000000000000000 0000000000000000'   result
000026D0  00000000 00000000
000026D8  01020304 05060708                         2233          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
000026E0  090A0B0C 0D0E0F10
000026E8  01020304 05060708                         2234          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000026F0  090A0B0C 0D0E0F10
                                                    2235
                                                    2236          VRR_C VRL,X'3',X'8',N
000026F8                                            2237+         DS    0FD
000026F8                         000026F8           2238+         USING *,R5               base for test data and test routine
000026F8  00002768                                  2239+T29      DC    A(X29)             address of test routine
000026FC  001D                                      2240+         DC    H'29'              test number
000026FE  00                                        2241+         DC    X'00'
000026FF  D5                                        2242+         DC    CL1'N'             S or Y = skip cross check
00002700  03                                        2243+         DC    X'3'                m4
00002701  08                                        2244+         DC    X'8'                m5
00002702  E5D9D340 40404040                         2245+         DC    CL8'VRL'           instruction name
0000270C  000027A0                                  2246+         DC    A(RE29+16)         address of v2 source
00002710  000027B0                                  2247+         DC    A(RE29+32)         address of v3 source
00002714  00000010                                  2248+         DC    A(16)              result length
00002718  00002790                                  2249+REA29    DC    A(RE29)            result address
00002720  00000000 00000000                         2250+         DS    FD                   gap
00002728  00000000 00000000                         2251+V1O29    DS    XL16               V1 output
00002730  00000000 00000000
00002738  00000000 00000000                         2252+         DS    FD                   gap
00002740  40404040 40404040                         2253+         DC    CL8' '             was cross check skipped?
00002748  00000000 00000000                         2254+         DS    FD                   gap
00002750  00000000 00000000                         2255+XCO29    DS    XL16               Cross check Output
00002758  00000000 00000000
00002760  00000000 00000000                         2256+         DS    FD                   gap
                                                    2257+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002768                                          2258+X29      DS    0F
00002768  E310 5014 0014                 00000014 2259+         LGF   R1,V2ADDR         load v2 source
0000276E  E721 0000 0006                 00000000 2260+         VL    v2,0(R1)
00002774  E310 5018 0014                 00000018 2261+         LGF   R1,V3ADDR         load v3 source
0000277A  E731 0000 0006                 00000000 2262+         VL    v3,0(R1)

00002780                                          2265+         DS    0H                      E7B1 VRL
00002780  E7                                      2266+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002781  12                                      2267+         DC    X'12'                    v1, v2
00002782  30                                      2268+         DC    X'30'                    v3, reserved
00002783  0830                                    2269+         DC    HL2'2096'                reserved, m5, m4, RXB
00002785  B1                                      2270+         DC    X'B1'

00002786  E710 5030 000E                 00002728 2272+         VST   V1,V1O29          save v1 output
0000278C  07FB                                    2273+         BR    R11               return
00002790                                          2274+RE29     DC    0F                xl16 expected result
00002790                                          2275+         DROP  R5
00002790  00000000 00000000                       2276          DC    XL16'0000000000000000 0000000000000001'   result
00002798  00000000 00000001
000027A0  FFFFFFFF FFFFFFFF                       2277          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
000027A8  FFFFFFFF FFFFFFFF
000027B0  00000000 00000001                       2278          DC    XL16'0000000000000001 0000000000000002'   v3
000027B8  00000000 00000002
                                                  2279
                                                  2280          VRR_C VRL,X'3',X'8',N
000027C0                                          2281+         DS    0FD
000027C0                      000027C0            2282+         USING *,R5              base for test data and test routine
000027C0  00002830                                2283+T30      DC    A(X30)            address of test routine
000027C4  001E                                    2284+         DC    H'30'             test number
000027C6  00                                      2285+         DC    X'00'
000027C7  D5                                      2286+         DC    CL1'N'            S or Y = skip cross check
000027C8  03                                      2287+         DC    X'3'               m4
000027C9  08                                      2288+         DC    X'8'               m5
000027CA  E5D9D340 40404040                       2289+         DC    CL8'VRL'          instruction name
000027D4  00002868                                2290+         DC    A(RE30+16)        address of v2 source
000027D8  00002878                                2291+         DC    A(RE30+32)        address of v3 source
000027DC  00000010                                2292+         DC    A(16)             result length
000027E0  00002858                                2293+REA30    DC    A(RE30)           result address
000027E8  00000000 00000000                       2294+         DS    FD                   gap
000027F0  00000000 00000000                       2295+V1O30    DS    XL16              V1 output
000027F8  00000000 00000000
00002800  00000000 00000000                       2296+         DS    FD                   gap
00002808  40404040 40404040                       2297+         DC    CL8' '            was cross check skipped?
00002810  00000000 00000000                       2298+         DS    FD                   gap
00002818  00000000 00000000                       2299+XCO30    DS    XL16              Cross check Output
00002820  00000000 00000000
00002828  00000000 00000000                       2300+         DS    FD                   gap
                                                  2301+*
00002830                                          2302+X30      DS    0F
00002830  E310 5014 0014                 00000014 2303+         LGF   R1,V2ADDR         load v2 source
00002836  E721 0000 0006                 00000000 2304+         VL    v2,0(R1)
0000283C  E310 5018 0014                 00000018 2305+         LGF   R1,V3ADDR         load v3 source
00002842  E731 0000 0006                 00000000 2306+         VL    v3,0(R1)

00002848                                          2309+         DS    0H                      E7B1 VRL
00002848  E7                                      2310+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002849  12                                      2311+         DC    X'12'                    v1, v2
0000284A  30                                      2312+         DC    X'30'                    v3, reserved
0000284B  0830                                    2313+         DC    HL2'2096'                reserved, m5, m4, RXB
0000284D  B1                                      2314+         DC    X'B1'

0000284E  E710 5030 000E                000027F0  2316+         VST   V1,V1O30          save v1 output
00002854  07FB                                    2317+         BR    R11               return
00002858                                          2318+RE30     DC    0F                xl16 expected result
00002858                                          2319+         DROP  R5
00002858  00000000 00000005                       2320          DC    XL16'0000000000000005 0000000000000000'   result
00002860  00000000 00000000
00002868  FFFFFFFF FFFFFFFF                       2321          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00002870  FFFFFFFF FFFFFFFF
00002878  00000000 0000000A                       2322          DC    XL16'000000000000000A 000000000000000F'   v3
00002880  00000000 0000000F
                                                  2323
                                                  2324          VRR_C VRL,X'3',X'8',N
00002888                                          2325+         DS    0FD
00002888                      00002888            2326+         USING *,R5              base for test data and test routine
00002888  000028F8                                2327+T31      DC    A(X31)            address of test routine
0000288C  001F                                    2328+         DC    H'31'             test number
0000288E  00                                      2329+         DC    X'00'
0000288F  D5                                      2330+         DC    CL1'N'            S or Y = skip cross check
00002890  03                                      2331+         DC    X'3'               m4
00002891  08                                      2332+         DC    X'8'               m5
00002892  E5D9D340 40404040                       2333+         DC    CL8'VRL'          instruction name
0000289C  00002930                                2334+         DC    A(RE31+16)        address of v2 source
000028A0  00002940                                2335+         DC    A(RE31+32)        address of v3 source
000028A4  00000010                                2336+         DC    A(16)             result length
000028A8  00002920                                2337+REA31    DC    A(RE31)           result address
000028B0  00000000 00000000                       2338+         DS    FD                   gap
000028B8  00000000 00000000                       2339+V1O31    DS    XL16              V1 output
000028C0  00000000 00000000
000028C8  00000000 00000000                       2340+         DS    FD                   gap
000028D0  40404040 40404040                       2341+         DC    CL8' '            was cross check skipped?
000028D8  00000000 00000000                       2342+         DS    FD                   gap
000028E0  00000000 00000000                       2343+XCO31    DS    XL16              Cross check Output
000028E8  00000000 00000000
000028F0  00000000 00000000                       2344+         DS    FD                   gap
                                                  2345+*
000028F8                                          2346+X31      DS    0F
000028F8  E310 5014 0014                00000014  2347+         LGF   R1,V2ADDR         load v2 source
000028FE  E721 0000 0006                00000000  2348+         VL    v2,0(R1)
00002904  E310 5018 0014                00000018  2349+         LGF   R1,V3ADDR         load v3 source
0000290A  E731 0000 0006                00000000  2350+         VL    v3,0(R1)

00002910                                          2353+         DS    0H                     E7B1 VRL
00002910  E7                                      2354+         DC    X'E7'                  - VECTOR REMAINDER LOGICAL
00002911  12                                      2355+         DC    X'12'                   v1, v2
00002912  30                                      2356+         DC    X'30'                   v3, reserved
00002913  0830                                    2357+         DC    HL2'2096'               reserved, m5, m4, RXB
00002915  B1                                      2358+         DC    X'B1'

00002916  E710 5030 000E                000028B8  2360+         VST   V1,V1O31          save v1 output
0000291C  07FB                                    2361+         BR    R11               return
00002920                                          2362+RE31     DC    0F                xl16 expected result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002920                                           2363+         DROP  R5
00002920  00008101 82028307                        2364          DC    XL16'0000810182028307 01736557493B2D1F'   result
00002928  01736557 493B2D1F
00002930  7FFFFFFF FFFFFFFF                        2365          DC    XL16'7FFFFFFFFFFFFFFF 7FFFFFFFFFFFFFFF'   v2
00002938  7FFFFFFF FFFFFFFF
00002940  01020304 05060708                        2366          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00002948  090A0B0C 0D0E0F10
                                                   2367
                                                   2368 *Quadword      M4:  4, m5= 8 (IDC=1)
                                                   2369
                                                   2370          VRR_C VRL,X'4',X'8',N
00002950                                           2371+         DS    0FD
00002950                       00002950            2372+         USING *,R5              base for test data and test routine
00002950  000029C0                                 2373+T32      DC    A(X32)            address of test routine
00002954  0020                                     2374+         DC    H'32'             test number
00002956  00                                       2375+         DC    X'00'
00002957  D5                                       2376+         DC    CL1'N'            S or Y = skip cross check
00002958  04                                       2377+         DC    X'4'               m4
00002959  08                                       2378+         DC    X'8'               m5
0000295A  E5D9D340 40404040                        2379+         DC    CL8'VRL'          instruction name
00002964  000029F8                                 2380+         DC    A(RE32+16)        address of v2 source
00002968  00002A08                                 2381+         DC    A(RE32+32)        address of v3 source
0000296C  00000010                                 2382+         DC    A(16)             result length
00002970  000029E8                                 2383+REA32    DC    A(RE32)           result address
00002978  00000000 00000000                        2384+         DS    FD                   gap
00002980  00000000 00000000                        2385+V1O32    DS    XL16              V1 output
00002988  00000000 00000000
00002990  00000000 00000000                        2386+         DS    FD                   gap
00002998  40404040 40404040                        2387+         DC    CL8' '            was cross check skipped?
000029A0  00000000 00000000                        2388+         DS    FD                   gap
000029A8  00000000 00000000                        2389+XCO32    DS    XL16              Cross check Output
000029B0  00000000 00000000
000029B8  00000000 00000000                        2390+         DS    FD                   gap
                                                   2391+*
000029C0                                           2392+X32      DS    0F
000029C0  E310 5014 0014                00000014   2393+         LGF   R1,V2ADDR         load v2 source
000029C6  E721 0000 0006                00000000   2394+         VL    v2,0(R1)
000029CC  E310 5018 0014                00000018   2395+         LGF   R1,V3ADDR         load v3 source
000029D2  E731 0000 0006                00000000   2396+         VL    v3,0(R1)

000029D8                                           2399+         DS    0H                     E7B1 VRL
000029D8  E7                                       2400+         DC    X'E7'                  - VECTOR REMAINDER LOGICAL
000029D9  12                                       2401+         DC    X'12'                   v1, v2
000029DA  30                                       2402+         DC    X'30'                   v3, reserved
000029DB  0840                                     2403+         DC    HL2'2112'               reserved, m5, m4, RXB
000029DD  B1                                       2404+         DC    X'B1'

000029DE  E710 5030 000E                00002980   2406+         VST   V1,V1O32          save v1 output
000029E4  07FB                                     2407+         BR    R11               return
000029E8                                           2408+RE32     DC    0F                xl16 expected result
000029E8                                           2409+         DROP  R5
000029E8  00000000 00000000                        2410          DC    XL16'0000000000000000 0000000000000000'   result
000029F0  00000000 00000000
000029F8  00000000 00000000                        2411          DC    XL16'0000000000000000 0000000000000000'   v2
00002A00  00000000 00000000
00002A08  01020304 05060708                        2412          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002A10  090A0B0C 0D0E0F10
                                                   2413
                                                   2414          VRR_C VRL,X'4',X'8',N
00002A18                                           2415+         DS    0FD
00002A18                       00002A18            2416+         USING *,R5              base for test data and test routine
00002A18  00002A88                                 2417+T33      DC    A(X33)            address of test routine
00002A1C  0021                                     2418+         DC    H'33'             test number
00002A1E  00                                       2419+         DC    X'00'
00002A1F  D5                                       2420+         DC    CL1'N'            S or Y = skip cross check
00002A20  04                                       2421+         DC    X'4'               m4
00002A21  08                                       2422+         DC    X'8'               m5
00002A22  E5D9D340 40404040                        2423+         DC    CL8'VRL'          instruction name
00002A2C  00002AC0                                 2424+         DC    A(RE33+16)        address of v2 source
00002A30  00002AD0                                 2425+         DC    A(RE33+32)        address of v3 source
00002A34  00000010                                 2426+         DC    A(16)             result length
00002A38  00002AB0                                 2427+REA33    DC    A(RE33)           result address
00002A40  00000000 00000000                        2428+         DS    FD                   gap
00002A48  00000000 00000000                        2429+V1O33    DS    XL16              V1 output
00002A50  00000000 00000000
00002A58  00000000 00000000                        2430+         DS    FD                   gap
00002A60  40404040 40404040                        2431+         DC    CL8' '            was cross check skipped?
00002A68  00000000 00000000                        2432+         DS    FD                   gap
00002A70  00000000 00000000                        2433+XCO33    DS    XL16              Cross check Output
00002A78  00000000 00000000
00002A80  00000000 00000000                        2434+         DS    FD                   gap
                                                   2435+*
00002A88                                           2436+X33      DS    0F
00002A88  E310 5014 0014                00000014   2437+         LGF   R1,V2ADDR         load v2 source
00002A8E  E721 0000 0006                00000000   2438+         VL    v2,0(R1)
00002A94  E310 5018 0014                00000018   2439+         LGF   R1,V3ADDR         load v3 source
00002A9A  E731 0000 0006                00000000   2440+         VL    v3,0(R1)

00002AA0                                           2443+         DS    0H                      E7B1 VRL
00002AA0  E7                                       2444+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002AA1  12                                       2445+         DC    X'12'                    v1, v2
00002AA2  30                                       2446+         DC    X'30'                    v3, reserved
00002AA3  0840                                     2447+         DC    HL2'2112'                reserved, m5, m4, RXB
00002AA5  B1                                       2448+         DC    X'B1'

00002AA6  E710 5030 000E                00002A48   2450+         VST   V1,V1O33          save v1 output
00002AAC  07FB                                     2451+         BR    R11               return
00002AB0                                           2452+RE33     DC    0F                xl16 expected result
00002AB0                                           2453+         DROP  R5
00002AB0  01020303 05060707                        2454          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   result
00002AB8  090A0B0B 0D0E0F0F
00002AC0  01020303 05060707                        2455          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00002AC8  090A0B0B 0D0E0F0F
00002AD0  01020304 05060708                        2456          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00002AD8  090A0B0C 0D0E0F10
                                                   2457
                                                   2458          VRR_C VRL,X'4',X'8',N
00002AE0                                           2459+         DS    0FD
00002AE0                       00002AE0            2460+         USING *,R5              base for test data and test routine
00002AE0  00002B50                                 2461+T34      DC    A(X34)            address of test routine
00002AE4  0022                                     2462+         DC    H'34'             test number
00002AE6  00                                       2463+         DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002AE7  D5                                       2464+         DC    CL1'N'            S or Y = skip cross check
00002AE8  04                                       2465+         DC    X'4'               m4
00002AE9  08                                       2466+         DC    X'8'               m5
00002AEA  E5D9D340 40404040                        2467+         DC    CL8'VRL'          instruction name
00002AF4  00002B88                                 2468+         DC    A(RE34+16)        address of v2 source
00002AF8  00002B98                                 2469+         DC    A(RE34+32)        address of v3 source
00002AFC  00000010                                 2470+         DC    A(16)             result length
00002B00  00002B78                                 2471+REA34    DC    A(RE34)           result address
00002B08  00000000 00000000                        2472+         DS    FD                   gap
00002B10  00000000 00000000                        2473+V1O34    DS    XL16              V1 output
00002B18  00000000 00000000
00002B20  00000000 00000000                        2474+         DS    FD                   gap
00002B28  40404040 40404040                        2475+         DC    CL8' '            was cross check skipped?
00002B30  00000000 00000000                        2476+         DS    FD                   gap
00002B38  00000000 00000000                        2477+XCO34    DS    XL16              Cross check Output
00002B40  00000000 00000000
00002B48  00000000 00000000                        2478+         DS    FD                   gap
                                                   2479+*
00002B50                                           2480+X34      DS    0F
00002B50  E310 5014 0014                00000014   2481+         LGF   R1,V2ADDR         load v2 source
00002B56  E721 0000 0006                00000000   2482+         VL    v2,0(R1)
00002B5C  E310 5018 0014                00000018   2483+         LGF   R1,V3ADDR         load v3 source
00002B62  E731 0000 0006                00000000   2484+         VL    v3,0(R1)

00002B68                                           2487+         DS    0H                     E7B1 VRL
00002B68  E7                                       2488+         DC    X'E7'                  - VECTOR REMAINDER LOGICAL
00002B69  12                                       2489+         DC    X'12'                    v1, v2
00002B6A  30                                       2490+         DC    X'30'                    v3, reserved
00002B6B  0840                                     2491+         DC    HL2'2112'                reserved, m5, m4, RXB
00002B6D  B1                                       2492+         DC    X'B1'

00002B6E  E710 5030 000E                00002B10   2494+         VST   V1,V1O34          save v1 output
00002B74  07FB                                     2495+         BR    R11               return
00002B78                                           2496+RE34     DC    0F                xl16 expected result
00002B78                                           2497+         DROP  R5
00002B78  00000000 00000000                        2498          DC    XL16'0000000000000000 0000000000000000'   result
00002B80  00000000 00000000
00002B88  01020304 05060708                        2499          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
00002B90  090A0B0C 0D0E0F10
00002B98  01020304 05060708                        2500          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00002BA0  090A0B0C 0D0E0F10
                                                   2501
                                                   2502          VRR_C VRL,X'4',X'8',N
00002BA8                                           2503+         DS    0FD
00002BA8                        00002BA8           2504+         USING *,R5              base for test data and test routine
00002BA8  00002C18                                 2505+T35      DC    A(X35)            address of test routine
00002BAC  0023                                     2506+         DC    H'35'             test number
00002BAE  00                                       2507+         DC    X'00'
00002BAF  D5                                       2508+         DC    CL1'N'            S or Y = skip cross check
00002BB0  04                                       2509+         DC    X'4'               m4
00002BB1  08                                       2510+         DC    X'8'               m5
00002BB2  E5D9D340 40404040                        2511+         DC    CL8'VRL'          instruction name
00002BBC  00002C50                                 2512+         DC    A(RE35+16)        address of v2 source
00002BC0  00002C60                                 2513+         DC    A(RE35+32)        address of v3 source
00002BC4  00000010                                 2514+         DC    A(16)             result length
00002BC8  00002C40                                 2515+REA35    DC    A(RE35)           result address
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002BD0  00000000 00000000                        2516+          DS    FD                    gap
00002BD8  00000000 00000000                        2517+V1O35     DS    XL16               V1 output
00002BE0  00000000 00000000
00002BE8  00000000 00000000                        2518+          DS    FD                    gap
00002BF0  40404040 40404040                        2519+          DC    CL8' '             was cross check skipped?
00002BF8  00000000 00000000                        2520+          DS    FD                    gap
00002C00  00000000 00000000                        2521+XCO35     DS    XL16               Cross check Output
00002C08  00000000 00000000
00002C10  00000000 00000000                        2522+          DS    FD                    gap
                                                   2523+*
00002C18                                           2524+X35       DS    0F
00002C18  E310 5014 0014                 00000014  2525+          LGF   R1,V2ADDR          load v2 source
00002C1E  E721 0000 0006                 00000000  2526+          VL    v2,0(R1)
00002C24  E310 5018 0014                 00000018  2527+          LGF   R1,V3ADDR          load v3 source
00002C2A  E731 0000 0006                 00000000  2528+          VL    v3,0(R1)

00002C30                                           2531+          DS    0H                       E7B1 VRL
00002C30  E7                                       2532+          DC    X'E7'                    - VECTOR REMAINDER LOGICAL
00002C31  12                                       2533+          DC    X'12'                     v1, v2
00002C32  30                                       2534+          DC    X'30'                     v3, reserved
00002C33  0840                                     2535+          DC    HL2'2112'                 reserved, m5, m4, RXB
00002C35  B1                                       2536+          DC    X'B1'

00002C36  E710 5030 000E                 00002BD8  2538+          VST   V1,V1O35           save v1 output
00002C3C  07FB                                     2539+          BR    R11                return
00002C40                                           2540+RE35      DC    0F                 xl16 expected result
00002C40                                           2541+          DROP  R5
00002C40  00000000 00000000                        2542           DC    XL16'0000000000000000 0000000000000000'   result
00002C48  00000000 00000000
00002C50  FFFFFFFF FFFFFFFF                        2543           DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00002C58  FFFFFFFF FFFFFFFF
00002C60  00000000 00000000                        2544           DC    XL16'0000000000000000 0000000000000001'   v3
00002C68  00000000 00000001
                                                   2545
                                                   2546           VRR_C VRL,X'4',X'8',N
00002C70                                           2547+          DS    0FD
00002C70                      00002C70             2548+          USING *,R5               base for test data and test routine
00002C70  00002CE0                                 2549+T36       DC    A(X36)             address of test routine
00002C74  0024                                     2550+          DC    H'36'              test number
00002C76  00                                       2551+          DC    X'00'
00002C77  D5                                       2552+          DC    CL1'N'             S or Y = skip cross check
00002C78  04                                       2553+          DC    X'4'                m4
00002C79  08                                       2554+          DC    X'8'                m5
00002C7A  E5D9D340 40404040                        2555+          DC    CL8'VRL'           instruction name
00002C84  00002D18                                 2556+          DC    A(RE36+16)         address of v2 source
00002C88  00002D28                                 2557+          DC    A(RE36+32)         address of v3 source
00002C8C  00000010                                 2558+          DC    A(16)              result length
00002C90  00002D08                                 2559+REA36     DC    A(RE36)            result address
00002C98  00000000 00000000                        2560+          DS    FD                    gap
00002CA0  00000000 00000000                        2561+V1O36     DS    XL16               V1 output
00002CA8  00000000 00000000
00002CB0  00000000 00000000                        2562+          DS    FD                    gap
00002CB8  40404040 40404040                        2563+          DC    CL8' '             was cross check skipped?
00002CC0  00000000 00000000                        2564+          DS    FD                    gap
00002CC8  00000000 00000000                        2565+XCO36     DS    XL16               Cross check Output
00002CD0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002CD8  00000000 00000000                        2566+         DS    FD                    gap
                                                   2567+*
00002CE0                                           2568+X36      DS    0F
00002CE0  E310 5014 0014                 00000014  2569+         LGF   R1,V2ADDR         load v2 source
00002CE6  E721 0000 0006                 00000000  2570+         VL    v2,0(R1)
00002CEC  E310 5018 0014                 00000018  2571+         LGF   R1,V3ADDR         load v3 source
00002CF2  E731 0000 0006                 00000000  2572+         VL    v3,0(R1)

00002CF8                                           2575+         DS    0H                      E7B1 VRL
00002CF8  E7                                       2576+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002CF9  12                                       2577+         DC    X'12'                    v1, v2
00002CFA  30                                       2578+         DC    X'30'                    v3, reserved
00002CFB  0840                                     2579+         DC    HL2'2112'                reserved, m5, m4, RXB
00002CFD  B1                                       2580+         DC    X'B1'

00002CFE  E710 5030 000E                 00002CA0  2582+         VST   V1,V1O36          save v1 output
00002D04  07FB                                     2583+         BR    R11               return
00002D08                                           2584+RE36     DC    0F                xl16 expected result
00002D08                                           2585+         DROP  R5
00002D08  00000000 00000000                        2586          DC    XL16'0000000000000000 0000000000000001'   result
00002D10  00000000 00000001
00002D18  FFFFFFFF FFFFFFFF                        2587          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00002D20  FFFFFFFF FFFFFFFF
00002D28  00000000 00000000                        2588          DC    XL16'0000000000000000 0000000000000002'   v3
00002D30  00000000 00000002
                                                   2589
                                                   2590          VRR_C VRL,X'4',X'8',N
00002D38                                           2591+         DS    0FD
00002D38                      00002D38             2592+         USING *,R5              base for test data and test routine
00002D38  00002DA8                                 2593+T37      DC    A(X37)            address of test routine
00002D3C  0025                                     2594+         DC    H'37'             test number
00002D3E  00                                       2595+         DC    X'00'
00002D3F  D5                                       2596+         DC    CL1'N'            S or Y = skip cross check
00002D40  04                                       2597+         DC    X'4'               m4
00002D41  08                                       2598+         DC    X'8'               m5
00002D42  E5D9D340 40404040                        2599+         DC    CL8'VRL'          instruction name
00002D4C  00002DE0                                 2600+         DC    A(RE37+16)        address of v2 source
00002D50  00002DF0                                 2601+         DC    A(RE37+32)        address of v3 source
00002D54  00000010                                 2602+         DC    A(16)             result length
00002D58  00002DD0                                 2603+REA37    DC    A(RE37)           result address
00002D60  00000000 00000000                        2604+         DS    FD                   gap
00002D68  00000000 00000000                        2605+V1O37    DS    XL16              V1 output
00002D70  00000000 00000000
00002D78  00000000 00000000                        2606+         DS    FD                   gap
00002D80  40404040 40404040                        2607+         DC    CL8' '            was cross check skipped?
00002D88  00000000 00000000                        2608+         DS    FD                   gap
00002D90  00000000 00000000                        2609+XCO37    DS    XL16              Cross check Output
00002D98  00000000 00000000
00002DA0  00000000 00000000                        2610+         DS    FD                   gap
                                                   2611+*
00002DA8                                           2612+X37      DS    0F
00002DA8  E310 5014 0014                 00000014  2613+         LGF   R1,V2ADDR         load v2 source
00002DAE  E721 0000 0006                 00000000  2614+         VL    v2,0(R1)
00002DB4  E310 5018 0014                 00000018  2615+         LGF   R1,V3ADDR         load v3 source
00002DBA  E731 0000 0006                 00000000  2616+         VL    v3,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002DC0                                           2619+         DS    0H                      E7B1 VRL
00002DC0  E7                                       2620+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002DC1  12                                       2621+         DC    X'12'                    v1, v2
00002DC2  30                                       2622+         DC    X'30'                    v3, reserved
00002DC3  0840                                     2623+         DC    HL2'2112'                reserved, m5, m4, RXB
00002DC5  B1                                       2624+         DC    X'B1'

00002DC6  E710 5030 000E                00002D68   2626+         VST   V1,V1O37          save v1 output
00002DCC  07FB                                     2627+         BR    R11               return
00002DD0                                           2628+RE37     DC    0F                xl16 expected result
00002DD0                                           2629+         DROP  R5
00002DD0  00000000 00000000                        2630          DC    XL16'0000000000000000 0000000000000005'   result
00002DD8  00000000 00000005
00002DE0  FFFFFFFF FFFFFFFF                        2631          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00002DE8  FFFFFFFF FFFFFFFF
00002DF0  00000000 00000000                        2632          DC    XL16'0000000000000000 000000000000000A'   v3
00002DF8  00000000 0000000A
                                                   2633
                                                   2634          VRR_C VRL,X'4',X'8',N
00002E00                                           2635+         DS    0FD
00002E00                        00002E00           2636+         USING *,R5              base for test data and test routine
00002E00  00002E70                                 2637+T38      DC    A(X38)            address of test routine
00002E04  0026                                     2638+         DC    H'38'             test number
00002E06  00                                       2639+         DC    X'00'
00002E07  D5                                       2640+         DC    CL1'N'            S or Y = skip cross check
00002E08  04                                       2641+         DC    X'4'               m4
00002E09  08                                       2642+         DC    X'8'               m5
00002E0A  E5D9D340 40404040                        2643+         DC    CL8'VRL'          instruction name
00002E14  00002EA8                                 2644+         DC    A(RE38+16)        address of v2 source
00002E18  00002EB8                                 2645+         DC    A(RE38+32)        address of v3 source
00002E1C  00000010                                 2646+         DC    A(16)             result length
00002E20  00002E98                                 2647+REA38    DC    A(RE38)           result address
00002E28  00000000 00000000                        2648+         DS    FD                   gap
00002E30  00000000 00000000                        2649+V1O38    DS    XL16              V1 output
00002E38  00000000 00000000
00002E40  00000000 00000000                        2650+         DS    FD                   gap
00002E48  40404040 40404040                        2651+         DC    CL8' '            was cross check skipped?
00002E50  00000000 00000000                        2652+         DS    FD                   gap
00002E58  00000000 00000000                        2653+XCO38    DS    XL16              Cross check Output
00002E60  00000000 00000000
00002E68  00000000 00000000                        2654+         DS    FD                   gap
                                                   2655+*
00002E70                                           2656+X38      DS    0F
00002E70  E310 5014 0014                00000014   2657+         LGF   R1,V2ADDR         load v2 source
00002E76  E721 0000 0006                00000000   2658+         VL    v2,0(R1)
00002E7C  E310 5018 0014                00000018   2659+         LGF   R1,V3ADDR         load v3 source
00002E82  E731 0000 0006                00000000   2660+         VL    v3,0(R1)

00002E88                                           2663+         DS    0H                      E7B1 VRL
00002E88  E7                                       2664+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00002E89  12                                       2665+         DC    X'12'                    v1, v2
00002E8A  30                                       2666+         DC    X'30'                    v3, reserved
00002E8B  0840                                     2667+         DC    HL2'2112'                reserved, m5, m4, RXB
00002E8D  B1                                       2668+         DC    X'B1'

00002E8E  E710 5030 000E                00002E30   2670+         VST   V1,V1O38          save v1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002E94  07FB                                     2671+         BR    R11                 return
00002E98                                           2672+RE38     DC    0F                  xl16 expected result
00002E98                                           2673+         DROP  R5
00002E98  00000000 00000000                        2674          DC    XL16'0000000000000000 0000000000000000'   result
00002EA0  00000000 00000000
00002EA8  FFFFFFFF FFFFFFFF                        2675          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00002EB0  FFFFFFFF FFFFFFFF
00002EB8  00000000 00000000                        2676          DC    XL16'0000000000000000 000000000000000F'   v3
00002EC0  00000000 0000000F
                                                   2677
                                                   2678          VRR_C VRL,X'4',X'8',N
00002EC8                                           2679+         DS    0FD
00002EC8                       00002EC8            2680+         USING *,R5                base for test data and test routine
00002EC8  00002F38                                 2681+T39      DC    A(X39)              address of test routine
00002ECC  0027                                     2682+         DC    H'39'               test number
00002ECE  00                                       2683+         DC    X'00'
00002ECF  D5                                       2684+         DC    CL1'N'              S or Y = skip cross check
00002ED0  04                                       2685+         DC    X'4'                 m4
00002ED1  08                                       2686+         DC    X'8'                 m5
00002ED2  E5D9D340 40404040                        2687+         DC    CL8'VRL'            instruction name
00002EDC  00002F70                                 2688+         DC    A(RE39+16)          address of v2 source
00002EE0  00002F80                                 2689+         DC    A(RE39+32)          address of v3 source
00002EE4  00000010                                 2690+         DC    A(16)               result length
00002EE8  00002F60                                 2691+REA39    DC    A(RE39)             result address
00002EF0  00000000 00000000                        2692+         DS    FD                     gap
00002EF8  00000000 00000000                        2693+V1O39    DS    XL16                V1 output
00002F00  00000000 00000000
00002F08  00000000 00000000                        2694+         DS    FD                     gap
00002F10  40404040 40404040                        2695+         DC    CL8' '              was cross check skipped?
00002F18  00000000 00000000                        2696+         DS    FD                     gap
00002F20  00000000 00000000                        2697+XCO39    DS    XL16                Cross check Output
00002F28  00000000 00000000
00002F30  00000000 00000000                        2698+         DS    FD                     gap
                                                   2699+*
00002F38                                           2700+X39      DS    0F
00002F38  E310 5014 0014                00000014   2701+         LGF   R1,V2ADDR           load v2 source
00002F3E  E721 0000 0006                00000000   2702+         VL    v2,0(R1)
00002F44  E310 5018 0014                00000018   2703+         LGF   R1,V3ADDR           load v3 source
00002F4A  E731 0000 0006                00000000   2704+         VL    v3,0(R1)

00002F50                                           2707+         DS    0H                        E7B1 VRL
00002F50  E7                                       2708+         DC    X'E7'                     - VECTOR REMAINDER LOGICAL
00002F51  12                                       2709+         DC    X'12'                      v1, v2
00002F52  30                                       2710+         DC    X'30'                      v3, reserved
00002F53  0840                                     2711+         DC    HL2'2112'                  reserved, m5, m4, RXB
00002F55  B1                                       2712+         DC    X'B1'

00002F56  E710 5030 000E                00002EF8   2714+         VST   V1,V1O39            save v1 output
00002F5C  07FB                                     2715+         BR    R11                 return
00002F60                                           2716+RE39     DC    0F                  xl16 expected result
00002F60                                           2717+         DROP  R5
00002F60  00008101 82028303                        2718          DC    XL16'0000810182028303 840485058606870F'   result
00002F68  84048505 8606870F
00002F70  7FFFFFFF FFFFFFFF                        2719          DC    XL16'7FFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00002F78  FFFFFFFF FFFFFFFF
00002F80  01020304 05060708                        2720          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002F88  090A0B0C 0D0E0F10
                                                   2721
                                                   2722          VRR_C VRL,X'4',X'8',N
00002F90                                           2723+         DS    0FD
00002F90                       00002F90            2724+         USING *,R5               base for test data and test routine
00002F90  00003000                                 2725+T40      DC    A(X40)             address of test routine
00002F94  0028                                     2726+         DC    H'40'              test number
00002F96  00                                       2727+         DC    X'00'
00002F97  D5                                       2728+         DC    CL1'N'             S or Y = skip cross check
00002F98  04                                       2729+         DC    X'4'                m4
00002F99  08                                       2730+         DC    X'8'                m5
00002F9A  E5D9D340 40404040                        2731+         DC    CL8'VRL'           instruction name
00002FA4  00003038                                 2732+         DC    A(RE40+16)         address of v2 source
00002FA8  00003048                                 2733+         DC    A(RE40+32)         address of v3 source
00002FAC  00000010                                 2734+         DC    A(16)              result length
00002FB0  00003028                                 2735+REA40    DC    A(RE40)            result address
00002FB8  00000000 00000000                        2736+         DS    FD                    gap
00002FC0  00000000 00000000                        2737+V1O40    DS    XL16               V1 output
00002FC8  00000000 00000000
00002FD0  00000000 00000000                        2738+         DS    FD                    gap
00002FD8  40404040 40404040                        2739+         DC    CL8' '             was cross check skipped?
00002FE0  00000000 00000000                        2740+         DS    FD                    gap
00002FE8  00000000 00000000                        2741+XCO40    DS    XL16               Cross check Output
00002FF0  00000000 00000000
00002FF8  00000000 00000000                        2742+         DS    FD                    gap
                                                   2743+*
00003000                                           2744+X40      DS    0F
00003000  E310 5014 0014                00000014   2745+         LGF   R1,V2ADDR          load v2 source
00003006  E721 0000 0006                00000000   2746+         VL    v2,0(R1)
0000300C  E310 5018 0014                00000018   2747+         LGF   R1,V3ADDR          load v3 source
00003012  E731 0000 0006                00000000   2748+         VL    v3,0(R1)

00003018                                           2751+         DS    0H                      E7B1 VRL
00003018  E7                                       2752+         DC    X'E7'                   - VECTOR REMAINDER LOGICAL
00003019  12                                       2753+         DC    X'12'                    v1, v2
0000301A  30                                       2754+         DC    X'30'                    v3, reserved
0000301B  0840                                     2755+         DC    HL2'2112'                reserved, m5, m4, RXB
0000301D  B1                                       2756+         DC    X'B1'

0000301E  E710 5030 000E                00002FC0   2758+         VST   V1,V1O40           save v1 output
00003024  07FB                                     2759+         BR    R11                return
00003028                                           2760+RE40     DC    0F                 xl16 expected result
00003028                                           2761+         DROP  R5
00003028  00000000 00000000                        2762          DC    XL16'0000000000000000 02CD81E99B55FC1F'   result
00003030  02CD81E9 9B55FC1F
00003038  7FFFFFFF FFFFFFFF                        2763          DC    XL16'7FFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00003040  FFFFFFFF FFFFFFFF
00003048  00000000 00000000                        2764          DC    XL16'0000000000000000 090A0B0C0D0E0F10'   v3
00003050  090A0B0C 0D0E0F10
                                                   2765
                                                   2766 *-------------------------------------------------------------------
                                                   2767 *   VD    - VECTOR DIVIDE
                                                   2768 *-------------------------------------------------------------------
                                                   2769 *Word    M4:  2, m5= 8 (IDC=1)
                                                   2770
                                                   2771          VRR_C VD,X'2',X'8',N
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003058                                           2772+         DS    0FD
00003058                     00003058              2773+         USING *,R5              base for test data and test routine
00003058  000030C8                                 2774+T41      DC    A(X41)            address of test routine
0000305C  0029                                     2775+         DC    H'41'             test number
0000305E  00                                       2776+         DC    X'00'
0000305F  D5                                       2777+         DC    CL1'N'            S or Y = skip cross check
00003060  02                                       2778+         DC    X'2'               m4
00003061  08                                       2779+         DC    X'8'               m5
00003062  E5C44040 40404040                        2780+         DC    CL8'VD'           instruction name
0000306C  00003100                                 2781+         DC    A(RE41+16)        address of v2 source
00003070  00003110                                 2782+         DC    A(RE41+32)        address of v3 source
00003074  00000010                                 2783+         DC    A(16)             result length
00003078  000030F0                                 2784+REA41    DC    A(RE41)           result address
00003080  00000000 00000000                        2785+         DS    FD                   gap
00003088  00000000 00000000                        2786+V1O41    DS    XL16              V1 output
00003090  00000000 00000000
00003098  00000000 00000000                        2787+         DS    FD                   gap
000030A0  40404040 40404040                        2788+         DC    CL8' '            was cross check skipped?
000030A8  00000000 00000000                        2789+         DS    FD                   gap
000030B0  00000000 00000000                        2790+XCO41    DS    XL16              Cross check Output
000030B8  00000000 00000000
000030C0  00000000 00000000                        2791+         DS    FD                   gap
                                                   2792+*
000030C8                                           2793+X41      DS    0F
000030C8  E310 5014 0014                 00000014  2794+         LGF   R1,V2ADDR         load v2 source
000030CE  E721 0000 0006                 00000000  2795+         VL    v2,0(R1)
000030D4  E310 5018 0014                 00000018  2796+         LGF   R1,V3ADDR         load v3 source
000030DA  E731 0000 0006                 00000000  2797+         VL    v3,0(R1)

000030E0                                           2800+         DS    0H                      E7B2 VD
000030E0  E7                                       2801+         DC    X'E7'                   - VECTOR DIVIDE
000030E1  12                                       2802+         DC    X'12'                    v1, v2
000030E2  30                                       2803+         DC    X'30'                    v3, reserved
000030E3  0820                                     2804+         DC    HL2'2080'                reserved, m5, m4, RXB
000030E5  B2                                       2805+         DC    X'B2'

000030E6  E710 5030 000E                 00003088  2807+         VST   V1,V1O41          save v1 output
000030EC  07FB                                     2808+         BR    R11               return
000030F0                                           2809+RE41     DC    0F                xl16 expected result
000030F0                                           2810+         DROP  R5
000030F0  00000000 00000000                        2811          DC    XL16'00000000 00000000 00000000 00000000'   result
000030F8  00000000 00000000
00003100  00000000 00000000                        2812          DC    XL16'00000000 00000000 00000000 00000000'   v2
00003108  00000000 00000000
00003110  01020304 05060708                        2813          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00003118  090A0B0C 0D0E0F10
                                                   2814
                                                   2815          VRR_C VD,X'2',X'8',N
00003120                                           2816+         DS    0FD
00003120                     00003120              2817+         USING *,R5              base for test data and test routine
00003120  00003190                                 2818+T42      DC    A(X42)            address of test routine
00003124  002A                                     2819+         DC    H'42'             test number
00003126  00                                       2820+         DC    X'00'
00003127  D5                                       2821+         DC    CL1'N'            S or Y = skip cross check
00003128  02                                       2822+         DC    X'2'               m4
00003129  08                                       2823+         DC    X'8'               m5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000312A  E5C44040 40404040                        2824+         DC    CL8'VD'           instruction name
00003134  000031C8                                 2825+         DC    A(RE42+16)        address of v2 source
00003138  000031D8                                 2826+         DC    A(RE42+32)        address of v3 source
0000313C  00000010                                 2827+         DC    A(16)             result length
00003140  000031B8                                 2828+REA42    DC    A(RE42)           result address
00003148  00000000 00000000                        2829+         DS    FD                   gap
00003150  00000000 00000000                        2830+V1O42    DS    XL16              V1 output
00003158  00000000 00000000
00003160  00000000 00000000                        2831+         DS    FD                   gap
00003168  40404040 40404040                        2832+         DC    CL8' '            was cross check skipped?
00003170  00000000 00000000                        2833+         DS    FD                   gap
00003178  00000000 00000000                        2834+XCO42    DS    XL16              Cross check Output
00003180  00000000 00000000
00003188  00000000 00000000                        2835+         DS    FD                   gap
                                                   2836+*
00003190                                           2837+X42      DS    0F
00003190  E310 5014 0014                00000014   2838+         LGF   R1,V2ADDR         load v2 source
00003196  E721 0000 0006                00000000   2839+         VL    v2,0(R1)
0000319C  E310 5018 0014                00000018   2840+         LGF   R1,V3ADDR         load v3 source
000031A2  E731 0000 0006                00000000   2841+         VL    v3,0(R1)

000031A8                                           2844+         DS    0H                      E7B2 VD
000031A8  E7                                       2845+         DC    X'E7'                   - VECTOR DIVIDE
000031A9  12                                       2846+         DC    X'12'                    v1, v2
000031AA  30                                       2847+         DC    X'30'                    v3, reserved
000031AB  0820                                     2848+         DC    HL2'2080'                reserved, m5, m4, RXB
000031AD  B2                                       2849+         DC    X'B2'

000031AE  E710 5030 000E                00003150   2851+         VST   V1,V1O42          save v1 output
000031B4  07FB                                     2852+         BR    R11               return
000031B8                                           2853+RE42     DC    0F                xl16 expected result
000031B8                                           2854+         DROP  R5
000031B8  00000000 00000000                        2855          DC    XL16'00000000 00000000 00000000 00000000'   result
000031C0  00000000 00000000
000031C8  01020303 05060707                        2856          DC    XL16'01020303 05060707 090A0B0B 0D0E0F0F'   v2
000031D0  090A0B0B 0D0E0F0F
000031D8  01020304 05060708                        2857          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000031E0  090A0B0C 0D0E0F10
                                                   2858
                                                   2859          VRR_C VD,X'2',X'8',N
000031E8                                           2860+         DS    0FD
000031E8                        000031E8           2861+         USING *,R5              base for test data and test routine
000031E8  00003258                                 2862+T43      DC    A(X43)            address of test routine
000031EC  002B                                     2863+         DC    H'43'             test number
000031EE  00                                       2864+         DC    X'00'
000031EF  D5                                       2865+         DC    CL1'N'            S or Y = skip cross check
000031F0  02                                       2866+         DC    X'2'               m4
000031F1  08                                       2867+         DC    X'8'               m5
000031F2  E5C44040 40404040                        2868+         DC    CL8'VD'           instruction name
000031FC  00003290                                 2869+         DC    A(RE43+16)        address of v2 source
00003200  000032A0                                 2870+         DC    A(RE43+32)        address of v3 source
00003204  00000010                                 2871+         DC    A(16)             result length
00003208  00003280                                 2872+REA43    DC    A(RE43)           result address
00003210  00000000 00000000                        2873+         DS    FD                   gap
00003218  00000000 00000000                        2874+V1O43    DS    XL16              V1 output
00003220  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003228  00000000 00000000                        2875+         DS    FD                    gap
00003230  40404040 40404040                        2876+         DC    CL8' '             was cross check skipped?
00003238  00000000 00000000                        2877+         DS    FD                    gap
00003240  00000000 00000000                        2878+XC043    DS    XL16               Cross check Output
00003248  00000000 00000000
00003250  00000000 00000000                        2879+         DS    FD                    gap
                                                   2880+*
00003258                                           2881+X43      DS    0F
00003258  E310 5014 0014                 00000014  2882+         LGF   R1,V2ADDR          load v2 source
0000325E  E721 0000 0006                 00000000  2883+         VL    v2,0(R1)
00003264  E310 5018 0014                 00000018  2884+         LGF   R1,V3ADDR          load v3 source
0000326A  E731 0000 0006                 00000000  2885+         VL    v3,0(R1)

00003270                                           2888+         DS    0H                      E7B2 VD
00003270  E7                                       2889+         DC    X'E7'                   - VECTOR DIVIDE
00003271  12                                       2890+         DC    X'12'                    v1, v2
00003272  30                                       2891+         DC    X'30'                    v3, reserved
00003273  0820                                     2892+         DC    HL2'2080'                reserved, m5, m4, RXB
00003275  B2                                       2893+         DC    X'B2'

00003276  E710 5030 000E                 00003218  2895+         VST   V1,V1O43           save v1 output
0000327C  07FB                                     2896+         BR    R11                return
00003280                                           2897+RE43     DC    0F                 xl16 expected result
00003280                                           2898+         DROP  R5
00003280  00000001 00000001                        2899          DC    XL16'00000001 00000001 00000001 00000001'   result
00003288  00000001 00000001
00003290  01020304 05060708                        2900          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v2
00003298  090A0B0C 0D0E0F10
000032A0  01020304 05060708                        2901          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000032A8  090A0B0C 0D0E0F10
                                                   2902
                                                   2903          VRR_C VD,X'2',X'8',N
000032B0                                           2904+         DS    0FD
000032B0                      000032B0             2905+         USING *,R5               base for test data and test routine
000032B0  00003320                                 2906+T44      DC    A(X44)             address of test routine
000032B4  002C                                     2907+         DC    H'44'              test number
000032B6  00                                       2908+         DC    X'00'
000032B7  D5                                       2909+         DC    CL1'N'             S or Y = skip cross check
000032B8  02                                       2910+         DC    X'2'                m4
000032B9  08                                       2911+         DC    X'8'                m5
000032BA  E5C44040 40404040                        2912+         DC    CL8'VD'            instruction name
000032C4  00003358                                 2913+         DC    A(RE44+16)         address of v2 source
000032C8  00003368                                 2914+         DC    A(RE44+32)         address of v3 source
000032CC  00000010                                 2915+         DC    A(16)              result length
000032D0  00003348                                 2916+REA44    DC    A(RE44)            result address
000032D8  00000000 00000000                        2917+         DS    FD                    gap
000032E0  00000000 00000000                        2918+V1O44    DS    XL16               V1 output
000032E8  00000000 00000000
000032F0  00000000 00000000                        2919+         DS    FD                    gap
000032F8  40404040 40404040                        2920+         DC    CL8' '             was cross check skipped?
00003300  00000000 00000000                        2921+         DS    FD                    gap
00003308  00000000 00000000                        2922+XC044    DS    XL16               Cross check Output
00003310  00000000 00000000
00003318  00000000 00000000                        2923+         DS    FD                    gap
                                                   2924+*
00003320                                           2925+X44      DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00003320  E310 5014 0014                00000014  2926+         LGF   R1,V2ADDR         load v2 source
00003326  E721 0000 0006                00000000  2927+         VL    v2,0(R1)
0000332C  E310 5018 0014                00000018  2928+         LGF   R1,V3ADDR         load v3 source
00003332  E731 0000 0006                00000000  2929+         VL    v3,0(R1)

00003338                                          2932+         DS    0H                      E7B2 VD
00003338  E7                                      2933+         DC    X'E7'                   - VECTOR DIVIDE
00003339  12                                      2934+         DC    X'12'                    v1, v2
0000333A  30                                      2935+         DC    X'30'                    v3, reserved
0000333B  0820                                    2936+         DC    HL2'2080'                reserved, m5, m4, RXB
0000333D  B2                                      2937+         DC    X'B2'

0000333E  E710 5030 000E                000032E0  2939+         VST   V1,V1O44          save v1 output
00003344  07FB                                    2940+         BR    R11               return
00003348                                          2941+RE44     DC    0F                xl16 expected result
00003348                                          2942+         DROP  R5
00003348  7FFFFFFF 3FFFFFFF                       2943          DC    XL16'7FFFFFFF 3FFFFFFF 0CCCCCCC 08888888'   result
00003350  0CCCCCCC 08888888
00003358  7FFFFFFF 7FFFFFFF                       2944          DC    XL16'7FFFFFFF 7FFFFFFF 7FFFFFFF 7FFFFFFF'   v2
00003360  7FFFFFFF 7FFFFFFF
00003368  00000001 00000002                       2945          DC    XL16'00000001 00000002 0000000A 0000000F'   v3
00003370  0000000A 0000000F
                                                  2946
                                                  2947          VRR_C VD,X'2',X'8',N
00003378                                          2948+         DS    0FD
00003378                        00003378          2949+         USING *,R5              base for test data and test routine
00003378  000033E8                                2950+T45      DC    A(X45)            address of test routine
0000337C  002D                                    2951+         DC    H'45'             test number
0000337E  00                                      2952+         DC    X'00'
0000337F  D5                                      2953+         DC    CL1'N'            S or Y = skip cross check
00003380  02                                      2954+         DC    X'2'               m4
00003381  08                                      2955+         DC    X'8'               m5
00003382  E5C44040 40404040                       2956+         DC    CL8'VD'           instruction name
0000338C  00003420                                2957+         DC    A(RE45+16)        address of v2 source
00003390  00003430                                2958+         DC    A(RE45+32)        address of v3 source
00003394  00000010                                2959+         DC    A(16)             result length
00003398  00003410                                2960+REA45    DC    A(RE45)           result address
000033A0  00000000 00000000                       2961+         DS    FD                   gap
000033A8  00000000 00000000                       2962+V1O45    DS    XL16              V1 output
000033B0  00000000 00000000
000033B8  00000000 00000000                       2963+         DS    FD                   gap
000033C0  40404040 40404040                       2964+         DC    CL8' '            was cross check skipped?
000033C8  00000000 00000000                       2965+         DS    FD                   gap
000033D0  00000000 00000000                       2966+XCO45    DS    XL16              Cross check Output
000033D8  00000000 00000000
000033E0  00000000 00000000                       2967+         DS    FD                   gap
                                                  2968+*
000033E8                                          2969+X45      DS    0F
000033E8  E310 5014 0014                00000014  2970+         LGF   R1,V2ADDR         load v2 source
000033EE  E721 0000 0006                00000000  2971+         VL    v2,0(R1)
000033F4  E310 5018 0014                00000018  2972+         LGF   R1,V3ADDR         load v3 source
000033FA  E731 0000 0006                00000000  2973+         VL    v3,0(R1)

00003400                                          2976+         DS    0H                      E7B2 VD
00003400  E7                                      2977+         DC    X'E7'                   - VECTOR DIVIDE
00003401  12                                      2978+         DC    X'12'                    v1, v2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003402  30                                       2979+         DC    X'30'                    v3, reserved
00003403  0820                                     2980+         DC    HL2'2080'                reserved, m5, m4, RXB
00003405  B2                                       2981+         DC    X'B2'

00003406  E710 5030 000E                 000033A8  2983+         VST   V1,V1O45          save v1 output
0000340C  07FB                                     2984+         BR    R11               return
00003410                                           2985+RE45     DC    0F                xl16 expected result
00003410                                           2986+         DROP  R5
00003410  0000007F 00000019                        2987          DC    XL16'0000007F 00000019 0000000E 00000009'   result
00003418  0000000E 00000009
00003420  7FFFFFFF 7FFFFFFF                        2988          DC    XL16'7FFFFFFF 7FFFFFFF 7FFFFFFF 7FFFFFFF'   v2
00003428  7FFFFFFF 7FFFFFFF
00003430  01020304 05060708                        2989          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00003438  090A0B0C 0D0E0F10
                                                   2990
                                                   2991          VRR_C VD,X'2',X'8',N
00003440                                           2992+         DS    0FD
00003440                       00003440            2993+         USING *,R5              base for test data and test routine
00003440  000034B0                                 2994+T46      DC    A(X46)            address of test routine
00003444  002E                                     2995+         DC    H'46'             test number
00003446  00                                       2996+         DC    X'00'
00003447  D5                                       2997+         DC    CL1'N'            S or Y = skip cross check
00003448  02                                       2998+         DC    X'2'               m4
00003449  08                                       2999+         DC    X'8'               m5
0000344A  E5C44040 40404040                        3000+         DC    CL8'VD'           instruction name
00003454  000034E8                                 3001+         DC    A(RE46+16)        address of v2 source
00003458  000034F8                                 3002+         DC    A(RE46+32)        address of v3 source
0000345C  00000010                                 3003+         DC    A(16)             result length
00003460  000034D8                                 3004+REA46    DC    A(RE46)           result address
00003468  00000000 00000000                        3005+         DS    FD                   gap
00003470  00000000 00000000                        3006+V1O46    DS    XL16              V1 output
00003478  00000000 00000000
00003480  00000000 00000000                        3007+         DS    FD                   gap
00003488  40404040 40404040                        3008+         DC    CL8' '            was cross check skipped?
00003490  00000000 00000000                        3009+         DS    FD                   gap
00003498  00000000 00000000                        3010+XCO46    DS    XL16              Cross check Output
000034A0  00000000 00000000
000034A8  00000000 00000000                        3011+         DS    FD                   gap
                                                   3012+*
000034B0                                           3013+X46      DS    0F
000034B0  E310 5014 0014                 00000014  3014+         LGF   R1,V2ADDR         load v2 source
000034B6  E721 0000 0006                 00000000  3015+         VL    v2,0(R1)
000034BC  E310 5018 0014                 00000018  3016+         LGF   R1,V3ADDR         load v3 source
000034C2  E731 0000 0006                 00000000  3017+         VL    v3,0(R1)

000034C8                                           3020+         DS    0H                     E7B2 VD
000034C8  E7                                       3021+         DC    X'E7'                  - VECTOR DIVIDE
000034C9  12                                       3022+         DC    X'12'                   v1, v2
000034CA  30                                       3023+         DC    X'30'                   v3, reserved
000034CB  0820                                     3024+         DC    HL2'2080'               reserved, m5, m4, RXB
000034CD  B2                                       3025+         DC    X'B2'

000034CE  E710 5030 000E                 00003470  3027+         VST   V1,V1O46          save v1 output
000034D4  07FB                                     3028+         BR    R11               return
000034D8                                           3029+RE46     DC    0F                xl16 expected result
000034D8                                           3030+         DROP  R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000034D8  FFFFFFFF 00000000                        3031          DC    XL16'FFFFFFFF 00000000 00000000 00000000'   result
000034E0  00000000 00000000
000034E8  FFFFFFFF FFFFFFFF                        3032          DC    XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'   v2
000034F0  FFFFFFFF FFFFFFFF
000034F8  00000001 00000002                        3033          DC    XL16'00000001 00000002 0000000A 0000000F'   v3
00003500  0000000A 0000000F
                                                   3034
                                                   3035          VRR_C VD,X'2',X'8',N
00003508                                           3036+         DS    0FD
00003508                      00003508             3037+         USING *,R5              base for test data and test routine
00003508  00003578                                 3038+T47      DC    A(X47)            address of test routine
0000350C  002F                                     3039+         DC    H'47'             test number
0000350E  00                                       3040+         DC    X'00'
0000350F  D5                                       3041+         DC    CL1'N'            S or Y = skip cross check
00003510  02                                       3042+         DC    X'2'               m4
00003511  08                                       3043+         DC    X'8'               m5
00003512  E5C44040 40404040                        3044+         DC    CL8'VD'           instruction name
0000351C  000035B0                                 3045+         DC    A(RE47+16)        address of v2 source
00003520  000035C0                                 3046+         DC    A(RE47+32)        address of v3 source
00003524  00000010                                 3047+         DC    A(16)             result length
00003528  000035A0                                 3048+REA47    DC    A(RE47)           result address
00003530  00000000 00000000                        3049+         DS    FD                   gap
00003538  00000000 00000000                        3050+V1O47    DS    XL16              V1 output
00003540  00000000 00000000
00003548  00000000 00000000                        3051+         DS    FD                   gap
00003550  40404040 40404040                        3052+         DC    CL8' '            was cross check skipped?
00003558  00000000 00000000                        3053+         DS    FD                   gap
00003560  00000000 00000000                        3054+XCO47    DS    XL16            Cross check Output
00003568  00000000 00000000
00003570  00000000 00000000                        3055+         DS    FD                   gap
                                                   3056+*
00003578                                           3057+X47      DS    0F
00003578  E310 5014 0014              00000014     3058+         LGF   R1,V2ADDR         load v2 source
0000357E  E721 0000 0006              00000000     3059+         VL    v2,0(R1)
00003584  E310 5018 0014              00000018     3060+         LGF   R1,V3ADDR         load v3 source
0000358A  E731 0000 0006              00000000     3061+         VL    v3,0(R1)

00003590                                           3064+         DS    0H                     E7B2 VD
00003590  E7                                       3065+         DC    X'E7'                  - VECTOR DIVIDE
00003591  12                                       3066+         DC    X'12'                   v1, v2
00003592  30                                       3067+         DC    X'30'                   v3, reserved
00003593  0820                                     3068+         DC    HL2'2080'               reserved, m5, m4, RXB
00003595  B2                                       3069+         DC    X'B2'

00003596  E710 5030 000E              00003538     3071+         VST   V1,V1O47          save v1 output
0000359C  07FB                                     3072+         BR    R11               return
000035A0                                           3073+RE47     DC    0F                xl16 expected result
000035A0                                           3074+         DROP  R5
000035A0  FFFFFF81 FFFFFFE7                        3075          DC    XL16'FFFFFF81 FFFFFFE7 FFFFFFF2 FFFFFFF7'   result
000035A8  FFFFFFF2 FFFFFFF7
000035B0  80000000 80000000                        3076          DC    XL16'80000000 80000000 80000000 80000000'   v2
000035B8  80000000 80000000
000035C0  01020304 05060708                        3077          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000035C8  090A0B0C 0D0E0F10
                                                   3078
                                                   3079          VRR_C VD,X'2',X'8',N
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000035D0                                           3080+         DS    0FD
000035D0                     000035D0              3081+         USING *,R5              base for test data and test routine
000035D0  00003640                                 3082+T48      DC    A(X48)            address of test routine
000035D4  0030                                     3083+         DC    H'48'             test number
000035D6  00                                       3084+         DC    X'00'
000035D7  D5                                       3085+         DC    CL1'N'            S or Y = skip cross check
000035D8  02                                       3086+         DC    X'2'               m4
000035D9  08                                       3087+         DC    X'8'               m5
000035DA  E5C44040 40404040                        3088+         DC    CL8'VD'           instruction name
000035E4  00003678                                 3089+         DC    A(RE48+16)        address of v2 source
000035E8  00003688                                 3090+         DC    A(RE48+32)        address of v3 source
000035EC  00000010                                 3091+         DC    A(16)             result length
000035F0  00003668                                 3092+REA48    DC    A(RE48)           result address
000035F8  00000000 00000000                        3093+         DS    FD                   gap
00003600  00000000 00000000                        3094+V1O48    DS    XL16              V1 output
00003608  00000000 00000000
00003610  00000000 00000000                        3095+         DS    FD                   gap
00003618  40404040 40404040                        3096+         DC    CL8' '            was cross check skipped?
00003620  00000000 00000000                        3097+         DS    FD                   gap
00003628  00000000 00000000                        3098+XCO48    DS    XL16              Cross check Output
00003630  00000000 00000000
00003638  00000000 00000000                        3099+         DS    FD                   gap
                                                   3100+*
00003640                                           3101+X48      DS    0F
00003640  E310 5014 0014                00000014   3102+         LGF   R1,V2ADDR         load v2 source
00003646  E721 0000 0006                00000000   3103+         VL    v2,0(R1)
0000364C  E310 5018 0014                00000018   3104+         LGF   R1,V3ADDR         load v3 source
00003652  E731 0000 0006                00000000   3105+         VL    v3,0(R1)

00003658                                           3108+         DS    0H                      E7B2 VD
00003658  E7                                       3109+         DC    X'E7'                   - VECTOR DIVIDE
00003659  12                                       3110+         DC    X'12'                    v1, v2
0000365A  30                                       3111+         DC    X'30'                    v3, reserved
0000365B  0820                                     3112+         DC    HL2'2080'                reserved, m5, m4, RXB
0000365D  B2                                       3113+         DC    X'B2'

0000365E  E710 5030 000E                00003600   3115+         VST   V1,V1O48          save v1 output
00003664  07FB                                     3116+         BR    R11               return
00003668                                           3117+RE48     DC    0F                xl16 expected result
00003668                                           3118+         DROP  R5
00003668  7FFFFFFF 3FFFFFFF                        3119          DC    XL16'7FFFFFFF 3FFFFFFF 0CCCCCCC 08888888'   result
00003670  0CCCCCCC 08888888
00003678  80000001 80000001                        3120          DC    XL16'80000001 80000001 80000001 80000001'   v2
00003680  80000001 80000001
00003688  FFFFFFFF FFFFFFFE                        3121          DC    XL16'FFFFFFFF FFFFFFFE FFFFFFF6 FFFFFFF1'   v3
00003690  FFFFFFF6 FFFFFFF1
                                                   3122
                                                   3123          VRR_C VD,X'2',X'8',N
00003698                                           3124+         DS    0FD
00003698                     00003698              3125+         USING *,R5              base for test data and test routine
00003698  00003708                                 3126+T49      DC    A(X49)            address of test routine
0000369C  0031                                     3127+         DC    H'49'             test number
0000369E  00                                       3128+         DC    X'00'
0000369F  D5                                       3129+         DC    CL1'N'            S or Y = skip cross check
000036A0  02                                       3130+         DC    X'2'               m4
000036A1  08                                       3131+         DC    X'8'               m5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000036A2  E5C44040 40404040                        3132+         DC    CL8'VD'           instruction name
000036AC  00003740                                 3133+         DC    A(RE49+16)        address of v2 source
000036B0  00003750                                 3134+         DC    A(RE49+32)        address of v3 source
000036B4  00000010                                 3135+         DC    A(16)             result length
000036B8  00003730                                 3136+REA49    DC    A(RE49)           result address
000036C0  00000000 00000000                        3137+         DS    FD                   gap
000036C8  00000000 00000000                        3138+V1O49    DS    XL16              V1 output
000036D0  00000000 00000000
000036D8  00000000 00000000                        3139+         DS    FD                   gap
000036E0  40404040 40404040                        3140+         DC    CL8' '            was cross check skipped?
000036E8  00000000 00000000                        3141+         DS    FD                   gap
000036F0  00000000 00000000                        3142+XCO49    DS    XL16              Cross check Output
000036F8  00000000 00000000
00003700  00000000 00000000                        3143+         DS    FD                   gap
                                                   3144+*
00003708                                           3145+X49      DS    0F
00003708  E310 5014 0014                 00000014  3146+         LGF   R1,V2ADDR         load v2 source
0000370E  E721 0000 0006                 00000000  3147+         VL    v2,0(R1)
00003714  E310 5018 0014                 00000018  3148+         LGF   R1,V3ADDR         load v3 source
0000371A  E731 0000 0006                 00000000  3149+         VL    v3,0(R1)

00003720                                           3152+         DS    0H                     E7B2 VD
00003720  E7                                       3153+         DC    X'E7'                  - VECTOR DIVIDE
00003721  12                                       3154+         DC    X'12'                   v1, v2
00003722  30                                       3155+         DC    X'30'                   v3, reserved
00003723  0820                                     3156+         DC    HL2'2080'               reserved, m5, m4, RXB
00003725  B2                                       3157+         DC    X'B2'

00003726  E710 5030 000E                 000036C8  3159+         VST   V1,V1O49          save v1 output
0000372C  07FB                                     3160+         BR    R11               return
00003730                                           3161+RE49     DC    0F                xl16 expected result
00003730                                           3162+         DROP  R5
00003730  00000001 00000001                        3163          DC    XL16'00000001 00000001 00000001 00000001'   result
00003738  00000001 00000001
00003740  80000000 80000000                        3164          DC    XL16'80000000 80000000 80000000 80000000'   v2
00003748  80000000 80000000
00003750  81020304 85060708                        3165          DC    XL16'81020304 85060708 890A0B0C 8D0E0F10'   v3
00003758  890A0B0C 8D0E0F10
                                                   3166
                                                   3167 * skip xc
                                                   3168          VRR_C VD,X'2',X'8',S
00003760                                           3169+         DS    0FD
00003760                      00003760             3170+         USING *,R5              base for test data and test routine
00003760  000037D0                                 3171+T50      DC    A(X50)            address of test routine
00003764  0032                                     3172+         DC    H'50'             test number
00003766  00                                       3173+         DC    X'00'
00003767  E2                                       3174+         DC    CL1'S'            S or Y = skip cross check
00003768  02                                       3175+         DC    X'2'               m4
00003769  08                                       3176+         DC    X'8'               m5
0000376A  E5C44040 40404040                        3177+         DC    CL8'VD'           instruction name
00003774  00003808                                 3178+         DC    A(RE50+16)        address of v2 source
00003778  00003818                                 3179+         DC    A(RE50+32)        address of v3 source
0000377C  00000010                                 3180+         DC    A(16)             result length
00003780  000037F8                                 3181+REA50    DC    A(RE50)           result address
00003788  00000000 00000000                        3182+         DS    FD                   gap
00003790  00000000 00000000                        3183+V1O50    DS    XL16              V1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003798  00000000 00000000
000037A0  00000000 00000000                         3184+         DS    FD                   gap
000037A8  40404040 40404040                         3185+         DC    CL8' '            was cross check skipped?
000037B0  00000000 00000000                         3186+         DS    FD                   gap
000037B8  00000000 00000000                         3187+XCO50    DS    XL16              Cross check Output
000037C0  00000000 00000000
000037C8  00000000 00000000                         3188+         DS    FD                   gap
                                                    3189+*
000037D0                                            3190+X50      DS    0F
000037D0  E310 5014 0014                 00000014   3191+         LGF   R1,V2ADDR         load v2 source
000037D6  E721 0000 0006                 00000000   3192+         VL    v2,0(R1)
000037DC  E310 5018 0014                 00000018   3193+         LGF   R1,V3ADDR         load v3 source
000037E2  E731 0000 0006                 00000000   3194+         VL    v3,0(R1)

000037E8                                            3197+         DS    0H                     E7B2 VD
000037E8  E7                                        3198+         DC    X'E7'                  - VECTOR DIVIDE
000037E9  12                                        3199+         DC    X'12'                   v1, v2
000037EA  30                                        3200+         DC    X'30'                   v3, reserved
000037EB  0820                                      3201+         DC    HL2'2080'               reserved, m5, m4, RXB
000037ED  B2                                        3202+         DC    X'B2'

000037EE  E710 5030 000E                 00003790   3204+         VST   V1,V1O50          save v1 output
000037F4  07FB                                      3205+         BR    R11               return
000037F8                                            3206+RE50     DC    0F                xl16 expected result
000037F8                                            3207+         DROP  R5
000037F8  00000000 40000000                         3208          DC    XL16'00000000 40000000 0CCCCCCC 08888888'   result
00003800  0CCCCCCC 08888888
00003808  80000000 80000000                         3209          DC    XL16'80000000 80000000 80000000 80000000'   v2
00003810  80000000 80000000
00003818  FFFFFFFF FFFFFFFE                         3210          DC    XL16'FFFFFFFF FFFFFFFE FFFFFFF6 FFFFFFF1'   v3
00003820  FFFFFFF6 FFFFFFF1
                                                    3211
                                                    3212 *Doubleword    M4:  3, m5= 8 (IDC=1)
                                                    3213
                                                    3214          VRR_C VD,X'3',X'8',N
00003828                                            3215+         DS    0FD
00003828                         00003828           3216+         USING *,R5              base for test data and test routine
00003828  00003898                                  3217+T51      DC    A(X51)            address of test routine
0000382C  0033                                      3218+         DC    H'51'             test number
0000382E  00                                        3219+         DC    X'00'
0000382F  D5                                        3220+         DC    CL1'N'            S or Y = skip cross check
00003830  03                                        3221+         DC    X'3'               m4
00003831  08                                        3222+         DC    X'8'               m5
00003832  E5C44040 40404040                         3223+         DC    CL8'VD'           instruction name
0000383C  000038D0                                  3224+         DC    A(RE51+16)        address of v2 source
00003840  000038E0                                  3225+         DC    A(RE51+32)        address of v3 source
00003844  00000010                                  3226+         DC    A(16)             result length
00003848  000038C0                                  3227+REA51    DC    A(RE51)           result address
00003850  00000000 00000000                         3228+         DS    FD                   gap
00003858  00000000 00000000                         3229+V1O51    DS    XL16              V1 output
00003860  00000000 00000000
00003868  00000000 00000000                         3230+         DS    FD                   gap
00003870  40404040 40404040                         3231+         DC    CL8' '            was cross check skipped?
00003878  00000000 00000000                         3232+         DS    FD                   gap
00003880  00000000 00000000                         3233+XCO51    DS    XL16              Cross check Output
00003888  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003890  00000000 00000000                        3234+         DS    FD                    gap
                                                   3235+*
00003898                                           3236+X51      DS    0F
00003898  E310 5014 0014                 00000014  3237+         LGF   R1,V2ADDR         load v2 source
0000389E  E721 0000 0006                 00000000  3238+         VL    v2,0(R1)
000038A4  E310 5018 0014                 00000018  3239+         LGF   R1,V3ADDR         load v3 source
000038AA  E731 0000 0006                 00000000  3240+         VL    v3,0(R1)

000038B0                                           3243+         DS    0H                      E7B2 VD
000038B0  E7                                       3244+         DC    X'E7'                   - VECTOR DIVIDE
000038B1  12                                       3245+         DC    X'12'                    v1, v2
000038B2  30                                       3246+         DC    X'30'                    v3, reserved
000038B3  0830                                     3247+         DC    HL2'2096'                reserved, m5, m4, RXB
000038B5  B2                                       3248+         DC    X'B2'

000038B6  E710 5030 000E                 00003858  3250+         VST   V1,V1O51          save v1 output
000038BC  07FB                                     3251+         BR    R11               return
000038C0                                           3252+RE51     DC    0F                xl16 expected result
000038C0                                           3253+         DROP  R5
000038C0  00000000 00000000                        3254          DC    XL16'0000000000000000 0000000000000000'   result
000038C8  00000000 00000000
000038D0  00000000 00000000                        3255          DC    XL16'0000000000000000 0000000000000000'   v2
000038D8  00000000 00000000
000038E0  01020304 05060708                        3256          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000038E8  090A0B0C 0D0E0F10
                                                   3257
                                                   3258          VRR_C VD,X'3',X'8',N
000038F0                                           3259+         DS    0FD
000038F0                      000038F0             3260+         USING *,R5              base for test data and test routine
000038F0  00003960                                 3261+T52      DC    A(X52)            address of test routine
000038F4  0034                                     3262+         DC    H'52'             test number
000038F6  00                                       3263+         DC    X'00'
000038F7  D5                                       3264+         DC    CL1'N'            S or Y = skip cross check
000038F8  03                                       3265+         DC    X'3'               m4
000038F9  08                                       3266+         DC    X'8'               m5
000038FA  E5C44040 40404040                        3267+         DC    CL8'VD'           instruction name
00003904  00003998                                 3268+         DC    A(RE52+16)        address of v2 source
00003908  000039A8                                 3269+         DC    A(RE52+32)        address of v3 source
0000390C  00000010                                 3270+         DC    A(16)             result length
00003910  00003988                                 3271+REA52    DC    A(RE52)           result address
00003918  00000000 00000000                        3272+         DS    FD                   gap
00003920  00000000 00000000                        3273+V1O52    DS    XL16              V1 output
00003928  00000000 00000000
00003930  00000000 00000000                        3274+         DS    FD                   gap
00003938  40404040 40404040                        3275+         DC    CL8' '            was cross check skipped?
00003940  00000000 00000000                        3276+         DS    FD                   gap
00003948  00000000 00000000                        3277+XCO52    DS    XL16              Cross check Output
00003950  00000000 00000000
00003958  00000000 00000000                        3278+         DS    FD                   gap
                                                   3279+*
00003960                                           3280+X52      DS    0F
00003960  E310 5014 0014                 00000014  3281+         LGF   R1,V2ADDR         load v2 source
00003966  E721 0000 0006                 00000000  3282+         VL    v2,0(R1)
0000396C  E310 5018 0014                 00000018  3283+         LGF   R1,V3ADDR         load v3 source
00003972  E731 0000 0006                 00000000  3284+         VL    v3,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003978                                           3287+         DS    0H                      E7B2 VD
00003978  E7                                       3288+         DC    X'E7'                   - VECTOR DIVIDE
00003979  12                                       3289+         DC    X'12'                    v1, v2
0000397A  30                                       3290+         DC    X'30'                    v3, reserved
0000397B  0830                                     3291+         DC    HL2'2096'                reserved, m5, m4, RXB
0000397D  B2                                       3292+         DC    X'B2'

0000397E  E710 5030 000E                00003920   3294+         VST   V1,V1O52          save v1 output
00003984  07FB                                     3295+         BR    R11               return
00003988                                           3296+RE52     DC    0F                xl16 expected result
00003988                                           3297+         DROP  R5
00003988  00000000 00000000                        3298          DC    XL16'0000000000000000 0000000000000000'   result
00003990  00000000 00000000
00003998  01020303 05060707                        3299          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
000039A0  090A0B0B 0D0E0F0F
000039A8  01020304 05060708                        3300          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000039B0  090A0B0C 0D0E0F10
                                                   3301
                                                   3302          VRR_C VD,X'3',X'8',N
000039B8                                           3303+         DS    0FD
000039B8                        000039B8           3304+         USING *,R5              base for test data and test routine
000039B8  00003A28                                 3305+T53      DC    A(X53)            address of test routine
000039BC  0035                                     3306+         DC    H'53'             test number
000039BE  00                                       3307+         DC    X'00'
000039BF  D5                                       3308+         DC    CL1'N'            S or Y = skip cross check
000039C0  03                                       3309+         DC    X'3'               m4
000039C1  08                                       3310+         DC    X'8'               m5
000039C2  E5C44040 40404040                        3311+         DC    CL8'VD'           instruction name
000039CC  00003A60                                 3312+         DC    A(RE53+16)        address of v2 source
000039D0  00003A70                                 3313+         DC    A(RE53+32)        address of v3 source
000039D4  00000010                                 3314+         DC    A(16)             result length
000039D8  00003A50                                 3315+REA53    DC    A(RE53)           result address
000039E0  00000000 00000000                        3316+         DS    FD                   gap
000039E8  00000000 00000000                        3317+V1O53    DS    XL16              V1 output
000039F0  00000000 00000000
000039F8  00000000 00000000                        3318+         DS    FD                   gap
00003A00  40404040 40404040                        3319+         DC    CL8' '            was cross check skipped?
00003A08  00000000 00000000                        3320+         DS    FD                   gap
00003A10  00000000 00000000                        3321+XCO53    DS    XL16              Cross check Output
00003A18  00000000 00000000
00003A20  00000000 00000000                        3322+         DS    FD                   gap
                                                   3323+*
00003A28                                           3324+X53      DS    0F
00003A28  E310 5014 0014                00000014   3325+         LGF   R1,V2ADDR         load v2 source
00003A2E  E721 0000 0006                00000000   3326+         VL    v2,0(R1)
00003A34  E310 5018 0014                00000018   3327+         LGF   R1,V3ADDR         load v3 source
00003A3A  E731 0000 0006                00000000   3328+         VL    v3,0(R1)

00003A40                                           3331+         DS    0H                      E7B2 VD
00003A40  E7                                       3332+         DC    X'E7'                   - VECTOR DIVIDE
00003A41  12                                       3333+         DC    X'12'                    v1, v2
00003A42  30                                       3334+         DC    X'30'                    v3, reserved
00003A43  0830                                     3335+         DC    HL2'2096'                reserved, m5, m4, RXB
00003A45  B2                                       3336+         DC    X'B2'

00003A46  E710 5030 000E                000039E8   3338+         VST   V1,V1O53          save v1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003A4C  07FB                                     3339+         BR    R11                 return
00003A50                                           3340+RE53     DC    0F                  xl16 expected result
00003A50                                           3341+         DROP  R5
00003A50  00000000 00000001                        3342          DC    XL16'0000000000000001 0000000000000001'   result
00003A58  00000000 00000001
00003A60  01020304 05060708                        3343          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
00003A68  090A0B0C 0D0E0F10
00003A70  01020304 05060708                        3344          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00003A78  090A0B0C 0D0E0F10
                                                   3345
                                                   3346          VRR_C VD,X'3',X'8',N
00003A80                                           3347+         DS    0FD
00003A80                      00003A80             3348+         USING *,R5                base for test data and test routine
00003A80  00003AF0                                 3349+T54      DC    A(X54)              address of test routine
00003A84  0036                                     3350+         DC    H'54'               test number
00003A86  00                                       3351+         DC    X'00'
00003A87  D5                                       3352+         DC    CL1'N'              S or Y = skip cross check
00003A88  03                                       3353+         DC    X'3'                 m4
00003A89  08                                       3354+         DC    X'8'                 m5
00003A8A  E5C44040 40404040                        3355+         DC    CL8'VD'             instruction name
00003A94  00003B28                                 3356+         DC    A(RE54+16)          address of v2 source
00003A98  00003B38                                 3357+         DC    A(RE54+32)          address of v3 source
00003A9C  00000010                                 3358+         DC    A(16)               result length
00003AA0  00003B18                                 3359+REA54    DC    A(RE54)             result address
00003AA8  00000000 00000000                        3360+         DS    FD                     gap
00003AB0  00000000 00000000                        3361+V1O54    DS    XL16                V1 output
00003AB8  00000000 00000000
00003AC0  00000000 00000000                        3362+         DS    FD                     gap
00003AC8  40404040 40404040                        3363+         DC    CL8' '              was cross check skipped?
00003AD0  00000000 00000000                        3364+         DS    FD                     gap
00003AD8  00000000 00000000                        3365+XCO54    DS    XL16                Cross check Output
00003AE0  00000000 00000000
00003AE8  00000000 00000000                        3366+         DS    FD                     gap
                                                   3367+*
00003AF0                                           3368+X54      DS    0F
00003AF0  E310 5014 0014                 00000014  3369+         LGF   R1,V2ADDR           load v2 source
00003AF6  E721 0000 0006                 00000000  3370+         VL    v2,0(R1)
00003AFC  E310 5018 0014                 00000018  3371+         LGF   R1,V3ADDR           load v3 source
00003B02  E731 0000 0006                 00000000  3372+         VL    v3,0(R1)

00003B08                                           3375+         DS    0H                        E7B2 VD
00003B08  E7                                       3376+         DC    X'E7'                     - VECTOR DIVIDE
00003B09  12                                       3377+         DC    X'12'                      v1, v2
00003B0A  30                                       3378+         DC    X'30'                      v3, reserved
00003B0B  0830                                     3379+         DC    HL2'2096'                  reserved, m5, m4, RXB
00003B0D  B2                                       3380+         DC    X'B2'

00003B0E  E710 5030 000E                 00003AB0  3382+         VST   V1,V1O54            save v1 output
00003B14  07FB                                     3383+         BR    R11                 return
00003B18                                           3384+RE54     DC    0F                  xl16 expected result
00003B18                                           3385+         DROP  R5
00003B18  7FFFFFFF 7FFFFFFF                        3386          DC    XL16'7FFFFFFF7FFFFFFF 3FFFFFFFBFFFFFFF'   result
00003B20  3FFFFFFF BFFFFFFF
00003B28  7FFFFFFF 7FFFFFFF                        3387          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00003B30  7FFFFFFF 7FFFFFFF
00003B38  00000000 00000001                        3388          DC    XL16'0000000000000001 0000000000000002'   v3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003B40  00000000 00000002
                                                   3389
                                                   3390          VRR_C VD,X'3',X'8',N
00003B48                                           3391+         DS    0FD
00003B48                       00003B48            3392+         USING *,R5              base for test data and test routine
00003B48  00003BB8                                 3393+T55      DC    A(X55)            address of test routine
00003B4C  0037                                     3394+         DC    H'55'             test number
00003B4E  00                                       3395+         DC    X'00'
00003B4F  D5                                       3396+         DC    CL1'N'            S or Y = skip cross check
00003B50  03                                       3397+         DC    X'3'               m4
00003B51  08                                       3398+         DC    X'8'               m5
00003B52  E5C44040 40404040                        3399+         DC    CL8'VD'           instruction name
00003B5C  00003BF0                                 3400+         DC    A(RE55+16)        address of v2 source
00003B60  00003C00                                 3401+         DC    A(RE55+32)        address of v3 source
00003B64  00000010                                 3402+         DC    A(16)             result length
00003B68  00003BE0                                 3403+REA55    DC    A(RE55)           result address
00003B70  00000000 00000000                        3404+         DS    FD                   gap
00003B78  00000000 00000000                        3405+V1O55    DS    XL16              V1 output
00003B80  00000000 00000000
00003B88  00000000 00000000                        3406+         DS    FD                   gap
00003B90  40404040 40404040                        3407+         DC    CL8' '            was cross check skipped?
00003B98  00000000 00000000                        3408+         DS    FD                   gap
00003BA0  00000000 00000000                        3409+XCO55    DS    XL16              Cross check Output
00003BA8  00000000 00000000
00003BB0  00000000 00000000                        3410+         DS    FD                   gap
                                                   3411+*
00003BB8                                           3412+X55      DS    0F
00003BB8  E310 5014 0014                 00000014  3413+         LGF   R1,V2ADDR         load v2 source
00003BBE  E721 0000 0006                 00000000  3414+         VL    v2,0(R1)
00003BC4  E310 5018 0014                 00000018  3415+         LGF   R1,V3ADDR         load v3 source
00003BCA  E731 0000 0006                 00000000  3416+         VL    v3,0(R1)

00003BD0                                           3419+         DS    0H                      E7B2 VD
00003BD0  E7                                       3420+         DC    X'E7'                   - VECTOR DIVIDE
00003BD1  12                                       3421+         DC    X'12'                    v1, v2
00003BD2  30                                       3422+         DC    X'30'                    v3, reserved
00003BD3  0830                                     3423+         DC    HL2'2096'                reserved, m5, m4, RXB
00003BD5  B2                                       3424+         DC    X'B2'

00003BD6  E710 5030 000E                 00003B78  3426+         VST   V1,V1O55          save v1 output
00003BDC  07FB                                     3427+         BR    R11               return
00003BE0                                           3428+RE55     DC    0F                xl16 expected result
00003BE0                                           3429+         DROP  R5
00003BE0  0CCCCCCC BFFFFFFF                        3430          DC    XL16'0CCCCCCCBFFFFFFF 088888887FFFFFFF'   result
00003BE8  08888888 7FFFFFFF
00003BF0  7FFFFFFF 7FFFFFFF                        3431          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00003BF8  7FFFFFFF 7FFFFFFF
00003C00  00000000 0000000A                        3432          DC    XL16'000000000000000A 000000000000000F'   v3
00003C08  00000000 0000000F
                                                   3433
                                                   3434          VRR_C VD,X'3',X'8',N
00003C10                                           3435+         DS    0FD
00003C10                       00003C10            3436+         USING *,R5              base for test data and test routine
00003C10  00003C80                                 3437+T56      DC    A(X56)            address of test routine
00003C14  0038                                     3438+         DC    H'56'             test number
00003C16  00                                       3439+         DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003C17  D5                                       3440+          DC    CL1'N'            S or Y = skip cross check
00003C18  03                                       3441+          DC    X'3'               m4
00003C19  08                                       3442+          DC    X'8'               m5
00003C1A  E5C44040 40404040                        3443+          DC    CL8'VD'           instruction name
00003C24  00003CB8                                 3444+          DC    A(RE56+16)        address of v2 source
00003C28  00003CC8                                 3445+          DC    A(RE56+32)        address of v3 source
00003C2C  00000010                                 3446+          DC    A(16)             result length
00003C30  00003CA8                                 3447+REA56     DC    A(RE56)           result address
00003C38  00000000 00000000                        3448+          DS    FD                   gap
00003C40  00000000 00000000                        3449+V1O56     DS    XL16              V1 output
00003C48  00000000 00000000
00003C50  00000000 00000000                        3450+          DS    FD                   gap
00003C58  40404040 40404040                        3451+          DC    CL8' '            was cross check skipped?
00003C60  00000000 00000000                        3452+          DS    FD                   gap
00003C68  00000000 00000000                        3453+XCO56     DS    XL16              Cross check Output
00003C70  00000000 00000000
00003C78  00000000 00000000                        3454+          DS    FD                   gap
                                                   3455+*
00003C80                                           3456+X56       DS    0F
00003C80  E310 5014 0014                 00000014  3457+          LGF   R1,V2ADDR         load v2 source
00003C86  E721 0000 0006                 00000000  3458+          VL    v2,0(R1)
00003C8C  E310 5018 0014                 00000018  3459+          LGF   R1,V3ADDR         load v3 source
00003C92  E731 0000 0006                 00000000  3460+          VL    v3,0(R1)

00003C98                                           3463+          DS    0H                     E7B2 VD
00003C98  E7                                       3464+          DC    X'E7'                  - VECTOR DIVIDE
00003C99  12                                       3465+          DC    X'12'                  v1, v2
00003C9A  30                                       3466+          DC    X'30'                  v3, reserved
00003C9B  0830                                     3467+          DC    HL2'2096'              reserved, m5, m4, RXB
00003C9D  B2                                       3468+          DC    X'B2'

00003C9E  E710 5030 000E                 00003C40  3470+          VST   V1,V1O56          save v1 output
00003CA4  07FB                                     3471+          BR    R11               return
00003CA8                                           3472+RE56      DC    0F                xl16 expected result
00003CA8                                           3473+          DROP  R5
00003CA8  00000000 0000007F                        3474           DC    XL16'000000000000007F 000000000000000E'   result
00003CB0  00000000 0000000E
00003CB8  7FFFFFFF 7FFFFFFF                        3475           DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00003CC0  7FFFFFFF 7FFFFFFF
00003CC8  01020304 05060708                        3476           DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00003CD0  090A0B0C 0D0E0F10
                                                   3477
                                                   3478           VRR_C VD,X'3',X'8',N
00003CD8                                           3479+          DS    0FD
00003CD8                      00003CD8             3480+          USING *,R5              base for test data and test routine
00003CD8  00003D48                                 3481+T57       DC    A(X57)            address of test routine
00003CDC  0039                                     3482+          DC    H'57'             test number
00003CDE  00                                       3483+          DC    X'00'
00003CDF  D5                                       3484+          DC    CL1'N'            S or Y = skip cross check
00003CE0  03                                       3485+          DC    X'3'               m4
00003CE1  08                                       3486+          DC    X'8'               m5
00003CE2  E5C44040 40404040                        3487+          DC    CL8'VD'           instruction name
00003CEC  00003D80                                 3488+          DC    A(RE57+16)        address of v2 source
00003CF0  00003D90                                 3489+          DC    A(RE57+32)        address of v3 source
00003CF4  00000010                                 3490+          DC    A(16)             result length
00003CF8  00003D70                                 3491+REA57     DC    A(RE57)           result address
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003D00  00000000 00000000                        3492+          DS    FD                    gap
00003D08  00000000 00000000                        3493+V1O57     DS    XL16               V1 output
00003D10  00000000 00000000
00003D18  00000000 00000000                        3494+          DS    FD                    gap
00003D20  40404040 40404040                        3495+          DC    CL8' '             was cross check skipped?
00003D28  00000000 00000000                        3496+          DS    FD                    gap
00003D30  00000000 00000000                        3497+XCO57     DS    XL16               Cross check Output
00003D38  00000000 00000000
00003D40  00000000 00000000                        3498+          DS    FD                    gap
                                                   3499+*
00003D48                                           3500+X57       DS    0F
00003D48  E310 5014 0014                 00000014  3501+          LGF   R1,V2ADDR          load v2 source
00003D4E  E721 0000 0006                 00000000  3502+          VL    v2,0(R1)
00003D54  E310 5018 0014                 00000018  3503+          LGF   R1,V3ADDR          load v3 source
00003D5A  E731 0000 0006                 00000000  3504+          VL    v3,0(R1)

00003D60                                           3507+          DS    0H                       E7B2 VD
00003D60  E7                                       3508+          DC    X'E7'                    - VECTOR DIVIDE
00003D61  12                                       3509+          DC    X'12'                     v1, v2
00003D62  30                                       3510+          DC    X'30'                     v3, reserved
00003D63  0830                                     3511+          DC    HL2'2096'                 reserved, m5, m4, RXB
00003D65  B2                                       3512+          DC    X'B2'

00003D66  E710 5030 000E                 00003D08  3514+          VST   V1,V1O57           save v1 output
00003D6C  07FB                                     3515+          BR    R11                return
00003D70                                           3516+RE57      DC    0F                 xl16 expected result
00003D70                                           3517+          DROP  R5
00003D70  FFFFFFFF FFFFFFFF                        3518           DC    XL16'FFFFFFFFFFFFFFFF 0000000000000000'   result
00003D78  00000000 00000000
00003D80  FFFFFFFF FFFFFFFF                        3519           DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00003D88  FFFFFFFF FFFFFFFF
00003D90  00000000 00000001                        3520           DC    XL16'0000000000000001 0000000000000002'   v3
00003D98  00000000 00000002
                                                   3521
                                                   3522           VRR_C VD,X'3',X'8',N
00003DA0                                           3523+          DS    0FD
00003DA0                      00003DA0             3524+          USING *,R5               base for test data and test routine
00003DA0  00003E10                                 3525+T58       DC    A(X58)             address of test routine
00003DA4  003A                                     3526+          DC    H'58'              test number
00003DA6  00                                       3527+          DC    X'00'
00003DA7  D5                                       3528+          DC    CL1'N'             S or Y = skip cross check
00003DA8  03                                       3529+          DC    X'3'                m4
00003DA9  08                                       3530+          DC    X'8'                m5
00003DAA  E5C44040 40404040                        3531+          DC    CL8'VD'            instruction name
00003DB4  00003E48                                 3532+          DC    A(RE58+16)         address of v2 source
00003DB8  00003E58                                 3533+          DC    A(RE58+32)         address of v3 source
00003DBC  00000010                                 3534+          DC    A(16)              result length
00003DC0  00003E38                                 3535+REA58     DC    A(RE58)            result address
00003DC8  00000000 00000000                        3536+          DS    FD                    gap
00003DD0  00000000 00000000                        3537+V1O58     DS    XL16               V1 output
00003DD8  00000000 00000000
00003DE0  00000000 00000000                        3538+          DS    FD                    gap
00003DE8  40404040 40404040                        3539+          DC    CL8' '             was cross check skipped?
00003DF0  00000000 00000000                        3540+          DS    FD                    gap
00003DF8  00000000 00000000                        3541+XCO58     DS    XL16               Cross check Output
00003E00  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003E08  00000000 00000000                        3542+         DS    FD                    gap
                                                   3543+*
00003E10                                           3544+X58      DS    0F
00003E10  E310 5014 0014                 00000014  3545+         LGF   R1,V2ADDR         load v2 source
00003E16  E721 0000 0006                 00000000  3546+         VL    v2,0(R1)
00003E1C  E310 5018 0014                 00000018  3547+         LGF   R1,V3ADDR         load v3 source
00003E22  E731 0000 0006                 00000000  3548+         VL    v3,0(R1)

00003E28                                           3551+         DS    0H                      E7B2 VD
00003E28  E7                                       3552+         DC    X'E7'                   - VECTOR DIVIDE
00003E29  12                                       3553+         DC    X'12'                    v1, v2
00003E2A  30                                       3554+         DC    X'30'                    v3, reserved
00003E2B  0830                                     3555+         DC    HL2'2096'                reserved, m5, m4, RXB
00003E2D  B2                                       3556+         DC    X'B2'

00003E2E  E710 5030 000E                 00003DD0  3558+         VST   V1,V1O58          save v1 output
00003E34  07FB                                     3559+         BR    R11               return
00003E38                                           3560+RE58     DC    0F                xl16 expected result
00003E38                                           3561+         DROP  R5
00003E38  FFFFFFFF FFFFFF83                        3562          DC    XL16'FFFFFFFFFFFFFF83 FFFFFFFFFFFFFFFF'   result
00003E40  FFFFFFFF FFFFFFFF
00003E48  FFFFFF81 FFFFFFE7                        3563          DC    XL16'FFFFFF81FFFFFFE7 FFFFFFF2FFFFFFF7'    v2
00003E50  FFFFFFF2 FFFFFFF7
00003E58  00000001 00000002                        3564          DC    XL16'0000000100000002 0000000A0000000F'   v3
00003E60  0000000A 0000000F
                                                   3565
                                                   3566          VRR_C VD,X'3',X'8',N
00003E68                                           3567+         DS    0FD
00003E68                         00003E68          3568+         USING *,R5              base for test data and test routine
00003E68  00003ED8                                 3569+T59      DC    A(X59)            address of test routine
00003E6C  003B                                     3570+         DC    H'59'             test number
00003E6E  00                                       3571+         DC    X'00'
00003E6F  D5                                       3572+         DC    CL1'N'            S or Y = skip cross check
00003E70  03                                       3573+         DC    X'3'               m4
00003E71  08                                       3574+         DC    X'8'               m5
00003E72  E5C44040 40404040                        3575+         DC    CL8'VD'           instruction name
00003E7C  00003F10                                 3576+         DC    A(RE59+16)        address of v2 source
00003E80  00003F20                                 3577+         DC    A(RE59+32)        address of v3 source
00003E84  00000010                                 3578+         DC    A(16)             result length
00003E88  00003F00                                 3579+REA59    DC    A(RE59)           result address
00003E90  00000000 00000000                        3580+         DS    FD                   gap
00003E98  00000000 00000000                        3581+V1O59    DS    XL16              V1 output
00003EA0  00000000 00000000
00003EA8  00000000 00000000                        3582+         DS    FD                   gap
00003EB0  40404040 40404040                        3583+         DC    CL8' '            was cross check skipped?
00003EB8  00000000 00000000                        3584+         DS    FD                   gap
00003EC0  00000000 00000000                        3585+XCO59    DS    XL16              Cross check Output
00003EC8  00000000 00000000
00003ED0  00000000 00000000                        3586+         DS    FD                   gap
                                                   3587+*
00003ED8                                           3588+X59      DS    0F
00003ED8  E310 5014 0014                 00000014  3589+         LGF   R1,V2ADDR         load v2 source
00003EDE  E721 0000 0006                 00000000  3590+         VL    v2,0(R1)
00003EE4  E310 5018 0014                 00000018  3591+         LGF   R1,V3ADDR         load v3 source
00003EEA  E731 0000 0006                 00000000  3592+         VL    v3,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003EF0                                           3595+         DS    0H                      E7B2 VD
00003EF0  E7                                       3596+         DC    X'E7'                   - VECTOR DIVIDE
00003EF1  12                                       3597+         DC    X'12'                    v1, v2
00003EF2  30                                       3598+         DC    X'30'                    v3, reserved
00003EF3  0830                                     3599+         DC    HL2'2096'                reserved, m5, m4, RXB
00003EF5  B2                                       3600+         DC    X'B2'

00003EF6  E710 5030 000E                00003E98   3602+         VST   V1,V1O59          save v1 output
00003EFC  07FB                                     3603+         BR    R11               return
00003F00                                           3604+RE59     DC    0F                xl16 expected result
00003F00                                           3605+         DROP  R5
00003F00  FFFFFFFF FFFFFF81                        3606          DC    XL16'FFFFFFFFFFFFFF81 FFFFFFFFFFFFFFF2'   result
00003F08  FFFFFFFF FFFFFFF2
00003F10  80000000 80000000                        3607          DC    XL16'8000000080000000 8000000080000000'   v2
00003F18  80000000 80000000
00003F20  01020304 05060708                        3608          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00003F28  090A0B0C 0D0E0F10
                                                   3609
                                                   3610          VRR_C VD,X'3',X'8',N
00003F30                                           3611+         DS    0FD
00003F30                        00003F30           3612+         USING *,R5              base for test data and test routine
00003F30  00003FA0                                 3613+T60      DC    A(X60)            address of test routine
00003F34  003C                                     3614+         DC    H'60'             test number
00003F36  00                                       3615+         DC    X'00'
00003F37  D5                                       3616+         DC    CL1'N'            S or Y = skip cross check
00003F38  03                                       3617+         DC    X'3'               m4
00003F39  08                                       3618+         DC    X'8'               m5
00003F3A  E5C44040 40404040                        3619+         DC    CL8'VD'           instruction name
00003F44  00003FD8                                 3620+         DC    A(RE60+16)        address of v2 source
00003F48  00003FE8                                 3621+         DC    A(RE60+32)        address of v3 source
00003F4C  00000010                                 3622+         DC    A(16)             result length
00003F50  00003FC8                                 3623+REA60    DC    A(RE60)           result address
00003F58  00000000 00000000                        3624+         DS    FD                   gap
00003F60  00000000 00000000                        3625+V1O60    DS    XL16              V1 output
00003F68  00000000 00000000
00003F70  00000000 00000000                        3626+         DS    FD                   gap
00003F78  40404040 40404040                        3627+         DC    CL8' '            was cross check skipped?
00003F80  00000000 00000000                        3628+         DS    FD                   gap
00003F88  00000000 00000000                        3629+XCO60    DS    XL16              Cross check Output
00003F90  00000000 00000000
00003F98  00000000 00000000                        3630+         DS    FD                   gap
                                                   3631+*
00003FA0                                           3632+X60      DS    0F
00003FA0  E310 5014 0014                00000014   3633+         LGF   R1,V2ADDR         load v2 source
00003FA6  E721 0000 0006                00000000   3634+         VL    v2,0(R1)
00003FAC  E310 5018 0014                00000018   3635+         LGF   R1,V3ADDR         load v3 source
00003FB2  E731 0000 0006                00000000   3636+         VL    v3,0(R1)

00003FB8                                           3639+         DS    0H                      E7B2 VD
00003FB8  E7                                       3640+         DC    X'E7'                   - VECTOR DIVIDE
00003FB9  12                                       3641+         DC    X'12'                    v1, v2
00003FBA  30                                       3642+         DC    X'30'                    v3, reserved
00003FBB  0830                                     3643+         DC    HL2'2096'                reserved, m5, m4, RXB
00003FBD  B2                                       3644+         DC    X'B2'

00003FBE  E710 5030 000E                00003F60   3646+         VST   V1,V1O60          save v1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003FC4  07FB                                     3647+         BR    R11                 return
00003FC8                                           3648+RE60     DC    0F                  xl16 expected result
00003FC8                                           3649+         DROP  R5
00003FC8  3FFFFFFF 3FFFFFFF                        3650          DC    XL16'3FFFFFFF3FFFFFFF 000000000E38E38D'   result
00003FD0  00000000 0E38E38D
00003FD8  80000001 80000001                        3651          DC    XL16'8000000180000001 8000000180000001'   v2
00003FE0  80000001 80000001
00003FE8  FFFFFFFF FFFFFFFE                        3652          DC    XL16'FFFFFFFFFFFFFFFE FFFFFFF6FFFFFFF1'   v3
00003FF0  FFFFFFF6 FFFFFFF1
                                                   3653
                                                   3654          VRR_C VD,X'3',X'8',N
00003FF8                                           3655+         DS    0FD
00003FF8                      00003FF8             3656+         USING *,R5                base for test data and test routine
00003FF8  00004068                                 3657+T61      DC    A(X61)              address of test routine
00003FFC  003D                                     3658+         DC    H'61'               test number
00003FFE  00                                       3659+         DC    X'00'
00003FFF  D5                                       3660+         DC    CL1'N'              S or Y = skip cross check
00004000  03                                       3661+         DC    X'3'                 m4
00004001  08                                       3662+         DC    X'8'                 m5
00004002  E5C44040 40404040                        3663+         DC    CL8'VD'             instruction name
0000400C  000040A0                                 3664+         DC    A(RE61+16)          address of v2 source
00004010  000040B0                                 3665+         DC    A(RE61+32)          address of v3 source
00004014  00000010                                 3666+         DC    A(16)               result length
00004018  00004090                                 3667+REA61    DC    A(RE61)             result address
00004020  00000000 00000000                        3668+         DS    FD                     gap
00004028  00000000 00000000                        3669+V1O61    DS    XL16                V1 output
00004030  00000000 00000000
00004038  00000000 00000000                        3670+         DS    FD                     gap
00004040  40404040 40404040                        3671+         DC    CL8' '              was cross check skipped?
00004048  00000000 00000000                        3672+         DS    FD                     gap
00004050  00000000 00000000                        3673+XCO61    DS    XL16                Cross check Output
00004058  00000000 00000000
00004060  00000000 00000000                        3674+         DS    FD                     gap
                                                   3675+*
00004068                                           3676+X61      DS    0F
00004068  E310 5014 0014                 00000014  3677+         LGF   R1,V2ADDR           load v2 source
0000406E  E721 0000 0006                 00000000  3678+         VL    v2,0(R1)
00004074  E310 5018 0014                 00000018  3679+         LGF   R1,V3ADDR           load v3 source
0000407A  E731 0000 0006                 00000000  3680+         VL    v3,0(R1)

00004080                                           3683+         DS    0H                        E7B2 VD
00004080  E7                                       3684+         DC    X'E7'                     - VECTOR DIVIDE
00004081  12                                       3685+         DC    X'12'                      v1, v2
00004082  30                                       3686+         DC    X'30'                      v3, reserved
00004083  0830                                     3687+         DC    HL2'2096'                  reserved, m5, m4, RXB
00004085  B2                                       3688+         DC    X'B2'

00004086  E710 5030 000E                 00004028  3690+         VST   V1,V1O61            save v1 output
0000408C  07FB                                     3691+         BR    R11                 return
00004090                                           3692+RE61     DC    0F                  xl16 expected result
00004090                                           3693+         DROP  R5
00004090  00000000 00000001                        3694          DC    XL16'0000000000000001 0000000000000001'   result
00004098  00000000 00000001
000040A0  80000000 80000000                        3695          DC    XL16'8000000080000000 8000000080000000'   v2
000040A8  80000000 80000000
000040B0  81020304 85060708                        3696          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   v3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000040B8  890A0B0C 8D0E0F10
                                                   3697
                                                   3698 * skip XC
                                                   3699          VRR_C VD,X'3',X'8',S
000040C0                                           3700+         DS    0FD
000040C0                     000040C0              3701+         USING *,R5              base for test data and test routine
000040C0  00004130                                 3702+T62      DC    A(X62)            address of test routine
000040C4  003E                                     3703+         DC    H'62'             test number
000040C6  00                                       3704+         DC    X'00'
000040C7  E2                                       3705+         DC    CL1'S'            S or Y = skip cross check
000040C8  03                                       3706+         DC    X'3'               m4
000040C9  08                                       3707+         DC    X'8'               m5
000040CA  E5C44040 40404040                        3708+         DC    CL8'VD'           instruction name
000040D4  00004168                                 3709+         DC    A(RE62+16)        address of v2 source
000040D8  00004178                                 3710+         DC    A(RE62+32)        address of v3 source
000040DC  00000010                                 3711+         DC    A(16)             result length
000040E0  00004158                                 3712+REA62    DC    A(RE62)           result address
000040E8  00000000 00000000                        3713+         DS    FD                   gap
000040F0  00000000 00000000                        3714+V1O62    DS    XL16              V1 output
000040F8  00000000 00000000
00004100  00000000 00000000                        3715+         DS    FD                   gap
00004108  40404040 40404040                        3716+         DC    CL8' '            was cross check skipped?
00004110  00000000 00000000                        3717+         DS    FD                   gap
00004118  00000000 00000000                        3718+XCO62    DS    XL16              Cross check Output
00004120  00000000 00000000
00004128  00000000 00000000                        3719+         DS    FD                   gap
                                                   3720+*
00004130                                           3721+X62      DS    0F
00004130  E310 5014 0014                 00000014  3722+         LGF   R1,V2ADDR         load v2 source
00004136  E721 0000 0006                 00000000  3723+         VL    v2,0(R1)
0000413C  E310 5018 0014                 00000018  3724+         LGF   R1,V3ADDR         load v3 source
00004142  E731 0000 0006                 00000000  3725+         VL    v3,0(R1)

00004148                                           3728+         DS    0H                     E7B2 VD
00004148  E7                                       3729+         DC    X'E7'                  - VECTOR DIVIDE
00004149  12                                       3730+         DC    X'12'                   v1, v2
0000414A  30                                       3731+         DC    X'30'                   v3, reserved
0000414B  0830                                     3732+         DC    HL2'2096'               reserved, m5, m4, RXB
0000414D  B2                                       3733+         DC    X'B2'

0000414E  E710 5030 000E                 000040F0  3735+         VST   V1,V1O62          save v1 output
00004154  07FB                                     3736+         BR    R11               return
00004158                                           3737+RE62     DC    0F                xl16 expected result
00004158                                           3738+         DROP  R5
00004158  00000000 00000000                        3739          DC    XL16'0000000000000000 0888888888888888'   result
00004160  08888888 88888888
00004168  80000000 00000000                        3740          DC    XL16'8000000000000000 8000000000000000'   v2
00004170  80000000 00000000
00004178  FFFFFFFF FFFFFFFF                        3741          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFF1'   v3
00004180  FFFFFFFF FFFFFFF1
                                                   3742
                                                   3743 *Quadwordword    M4:  4, m5= 8 (IDC=1)
                                                   3744
                                                   3745          VRR_C VD,X'4',X'8',N
00004188                                           3746+         DS    0FD
00004188                     00004188              3747+         USING *,R5              base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004188  000041F8                                 3748+T63      DC    A(X63)            address of test routine
0000418C  003F                                     3749+         DC    H'63'             test number
0000418E  00                                       3750+         DC    X'00'
0000418F  D5                                       3751+         DC    CL1'N'            S or Y = skip cross check
00004190  04                                       3752+         DC    X'4'               m4
00004191  08                                       3753+         DC    X'8'               m5
00004192  E5C44040 40404040                        3754+         DC    CL8'VD'           instruction name
0000419C  00004230                                 3755+         DC    A(RE63+16)        address of v2 source
000041A0  00004240                                 3756+         DC    A(RE63+32)        address of v3 source
000041A4  00000010                                 3757+         DC    A(16)             result length
000041A8  00004220                                 3758+REA63    DC    A(RE63)           result address
000041B0  00000000 00000000                        3759+         DS    FD                   gap
000041B8  00000000 00000000                        3760+V1O63    DS    XL16              V1 output
000041C0  00000000 00000000
000041C8  00000000 00000000                        3761+         DS    FD                   gap
000041D0  40404040 40404040                        3762+         DC    CL8' '            was cross check skipped?
000041D8  00000000 00000000                        3763+         DS    FD                   gap
000041E0  00000000 00000000                        3764+XCO63    DS    XL16              Cross check Output
000041E8  00000000 00000000
000041F0  00000000 00000000                        3765+         DS    FD                   gap
                                                   3766+*
000041F8                                           3767+X63      DS    0F
000041F8  E310 5014 0014                 00000014  3768+         LGF   R1,V2ADDR         load v2 source
000041FE  E721 0000 0006                 00000000  3769+         VL    v2,0(R1)
00004204  E310 5018 0014                 00000018  3770+         LGF   R1,V3ADDR         load v3 source
0000420A  E731 0000 0006                 00000000  3771+         VL    v3,0(R1)

00004210                                           3774+         DS    0H                      E7B2 VD
00004210  E7                                       3775+         DC    X'E7'                   - VECTOR DIVIDE
00004211  12                                       3776+         DC    X'12'                    v1, v2
00004212  30                                       3777+         DC    X'30'                    v3, reserved
00004213  0840                                     3778+         DC    HL2'2112'                reserved, m5, m4, RXB
00004215  B2                                       3779+         DC    X'B2'

00004216  E710 5030 000E                 000041B8  3781+         VST   V1,V1O63          save v1 output
0000421C  07FB                                     3782+         BR    R11               return
00004220                                           3783+RE63     DC    0F                xl16 expected result
00004220                                           3784+         DROP  R5
00004220  00000000 00000000                        3785          DC    XL16'0000000000000000 0000000000000000'   result
00004228  00000000 00000000
00004230  00000000 00000000                        3786          DC    XL16'0000000000000000 0000000000000000'   v2
00004238  00000000 00000000
00004240  01020304 05060708                        3787          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00004248  090A0B0C 0D0E0F10
                                                   3788
                                                   3789          VRR_C VD,X'4',X'8',N
00004250                                           3790+         DS    0FD
00004250                      00004250             3791+         USING *,R5              base for test data and test routine
00004250  000042C0                                 3792+T64      DC    A(X64)            address of test routine
00004254  0040                                     3793+         DC    H'64'             test number
00004256  00                                       3794+         DC    X'00'
00004257  D5                                       3795+         DC    CL1'N'            S or Y = skip cross check
00004258  04                                       3796+         DC    X'4'               m4
00004259  08                                       3797+         DC    X'8'               m5
0000425A  E5C44040 40404040                        3798+         DC    CL8'VD'           instruction name
00004264  000042F8                                 3799+         DC    A(RE64+16)        address of v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004268  00004308                                 3800+         DC    A(RE64+32)        address of v3 source
0000426C  00000010                                 3801+         DC    A(16)             result length
00004270  000042E8                                 3802+REA64    DC    A(RE64)           result address
00004278  00000000 00000000                        3803+         DS    FD                   gap
00004280  00000000 00000000                        3804+V1O64    DS    XL16              V1 output
00004288  00000000 00000000
00004290  00000000 00000000                        3805+         DS    FD                   gap
00004298  40404040 40404040                        3806+         DC    CL8' '            was cross check skipped?
000042A0  00000000 00000000                        3807+         DS    FD                   gap
000042A8  00000000 00000000                        3808+XCO64    DS    XL16              Cross check Output
000042B0  00000000 00000000
000042B8  00000000 00000000                        3809+         DS    FD                   gap
                                                   3810+*
000042C0                                           3811+X64      DS    0F
000042C0  E310 5014 0014                 00000014  3812+         LGF   R1,V2ADDR         load v2 source
000042C6  E721 0000 0006                 00000000  3813+         VL    v2,0(R1)
000042CC  E310 5018 0014                 00000018  3814+         LGF   R1,V3ADDR         load v3 source
000042D2  E731 0000 0006                 00000000  3815+         VL    v3,0(R1)

000042D8                                           3818+         DS    0H                      E7B2 VD
000042D8  E7                                       3819+         DC    X'E7'                   - VECTOR DIVIDE
000042D9  12                                       3820+         DC    X'12'                    v1, v2
000042DA  30                                       3821+         DC    X'30'                    v3, reserved
000042DB  0840                                     3822+         DC    HL2'2112'                reserved, m5, m4, RXB
000042DD  B2                                       3823+         DC    X'B2'

000042DE  E710 5030 000E                 00004280  3825+         VST   V1,V1O64          save v1 output
000042E4  07FB                                     3826+         BR    R11               return
000042E8                                           3827+RE64     DC    0F                xl16 expected result
000042E8                                           3828+         DROP  R5
000042E8  00000000 00000000                        3829          DC    XL16'0000000000000000 0000000000000000'   result
000042F0  00000000 00000000
000042F8  01020303 05060707                        3830          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00004300  090A0B0B 0D0E0F0F
00004308  01020304 05060708                        3831          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00004310  090A0B0C 0D0E0F10
                                                   3832
                                                   3833          VRR_C VD,X'4',X'8',N
00004318                                           3834+         DS    0FD
00004318                      00004318             3835+         USING *,R5              base for test data and test routine
00004318  00004388                                 3836+T65      DC    A(X65)            address of test routine
0000431C  0041                                     3837+         DC    H'65'             test number
0000431E  00                                       3838+         DC    X'00'
0000431F  D5                                       3839+         DC    CL1'N'            S or Y = skip cross check
00004320  04                                       3840+         DC    X'4'               m4
00004321  08                                       3841+         DC    X'8'               m5
00004322  E5C44040 40404040                        3842+         DC    CL8'VD'           instruction name
0000432C  000043C0                                 3843+         DC    A(RE65+16)        address of v2 source
00004330  000043D0                                 3844+         DC    A(RE65+32)        address of v3 source
00004334  00000010                                 3845+         DC    A(16)             result length
00004338  000043B0                                 3846+REA65    DC    A(RE65)           result address
00004340  00000000 00000000                        3847+         DS    FD                   gap
00004348  00000000 00000000                        3848+V1O65    DS    XL16              V1 output
00004350  00000000 00000000
00004358  00000000 00000000                        3849+         DS    FD                   gap
00004360  40404040 40404040                        3850+         DC    CL8' '            was cross check skipped?
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004368  00000000 00000000                        3851+         DS    FD                   gap
00004370  00000000 00000000                        3852+XCO65    DS    XL16               Cross check Output
00004378  00000000 00000000
00004380  00000000 00000000                        3853+         DS    FD                   gap
                                                   3854+*
00004388                                           3855+X65      DS    0F
00004388  E310 5014 0014                 00000014  3856+         LGF   R1,V2ADDR          load v2 source
0000438E  E721 0000 0006                 00000000  3857+         VL    v2,0(R1)
00004394  E310 5018 0014                 00000018  3858+         LGF   R1,V3ADDR          load v3 source
0000439A  E731 0000 0006                 00000000  3859+         VL    v3,0(R1)

000043A0                                           3862+         DS    0H                     E7B2 VD
000043A0  E7                                       3863+         DC    X'E7'                  - VECTOR DIVIDE
000043A1  12                                       3864+         DC    X'12'                   v1, v2
000043A2  30                                       3865+         DC    X'30'                   v3, reserved
000043A3  0840                                     3866+         DC    HL2'2112'               reserved, m5, m4, RXB
000043A5  B2                                       3867+         DC    X'B2'

000043A6  E710 5030 000E                 00004348  3869+         VST   V1,V1O65           save v1 output
000043AC  07FB                                     3870+         BR    R11                return
000043B0                                           3871+RE65     DC    0F                 xl16 expected result
000043B0                                           3872+         DROP  R5
000043B0  00000000 00000000                        3873          DC    XL16'0000000000000000 0000000000000001'   result
000043B8  00000000 00000001
000043C0  01020304 05060708                        3874          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
000043C8  090A0B0C 0D0E0F10
000043D0  01020304 05060708                        3875          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000043D8  090A0B0C 0D0E0F10
                                                   3876
                                                   3877          VRR_C VD,X'4',X'8',N
000043E0                                           3878+         DS    0FD
000043E0                         000043E0          3879+         USING *,R5               base for test data and test routine
000043E0  00004450                                 3880+T66      DC    A(X66)             address of test routine
000043E4  0042                                     3881+         DC    H'66'              test number
000043E6  00                                       3882+         DC    X'00'
000043E7  D5                                       3883+         DC    CL1'N'             S or Y = skip cross check
000043E8  04                                       3884+         DC    X'4'                m4
000043E9  08                                       3885+         DC    X'8'                m5
000043EA  E5C44040 40404040                        3886+         DC    CL8'VD'            instruction name
000043F4  00004488                                 3887+         DC    A(RE66+16)         address of v2 source
000043F8  00004498                                 3888+         DC    A(RE66+32)         address of v3 source
000043FC  00000010                                 3889+         DC    A(16)              result length
00004400  00004478                                 3890+REA66    DC    A(RE66)            result address
00004408  00000000 00000000                        3891+         DS    FD                   gap
00004410  00000000 00000000                        3892+V1O66    DS    XL16               V1 output
00004418  00000000 00000000
00004420  00000000 00000000                        3893+         DS    FD                   gap
00004428  40404040 40404040                        3894+         DC    CL8' '             was cross check skipped?
00004430  00000000 00000000                        3895+         DS    FD                   gap
00004438  00000000 00000000                        3896+XCO66    DS    XL16               Cross check Output
00004440  00000000 00000000
00004448  00000000 00000000                        3897+         DS    FD                   gap
                                                   3898+*
00004450                                           3899+X66      DS    0F
00004450  E310 5014 0014                 00000014  3900+         LGF   R1,V2ADDR          load v2 source
00004456  E721 0000 0006                 00000000  3901+         VL    v2,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000445C  E310 5018 0014                00000018  3902+         LGF   R1,V3ADDR         load v3 source
00004462  E731 0000 0006                00000000  3903+         VL    v3,0(R1)

00004468                                          3906+         DS    0H                      E7B2 VD
00004468  E7                                      3907+         DC    X'E7'                   - VECTOR DIVIDE
00004469  12                                      3908+         DC    X'12'                    v1, v2
0000446A  30                                      3909+         DC    X'30'                    v3, reserved
0000446B  0840                                    3910+         DC    HL2'2112'                reserved, m5, m4, RXB
0000446D  B2                                      3911+         DC    X'B2'

0000446E  E710 5030 000E                00004410  3913+         VST   V1,V1O66          save v1 output
00004474  07FB                                    3914+         BR    R11               return
00004478                                          3915+RE66     DC    0F                xl16 expected result
00004478                                          3916+         DROP  R5
00004478  7FFFFFFF 7FFFFFFF                       3917          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   result
00004480  7FFFFFFF 7FFFFFFF
00004488  7FFFFFFF 7FFFFFFF                       3918          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00004490  7FFFFFFF 7FFFFFFF
00004498  00000000 00000000                       3919          DC    XL16'0000000000000000 0000000000000001'   v3
000044A0  00000000 00000001
                                                  3920
                                                  3921          VRR_C VD,X'4',X'8',N
000044A8                                          3922+         DS    0FD
000044A8                     000044A8             3923+         USING *,R5              base for test data and test routine
000044A8  00004518                                3924+T67      DC    A(X67)            address of test routine
000044AC  0043                                    3925+         DC    H'67'             test number
000044AE  00                                      3926+         DC    X'00'
000044AF  D5                                      3927+         DC    CL1'N'            S or Y = skip cross check
000044B0  04                                      3928+         DC    X'4'               m4
000044B1  08                                      3929+         DC    X'8'               m5
000044B2  E5C44040 40404040                       3930+         DC    CL8'VD'           instruction name
000044BC  00004550                                3931+         DC    A(RE67+16)        address of v2 source
000044C0  00004560                                3932+         DC    A(RE67+32)        address of v3 source
000044C4  00000010                                3933+         DC    A(16)             result length
000044C8  00004540                                3934+REA67    DC    A(RE67)           result address
000044D0  00000000 00000000                       3935+         DS    FD                   gap
000044D8  00000000 00000000                       3936+V1O67    DS    XL16              V1 output
000044E0  00000000 00000000
000044E8  00000000 00000000                       3937+         DS    FD                   gap
000044F0  40404040 40404040                       3938+         DC    CL8' '            was cross check skipped?
000044F8  00000000 00000000                       3939+         DS    FD                   gap
00004500  00000000 00000000                       3940+XCO67    DS    XL16              Cross check Output
00004508  00000000 00000000
00004510  00000000 00000000                       3941+         DS    FD                   gap
                                                  3942+*
00004518                                          3943+X67      DS    0F
00004518  E310 5014 0014                00000014  3944+         LGF   R1,V2ADDR         load v2 source
0000451E  E721 0000 0006                00000000  3945+         VL    v2,0(R1)
00004524  E310 5018 0014                00000018  3946+         LGF   R1,V3ADDR         load v3 source
0000452A  E731 0000 0006                00000000  3947+         VL    v3,0(R1)

00004530                                          3950+         DS    0H                      E7B2 VD
00004530  E7                                      3951+         DC    X'E7'                   - VECTOR DIVIDE
00004531  12                                      3952+         DC    X'12'                    v1, v2
00004532  30                                      3953+         DC    X'30'                    v3, reserved
00004533  0840                                    3954+         DC    HL2'2112'                reserved, m5, m4, RXB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004535  B2                                       3955+         DC    X'B2'

00004536  E710 5030 000E                 000044D8  3957+         VST   V1,V1O67          save v1 output
0000453C  07FB                                     3958+         BR    R11               return
00004540                                           3959+RE67     DC    0F                xl16 expected result
00004540                                           3960+         DROP  R5
00004540  3FFFFFFF BFFFFFFF                        3961          DC    XL16'3FFFFFFFBFFFFFFF BFFFFFFFBFFFFFFF'   result
00004548  BFFFFFFF BFFFFFFF
00004550  7FFFFFFF 7FFFFFFF                        3962          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00004558  7FFFFFFF 7FFFFFFF
00004560  00000000 00000000                        3963          DC    XL16'0000000000000000 0000000000000002'   v3
00004568  00000000 00000002
                                                   3964
                                                   3965          VRR_C VD,X'4',X'8',N
00004570                                           3966+         DS    0FD
00004570                      00004570             3967+         USING *,R5              base for test data and test routine
00004570  000045E0                                 3968+T68      DC    A(X68)            address of test routine
00004574  0044                                     3969+         DC    H'68'             test number
00004576  00                                       3970+         DC    X'00'
00004577  D5                                       3971+         DC    CL1'N'            S or Y = skip cross check
00004578  04                                       3972+         DC    X'4'               m4
00004579  08                                       3973+         DC    X'8'               m5
0000457A  E5C44040 40404040                        3974+         DC    CL8'VD'           instruction name
00004584  00004618                                 3975+         DC    A(RE68+16)        address of v2 source
00004588  00004628                                 3976+         DC    A(RE68+32)        address of v3 source
0000458C  00000010                                 3977+         DC    A(16)             result length
00004590  00004608                                 3978+REA68    DC    A(RE68)           result address
00004598  00000000 00000000                        3979+         DS    FD                   gap
000045A0  00000000 00000000                        3980+V1O68    DS    XL16              V1 output
000045A8  00000000 00000000
000045B0  00000000 00000000                        3981+         DS    FD                   gap
000045B8  40404040 40404040                        3982+         DC    CL8' '            was cross check skipped?
000045C0  00000000 00000000                        3983+         DS    FD                   gap
000045C8  00000000 00000000                        3984+XCO68    DS    XL16              Cross check Output
000045D0  00000000 00000000
000045D8  00000000 00000000                        3985+         DS    FD                   gap
                                                   3986+*
000045E0                                           3987+X68      DS    0F
000045E0  E310 5014 0014                 00000014  3988+         LGF   R1,V2ADDR         load v2 source
000045E6  E721 0000 0006                 00000000  3989+         VL    v2,0(R1)
000045EC  E310 5018 0014                 00000018  3990+         LGF   R1,V3ADDR         load v3 source
000045F2  E731 0000 0006                 00000000  3991+         VL    v3,0(R1)

000045F8                                           3994+         DS    0H                     E7B2 VD
000045F8  E7                                       3995+         DC    X'E7'                  - VECTOR DIVIDE
000045F9  12                                       3996+         DC    X'12'                   v1, v2
000045FA  30                                       3997+         DC    X'30'                   v3, reserved
000045FB  0840                                     3998+         DC    HL2'2112'               reserved, m5, m4, RXB
000045FD  B2                                       3999+         DC    X'B2'

000045FE  E710 5030 000E                 000045A0  4001+         VST   V1,V1O68          save v1 output
00004604  07FB                                     4002+         BR    R11               return
00004608                                           4003+RE68     DC    0F                xl16 expected result
00004608                                           4004+         DROP  R5
00004608  0CCCCCCC BFFFFFFF                        4005          DC    XL16'0CCCCCCCBFFFFFFF F333333326666666'   result
00004610  F3333333 26666666
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004618  7FFFFFFF 7FFFFFFF                        4006          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00004620  7FFFFFFF 7FFFFFFF
00004628  00000000 00000000                        4007          DC    XL16'0000000000000000 000000000000000A'   v3
00004630  00000000 0000000A
                                                   4008
                                                   4009          VRR_C VD,X'4',X'8',N
00004638                                           4010+         DS    0FD
00004638                       00004638            4011+         USING *,R5              base for test data and test routine
00004638  000046A8                                 4012+T69      DC    A(X69)            address of test routine
0000463C  0045                                     4013+         DC    H'69'             test number
0000463E  00                                       4014+         DC    X'00'
0000463F  D5                                       4015+         DC    CL1'N'            S or Y = skip cross check
00004640  04                                       4016+         DC    X'4'               m4
00004641  08                                       4017+         DC    X'8'               m5
00004642  E5C44040 40404040                        4018+         DC    CL8'VD'           instruction name
0000464C  000046E0                                 4019+         DC    A(RE69+16)        address of v2 source
00004650  000046F0                                 4020+         DC    A(RE69+32)        address of v3 source
00004654  00000010                                 4021+         DC    A(16)             result length
00004658  000046D0                                 4022+REA69    DC    A(RE69)           result address
00004660  00000000 00000000                        4023+         DS    FD                   gap
00004668  00000000 00000000                        4024+V1O69    DS    XL16              V1 output
00004670  00000000 00000000
00004678  00000000 00000000                        4025+         DS    FD                   gap
00004680  40404040 40404040                        4026+         DC    CL8' '            was cross check skipped?
00004688  00000000 00000000                        4027+         DS    FD                   gap
00004690  00000000 00000000                        4028+XCO69    DS    XL16              Cross check Output
00004698  00000000 00000000
000046A0  00000000 00000000                        4029+         DS    FD                   gap
                                                   4030+*
000046A8                                           4031+X69      DS    0F
000046A8  E310 5014 0014                00000014   4032+         LGF   R1,V2ADDR         load v2 source
000046AE  E721 0000 0006                00000000   4033+         VL    v2,0(R1)
000046B4  E310 5018 0014                00000018   4034+         LGF   R1,V3ADDR         load v3 source
000046BA  E731 0000 0006                00000000   4035+         VL    v3,0(R1)

000046C0                                           4038+         DS    0H                     E7B2 VD
000046C0  E7                                       4039+         DC    X'E7'                  - VECTOR DIVIDE
000046C1  12                                       4040+         DC    X'12'                   v1, v2
000046C2  30                                       4041+         DC    X'30'                   v3, reserved
000046C3  0840                                     4042+         DC    HL2'2112'               reserved, m5, m4, RXB
000046C5  B2                                       4043+         DC    X'B2'

000046C6  E710 5030 000E                00004668   4045+         VST   V1,V1O69          save v1 output
000046CC  07FB                                     4046+         BR    R11               return
000046D0                                           4047+RE69     DC    0F                xl16 expected result
000046D0                                           4048+         DROP  R5
000046D0  08888888 7FFFFFFF                        4049          DC    XL16'088888887FFFFFFF F77777776EEEEEEE'   result
000046D8  F7777777 6EEEEEEE
000046E0  7FFFFFFF 7FFFFFFF                        4050          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
000046E8  7FFFFFFF 7FFFFFFF
000046F0  00000000 00000000                        4051          DC    XL16'0000000000000000 000000000000000F'   v3
000046F8  00000000 0000000F
                                                   4052
                                                   4053          VRR_C VD,X'4',X'8',N
00004700                                           4054+         DS    0FD
00004700                       00004700            4055+         USING *,R5              base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004700  00004770                                 4056+T70      DC    A(X70)            address of test routine
00004704  0046                                     4057+         DC    H'70'             test number
00004706  00                                       4058+         DC    X'00'
00004707  D5                                       4059+         DC    CL1'N'            S or Y = skip cross check
00004708  04                                       4060+         DC    X'4'               m4
00004709  08                                       4061+         DC    X'8'               m5
0000470A  E5C44040 40404040                        4062+         DC    CL8'VD'           instruction name
00004714  000047A8                                 4063+         DC    A(RE70+16)        address of v2 source
00004718  000047B8                                 4064+         DC    A(RE70+32)        address of v3 source
0000471C  00000010                                 4065+         DC    A(16)             result length
00004720  00004798                                 4066+REA70    DC    A(RE70)           result address
00004728  00000000 00000000                        4067+         DS    FD                   gap
00004730  00000000 00000000                        4068+V1O70    DS    XL16              V1 output
00004738  00000000 00000000
00004740  00000000 00000000                        4069+         DS    FD                   gap
00004748  40404040 40404040                        4070+         DC    CL8' '            was cross check skipped?
00004750  00000000 00000000                        4071+         DS    FD                   gap
00004758  00000000 00000000                        4072+XCO70    DS    XL16              Cross check Output
00004760  00000000 00000000
00004768  00000000 00000000                        4073+         DS    FD                   gap
                                                   4074+*
00004770                                           4075+X70      DS    0F
00004770  E310 5014 0014                00000014   4076+         LGF   R1,V2ADDR         load v2 source
00004776  E721 0000 0006                00000000   4077+         VL    v2,0(R1)
0000477C  E310 5018 0014                00000018   4078+         LGF   R1,V3ADDR         load v3 source
00004782  E731 0000 0006                00000000   4079+         VL    v3,0(R1)

00004788                                           4082+         DS    0H                      E7B2 VD
00004788  E7                                       4083+         DC    X'E7'                   - VECTOR DIVIDE
00004789  12                                       4084+         DC    X'12'                    v1, v2
0000478A  30                                       4085+         DC    X'30'                    v3, reserved
0000478B  0840                                     4086+         DC    HL2'2112'                reserved, m5, m4, RXB
0000478D  B2                                       4087+         DC    X'B2'

0000478E  E710 5030 000E                00004730   4089+         VST   V1,V1O70          save v1 output
00004794  07FB                                     4090+         BR    R11               return
00004798                                           4091+RE70     DC    0F                xl16 expected result
00004798                                           4092+         DROP  R5
00004798  00000000 00000000                        4093          DC    XL16'0000000000000000 000000000000007F'   result
000047A0  00000000 0000007F
000047A8  7FFFFFFF 7FFFFFFF                        4094          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
000047B0  7FFFFFFF 7FFFFFFF
000047B8  01020304 05060708                        4095          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000047C0  090A0B0C 0D0E0F10
                                                   4096
                                                   4097          VRR_C VD,X'4',X'8',N
000047C8                                           4098+         DS    0FD
000047C8                        000047C8           4099+         USING *,R5              base for test data and test routine
000047C8  00004838                                 4100+T71      DC    A(X71)            address of test routine
000047CC  0047                                     4101+         DC    H'71'             test number
000047CE  00                                       4102+         DC    X'00'
000047CF  D5                                       4103+         DC    CL1'N'            S or Y = skip cross check
000047D0  04                                       4104+         DC    X'4'               m4
000047D1  08                                       4105+         DC    X'8'               m5
000047D2  E5C44040 40404040                        4106+         DC    CL8'VD'           instruction name
000047DC  00004870                                 4107+         DC    A(RE71+16)        address of v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000047E0  00004880                                 4108+         DC    A(RE71+32)        address of v3 source
000047E4  00000010                                 4109+         DC    A(16)             result length
000047E8  00004860                                 4110+REA71    DC    A(RE71)           result address
000047F0  00000000 00000000                        4111+         DS    FD                   gap
000047F8  00000000 00000000                        4112+V1O71    DS    XL16              V1 output
00004800  00000000 00000000
00004808  00000000 00000000                        4113+         DS    FD                   gap
00004810  40404040 40404040                        4114+         DC    CL8' '            was cross check skipped?
00004818  00000000 00000000                        4115+         DS    FD                   gap
00004820  00000000 00000000                        4116+XCO71    DS    XL16              Cross check Output
00004828  00000000 00000000
00004830  00000000 00000000                        4117+         DS    FD                   gap
                                                   4118+*
00004838                                           4119+X71      DS    0F
00004838  E310 5014 0014                 00000014  4120+         LGF   R1,V2ADDR         load v2 source
0000483E  E721 0000 0006                 00000000  4121+         VL    v2,0(R1)
00004844  E310 5018 0014                 00000018  4122+         LGF   R1,V3ADDR         load v3 source
0000484A  E731 0000 0006                 00000000  4123+         VL    v3,0(R1)

00004850                                           4126+         DS    0H                      E7B2 VD
00004850  E7                                       4127+         DC    X'E7'                   - VECTOR DIVIDE
00004851  12                                       4128+         DC    X'12'                    v1, v2
00004852  30                                       4129+         DC    X'30'                    v3, reserved
00004853  0840                                     4130+         DC    HL2'2112'                reserved, m5, m4, RXB
00004855  B2                                       4131+         DC    X'B2'

00004856  E710 5030 000E                 000047F8  4133+         VST   V1,V1O71          save v1 output
0000485C  07FB                                     4134+         BR    R11               return
00004860                                           4135+RE71     DC    0F                xl16 expected result
00004860                                           4136+         DROP  R5
00004860  FFFFFFFF FFFFFFFF                        4137          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   result
00004868  FFFFFFFF FFFFFF00
00004870  FFFFFFFF FFFFFFFF                        4138          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
00004878  FFFFFFFF FFFFFF00
00004880  00000000 00000000                        4139          DC    XL16'0000000000000000 0000000000000001'   v3
00004888  00000000 00000001
                                                   4140
                                                   4141          VRR_C VD,X'4',X'8',N
00004890                                           4142+         DS    0FD
00004890                      00004890             4143+         USING *,R5              base for test data and test routine
00004890  00004900                                 4144+T72      DC    A(X72)            address of test routine
00004894  0048                                     4145+         DC    H'72'             test number
00004896  00                                       4146+         DC    X'00'
00004897  D5                                       4147+         DC    CL1'N'            S or Y = skip cross check
00004898  04                                       4148+         DC    X'4'               m4
00004899  08                                       4149+         DC    X'8'               m5
0000489A  E5C44040 40404040                        4150+         DC    CL8'VD'           instruction name
000048A4  00004938                                 4151+         DC    A(RE72+16)        address of v2 source
000048A8  00004948                                 4152+         DC    A(RE72+32)        address of v3 source
000048AC  00000010                                 4153+         DC    A(16)             result length
000048B0  00004928                                 4154+REA72    DC    A(RE72)           result address
000048B8  00000000 00000000                        4155+         DS    FD                   gap
000048C0  00000000 00000000                        4156+V1O72    DS    XL16              V1 output
000048C8  00000000 00000000
000048D0  00000000 00000000                        4157+         DS    FD                   gap
000048D8  40404040 40404040                        4158+         DC    CL8' '            was cross check skipped?
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000048E0  00000000 00000000                        4159+          DS    FD                    gap
000048E8  00000000 00000000                        4160+XCO72     DS    XL16               Cross check Output
000048F0  00000000 00000000
000048F8  00000000 00000000                        4161+          DS    FD                    gap
                                                   4162+*
00004900                                           4163+X72       DS    0F
00004900  E310 5014 0014                 00000014  4164+          LGF   R1,V2ADDR          load v2 source
00004906  E721 0000 0006                 00000000  4165+          VL    v2,0(R1)
0000490C  E310 5018 0014                 00000018  4166+          LGF   R1,V3ADDR          load v3 source
00004912  E731 0000 0006                 00000000  4167+          VL    v3,0(R1)

00004918                                           4170+          DS    0H                       E7B2 VD
00004918  E7                                       4171+          DC    X'E7'                    - VECTOR DIVIDE
00004919  12                                       4172+          DC    X'12'                     v1, v2
0000491A  30                                       4173+          DC    X'30'                     v3, reserved
0000491B  0840                                     4174+          DC    HL2'2112'                 reserved, m5, m4, RXB
0000491D  B2                                       4175+          DC    X'B2'

0000491E  E710 5030 000E                 000048C0  4177+          VST   V1,V1O72           save v1 output
00004924  07FB                                     4178+          BR    R11                return
00004928                                           4179+RE72      DC    0F                 xl16 expected result
00004928                                           4180+          DROP  R5
00004928  FFFFFFFF FFFFFFFF                        4181           DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF80'   result
00004930  FFFFFFFF FFFFFF80
00004938  FFFFFFFF FFFFFFFF                        4182           DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
00004940  FFFFFFFF FFFFFF00
00004948  00000000 00000000                        4183           DC    XL16'0000000000000000 0000000000000002'   v3
00004950  00000000 00000002
                                                   4184
                                                   4185           VRR_C VD,X'4',X'8',N
00004958                                           4186+          DS    0FD
00004958                         00004958          4187+          USING *,R5               base for test data and test routine
00004958  000049C8                                 4188+T73       DC    A(X73)             address of test routine
0000495C  0049                                     4189+          DC    H'73'              test number
0000495E  00                                       4190+          DC    X'00'
0000495F  D5                                       4191+          DC    CL1'N'             S or Y = skip cross check
00004960  04                                       4192+          DC    X'4'                m4
00004961  08                                       4193+          DC    X'8'                m5
00004962  E5C44040 40404040                        4194+          DC    CL8'VD'            instruction name
0000496C  00004A00                                 4195+          DC    A(RE73+16)         address of v2 source
00004970  00004A10                                 4196+          DC    A(RE73+32)         address of v3 source
00004974  00000010                                 4197+          DC    A(16)              result length
00004978  000049F0                                 4198+REA73     DC    A(RE73)            result address
00004980  00000000 00000000                        4199+          DS    FD                    gap
00004988  00000000 00000000                        4200+V1O73     DS    XL16               V1 output
00004990  00000000 00000000
00004998  00000000 00000000                        4201+          DS    FD                    gap
000049A0  40404040 40404040                        4202+          DC    CL8' '             was cross check skipped?
000049A8  00000000 00000000                        4203+          DS    FD                    gap
000049B0  00000000 00000000                        4204+XCO73     DS    XL16               Cross check Output
000049B8  00000000 00000000
000049C0  00000000 00000000                        4205+          DS    FD                    gap
                                                   4206+*
000049C8                                           4207+X73       DS    0F
000049C8  E310 5014 0014                 00000014  4208+          LGF   R1,V2ADDR          load v2 source
000049CE  E721 0000 0006                 00000000  4209+          VL    v2,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000049D4  E310 5018 0014                00000018  4210+         LGF   R1,V3ADDR         load v3 source
000049DA  E731 0000 0006                00000000  4211+         VL    v3,0(R1)

000049E0                                          4214+         DS    0H                      E7B2 VD
000049E0  E7                                      4215+         DC    X'E7'                   - VECTOR DIVIDE
000049E1  12                                      4216+         DC    X'12'                    v1, v2
000049E2  30                                      4217+         DC    X'30'                    v3, reserved
000049E3  0840                                    4218+         DC    HL2'2112'                reserved, m5, m4, RXB
000049E5  B2                                      4219+         DC    X'B2'

000049E6  E710 5030 000E                00004988  4221+         VST   V1,V1O73          save v1 output
000049EC  07FB                                    4222+         BR    R11               return
000049F0                                          4223+RE73     DC    0F                xl16 expected result
000049F0                                          4224+         DROP  R5
000049F0  FFFFFFFF FFFFFFFF                       4225          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFE7'   result
000049F8  FFFFFFFF FFFFFFE7
00004A00  FFFFFFFF FFFFFFFF                       4226          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
00004A08  FFFFFFFF FFFFFF00
00004A10  00000000 00000000                       4227          DC    XL16'0000000000000000 000000000000000A'   v3
00004A18  00000000 0000000A
                                                  4228
                                                  4229          VRR_C VD,X'4',X'8',N
00004A20                                          4230+         DS    0FD
00004A20                      00004A20            4231+         USING *,R5              base for test data and test routine
00004A20  00004A90                                4232+T74      DC    A(X74)            address of test routine
00004A24  004A                                    4233+         DC    H'74'             test number
00004A26  00                                      4234+         DC    X'00'
00004A27  D5                                      4235+         DC    CL1'N'            S or Y = skip cross check
00004A28  04                                      4236+         DC    X'4'               m4
00004A29  08                                      4237+         DC    X'8'               m5
00004A2A  E5C44040 40404040                       4238+         DC    CL8'VD'           instruction name
00004A34  00004AC8                                4239+         DC    A(RE74+16)        address of v2 source
00004A38  00004AD8                                4240+         DC    A(RE74+32)        address of v3 source
00004A3C  00000010                                4241+         DC    A(16)             result length
00004A40  00004AB8                                4242+REA74    DC    A(RE74)           result address
00004A48  00000000 00000000                       4243+         DS    FD                   gap
00004A50  00000000 00000000                       4244+V1O74    DS    XL16              V1 output
00004A58  00000000 00000000
00004A60  00000000 00000000                       4245+         DS    FD                   gap
00004A68  40404040 40404040                       4246+         DC    CL8' '            was cross check skipped?
00004A70  00000000 00000000                       4247+         DS    FD                   gap
00004A78  00000000 00000000                       4248+XCO74    DS    XL16              Cross check Output
00004A80  00000000 00000000
00004A88  00000000 00000000                       4249+         DS    FD                   gap
                                                  4250+*
00004A90                                          4251+X74      DS    0F
00004A90  E310 5014 0014                00000014  4252+         LGF   R1,V2ADDR         load v2 source
00004A96  E721 0000 0006                00000000  4253+         VL    v2,0(R1)
00004A9C  E310 5018 0014                00000018  4254+         LGF   R1,V3ADDR         load v3 source
00004AA2  E731 0000 0006                00000000  4255+         VL    v3,0(R1)

00004AA8                                          4258+         DS    0H                      E7B2 VD
00004AA8  E7                                      4259+         DC    X'E7'                   - VECTOR DIVIDE
00004AA9  12                                      4260+         DC    X'12'                    v1, v2
00004AAA  30                                      4261+         DC    X'30'                    v3, reserved
00004AAB  0840                                    4262+         DC    HL2'2112'                reserved, m5, m4, RXB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004AAD  B2                                      4263+         DC    X'B2'

00004AAE  E710 5030 000E                00004A50  4265+         VST   V1,V1O74          save v1 output
00004AB4  07FB                                    4266+         BR    R11               return
00004AB8                                          4267+RE74     DC    0F                xl16 expected result
00004AB8                                          4268+         DROP  R5
00004AB8  FFFFFFFF FFFFFFFF                       4269          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFEF'   result
00004AC0  FFFFFFFF FFFFFFEF
00004AC8  FFFFFFFF FFFFFFFF                       4270          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
00004AD0  FFFFFFFF FFFFFF00
00004AD8  00000000 00000000                       4271          DC    XL16'0000000000000000 000000000000000F'   v3
00004AE0  00000000 0000000F
                                                  4272
                                                  4273          VRR_C VD,X'4',X'8',N
00004AE8                                          4274+         DS    0FD
00004AE8                      00004AE8            4275+         USING *,R5              base for test data and test routine
00004AE8  00004B58                                4276+T75      DC    A(X75)            address of test routine
00004AEC  004B                                    4277+         DC    H'75'             test number
00004AEE  00                                      4278+         DC    X'00'
00004AEF  D5                                      4279+         DC    CL1'N'            S or Y = skip cross check
00004AF0  04                                      4280+         DC    X'4'               m4
00004AF1  08                                      4281+         DC    X'8'               m5
00004AF2  E5C44040 40404040                       4282+         DC    CL8'VD'           instruction name
00004AFC  00004B90                                4283+         DC    A(RE75+16)        address of v2 source
00004B00  00004BA0                                4284+         DC    A(RE75+32)        address of v3 source
00004B04  00000010                                4285+         DC    A(16)             result length
00004B08  00004B80                                4286+REA75    DC    A(RE75)           result address
00004B10  00000000 00000000                       4287+         DS    FD                   gap
00004B18  00000000 00000000                       4288+V1O75    DS    XL16              V1 output
00004B20  00000000 00000000
00004B28  00000000 00000000                       4289+         DS    FD                   gap
00004B30  40404040 40404040                       4290+         DC    CL8' '            was cross check skipped?
00004B38  00000000 00000000                       4291+         DS    FD                   gap
00004B40  00000000 00000000                       4292+XCO75    DS    XL16              Cross check Output
00004B48  00000000 00000000
00004B50  00000000 00000000                       4293+         DS    FD                   gap
                                                  4294+*
00004B58                                          4295+X75      DS    0F
00004B58  E310 5014 0014                00000014  4296+         LGF   R1,V2ADDR         load v2 source
00004B5E  E721 0000 0006                00000000  4297+         VL    v2,0(R1)
00004B64  E310 5018 0014                00000018  4298+         LGF   R1,V3ADDR         load v3 source
00004B6A  E731 0000 0006                00000000  4299+         VL    v3,0(R1)

00004B70                                          4302+         DS    0H                     E7B2 VD
00004B70  E7                                      4303+         DC    X'E7'                  - VECTOR DIVIDE
00004B71  12                                      4304+         DC    X'12'                   v1, v2
00004B72  30                                      4305+         DC    X'30'                   v3, reserved
00004B73  0840                                    4306+         DC    HL2'2112'               reserved, m5, m4, RXB
00004B75  B2                                      4307+         DC    X'B2'

00004B76  E710 5030 000E                00004B18  4309+         VST   V1,V1O75          save v1 output
00004B7C  07FB                                    4310+         BR    R11               return
00004B80                                          4311+RE75     DC    0F                xl16 expected result
00004B80                                          4312+         DROP  R5
00004B80  FFFFFFFF FFFFFFFF                       4313          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF81'   result
00004B88  FFFFFFFF FFFFFF81
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004B90  80000000 00000000                        4314          DC    XL16'8000000000000000 00000000000000FF'   v2
00004B98  00000000 000000FF
00004BA0  01020304 05060708                        4315          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00004BA8  090A0B0C 0D0E0F10
                                                   4316
                                                   4317          VRR_C VD,X'4',X'8',N
00004BB0                                           4318+         DS    0FD
00004BB0                       00004BB0            4319+         USING *,R5              base for test data and test routine
00004BB0  00004C20                                 4320+T76      DC    A(X76)            address of test routine
00004BB4  004C                                     4321+         DC    H'76'             test number
00004BB6  00                                       4322+         DC    X'00'
00004BB7  D5                                       4323+         DC    CL1'N'            S or Y = skip cross check
00004BB8  04                                       4324+         DC    X'4'               m4
00004BB9  08                                       4325+         DC    X'8'               m5
00004BBA  E5C44040 40404040                        4326+         DC    CL8'VD'           instruction name
00004BC4  00004C58                                 4327+         DC    A(RE76+16)        address of v2 source
00004BC8  00004C68                                 4328+         DC    A(RE76+32)        address of v3 source
00004BCC  00000010                                 4329+         DC    A(16)             result length
00004BD0  00004C48                                 4330+REA76    DC    A(RE76)           result address
00004BD8  00000000 00000000                        4331+         DS    FD                   gap
00004BE0  00000000 00000000                        4332+V1O76    DS    XL16              V1 output
00004BE8  00000000 00000000
00004BF0  00000000 00000000                        4333+         DS    FD                   gap
00004BF8  40404040 40404040                        4334+         DC    CL8' '            was cross check skipped?
00004C00  00000000 00000000                        4335+         DS    FD                   gap
00004C08  00000000 00000000                        4336+XCO76    DS    XL16              Cross check Output
00004C10  00000000 00000000
00004C18  00000000 00000000                        4337+         DS    FD                   gap
                                                   4338+*
00004C20                                           4339+X76      DS    0F
00004C20  E310 5014 0014                00000014   4340+         LGF   R1,V2ADDR         load v2 source
00004C26  E721 0000 0006                00000000   4341+         VL    v2,0(R1)
00004C2C  E310 5018 0014                00000018   4342+         LGF   R1,V3ADDR         load v3 source
00004C32  E731 0000 0006                00000000   4343+         VL    v3,0(R1)

00004C38                                           4346+         DS    0H                     E7B2 VD
00004C38  E7                                       4347+         DC    X'E7'                  - VECTOR DIVIDE
00004C39  12                                       4348+         DC    X'12'                   v1, v2
00004C3A  30                                       4349+         DC    X'30'                   v3, reserved
00004C3B  0840                                     4350+         DC    HL2'2112'               reserved, m5, m4, RXB
00004C3D  B2                                       4351+         DC    X'B2'

00004C3E  E710 5030 000E                00004BE0   4353+         VST   V1,V1O76          save v1 output
00004C44  07FB                                     4354+         BR    R11               return
00004C48                                           4355+RE76     DC    0F                xl16 expected result
00004C48                                           4356+         DROP  R5
00004C48  007FFFFF FFFFFFFF                        4357          DC    XL16'007FFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   result
00004C50  FFFFFFFF FFFFFFFF
00004C58  80000000 00000000                        4358          DC    XL16'8000000000000000 00000000000000FF'   v2
00004C60  00000000 000000FF
00004C68  FFFFFFFF FFFFFFFF                        4359          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v3
00004C70  FFFFFFFF FFFFFF00
                                                   4360
                                                   4361          VRR_C VD,X'4',X'8',N
00004C78                                           4362+         DS    0FD
00004C78                       00004C78            4363+         USING *,R5              base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004C78  00004CE8                                 4364+T77      DC    A(X77)            address of test routine
00004C7C  004D                                     4365+         DC    H'77'             test number
00004C7E  00                                       4366+         DC    X'00'
00004C7F  D5                                       4367+         DC    CL1'N'            S or Y = skip cross check
00004C80  04                                       4368+         DC    X'4'               m4
00004C81  08                                       4369+         DC    X'8'               m5
00004C82  E5C44040 40404040                        4370+         DC    CL8'VD'           instruction name
00004C8C  00004D20                                 4371+         DC    A(RE77+16)        address of v2 source
00004C90  00004D30                                 4372+         DC    A(RE77+32)        address of v3 source
00004C94  00000010                                 4373+         DC    A(16)             result length
00004C98  00004D10                                 4374+REA77    DC    A(RE77)           result address
00004CA0  00000000 00000000                        4375+         DS    FD                   gap
00004CA8  00000000 00000000                        4376+V1O77    DS    XL16              V1 output
00004CB0  00000000 00000000
00004CB8  00000000 00000000                        4377+         DS    FD                   gap
00004CC0  40404040 40404040                        4378+         DC    CL8' '            was cross check skipped?
00004CC8  00000000 00000000                        4379+         DS    FD                   gap
00004CD0  00000000 00000000                        4380+XCO77    DS    XL16              Cross check Output
00004CD8  00000000 00000000
00004CE0  00000000 00000000                        4381+         DS    FD                   gap
                                                   4382+*
00004CE8                                           4383+X77      DS    0F
00004CE8  E310 5014 0014                 00000014  4384+         LGF   R1,V2ADDR         load v2 source
00004CEE  E721 0000 0006                 00000000  4385+         VL    v2,0(R1)
00004CF4  E310 5018 0014                 00000018  4386+         LGF   R1,V3ADDR         load v3 source
00004CFA  E731 0000 0006                 00000000  4387+         VL    v3,0(R1)

00004D00                                           4390+         DS    0H                      E7B2 VD
00004D00  E7                                       4391+         DC    X'E7'                   - VECTOR DIVIDE
00004D01  12                                       4392+         DC    X'12'                    v1, v2
00004D02  30                                       4393+         DC    X'30'                    v3, reserved
00004D03  0840                                     4394+         DC    HL2'2112'                reserved, m5, m4, RXB
00004D05  B2                                       4395+         DC    X'B2'

00004D06  E710 5030 000E                 00004CA8  4397+         VST   V1,V1O77          save v1 output
00004D0C  07FB                                     4398+         BR    R11               return
00004D10                                           4399+RE77     DC    0F                xl16 expected result
00004D10                                           4400+         DROP  R5
00004D10  00000000 00000000                        4401          DC    XL16'0000000000000000 0000000000000001'   result
00004D18  00000000 00000001
00004D20  80000000 00000000                        4402          DC    XL16'8000000000000000 00000000000000FF'   v2
00004D28  00000000 000000FF
00004D30  81020304 85060708                        4403          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   v3
00004D38  890A0B0C 8D0E0F10
                                                   4404
                                                   4405          VRR_C VD,X'4',X'8',N
00004D40                                           4406+         DS    0FD
00004D40                         00004D40          4407+         USING *,R5              base for test data and test routine
00004D40  00004DB0                                 4408+T78      DC    A(X78)            address of test routine
00004D44  004E                                     4409+         DC    H'78'             test number
00004D46  00                                       4410+         DC    X'00'
00004D47  D5                                       4411+         DC    CL1'N'            S or Y = skip cross check
00004D48  04                                       4412+         DC    X'4'               m4
00004D49  08                                       4413+         DC    X'8'               m5
00004D4A  E5C44040 40404040                        4414+         DC    CL8'VD'           instruction name
00004D54  00004DE8                                 4415+         DC    A(RE78+16)        address of v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004D58  00004DF8                                 4416+         DC    A(RE78+32)        address of v3 source
00004D5C  00000010                                 4417+         DC    A(16)             result length
00004D60  00004DD8                                 4418+REA78    DC    A(RE78)           result address
00004D68  00000000 00000000                        4419+         DS    FD                   gap
00004D70  00000000 00000000                        4420+V1O78    DS    XL16              V1 output
00004D78  00000000 00000000
00004D80  00000000 00000000                        4421+         DS    FD                   gap
00004D88  40404040 40404040                        4422+         DC    CL8' '            was cross check skipped?
00004D90  00000000 00000000                        4423+         DS    FD                   gap
00004D98  00000000 00000000                        4424+XCO78    DS    XL16              Cross check Output
00004DA0  00000000 00000000
00004DA8  00000000 00000000                        4425+         DS    FD                   gap
                                                   4426+*
00004DB0                                           4427+X78      DS    0F
00004DB0  E310 5014 0014                 00000014  4428+         LGF   R1,V2ADDR         load v2 source
00004DB6  E721 0000 0006                 00000000  4429+         VL    v2,0(R1)
00004DBC  E310 5018 0014                 00000018  4430+         LGF   R1,V3ADDR         load v3 source
00004DC2  E731 0000 0006                 00000000  4431+         VL    v3,0(R1)

00004DC8                                           4434+         DS    0H                     E7B2 VD
00004DC8  E7                                       4435+         DC    X'E7'                  - VECTOR DIVIDE
00004DC9  12                                       4436+         DC    X'12'                   v1, v2
00004DCA  30                                       4437+         DC    X'30'                   v3, reserved
00004DCB  0840                                     4438+         DC    HL2'2112'               reserved, m5, m4, RXB
00004DCD  B2                                       4439+         DC    X'B2'

00004DCE  E710 5030 000E                 00004D70  4441+         VST   V1,V1O78          save v1 output
00004DD4  07FB                                     4442+         BR    R11               return
00004DD8                                           4443+RE78     DC    0F                xl16 expected result
00004DD8                                           4444+         DROP  R5
00004DD8  00000000 00000000                        4445          DC    XL16'0000000000000000 0000000000000000'   result
00004DE0  00000000 00000000
00004DE8  81020304 85060708                        4446          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   v2
00004DF0  890A0B0C 8D0E0F10
00004DF8  80000000 00000000                        4447          DC    XL16'8000000000000000 00000000000000FF'   v3
00004E00  00000000 000000FF
                                                   4448
                                                   4449 * skip XC
                                                   4450          VRR_C VD,X'4',X'8',S
00004E08                                           4451+         DS    0FD
00004E08                         00004E08          4452+         USING *,R5              base for test data and test routine
00004E08  00004E78                                 4453+T79      DC    A(X79)            address of test routine
00004E0C  004F                                     4454+         DC    H'79'             test number
00004E0E  00                                       4455+         DC    X'00'
00004E0F  E2                                       4456+         DC    CL1'S'            S or Y = skip cross check
00004E10  04                                       4457+         DC    X'4'               m4
00004E11  08                                       4458+         DC    X'8'               m5
00004E12  E5C44040 40404040                        4459+         DC    CL8'VD'           instruction name
00004E1C  00004EB0                                 4460+         DC    A(RE79+16)        address of v2 source
00004E20  00004EC0                                 4461+         DC    A(RE79+32)        address of v3 source
00004E24  00000010                                 4462+         DC    A(16)             result length
00004E28  00004EA0                                 4463+REA79    DC    A(RE79)           result address
00004E30  00000000 00000000                        4464+         DS    FD                   gap
00004E38  00000000 00000000                        4465+V1O79    DS    XL16              V1 output
00004E40  00000000 00000000
00004E48  00000000 00000000                        4466+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004E50  40404040 40404040                        4467+         DC    CL8' '            was cross check skipped?
00004E58  00000000 00000000                        4468+         DS    FD                   gap
00004E60  00000000 00000000                        4469+XCO79    DS    XL16              Cross check Output
00004E68  00000000 00000000
00004E70  00000000 00000000                        4470+         DS    FD                   gap
                                                   4471+*
00004E78                                           4472+X79      DS    0F
00004E78  E310 5014 0014                 00000014  4473+         LGF   R1,V2ADDR         load v2 source
00004E7E  E721 0000 0006                 00000000  4474+         VL    v2,0(R1)
00004E84  E310 5018 0014                 00000018  4475+         LGF   R1,V3ADDR         load v3 source
00004E8A  E731 0000 0006                 00000000  4476+         VL    v3,0(R1)

00004E90                                           4479+         DS    0H                      E7B2 VD
00004E90  E7                                       4480+         DC    X'E7'                   - VECTOR DIVIDE
00004E91  12                                       4481+         DC    X'12'                    v1, v2
00004E92  30                                       4482+         DC    X'30'                    v3, reserved
00004E93  0840                                     4483+         DC    HL2'2112'                reserved, m5, m4, RXB
00004E95  B2                                       4484+         DC    X'B2'

00004E96  E710 5030 000E                 00004E38  4486+         VST   V1,V1O79          save v1 output
00004E9C  07FB                                     4487+         BR    R11               return
00004EA0                                           4488+RE79     DC    0F                xl16 expected result
00004EA0                                           4489+         DROP  R5
00004EA0  00000000 00000000                        4490          DC    XL16'0000000000000000 0000000000000000'   result
00004EA8  00000000 00000000
00004EB0  80000000 00000000                        4491          DC    XL16'8000000000000000 0000000000000000'   v2
00004EB8  00000000 00000000
00004EC0  FFFFFFFF FFFFFFFF                        4492          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v3
00004EC8  FFFFFFFF FFFFFFFF
                                                   4493
                                                   4494 *-----------------------------------------------------------------
                                                   4495 *   VR     - VECTOR REMAINDER
                                                   4496 *-----------------------------------------------------------------
                                                   4497 *Word    M4:  2, m5= 8 (IDC=1)
                                                   4498
                                                   4499          VRR_C VR,X'2',X'8',N
00004ED0                                           4500+         DS    0FD
00004ED0                         00004ED0          4501+         USING *,R5              base for test data and test routine
00004ED0  00004F40                                 4502+T80      DC    A(X80)            address of test routine
00004ED4  0050                                     4503+         DC    H'80'             test number
00004ED6  00                                       4504+         DC    X'00'
00004ED7  D5                                       4505+         DC    CL1'N'            S or Y = skip cross check
00004ED8  02                                       4506+         DC    X'2'               m4
00004ED9  08                                       4507+         DC    X'8'               m5
00004EDA  E5D94040 40404040                        4508+         DC    CL8'VR'           instruction name
00004EE4  00004F78                                 4509+         DC    A(RE80+16)        address of v2 source
00004EE8  00004F88                                 4510+         DC    A(RE80+32)        address of v3 source
00004EEC  00000010                                 4511+         DC    A(16)             result length
00004EF0  00004F68                                 4512+REA80    DC    A(RE80)           result address
00004EF8  00000000 00000000                        4513+         DS    FD                   gap
00004F00  00000000 00000000                        4514+V1O80    DS    XL16              V1 output
00004F08  00000000 00000000
00004F10  00000000 00000000                        4515+         DS    FD                   gap
00004F18  40404040 40404040                        4516+         DC    CL8' '            was cross check skipped?
00004F20  00000000 00000000                        4517+         DS    FD                   gap
00004F28  00000000 00000000                        4518+XCO80    DS    XL16              Cross check Output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004F30  00000000 00000000
00004F38  00000000 00000000                         4519+         DS    FD                    gap
                                                    4520+*
00004F40                                            4521+X80      DS    0F
00004F40  E310 5014 0014                 00000014   4522+         LGF   R1,V2ADDR         load v2 source
00004F46  E721 0000 0006                 00000000   4523+         VL    v2,0(R1)
00004F4C  E310 5018 0014                 00000018   4524+         LGF   R1,V3ADDR         load v3 source
00004F52  E731 0000 0006                 00000000   4525+         VL    v3,0(R1)

00004F58                                            4528+         DS    0H                      E7B3 VR
00004F58  E7                                        4529+         DC    X'E7'                   - VECTOR REMAINDER
00004F59  12                                        4530+         DC    X'12'                    v1, v2
00004F5A  30                                        4531+         DC    X'30'                    v3, reserved
00004F5B  0820                                      4532+         DC    HL2'2080'                reserved, m5, m4, RXB
00004F5D  B3                                        4533+         DC    X'B3'

00004F5E  E710 5030 000E                 00004F00   4535+         VST   V1,V1O80          save v1 output
00004F64  07FB                                      4536+         BR    R11               return
00004F68                                            4537+RE80     DC    0F                xl16 expected result
00004F68                                            4538+         DROP  R5
00004F68  00000000 00000000                         4539          DC    XL16'00000000 00000000 00000000 00000000'   result
00004F70  00000000 00000000
00004F78  00000000 00000000                         4540          DC    XL16'00000000 00000000 00000000 00000000'   v2
00004F80  00000000 00000000
00004F88  01020304 05060708                         4541          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00004F90  090A0B0C 0D0E0F10
                                                    4542
                                                    4543          VRR_C VR,X'2',X'8',N
00004F98                                            4544+         DS    0FD
00004F98                         00004F98           4545+         USING *,R5              base for test data and test routine
00004F98  00005008                                  4546+T81      DC    A(X81)            address of test routine
00004F9C  0051                                      4547+         DC    H'81'             test number
00004F9E  00                                        4548+         DC    X'00'
00004F9F  D5                                        4549+         DC    CL1'N'            S or Y = skip cross check
00004FA0  02                                        4550+         DC    X'2'               m4
00004FA1  08                                        4551+         DC    X'8'               m5
00004FA2  E5D94040 40404040                         4552+         DC    CL8'VR'           instruction name
00004FAC  00005040                                  4553+         DC    A(RE81+16)        address of v2 source
00004FB0  00005050                                  4554+         DC    A(RE81+32)        address of v3 source
00004FB4  00000010                                  4555+         DC    A(16)             result length
00004FB8  00005030                                  4556+REA81    DC    A(RE81)           result address
00004FC0  00000000 00000000                         4557+         DS    FD                   gap
00004FC8  00000000 00000000                         4558+V1O81    DS    XL16              V1 output
00004FD0  00000000 00000000
00004FD8  00000000 00000000                         4559+         DS    FD                   gap
00004FE0  40404040 40404040                         4560+         DC    CL8' '            was cross check skipped?
00004FE8  00000000 00000000                         4561+         DS    FD                   gap
00004FF0  00000000 00000000                         4562+XCO81    DS    XL16              Cross check Output
00004FF8  00000000 00000000
00005000  00000000 00000000                         4563+         DS    FD                   gap
                                                    4564+*
00005008                                            4565+X81      DS    0F
00005008  E310 5014 0014                 00000014   4566+         LGF   R1,V2ADDR         load v2 source
0000500E  E721 0000 0006                 00000000   4567+         VL    v2,0(R1)
00005014  E310 5018 0014                 00000018   4568+         LGF   R1,V3ADDR         load v3 source
0000501A  E731 0000 0006                 00000000   4569+         VL    v3,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005020                                           4572+         DS    0H                      E7B3 VR
00005020  E7                                       4573+         DC    X'E7'                   - VECTOR REMAINDER
00005021  12                                       4574+         DC    X'12'                    v1, v2
00005022  30                                       4575+         DC    X'30'                    v3, reserved
00005023  0820                                     4576+         DC    HL2'2080'                reserved, m5, m4, RXB
00005025  B3                                       4577+         DC    X'B3'

00005026  E710 5030 000E                00004FC8   4579+         VST   V1,V1O81          save v1 output
0000502C  07FB                                     4580+         BR    R11               return
00005030                                           4581+RE81     DC    0F                xl16 expected result
00005030                                           4582+         DROP  R5
00005030  01020303 05060707                        4583          DC    XL16'01020303 05060707 090A0B0B 0D0E0F0F'   result
00005038  090A0B0B 0D0E0F0F
00005040  01020303 05060707                        4584          DC    XL16'01020303 05060707 090A0B0B 0D0E0F0F'   v2
00005048  090A0B0B 0D0E0F0F
00005050  01020304 05060708                        4585          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00005058  090A0B0C 0D0E0F10
                                                   4586
                                                   4587          VRR_C VR,X'2',X'8',N
00005060                                           4588+         DS    0FD
00005060                        00005060           4589+         USING *,R5              base for test data and test routine
00005060  000050D0                                 4590+T82      DC    A(X82)            address of test routine
00005064  0052                                     4591+         DC    H'82'             test number
00005066  00                                       4592+         DC    X'00'
00005067  D5                                       4593+         DC    CL1'N'            S or Y = skip cross check
00005068  02                                       4594+         DC    X'2'               m4
00005069  08                                       4595+         DC    X'8'               m5
0000506A  E5D94040 40404040                        4596+         DC    CL8'VR'           instruction name
00005074  00005108                                 4597+         DC    A(RE82+16)        address of v2 source
00005078  00005118                                 4598+         DC    A(RE82+32)        address of v3 source
0000507C  00000010                                 4599+         DC    A(16)             result length
00005080  000050F8                                 4600+REA82    DC    A(RE82)           result address
00005088  00000000 00000000                        4601+         DS    FD                   gap
00005090  00000000 00000000                        4602+V1O82    DS    XL16              V1 output
00005098  00000000 00000000
000050A0  00000000 00000000                        4603+         DS    FD                   gap
000050A8  40404040 40404040                        4604+         DC    CL8' '            was cross check skipped?
000050B0  00000000 00000000                        4605+         DS    FD                   gap
000050B8  00000000 00000000                        4606+XCO82    DS    XL16              Cross check Output
000050C0  00000000 00000000
000050C8  00000000 00000000                        4607+         DS    FD                   gap
                                                   4608+*
000050D0                                           4609+X82      DS    0F
000050D0  E310 5014 0014                00000014   4610+         LGF   R1,V2ADDR         load v2 source
000050D6  E721 0000 0006                00000000   4611+         VL    v2,0(R1)
000050DC  E310 5018 0014                00000018   4612+         LGF   R1,V3ADDR         load v3 source
000050E2  E731 0000 0006                00000000   4613+         VL    v3,0(R1)

000050E8                                           4616+         DS    0H                      E7B3 VR
000050E8  E7                                       4617+         DC    X'E7'                   - VECTOR REMAINDER
000050E9  12                                       4618+         DC    X'12'                    v1, v2
000050EA  30                                       4619+         DC    X'30'                    v3, reserved
000050EB  0820                                     4620+         DC    HL2'2080'                reserved, m5, m4, RXB
000050ED  B3                                       4621+         DC    X'B3'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000050EE  E710 5030 000E                00005090  4623+         VST   V1,V1O82          save v1 output
000050F4  07FB                                    4624+         BR    R11               return
000050F8                                          4625+RE82     DC    0F                xl16 expected result
000050F8                                          4626+         DROP  R5
000050F8  00000000 00000000                       4627          DC    XL16'00000000 00000000 00000000 00000000'   result
00005100  00000000 00000000
00005108  01020304 05060708                       4628          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v2
00005110  090A0B0C 0D0E0F10
00005118  01020304 05060708                       4629          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00005120  090A0B0C 0D0E0F10
                                                  4630
                                                  4631          VRR_C VR,X'2',X'8',N
00005128                                          4632+         DS    0FD
00005128                      00005128            4633+         USING *,R5              base for test data and test routine
00005128  00005198                                4634+T83      DC    A(X83)            address of test routine
0000512C  0053                                    4635+         DC    H'83'             test number
0000512E  00                                      4636+         DC    X'00'
0000512F  D5                                      4637+         DC    CL1'N'            S or Y = skip cross check
00005130  02                                      4638+         DC    X'2'               m4
00005131  08                                      4639+         DC    X'8'               m5
00005132  E5D94040 40404040                       4640+         DC    CL8'VR'           instruction name
0000513C  000051D0                                4641+         DC    A(RE83+16)        address of v2 source
00005140  000051E0                                4642+         DC    A(RE83+32)        address of v3 source
00005144  00000010                                4643+         DC    A(16)             result length
00005148  000051C0                                4644+REA83    DC    A(RE83)           result address
00005150  00000000 00000000                       4645+         DS    FD                   gap
00005158  00000000 00000000                       4646+V1O83    DS    XL16              V1 output
00005160  00000000 00000000
00005168  00000000 00000000                       4647+         DS    FD                   gap
00005170  40404040 40404040                       4648+         DC    CL8' '            was cross check skipped?
00005178  00000000 00000000                       4649+         DS    FD                   gap
00005180  00000000 00000000                       4650+XCO83    DS    XL16              Cross check Output
00005188  00000000 00000000
00005190  00000000 00000000                       4651+         DS    FD                   gap
                                                  4652+*
00005198                                          4653+X83      DS    0F
00005198  E310 5014 0014                00000014  4654+         LGF   R1,V2ADDR         load v2 source
0000519E  E721 0000 0006                00000000  4655+         VL    v2,0(R1)
000051A4  E310 5018 0014                00000018  4656+         LGF   R1,V3ADDR         load v3 source
000051AA  E731 0000 0006                00000000  4657+         VL    v3,0(R1)

000051B0                                          4660+         DS    0H                     E7B3 VR
000051B0  E7                                      4661+         DC    X'E7'                  - VECTOR REMAINDER
000051B1  12                                      4662+         DC    X'12'                   v1, v2
000051B2  30                                      4663+         DC    X'30'                   v3, reserved
000051B3  0820                                    4664+         DC    HL2'2080'               reserved, m5, m4, RXB
000051B5  B3                                      4665+         DC    X'B3'

000051B6  E710 5030 000E                00005158  4667+         VST   V1,V1O83          save v1 output
000051BC  07FB                                    4668+         BR    R11               return
000051C0                                          4669+RE83     DC    0F                xl16 expected result
000051C0                                          4670+         DROP  R5
000051C0  00000000 00000001                       4671          DC    XL16'00000000 00000001 00000007 00000007'   result
000051C8  00000007 00000007
000051D0  7FFFFFFF 7FFFFFFF                       4672          DC    XL16'7FFFFFFF 7FFFFFFF 7FFFFFFF 7FFFFFFF'   v2
000051D8  7FFFFFFF 7FFFFFFF
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000051E0  00000001 00000002                        4673          DC    XL16'00000001 00000002 0000000A 0000000F'   v3
000051E8  0000000A 0000000F
                                                   4674
                                                   4675          VRR_C VR,X'2',X'8',N
000051F0                                           4676+         DS    0FD
000051F0                      000051F0             4677+         USING *,R5              base for test data and test routine
000051F0  00005260                                 4678+T84      DC    A(X84)            address of test routine
000051F4  0054                                     4679+         DC    H'84'             test number
000051F6  00                                       4680+         DC    X'00'
000051F7  D5                                       4681+         DC    CL1'N'            S or Y = skip cross check
000051F8  02                                       4682+         DC    X'2'               m4
000051F9  08                                       4683+         DC    X'8'               m5
000051FA  E5D94040 40404040                        4684+         DC    CL8'VR'           instruction name
00005204  00005298                                 4685+         DC    A(RE84+16)        address of v2 source
00005208  000052A8                                 4686+         DC    A(RE84+32)        address of v3 source
0000520C  00000010                                 4687+         DC    A(16)             result length
00005210  00005288                                 4688+REA84    DC    A(RE84)           result address
00005218  00000000 00000000                        4689+         DS    FD                   gap
00005220  00000000 00000000                        4690+V1O84    DS    XL16              V1 output
00005228  00000000 00000000
00005230  00000000 00000000                        4691+         DS    FD                   gap
00005238  40404040 40404040                        4692+         DC    CL8' '            was cross check skipped?
00005240  00000000 00000000                        4693+         DS    FD                   gap
00005248  00000000 00000000                        4694+XCO84    DS    XL16              Cross check Output
00005250  00000000 00000000
00005258  00000000 00000000                        4695+         DS    FD                   gap
                                                   4696+*
00005260                                           4697+X84      DS    0F
00005260  E310 5014 0014                 00000014  4698+         LGF   R1,V2ADDR         load v2 source
00005266  E721 0000 0006                 00000000  4699+         VL    v2,0(R1)
0000526C  E310 5018 0014                 00000018  4700+         LGF   R1,V3ADDR         load v3 source
00005272  E731 0000 0006                 00000000  4701+         VL    v3,0(R1)

00005278                                           4704+         DS    0H                      E7B3 VR
00005278  E7                                       4705+         DC    X'E7'                   - VECTOR REMAINDER
00005279  12                                       4706+         DC    X'12'                    v1, v2
0000527A  30                                       4707+         DC    X'30'                    v3, reserved
0000527B  0820                                     4708+         DC    HL2'2080'                reserved, m5, m4, RXB
0000527D  B3                                       4709+         DC    X'B3'

0000527E  E710 5030 000E                 00005220  4711+         VST   V1,V1O84          save v1 output
00005284  07FB                                     4712+         BR    R11               return
00005288                                           4713+RE84     DC    0F                xl16 expected result
00005288                                           4714+         DROP  R5
00005288  00008103 02695037                        4715          DC    XL16'00008103 02695037 01736557 0A81786F'   result
00005290  01736557 0A81786F
00005298  7FFFFFFF 7FFFFFFF                        4716          DC    XL16'7FFFFFFF 7FFFFFFF 7FFFFFFF 7FFFFFFF'   v2
000052A0  7FFFFFFF 7FFFFFFF
000052A8  01020304 05060708                        4717          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
000052B0  090A0B0C 0D0E0F10
                                                   4718
                                                   4719          VRR_C VR,X'2',X'8',N
000052B8                                           4720+         DS    0FD
000052B8                      000052B8             4721+         USING *,R5              base for test data and test routine
000052B8  00005328                                 4722+T85      DC    A(X85)            address of test routine
000052BC  0055                                     4723+         DC    H'85'             test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000052BE  00                                       4724+         DC    X'00'
000052BF  D5                                       4725+         DC    CL1'N'            S or Y = skip cross check
000052C0  02                                       4726+         DC    X'2'               m4
000052C1  08                                       4727+         DC    X'8'               m5
000052C2  E5D94040 40404040                        4728+         DC    CL8'VR'           instruction name
000052CC  00005360                                 4729+         DC    A(RE85+16)        address of v2 source
000052D0  00005370                                 4730+         DC    A(RE85+32)        address of v3 source
000052D4  00000010                                 4731+         DC    A(16)             result length
000052D8  00005350                                 4732+REA85    DC    A(RE85)           result address
000052E0  00000000 00000000                        4733+         DS    FD                   gap
000052E8  00000000 00000000                        4734+V1O85    DS    XL16              V1 output
000052F0  00000000 00000000
000052F8  00000000 00000000                        4735+         DS    FD                   gap
00005300  40404040 40404040                        4736+         DC    CL8' '            was cross check skipped?
00005308  00000000 00000000                        4737+         DS    FD                   gap
00005310  00000000 00000000                        4738+XCO85    DS    XL16              Cross check Output
00005318  00000000 00000000
00005320  00000000 00000000                        4739+         DS    FD                   gap
                                                   4740+*
00005328                                           4741+X85      DS    0F
00005328  E310 5014 0014                00000014   4742+         LGF   R1,V2ADDR         load v2 source
0000532E  E721 0000 0006                00000000   4743+         VL    v2,0(R1)
00005334  E310 5018 0014                00000018   4744+         LGF   R1,V3ADDR         load v3 source
0000533A  E731 0000 0006                00000000   4745+         VL    v3,0(R1)

00005340                                           4748+         DS    0H                     E7B3 VR
00005340  E7                                       4749+         DC    X'E7'                - VECTOR REMAINDER
00005341  12                                       4750+         DC    X'12'                 v1, v2
00005342  30                                       4751+         DC    X'30'                 v3, reserved
00005343  0820                                     4752+         DC    HL2'2080'             reserved, m5, m4, RXB
00005345  B3                                       4753+         DC    X'B3'

00005346  E710 5030 000E                000052E8   4755+         VST   V1,V1O85          save v1 output
0000534C  07FB                                     4756+         BR    R11               return
00005350                                           4757+RE85     DC    0F                xl16 expected result
00005350                                           4758+         DROP  R5
00005350  00000000 FFFFFFFF                        4759          DC    XL16'00000000 FFFFFFFF FFFFFFFF FFFFFFFF'   result
00005358  FFFFFFFF FFFFFFFF
00005360  FFFFFFFF FFFFFFFF                        4760          DC    XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'   v2
00005368  FFFFFFFF FFFFFFFF
00005370  00000001 00000002                        4761          DC    XL16'00000001 00000002 0000000A 0000000F'   v3
00005378  0000000A 0000000F
                                                   4762
                                                   4763          VRR_C VR,X'2',X'8',N
00005380                                           4764+         DS    0FD
00005380                        00005380           4765+         USING *,R5              base for test data and test routine
00005380  000053F0                                 4766+T86      DC    A(X86)            address of test routine
00005384  0056                                     4767+         DC    H'86'             test number
00005386  00                                       4768+         DC    X'00'
00005387  D5                                       4769+         DC    CL1'N'            S or Y = skip cross check
00005388  02                                       4770+         DC    X'2'               m4
00005389  08                                       4771+         DC    X'8'               m5
0000538A  E5D94040 40404040                        4772+         DC    CL8'VR'           instruction name
00005394  00005428                                 4773+         DC    A(RE86+16)        address of v2 source
00005398  00005438                                 4774+         DC    A(RE86+32)        address of v3 source
0000539C  00000010                                 4775+         DC    A(16)             result length
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000053A0  00005418                                 4776+REA86    DC    A(RE86)           result address
000053A8  00000000 00000000                        4777+         DS    FD                   gap
000053B0  00000000 00000000                        4778+V1O86    DS    XL16              V1 output
000053B8  00000000 00000000
000053C0  00000000 00000000                        4779+         DS    FD                   gap
000053C8  40404040 40404040                        4780+         DC    CL8' '            was cross check skipped?
000053D0  00000000 00000000                        4781+         DS    FD                   gap
000053D8  00000000 00000000                        4782+XCO86    DS    XL16              Cross check Output
000053E0  00000000 00000000
000053E8  00000000 00000000                        4783+         DS    FD                   gap
                                                   4784+*
000053F0                                           4785+X86      DS    0F
000053F0  E310 5014 0014                 00000014  4786+         LGF   R1,V2ADDR         load v2 source
000053F6  E721 0000 0006                 00000000  4787+         VL    v2,0(R1)
000053FC  E310 5018 0014                 00000018  4788+         LGF   R1,V3ADDR         load v3 source
00005402  E731 0000 0006                 00000000  4789+         VL    v3,0(R1)

00005408                                           4792+         DS    0H                     E7B3 VR
00005408  E7                                       4793+         DC    X'E7'                  - VECTOR REMAINDER
00005409  12                                       4794+         DC    X'12'                   v1, v2
0000540A  30                                       4795+         DC    X'30'                   v3, reserved
0000540B  0820                                     4796+         DC    HL2'2080'               reserved, m5, m4, RXB
0000540D  B3                                       4797+         DC    X'B3'

0000540E  E710 5030 000E                 000053B0  4799+         VST   V1,V1O86          save v1 output
00005414  07FB                                     4800+         BR    R11               return
00005418                                           4801+RE86     DC    0F                xl16 expected result
00005418                                           4802+         DROP  R5
00005418  FFFF7EFC FD96AFC8                        4803          DC    XL16'FFFF7EFC FD96AFC8 FE8C9AA8 F57E8790'   result
00005420  FE8C9AA8 F57E8790
00005428  80000000 80000000                        4804          DC    XL16'80000000 80000000 80000000 80000000'   v2
00005430  80000000 80000000
00005438  01020304 05060708                        4805          DC    XL16'01020304 05060708 090A0B0C 0D0E0F10'   v3
00005440  090A0B0C 0D0E0F10
                                                   4806
                                                   4807          VRR_C VR,X'2',X'8',N
00005448                                           4808+         DS    0FD
00005448                       00005448            4809+         USING *,R5              base for test data and test routine
00005448  000054B8                                 4810+T87      DC    A(X87)            address of test routine
0000544C  0057                                     4811+         DC    H'87'             test number
0000544E  00                                       4812+         DC    X'00'
0000544F  D5                                       4813+         DC    CL1'N'            S or Y = skip cross check
00005450  02                                       4814+         DC    X'2'               m4
00005451  08                                       4815+         DC    X'8'               m5
00005452  E5D94040 40404040                        4816+         DC    CL8'VR'           instruction name
0000545C  000054F0                                 4817+         DC    A(RE87+16)        address of v2 source
00005460  00005500                                 4818+         DC    A(RE87+32)        address of v3 source
00005464  00000010                                 4819+         DC    A(16)             result length
00005468  000054E0                                 4820+REA87    DC    A(RE87)           result address
00005470  00000000 00000000                        4821+         DS    FD                   gap
00005478  00000000 00000000                        4822+V1O87    DS    XL16              V1 output
00005480  00000000 00000000
00005488  00000000 00000000                        4823+         DS    FD                   gap
00005490  40404040 40404040                        4824+         DC    CL8' '            was cross check skipped?
00005498  00000000 00000000                        4825+         DS    FD                   gap
000054A0  00000000 00000000                        4826+XCO87    DS    XL16              Cross check Output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000054A8  00000000 00000000
000054B0  00000000 00000000                        4827+         DS    FD                    gap
                                                   4828+*
000054B8                                           4829+X87      DS    0F
000054B8  E310 5014 0014                00000014   4830+         LGF   R1,V2ADDR         load v2 source
000054BE  E721 0000 0006                00000000   4831+         VL    v2,0(R1)
000054C4  E310 5018 0014                00000018   4832+         LGF   R1,V3ADDR         load v3 source
000054CA  E731 0000 0006                00000000   4833+         VL    v3,0(R1)

000054D0                                           4836+         DS    0H                      E7B3 VR
000054D0  E7                                       4837+         DC    X'E7'                   - VECTOR REMAINDER
000054D1  12                                       4838+         DC    X'12'                    v1, v2
000054D2  30                                       4839+         DC    X'30'                    v3, reserved
000054D3  0820                                     4840+         DC    HL2'2080'                reserved, m5, m4, RXB
000054D5  B3                                       4841+         DC    X'B3'

000054D6  E710 5030 000E                00005478   4843+         VST   V1,V1O87          save v1 output
000054DC  07FB                                     4844+         BR    R11               return
000054E0                                           4845+RE87     DC    0F                xl16 expected result
000054E0                                           4846+         DROP  R5
000054E0  00000000 FFFFFFFF                        4847          DC    XL16'00000000 FFFFFFFF FFFFFFF9 FFFFFFF9'   result
000054E8  FFFFFFF9 FFFFFFF9
000054F0  80000001 80000001                        4848          DC    XL16'80000001 80000001 80000001 80000001'   v2
000054F8  80000001 80000001
00005500  FFFFFFFF FFFFFFFE                        4849          DC    XL16'FFFFFFFF FFFFFFFE FFFFFFF6 FFFFFFF1'   v3
00005508  FFFFFFF6 FFFFFFF1
                                                   4850
                                                   4851          VRR_C VR,X'2',X'8',N
00005510                                           4852+         DS    0FD
00005510                        00005510           4853+         USING *,R5              base for test data and test routine
00005510  00005580                                 4854+T88      DC    A(X88)            address of test routine
00005514  0058                                     4855+         DC    H'88'             test number
00005516  00                                       4856+         DC    X'00'
00005517  D5                                       4857+         DC    CL1'N'            S or Y = skip cross check
00005518  02                                       4858+         DC    X'2'               m4
00005519  08                                       4859+         DC    X'8'               m5
0000551A  E5D94040 40404040                        4860+         DC    CL8'VR'           instruction name
00005524  000055B8                                 4861+         DC    A(RE88+16)        address of v2 source
00005528  000055C8                                 4862+         DC    A(RE88+32)        address of v3 source
0000552C  00000010                                 4863+         DC    A(16)             result length
00005530  000055A8                                 4864+REA88    DC    A(RE88)           result address
00005538  00000000 00000000                        4865+         DS    FD                   gap
00005540  00000000 00000000                        4866+V1O88    DS    XL16              V1 output
00005548  00000000 00000000
00005550  00000000 00000000                        4867+         DS    FD                   gap
00005558  40404040 40404040                        4868+         DC    CL8' '            was cross check skipped?
00005560  00000000 00000000                        4869+         DS    FD                   gap
00005568  00000000 00000000                        4870+XCO88    DS    XL16              Cross check Output
00005570  00000000 00000000
00005578  00000000 00000000                        4871+         DS    FD                   gap
                                                   4872+*
00005580                                           4873+X88      DS    0F
00005580  E310 5014 0014                00000014   4874+         LGF   R1,V2ADDR         load v2 source
00005586  E721 0000 0006                00000000   4875+         VL    v2,0(R1)
0000558C  E310 5018 0014                00000018   4876+         LGF   R1,V3ADDR         load v3 source
00005592  E731 0000 0006                00000000   4877+         VL    v3,0(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005598                                           4880+         DS    0H                      E7B3 VR
00005598  E7                                       4881+         DC    X'E7'                   - VECTOR REMAINDER
00005599  12                                       4882+         DC    X'12'                    v1, v2
0000559A  30                                       4883+         DC    X'30'                    v3, reserved
0000559B  0820                                     4884+         DC    HL2'2080'                reserved, m5, m4, RXB
0000559D  B3                                       4885+         DC    X'B3'

0000559E  E710 5030 000E                00005540   4887+         VST   V1,V1O88          save v1 output
000055A4  07FB                                     4888+         BR    R11               return
000055A8                                           4889+RE88     DC    0F                xl16 expected result
000055A8                                           4890+         DROP  R5
000055A8  FEFDFCFC FAF9F8F8                        4891          DC    XL16'FEFDFCFC FAF9F8F8 F6F5F4F4 F2F1F0F0'   result
000055B0  F6F5F4F4 F2F1F0F0
000055B8  80000000 80000000                        4892          DC    XL16'80000000 80000000 80000000 80000000'   v2
000055C0  80000000 80000000
000055C8  81020304 85060708                        4893          DC    XL16'81020304 85060708 890A0B0C 8D0E0F10'   v3
000055D0  890A0B0C 8D0E0F10
                                                   4894
                                                   4895 * skip xc
                                                   4896          VRR_C VR,X'2',X'8',S
000055D8                                           4897+         DS    0FD
000055D8                        000055D8           4898+         USING *,R5              base for test data and test routine
000055D8  00005648                                 4899+T89      DC    A(X89)            address of test routine
000055DC  0059                                     4900+         DC    H'89'             test number
000055DE  00                                       4901+         DC    X'00'
000055DF  E2                                       4902+         DC    CL1'S'            S or Y = skip cross check
000055E0  02                                       4903+         DC    X'2'               m4
000055E1  08                                       4904+         DC    X'8'               m5
000055E2  E5D94040 40404040                        4905+         DC    CL8'VR'           instruction name
000055EC  00005680                                 4906+         DC    A(RE89+16)        address of v2 source
000055F0  00005690                                 4907+         DC    A(RE89+32)        address of v3 source
000055F4  00000010                                 4908+         DC    A(16)             result length
000055F8  00005670                                 4909+REA89    DC    A(RE89)           result address
00005600  00000000 00000000                        4910+         DS    FD                   gap
00005608  00000000 00000000                        4911+V1O89    DS    XL16              V1 output
00005610  00000000 00000000
00005618  00000000 00000000                        4912+         DS    FD                   gap
00005620  40404040 40404040                        4913+         DC    CL8' '            was cross check skipped?
00005628  00000000 00000000                        4914+         DS    FD                   gap
00005630  00000000 00000000                        4915+XCO89    DS    XL16              Cross check Output
00005638  00000000 00000000
00005640  00000000 00000000                        4916+         DS    FD                   gap
                                                   4917+*
00005648                                           4918+X89      DS    0F
00005648  E310 5014 0014                00000014   4919+         LGF   R1,V2ADDR         load v2 source
0000564E  E721 0000 0006                00000000   4920+         VL    v2,0(R1)
00005654  E310 5018 0014                00000018   4921+         LGF   R1,V3ADDR         load v3 source
0000565A  E731 0000 0006                00000000   4922+         VL    v3,0(R1)

00005660                                           4925+         DS    0H                      E7B3 VR
00005660  E7                                       4926+         DC    X'E7'                   - VECTOR REMAINDER
00005661  12                                       4927+         DC    X'12'                    v1, v2
00005662  30                                       4928+         DC    X'30'                    v3, reserved
00005663  0820                                     4929+         DC    HL2'2080'                reserved, m5, m4, RXB
00005665  B3                                       4930+         DC    X'B3'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00005666  E710 5030 000E                00005608  4932+         VST   V1,V1O89          save v1 output
0000566C  07FB                                    4933+         BR    R11               return
00005670                                          4934+RE89     DC    0F                xl16 expected result
00005670                                          4935+         DROP  R5
00005670  00000000 00000000                       4936          DC    XL16'00000000 00000000 FFFFFFF8 FFFFFFF8'   result
00005678  FFFFFFF8 FFFFFFF8
00005680  80000000 80000000                       4937          DC    XL16'80000000 80000000 80000000 80000000'   v2
00005688  80000000 80000000
00005690  FFFFFFFF FFFFFFFE                       4938          DC    XL16'FFFFFFFF FFFFFFFE FFFFFFF6 FFFFFFF1'   v3
00005698  FFFFFFF6 FFFFFFF1
                                                  4939
                                                  4940 *Doubleword    M4:  3, m5= 8 (IDC=1)
                                                  4941
                                                  4942          VRR_C VR,X'3',X'8',N
000056A0                                          4943+         DS    0FD
000056A0                        000056A0          4944+         USING *,R5              base for test data and test routine
000056A0  00005710                                4945+T90      DC    A(X90)            address of test routine
000056A4  005A                                    4946+         DC    H'90'             test number
000056A6  00                                      4947+         DC    X'00'
000056A7  D5                                      4948+         DC    CL1'N'            S or Y = skip cross check
000056A8  03                                      4949+         DC    X'3'               m4
000056A9  08                                      4950+         DC    X'8'               m5
000056AA  E5D94040 40404040                       4951+         DC    CL8'VR'           instruction name
000056B4  00005748                                4952+         DC    A(RE90+16)        address of v2 source
000056B8  00005758                                4953+         DC    A(RE90+32)        address of v3 source
000056BC  00000010                                4954+         DC    A(16)             result length
000056C0  00005738                                4955+REA90    DC    A(RE90)           result address
000056C8  00000000 00000000                       4956+         DS    FD                   gap
000056D0  00000000 00000000                       4957+V1O90    DS    XL16              V1 output
000056D8  00000000 00000000
000056E0  00000000 00000000                       4958+         DS    FD                   gap
000056E8  40404040 40404040                       4959+         DC    CL8' '            was cross check skipped?
000056F0  00000000 00000000                       4960+         DS    FD                   gap
000056F8  00000000 00000000                       4961+XCO90    DS    XL16              Cross check Output
00005700  00000000 00000000
00005708  00000000 00000000                       4962+         DS    FD                   gap
                                                  4963+*
00005710                                          4964+X90      DS    0F
00005710  E310 5014 0014                00000014  4965+         LGF   R1,V2ADDR         load v2 source
00005716  E721 0000 0006                00000000  4966+         VL    v2,0(R1)
0000571C  E310 5018 0014                00000018  4967+         LGF   R1,V3ADDR         load v3 source
00005722  E731 0000 0006                00000000  4968+         VL    v3,0(R1)

00005728                                          4971+         DS    0H                     E7B3 VR
00005728  E7                                      4972+         DC    X'E7'                  - VECTOR REMAINDER
00005729  12                                      4973+         DC    X'12'                   v1, v2
0000572A  30                                      4974+         DC    X'30'                   v3, reserved
0000572B  0830                                    4975+         DC    HL2'2096'               reserved, m5, m4, RXB
0000572D  B3                                      4976+         DC    X'B3'

0000572E  E710 5030 000E                000056D0  4978+         VST   V1,V1O90          save v1 output
00005734  07FB                                    4979+         BR    R11               return
00005738                                          4980+RE90     DC    0F                xl16 expected result
00005738                                          4981+         DROP  R5
00005738  00000000 00000000                       4982          DC    XL16'0000000000000000 0000000000000000'   result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005740  00000000 00000000
00005748  00000000 00000000                         4983          DC    XL16'0000000000000000 0000000000000000'   v2
00005750  00000000 00000000
00005758  01020304 05060708                         4984          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00005760  090A0B0C 0D0E0F10
                                                    4985
                                                    4986          VRR_C VR,X'3',X'8',N
00005768                                            4987+         DS    0FD
00005768                      00005768              4988+         USING *,R5                base for test data and test routine
00005768  000057D8                                  4989+T91      DC    A(X91)              address of test routine
0000576C  005B                                      4990+         DC    H'91'               test number
0000576E  00                                        4991+         DC    X'00'
0000576F  D5                                        4992+         DC    CL1'N'              S or Y = skip cross check
00005770  03                                        4993+         DC    X'3'                 m4
00005771  08                                        4994+         DC    X'8'                 m5
00005772  E5D94040 40404040                         4995+         DC    CL8'VR'             instruction name
0000577C  00005810                                  4996+         DC    A(RE91+16)          address of v2 source
00005780  00005820                                  4997+         DC    A(RE91+32)          address of v3 source
00005784  00000010                                  4998+         DC    A(16)               result length
00005788  00005800                                  4999+REA91    DC    A(RE91)             result address
00005790  00000000 00000000                         5000+         DS    FD                     gap
00005798  00000000 00000000                         5001+V1O91    DS    XL16                V1 output
000057A0  00000000 00000000
000057A8  00000000 00000000                         5002+         DS    FD                     gap
000057B0  40404040 40404040                         5003+         DC    CL8' '              was cross check skipped?
000057B8  00000000 00000000                         5004+         DS    FD                     gap
000057C0  00000000 00000000                         5005+XCO91    DS    XL16                Cross check Output
000057C8  00000000 00000000
000057D0  00000000 00000000                         5006+         DS    FD                     gap
                                                    5007+*
000057D8                                            5008+X91      DS    0F
000057D8  E310 5014 0014                 00000014   5009+         LGF   R1,V2ADDR           load v2 source
000057DE  E721 0000 0006                 00000000   5010+         VL    v2,0(R1)
000057E4  E310 5018 0014                 00000018   5011+         LGF   R1,V3ADDR           load v3 source
000057EA  E731 0000 0006                 00000000   5012+         VL    v3,0(R1)

000057F0                                            5015+         DS    0H                        E7B3 VR
000057F0  E7                                        5016+         DC    X'E7'                     - VECTOR REMAINDER
000057F1  12                                        5017+         DC    X'12'                      v1, v2
000057F2  30                                        5018+         DC    X'30'                      v3, reserved
000057F3  0830                                      5019+         DC    HL2'2096'                  reserved, m5, m4, RXB
000057F5  B3                                        5020+         DC    X'B3'

000057F6  E710 5030 000E                 00005798   5022+         VST   V1,V1O91            save v1 output
000057FC  07FB                                      5023+         BR    R11                 return
00005800                                            5024+RE91     DC    0F                  xl16 expected result
00005800                                            5025+         DROP  R5
00005800  01020303 05060707                         5026          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   result
00005808  090A0B0B 0D0E0F0F
00005810  01020303 05060707                         5027          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00005818  090A0B0B 0D0E0F0F
00005820  01020304 05060708                         5028          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00005828  090A0B0C 0D0E0F10
                                                    5029
                                                    5030          VRR_C VR,X'3',X'8',N
00005830                                            5031+         DS    0FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005830                     00005830             5032+         USING *,R5              base for test data and test routine
00005830  000058A0                                5033+T92      DC    A(X92)            address of test routine
00005834  005C                                    5034+         DC    H'92'             test number
00005836  00                                      5035+         DC    X'00'
00005837  D5                                      5036+         DC    CL1'N'            S or Y = skip cross check
00005838  03                                      5037+         DC    X'3'               m4
00005839  08                                      5038+         DC    X'8'               m5
0000583A  E5D94040 40404040                       5039+         DC    CL8'VR'           instruction name
00005844  000058D8                                5040+         DC    A(RE92+16)        address of v2 source
00005848  000058E8                                5041+         DC    A(RE92+32)        address of v3 source
0000584C  00000010                                5042+         DC    A(16)             result length
00005850  000058C8                                5043+REA92    DC    A(RE92)           result address
00005858  00000000 00000000                       5044+         DS    FD                   gap
00005860  00000000 00000000                       5045+V1O92    DS    XL16              V1 output
00005868  00000000 00000000
00005870  00000000 00000000                       5046+         DS    FD                   gap
00005878  40404040 40404040                       5047+         DC    CL8' '            was cross check skipped?
00005880  00000000 00000000                       5048+         DS    FD                   gap
00005888  00000000 00000000                       5049+XCO92    DS    XL16              Cross check Output
00005890  00000000 00000000
00005898  00000000 00000000                       5050+         DS    FD                   gap
                                                  5051+*
000058A0                                          5052+X92      DS    0F
000058A0  E310 5014 0014                 00000014 5053+         LGF   R1,V2ADDR         load v2 source
000058A6  E721 0000 0006                 00000000 5054+         VL    v2,0(R1)
000058AC  E310 5018 0014                 00000018 5055+         LGF   R1,V3ADDR         load v3 source
000058B2  E731 0000 0006                 00000000 5056+         VL    v3,0(R1)

000058B8                                          5059+         DS    0H                      E7B3 VR
000058B8  E7                                      5060+         DC    X'E7'                   - VECTOR REMAINDER
000058B9  12                                      5061+         DC    X'12'                    v1, v2
000058BA  30                                      5062+         DC    X'30'                    v3, reserved
000058BB  0830                                    5063+         DC    HL2'2096'                reserved, m5, m4, RXB
000058BD  B3                                      5064+         DC    X'B3'

000058BE  E710 5030 000E                 00005860 5066+         VST   V1,V1O92          save v1 output
000058C4  07FB                                    5067+         BR    R11               return
000058C8                                          5068+RE92     DC    0F                xl16 expected result
000058C8                                          5069+         DROP  R5
000058C8  00000000 00000000                       5070          DC    XL16'0000000000000000 0000000000000000'   result
000058D0  00000000 00000000
000058D8  01020304 05060708                       5071          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
000058E0  090A0B0C 0D0E0F10
000058E8  01020304 05060708                       5072          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000058F0  090A0B0C 0D0E0F10
                                                  5073
                                                  5074          VRR_C VR,X'3',X'8',N
000058F8                                          5075+         DS    0FD
000058F8                     000058F8             5076+         USING *,R5              base for test data and test routine
000058F8  00005968                                5077+T93      DC    A(X93)            address of test routine
000058FC  005D                                    5078+         DC    H'93'             test number
000058FE  00                                      5079+         DC    X'00'
000058FF  D5                                      5080+         DC    CL1'N'            S or Y = skip cross check
00005900  03                                      5081+         DC    X'3'               m4
00005901  08                                      5082+         DC    X'8'               m5
00005902  E5D94040 40404040                       5083+         DC    CL8'VR'           instruction name
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000590C  000059A0                                 5084+         DC    A(RE93+16)        address of v2 source
00005910  000059B0                                 5085+         DC    A(RE93+32)        address of v3 source
00005914  00000010                                 5086+         DC    A(16)             result length
00005918  00005990                                 5087+REA93    DC    A(RE93)           result address
00005920  00000000 00000000                        5088+         DS    FD                   gap
00005928  00000000 00000000                        5089+V1O93    DS    XL16              V1 output
00005930  00000000 00000000
00005938  00000000 00000000                        5090+         DS    FD                   gap
00005940  40404040 40404040                        5091+         DC    CL8' '            was cross check skipped?
00005948  00000000 00000000                        5092+         DS    FD                   gap
00005950  00000000 00000000                        5093+XCO93    DS    XL16              Cross check Output
00005958  00000000 00000000
00005960  00000000 00000000                        5094+         DS    FD                   gap
                                                   5095+*
00005968                                           5096+X93      DS    0F
00005968  E310 5014 0014                00000014   5097+         LGF   R1,V2ADDR         load v2 source
0000596E  E721 0000 0006                00000000   5098+         VL    v2,0(R1)
00005974  E310 5018 0014                00000018   5099+         LGF   R1,V3ADDR         load v3 source
0000597A  E731 0000 0006                00000000   5100+         VL    v3,0(R1)

00005980                                           5103+         DS    0H                     E7B3 VR
00005980  E7                                       5104+         DC    X'E7'                  - VECTOR REMAINDER
00005981  12                                       5105+         DC    X'12'                   v1, v2
00005982  30                                       5106+         DC    X'30'                   v3, reserved
00005983  0830                                     5107+         DC    HL2'2096'               reserved, m5, m4, RXB
00005985  B3                                       5108+         DC    X'B3'

00005986  E710 5030 000E                00005928   5110+         VST   V1,V1O93          save v1 output
0000598C  07FB                                     5111+         BR    R11               return
00005990                                           5112+RE93     DC    0F                xl16 expected result
00005990                                           5113+         DROP  R5
00005990  00000000 00000000                        5114          DC    XL16'0000000000000000 0000000000000001'   result
00005998  00000000 00000001
000059A0  7FFFFFFF 7FFFFFFF                        5115          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
000059A8  7FFFFFFF 7FFFFFFF
000059B0  00000000 00000001                        5116          DC    XL16'0000000000000001 0000000000000002'   v3
000059B8  00000000 00000002
                                                   5117
                                                   5118          VRR_C VR,X'3',X'8',N
000059C0                                           5119+         DS    0FD
000059C0                        000059C0           5120+         USING *,R5              base for test data and test routine
000059C0  00005A30                                 5121+T94      DC    A(X94)            address of test routine
000059C4  005E                                     5122+         DC    H'94'             test number
000059C6  00                                       5123+         DC    X'00'
000059C7  D5                                       5124+         DC    CL1'N'            S or Y = skip cross check
000059C8  03                                       5125+         DC    X'3'               m4
000059C9  08                                       5126+         DC    X'8'               m5
000059CA  E5D94040 40404040                        5127+         DC    CL8'VR'           instruction name
000059D4  00005A68                                 5128+         DC    A(RE94+16)        address of v2 source
000059D8  00005A78                                 5129+         DC    A(RE94+32)        address of v3 source
000059DC  00000010                                 5130+         DC    A(16)             result length
000059E0  00005A58                                 5131+REA94    DC    A(RE94)           result address
000059E8  00000000 00000000                        5132+         DS    FD                   gap
000059F0  00000000 00000000                        5133+V1O94    DS    XL16              V1 output
000059F8  00000000 00000000
00005A00  00000000 00000000                        5134+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005A08  40404040 40404040                        5135+         DC    CL8' '            was cross check skipped?
00005A10  00000000 00000000                        5136+         DS    FD                   gap
00005A18  00000000 00000000                        5137+XCO94    DS    XL16              Cross check Output
00005A20  00000000 00000000
00005A28  00000000 00000000                        5138+         DS    FD                   gap
                                                   5139+*
00005A30                                           5140+X94      DS    0F
00005A30  E310 5014 0014                 00000014  5141+         LGF   R1,V2ADDR         load v2 source
00005A36  E721 0000 0006                 00000000  5142+         VL    v2,0(R1)
00005A3C  E310 5018 0014                 00000018  5143+         LGF   R1,V3ADDR         load v3 source
00005A42  E731 0000 0006                 00000000  5144+         VL    v3,0(R1)

00005A48                                           5147+         DS    0H                     E7B3 VR
00005A48  E7                                       5148+         DC    X'E7'                  - VECTOR REMAINDER
00005A49  12                                       5149+         DC    X'12'                   v1, v2
00005A4A  30                                       5150+         DC    X'30'                   v3, reserved
00005A4B  0830                                     5151+         DC    HL2'2096'               reserved, m5, m4, RXB
00005A4D  B3                                       5152+         DC    X'B3'

00005A4E  E710 5030 000E                 000059F0  5154+         VST   V1,V1O94          save v1 output
00005A54  07FB                                     5155+         BR    R11               return
00005A58                                           5156+RE94     DC    0F                xl16 expected result
00005A58                                           5157+         DROP  R5
00005A58  00000000 00000009                        5158          DC    XL16'0000000000000009 000000000000000E'   result
00005A60  00000000 0000000E
00005A68  7FFFFFFF 7FFFFFFF                        5159          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00005A70  7FFFFFFF 7FFFFFFF
00005A78  00000000 0000000A                        5160          DC    XL16'000000000000000A 000000000000000F'   v3
00005A80  00000000 0000000F
                                                   5161
                                                   5162          VRR_C VR,X'3',X'8',N
00005A88                                           5163+         DS    0FD
00005A88                      00005A88             5164+         USING *,R5              base for test data and test routine
00005A88  00005AF8                                 5165+T95      DC    A(X95)            address of test routine
00005A8C  005F                                     5166+         DC    H'95'             test number
00005A8E  00                                       5167+         DC    X'00'
00005A8F  D5                                       5168+         DC    CL1'N'            S or Y = skip cross check
00005A90  03                                       5169+         DC    X'3'               m4
00005A91  08                                       5170+         DC    X'8'               m5
00005A92  E5D94040 40404040                        5171+         DC    CL8'VR'           instruction name
00005A9C  00005B30                                 5172+         DC    A(RE95+16)        address of v2 source
00005AA0  00005B40                                 5173+         DC    A(RE95+32)        address of v3 source
00005AA4  00000010                                 5174+         DC    A(16)             result length
00005AA8  00005B20                                 5175+REA95    DC    A(RE95)           result address
00005AB0  00000000 00000000                        5176+         DS    FD                   gap
00005AB8  00000000 00000000                        5177+V1O95    DS    XL16              V1 output
00005AC0  00000000 00000000
00005AC8  00000000 00000000                        5178+         DS    FD                   gap
00005AD0  40404040 40404040                        5179+         DC    CL8' '            was cross check skipped?
00005AD8  00000000 00000000                        5180+         DS    FD                   gap
00005AE0  00000000 00000000                        5181+XCO95    DS    XL16              Cross check Output
00005AE8  00000000 00000000
00005AF0  00000000 00000000                        5182+         DS    FD                   gap
                                                   5183+*
00005AF8                                           5184+X95      DS    0F
00005AF8  E310 5014 0014                 00000014  5185+         LGF   R1,V2ADDR         load v2 source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00005AFE  E721 0000 0006                00000000  5186+         VL    v2,0(R1)
00005B04  E310 5018 0014                00000018  5187+         LGF   R1,V3ADDR         load v3 source
00005B0A  E731 0000 0006                00000000  5188+         VL    v3,0(R1)

00005B10                                          5191+         DS    0H                      E7B3 VR
00005B10  E7                                      5192+         DC    X'E7'                   - VECTOR REMAINDER
00005B11  12                                      5193+         DC    X'12'                    v1, v2
00005B12  30                                      5194+         DC    X'30'                    v3, reserved
00005B13  0830                                    5195+         DC    HL2'2096'                reserved, m5, m4, RXB
00005B15  B3                                      5196+         DC    X'B3'

00005B16  E710 5030 000E                00005AB8  5198+         VST   V1,V1O95          save v1 output
00005B1C  07FB                                    5199+         BR    R11               return
00005B20                                          5200+RE95     DC    0F                xl16 expected result
00005B20                                          5201+         DROP  R5
00005B20  00008101 02028307                       5202          DC    XL16'0000810102028307 01736556C93B2D1F'   result
00005B28  01736556 C93B2D1F
00005B30  7FFFFFFF 7FFFFFFF                       5203          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00005B38  7FFFFFFF 7FFFFFFF
00005B40  01020304 05060708                       5204          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00005B48  090A0B0C 0D0E0F10
                                                  5205
                                                  5206          VRR_C VR,X'3',X'8',N
00005B50                                          5207+         DS    0FD
00005B50                     00005B50             5208+         USING *,R5              base for test data and test routine
00005B50  00005BC0                                5209+T96      DC    A(X96)            address of test routine
00005B54  0060                                    5210+         DC    H'96'             test number
00005B56  00                                      5211+         DC    X'00'
00005B57  D5                                      5212+         DC    CL1'N'            S or Y = skip cross check
00005B58  03                                      5213+         DC    X'3'               m4
00005B59  08                                      5214+         DC    X'8'               m5
00005B5A  E5D94040 40404040                       5215+         DC    CL8'VR'           instruction name
00005B64  00005BF8                                5216+         DC    A(RE96+16)        address of v2 source
00005B68  00005C08                                5217+         DC    A(RE96+32)        address of v3 source
00005B6C  00000010                                5218+         DC    A(16)             result length
00005B70  00005BE8                                5219+REA96    DC    A(RE96)           result address
00005B78  00000000 00000000                       5220+         DS    FD                   gap
00005B80  00000000 00000000                       5221+V1O96    DS    XL16              V1 output
00005B88  00000000 00000000
00005B90  00000000 00000000                       5222+         DS    FD                   gap
00005B98  40404040 40404040                       5223+         DC    CL8' '            was cross check skipped?
00005BA0  00000000 00000000                       5224+         DS    FD                   gap
00005BA8  00000000 00000000                       5225+XCO96    DS    XL16              Cross check Output
00005BB0  00000000 00000000
00005BB8  00000000 00000000                       5226+         DS    FD                   gap
                                                  5227+*
00005BC0                                          5228+X96      DS    0F
00005BC0  E310 5014 0014                00000014  5229+         LGF   R1,V2ADDR         load v2 source
00005BC6  E721 0000 0006                00000000  5230+         VL    v2,0(R1)
00005BCC  E310 5018 0014                00000018  5231+         LGF   R1,V3ADDR         load v3 source
00005BD2  E731 0000 0006                00000000  5232+         VL    v3,0(R1)

00005BD8                                          5235+         DS    0H                      E7B3 VR
00005BD8  E7                                      5236+         DC    X'E7'                   - VECTOR REMAINDER
00005BD9  12                                      5237+         DC    X'12'                    v1, v2
00005BDA  30                                      5238+         DC    X'30'                    v3, reserved
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005BDB  0830                                    5239+         DC    HL2'2096'                reserved, m5, m4, RXB
00005BDD  B3                                      5240+         DC    X'B3'

00005BDE  E710 5030 000E                00005B80  5242+         VST   V1,V1O96          save v1 output
00005BE4  07FB                                    5243+         BR    R11               return
00005BE8                                          5244+RE96     DC    0F                xl16 expected result
00005BE8                                          5245+         DROP  R5
00005BE8  00000000 00000000                       5246          DC    XL16'0000000000000000 FFFFFFFFFFFFFFFF'   result
00005BF0  FFFFFFFF FFFFFFFF
00005BF8  FFFFFFFF FFFFFFFF                       5247          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v2
00005C00  FFFFFFFF FFFFFFFF
00005C08  00000000 00000001                       5248          DC    XL16'0000000000000001 0000000000000002'   v3
00005C10  00000000 00000002
                                                  5249
                                                  5250          VRR_C VR,X'3',X'8',N
00005C18                                          5251+         DS    0FD
00005C18                        00005C18          5252+         USING *,R5              base for test data and test routine
00005C18  00005C88                                5253+T97      DC    A(X97)            address of test routine
00005C1C  0061                                    5254+         DC    H'97'             test number
00005C1E  00                                      5255+         DC    X'00'
00005C1F  D5                                      5256+         DC    CL1'N'            S or Y = skip cross check
00005C20  03                                      5257+         DC    X'3'               m4
00005C21  08                                      5258+         DC    X'8'               m5
00005C22  E5D94040 40404040                       5259+         DC    CL8'VR'           instruction name
00005C2C  00005CC0                                5260+         DC    A(RE97+16)        address of v2 source
00005C30  00005CD0                                5261+         DC    A(RE97+32)        address of v3 source
00005C34  00000010                                5262+         DC    A(16)             result length
00005C38  00005CB0                                5263+REA97    DC    A(RE97)           result address
00005C40  00000000 00000000                       5264+         DS    FD                   gap
00005C48  00000000 00000000                       5265+V1O97    DS    XL16              V1 output
00005C50  00000000 00000000
00005C58  00000000 00000000                       5266+         DS    FD                   gap
00005C60  40404040 40404040                       5267+         DC    CL8' '            was cross check skipped?
00005C68  00000000 00000000                       5268+         DS    FD                   gap
00005C70  00000000 00000000                       5269+XCO97    DS    XL16              Cross check Output
00005C78  00000000 00000000
00005C80  00000000 00000000                       5270+         DS    FD                   gap
                                                  5271+*
00005C88                                          5272+X97      DS    0F
00005C88  E310 5014 0014                00000014  5273+         LGF   R1,V2ADDR         load v2 source
00005C8E  E721 0000 0006                00000000  5274+         VL    v2,0(R1)
00005C94  E310 5018 0014                00000018  5275+         LGF   R1,V3ADDR         load v3 source
00005C9A  E731 0000 0006                00000000  5276+         VL    v3,0(R1)

00005CA0                                          5279+         DS    0H                     E7B3 VR
00005CA0  E7                                      5280+         DC    X'E7'                  - VECTOR REMAINDER
00005CA1  12                                      5281+         DC    X'12'                   v1, v2
00005CA2  30                                      5282+         DC    X'30'                   v3, reserved
00005CA3  0830                                    5283+         DC    HL2'2096'               reserved, m5, m4, RXB
00005CA5  B3                                      5284+         DC    X'B3'

00005CA6  E710 5030 000E                00005C48  5286+         VST   V1,V1O97          save v1 output
00005CAC  07FB                                    5287+         BR    R11               return
00005CB0                                          5288+RE97     DC    0F                xl16 expected result
00005CB0                                          5289+         DROP  R5
00005CB0  FFFFFFFF FFFFFFFA                       5290          DC    XL16'FFFFFFFFFFFFFFFA FFFFFFFFFFFFFFFF'   result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005CB8  FFFFFFFF FFFFFFFF
00005CC0  FFFFFFFF FFFFFF00                         5291          DC    XL16'FFFFFFFFFFFFFF00 FFFFFFFFFFFFFF00'    v2
00005CC8  FFFFFFFF FFFFFF00
00005CD0  00000000 0000000A                         5292          DC    XL16'000000000000000A 000000000000000F'   v3
00005CD8  00000000 0000000F
                                                    5293
                                                    5294          VRR_C VR,X'3',X'8',N
00005CE0                                            5295+         DS    0FD
00005CE0                      00005CE0              5296+         USING *,R5              base for test data and test routine
00005CE0  00005D50                                  5297+T98      DC    A(X98)            address of test routine
00005CE4  0062                                      5298+         DC    H'98'             test number
00005CE6  00                                        5299+         DC    X'00'
00005CE7  D5                                        5300+         DC    CL1'N'            S or Y = skip cross check
00005CE8  03                                        5301+         DC    X'3'               m4
00005CE9  08                                        5302+         DC    X'8'               m5
00005CEA  E5D94040 40404040                         5303+         DC    CL8'VR'           instruction name
00005CF4  00005D88                                  5304+         DC    A(RE98+16)        address of v2 source
00005CF8  00005D98                                  5305+         DC    A(RE98+32)        address of v3 source
00005CFC  00000010                                  5306+         DC    A(16)             result length
00005D00  00005D78                                  5307+REA98    DC    A(RE98)           result address
00005D08  00000000 00000000                         5308+         DS    FD                   gap
00005D10  00000000 00000000                         5309+V1O98    DS    XL16              V1 output
00005D18  00000000 00000000
00005D20  00000000 00000000                         5310+         DS    FD                   gap
00005D28  40404040 40404040                         5311+         DC    CL8' '            was cross check skipped?
00005D30  00000000 00000000                         5312+         DS    FD                   gap
00005D38  00000000 00000000                         5313+XCO98    DS    XL16              Cross check Output
00005D40  00000000 00000000
00005D48  00000000 00000000                         5314+         DS    FD                   gap
                                                    5315+*
00005D50                                            5316+X98      DS    0F
00005D50  E310 5014 0014                 00000014   5317+         LGF   R1,V2ADDR         load v2 source
00005D56  E721 0000 0006                 00000000   5318+         VL    v2,0(R1)
00005D5C  E310 5018 0014                 00000018   5319+         LGF   R1,V3ADDR         load v3 source
00005D62  E731 0000 0006                 00000000   5320+         VL    v3,0(R1)

00005D68                                            5323+         DS    0H                     E7B3 VR
00005D68  E7                                        5324+         DC    X'E7'                  - VECTOR REMAINDER
00005D69  12                                        5325+         DC    X'12'                   v1, v2
00005D6A  30                                        5326+         DC    X'30'                   v3, reserved
00005D6B  0830                                      5327+         DC    HL2'2096'               reserved, m5, m4, RXB
00005D6D  B3                                        5328+         DC    X'B3'

00005D6E  E710 5030 000E                 00005D10   5330+         VST   V1,V1O98          save v1 output
00005D74  07FB                                      5331+         BR    R11               return
00005D78                                            5332+RE98     DC    0F                xl16 expected result
00005D78                                            5333+         DROP  R5
00005D78  FFFF7EFE FDFD7CF8                         5334          DC    XL16'FFFF7EFEFDFD7CF8 FE8C9AA936C4D2E0'   result
00005D80  FE8C9AA9 36C4D2E0
00005D88  80000000 80000000                         5335          DC    XL16'8000000080000000 8000000080000000'   v2
00005D90  80000000 80000000
00005D98  01020304 05060708                         5336          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00005DA0  090A0B0C 0D0E0F10
                                                    5337
                                                    5338          VRR_C VR,X'3',X'8',N
00005DA8                                            5339+         DS    0FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005DA8                     00005DA8             5340+         USING *,R5              base for test data and test routine
00005DA8  00005E18                                5341+T99      DC    A(X99)            address of test routine
00005DAC  0063                                    5342+         DC    H'99'             test number
00005DAE  00                                      5343+         DC    X'00'
00005DAF  D5                                      5344+         DC    CL1'N'            S or Y = skip cross check
00005DB0  03                                      5345+         DC    X'3'               m4
00005DB1  08                                      5346+         DC    X'8'               m5
00005DB2  E5D94040 40404040                       5347+         DC    CL8'VR'           instruction name
00005DBC  00005E50                                5348+         DC    A(RE99+16)        address of v2 source
00005DC0  00005E60                                5349+         DC    A(RE99+32)        address of v3 source
00005DC4  00000010                                5350+         DC    A(16)             result length
00005DC8  00005E40                                5351+REA99    DC    A(RE99)           result address
00005DD0  00000000 00000000                       5352+         DS    FD                   gap
00005DD8  00000000 00000000                       5353+V1O99    DS    XL16              V1 output
00005DE0  00000000 00000000
00005DE8  00000000 00000000                       5354+         DS    FD                   gap
00005DF0  40404040 40404040                       5355+         DC    CL8' '            was cross check skipped?
00005DF8  00000000 00000000                       5356+         DS    FD                   gap
00005E00  00000000 00000000                       5357+XCO99    DS    XL16              Cross check Output
00005E08  00000000 00000000
00005E10  00000000 00000000                       5358+         DS    FD                   gap
                                                  5359+*
00005E18                                          5360+X99      DS    0F
00005E18  E310 5014 0014                 00000014 5361+         LGF   R1,V2ADDR         load v2 source
00005E1E  E721 0000 0006                 00000000 5362+         VL    v2,0(R1)
00005E24  E310 5018 0014                 00000018 5363+         LGF   R1,V3ADDR         load v3 source
00005E2A  E731 0000 0006                 00000000 5364+         VL    v3,0(R1)

00005E30                                          5367+         DS    0H                      E7B3 VR
00005E30  E7                                      5368+         DC    X'E7'                   - VECTOR REMAINDER
00005E31  12                                      5369+         DC    X'12'                    v1, v2
00005E32  30                                      5370+         DC    X'30'                    v3, reserved
00005E33  0830                                    5371+         DC    HL2'2096'                reserved, m5, m4, RXB
00005E35  B3                                      5372+         DC    X'B3'

00005E36  E710 5030 000E                 00005DD8 5374+         VST   V1,V1O99          save v1 output
00005E3C  07FB                                    5375+         BR    R11               return
00005E40                                          5376+RE99     DC    0F                xl16 expected result
00005E40                                          5377+         DROP  R5
00005E40  FFFFFFFF FFFFFFFF                       5378          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFF9'   result
00005E48  FFFFFFFF FFFFFFF9
00005E50  80000000 00000001                       5379          DC    XL16'8000000000000001 8000000000000001'   v2
00005E58  80000000 00000001
00005E60  FFFFFFFF FFFFFFFE                       5380          DC    XL16'FFFFFFFFFFFFFFFE FFFFFFFFFFFFFFF1'   v3
00005E68  FFFFFFFF FFFFFFF1
                                                  5381
                                                  5382          VRR_C VR,X'3',X'8',N
00005E70                                          5383+         DS    0FD
00005E70                     00005E70             5384+         USING *,R5              base for test data and test routine
00005E70  00005EE0                                5385+T100     DC    A(X100)           address of test routine
00005E74  0064                                    5386+         DC    H'100'            test number
00005E76  00                                      5387+         DC    X'00'
00005E77  D5                                      5388+         DC    CL1'N'            S or Y = skip cross check
00005E78  03                                      5389+         DC    X'3'               m4
00005E79  08                                      5390+         DC    X'8'               m5
00005E7A  E5D94040 40404040                       5391+         DC    CL8'VR'           instruction name
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005E84  00005F18                                5392+         DC    A(RE100+16)       address of v2 source
00005E88  00005F28                                5393+         DC    A(RE100+32)       address of v3 source
00005E8C  00000010                                5394+         DC    A(16)             result length
00005E90  00005F08                                5395+REA100   DC    A(RE100)          result address
00005E98  00000000 00000000                       5396+         DS    FD                   gap
00005EA0  00000000 00000000                       5397+V1O100   DS    XL16              V1 output
00005EA8  00000000 00000000
00005EB0  00000000 00000000                       5398+         DS    FD                   gap
00005EB8  40404040 40404040                       5399+         DC    CL8' '            was cross check skipped?
00005EC0  00000000 00000000                       5400+         DS    FD                   gap
00005EC8  00000000 00000000                       5401+XCO100   DS    XL16              Cross check Output
00005ED0  00000000 00000000
00005ED8  00000000 00000000                       5402+         DS    FD                   gap
                                                  5403+*
00005EE0                                          5404+X100     DS    0F
00005EE0  E310 5014 0014                00000014  5405+         LGF   R1,V2ADDR         load v2 source
00005EE6  E721 0000 0006                00000000  5406+         VL    v2,0(R1)
00005EEC  E310 5018 0014                00000018  5407+         LGF   R1,V3ADDR         load v3 source
00005EF2  E731 0000 0006                00000000  5408+         VL    v3,0(R1)

00005EF8                                          5411+         DS    0H                     E7B3 VR
00005EF8  E7                                      5412+         DC    X'E7'                  - VECTOR REMAINDER
00005EF9  12                                      5413+         DC    X'12'                   v1, v2
00005EFA  30                                      5414+         DC    X'30'                   v3, reserved
00005EFB  0830                                    5415+         DC    HL2'2096'               reserved, m5, m4, RXB
00005EFD  B3                                      5416+         DC    X'B3'

00005EFE  E710 5030 000E                00005EA0  5418+         VST   V1,V1O100         save v1 output
00005F04  07FB                                    5419+         BR    R11               return
00005F08                                          5420+RE100    DC    0F                xl16 expected result
00005F08                                          5421+         DROP  R5
00005F08  FEFDFCFB FAF9F8F8                       5422          DC    XL16'FEFDFCFBFAF9F8F8 F6F5F4F3F2F1F0F0'   result
00005F10  F6F5F4F3 F2F1F0F0
00005F18  80000000 80000000                       5423          DC    XL16'8000000080000000 8000000080000000'   v2
00005F20  80000000 80000000
00005F28  81020304 85060708                       5424          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   v3
00005F30  890A0B0C 8D0E0F10
                                                  5425
                                                  5426 * skip XC
                                                  5427          VRR_C VR,X'3',X'8',S
00005F38                                          5428+         DS    0FD
00005F38                      00005F38            5429+         USING *,R5              base for test data and test routine
00005F38  00005FA8                                5430+T101     DC    A(X101)           address of test routine
00005F3C  0065                                    5431+         DC    H'101'            test number
00005F3E  00                                      5432+         DC    X'00'
00005F3F  E2                                      5433+         DC    CL1'S'            S or Y = skip cross check
00005F40  03                                      5434+         DC    X'3'               m4
00005F41  08                                      5435+         DC    X'8'               m5
00005F42  E5D94040 40404040                       5436+         DC    CL8'VR'           instruction name
00005F4C  00005FE0                                5437+         DC    A(RE101+16)       address of v2 source
00005F50  00005FF0                                5438+         DC    A(RE101+32)       address of v3 source
00005F54  00000010                                5439+         DC    A(16)             result length
00005F58  00005FD0                                5440+REA101   DC    A(RE101)          result address
00005F60  00000000 00000000                       5441+         DS    FD                   gap
00005F68  00000000 00000000                       5442+V1O101   DS    XL16              V1 output
00005F70  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00005F78  00000000 00000000                        5443+         DS    FD                    gap
00005F80  40404040 40404040                        5444+         DC    CL8' '             was cross check skipped?
00005F88  00000000 00000000                        5445+         DS    FD                    gap
00005F90  00000000 00000000                        5446+XCO101   DS    XL16               Cross check Output
00005F98  00000000 00000000
00005FA0  00000000 00000000                        5447+         DS    FD                    gap
                                                   5448+*
00005FA8                                           5449+X101     DS    0F
00005FA8  E310 5014 0014                 00000014  5450+         LGF   R1,V2ADDR          load v2 source
00005FAE  E721 0000 0006                 00000000  5451+         VL    v2,0(R1)
00005FB4  E310 5018 0014                 00000018  5452+         LGF   R1,V3ADDR          load v3 source
00005FBA  E731 0000 0006                 00000000  5453+         VL    v3,0(R1)

00005FC0                                           5456+         DS    0H                       E7B3 VR
00005FC0  E7                                       5457+         DC    X'E7'                    - VECTOR REMAINDER
00005FC1  12                                       5458+         DC    X'12'                     v1, v2
00005FC2  30                                       5459+         DC    X'30'                     v3, reserved
00005FC3  0830                                     5460+         DC    HL2'2096'                 reserved, m5, m4, RXB
00005FC5  B3                                       5461+         DC    X'B3'

00005FC6  E710 5030 000E                 00005F68  5463+         VST   V1,V1O101          save v1 output
00005FCC  07FB                                     5464+         BR    R11                return
00005FD0                                           5465+RE101    DC    0F                 xl16 expected result
00005FD0                                           5466+         DROP  R5
00005FD0  00000000 00000000                        5467          DC    XL16'0000000000000000 0000000000000000'   result
00005FD8  00000000 00000000
00005FE0  80000000 00000000                        5468          DC    XL16'8000000000000000 8000000000000000'   v2
00005FE8  80000000 00000000
00005FF0  FFFFFFFF FFFFFFFF                        5469          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v3
00005FF8  FFFFFFFF FFFFFFFF
                                                   5470
                                                   5471 *Quadwordword    M4:  4, m5= 8 (IDC=1)
                                                   5472
                                                   5473          VRR_C VR,X'4',X'8',N
00006000                                           5474+         DS    0FD
00006000                         00006000          5475+         USING *,R5               base for test data and test routine
00006000  00006070                                 5476+T102     DC    A(X102)            address of test routine
00006004  0066                                     5477+         DC    H'102'             test number
00006006  00                                       5478+         DC    X'00'
00006007  D5                                       5479+         DC    CL1'N'             S or Y = skip cross check
00006008  04                                       5480+         DC    X'4'                m4
00006009  08                                       5481+         DC    X'8'                m5
0000600A  E5D94040 40404040                        5482+         DC    CL8'VR'            instruction name
00006014  000060A8                                 5483+         DC    A(RE102+16)        address of v2 source
00006018  000060B8                                 5484+         DC    A(RE102+32)        address of v3 source
0000601C  00000010                                 5485+         DC    A(16)              result length
00006020  00006098                                 5486+REA102   DC    A(RE102)           result address
00006028  00000000 00000000                        5487+         DS    FD                    gap
00006030  00000000 00000000                        5488+V1O102   DS    XL16               V1 output
00006038  00000000 00000000
00006040  00000000 00000000                        5489+         DS    FD                    gap
00006048  40404040 40404040                        5490+         DC    CL8' '             was cross check skipped?
00006050  00000000 00000000                        5491+         DS    FD                    gap
00006058  00000000 00000000                        5492+XCO102   DS    XL16               Cross check Output
00006060  00000000 00000000
00006068  00000000 00000000                        5493+         DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   5494+*
00006070                                           5495+X102     DS    0F
00006070  E310 5014 0014                 00000014  5496+         LGF   R1,V2ADDR         load v2 source
00006076  E721 0000 0006                 00000000  5497+         VL    v2,0(R1)
0000607C  E310 5018 0014                 00000018  5498+         LGF   R1,V3ADDR         load v3 source
00006082  E731 0000 0006                 00000000  5499+         VL    v3,0(R1)

00006088                                           5502+         DS    0H                      E7B3 VR
00006088  E7                                       5503+         DC    X'E7'                   - VECTOR REMAINDER
00006089  12                                       5504+         DC    X'12'                    v1, v2
0000608A  30                                       5505+         DC    X'30'                    v3, reserved
0000608B  0840                                     5506+         DC    HL2'2112'                reserved, m5, m4, RXB
0000608D  B3                                       5507+         DC    X'B3'

0000608E  E710 5030 000E                 00006030  5509+         VST   V1,V1O102         save v1 output
00006094  07FB                                     5510+         BR    R11               return
00006098                                           5511+RE102    DC    0F                xl16 expected result
00006098                                           5512+         DROP  R5
00006098  00000000 00000000                        5513          DC    XL16'0000000000000000 0000000000000000'   result
000060A0  00000000 00000000
000060A8  00000000 00000000                        5514          DC    XL16'0000000000000000 0000000000000000'   v2
000060B0  00000000 00000000
000060B8  01020304 05060708                        5515          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
000060C0  090A0B0C 0D0E0F10
                                                   5516
                                                   5517          VRR_C VR,X'4',X'8',N
000060C8                                           5518+         DS    0FD
000060C8                      000060C8             5519+         USING *,R5              base for test data and test routine
000060C8  00006138                                 5520+T103     DC    A(X103)           address of test routine
000060CC  0067                                     5521+         DC    H'103'            test number
000060CE  00                                       5522+         DC    X'00'
000060CF  D5                                       5523+         DC    CL1'N'            S or Y = skip cross check
000060D0  04                                       5524+         DC    X'4'               m4
000060D1  08                                       5525+         DC    X'8'               m5
000060D2  E5D94040 40404040                        5526+         DC    CL8'VR'           instruction name
000060DC  00006170                                 5527+         DC    A(RE103+16)       address of v2 source
000060E0  00006180                                 5528+         DC    A(RE103+32)       address of v3 source
000060E4  00000010                                 5529+         DC    A(16)             result length
000060E8  00006160                                 5530+REA103   DC    A(RE103)          result address
000060F0  00000000 00000000                        5531+         DS    FD                   gap
000060F8  00000000 00000000                        5532+V1O103   DS    XL16              V1 output
00006100  00000000 00000000
00006108  00000000 00000000                        5533+         DS    FD                   gap
00006110  40404040 40404040                        5534+         DC    CL8' '            was cross check skipped?
00006118  00000000 00000000                        5535+         DS    FD                   gap
00006120  00000000 00000000                        5536+XCO103   DS    XL16              Cross check Output
00006128  00000000 00000000
00006130  00000000 00000000                        5537+         DS    FD                   gap
                                                   5538+*
00006138                                           5539+X103     DS    0F
00006138  E310 5014 0014                 00000014  5540+         LGF   R1,V2ADDR         load v2 source
0000613E  E721 0000 0006                 00000000  5541+         VL    v2,0(R1)
00006144  E310 5018 0014                 00000018  5542+         LGF   R1,V3ADDR         load v3 source
0000614A  E731 0000 0006                 00000000  5543+         VL    v3,0(R1)

00006150                                           5546+         DS    0H                      E7B3 VR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006150  E7                                      5547+         DC    X'E7'                    - VECTOR REMAINDER
00006151  12                                      5548+         DC    X'12'                     v1, v2
00006152  30                                      5549+         DC    X'30'                     v3, reserved
00006153  0840                                    5550+         DC    HL2'2112'                 reserved, m5, m4, RXB
00006155  B3                                      5551+         DC    X'B3'

00006156  E710 5030 000E                000060F8  5553+         VST   V1,V1O103         save v1 output
0000615C  07FB                                    5554+         BR    R11               return
00006160                                          5555+RE103    DC    0F                xl16 expected result
00006160                                          5556+         DROP  R5
00006160  01020303 05060707                       5557          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   result
00006168  090A0B0B 0D0E0F0F
00006170  01020303 05060707                       5558          DC    XL16'0102030305060707 090A0B0B0D0E0F0F'   v2
00006178  090A0B0B 0D0E0F0F
00006180  01020304 05060708                       5559          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00006188  090A0B0C 0D0E0F10
                                                  5560
                                                  5561          VRR_C VR,X'4',X'8',N
00006190                                          5562+         DS    0FD
00006190                        00006190          5563+         USING *,R5              base for test data and test routine
00006190  00006200                                5564+T104     DC    A(X104)           address of test routine
00006194  0068                                    5565+         DC    H'104'            test number
00006196  00                                      5566+         DC    X'00'
00006197  D5                                      5567+         DC    CL1'N'            S or Y = skip cross check
00006198  04                                      5568+         DC    X'4'               m4
00006199  08                                      5569+         DC    X'8'               m5
0000619A  E5D94040 40404040                       5570+         DC    CL8'VR'           instruction name
000061A4  00006238                                5571+         DC    A(RE104+16)       address of v2 source
000061A8  00006248                                5572+         DC    A(RE104+32)       address of v3 source
000061AC  00000010                                5573+         DC    A(16)             result length
000061B0  00006228                                5574+REA104   DC    A(RE104)          result address
000061B8  00000000 00000000                       5575+         DS    FD                   gap
000061C0  00000000 00000000                       5576+V1O104   DS    XL16              V1 output
000061C8  00000000 00000000
000061D0  00000000 00000000                       5577+         DS    FD                   gap
000061D8  40404040 40404040                       5578+         DC    CL8' '            was cross check skipped?
000061E0  00000000 00000000                       5579+         DS    FD                   gap
000061E8  00000000 00000000                       5580+XCO104   DS    XL16              Cross check Output
000061F0  00000000 00000000
000061F8  00000000 00000000                       5581+         DS    FD                   gap
                                                  5582+*
00006200                                          5583+X104     DS    0F
00006200  E310 5014 0014                00000014  5584+         LGF   R1,V2ADDR         load v2 source
00006206  E721 0000 0006                00000000  5585+         VL    v2,0(R1)
0000620C  E310 5018 0014                00000018  5586+         LGF   R1,V3ADDR         load v3 source
00006212  E731 0000 0006                00000000  5587+         VL    v3,0(R1)

00006218                                          5590+         DS    0H                       E7B3 VR
00006218  E7                                      5591+         DC    X'E7'                    - VECTOR REMAINDER
00006219  12                                      5592+         DC    X'12'                     v1, v2
0000621A  30                                      5593+         DC    X'30'                     v3, reserved
0000621B  0840                                    5594+         DC    HL2'2112'                 reserved, m5, m4, RXB
0000621D  B3                                      5595+         DC    X'B3'

0000621E  E710 5030 000E                000061C0  5597+         VST   V1,V1O104         save v1 output
00006224  07FB                                    5598+         BR    R11               return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006228                                           5599+RE104    DC    0F                xl16 expected result
00006228                                           5600+         DROP  R5
00006228  00000000 00000000                        5601          DC    XL16'0000000000000000 0000000000000000'   result
00006230  00000000 00000000
00006238  01020304 05060708                        5602          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v2
00006240  090A0B0C 0D0E0F10
00006248  01020304 05060708                        5603          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00006250  090A0B0C 0D0E0F10
                                                   5604
                                                   5605          VRR_C VR,X'4',X'8',N
00006258                                           5606+         DS    0FD
00006258                      00006258             5607+         USING *,R5              base for test data and test routine
00006258  000062C8                                 5608+T105     DC    A(X105)           address of test routine
0000625C  0069                                     5609+         DC    H'105'            test number
0000625E  00                                       5610+         DC    X'00'
0000625F  D5                                       5611+         DC    CL1'N'            S or Y = skip cross check
00006260  04                                       5612+         DC    X'4'               m4
00006261  08                                       5613+         DC    X'8'               m5
00006262  E5D94040 40404040                        5614+         DC    CL8'VR'           instruction name
0000626C  00006300                                 5615+         DC    A(RE105+16)       address of v2 source
00006270  00006310                                 5616+         DC    A(RE105+32)       address of v3 source
00006274  00000010                                 5617+         DC    A(16)             result length
00006278  000062F0                                 5618+REA105   DC    A(RE105)          result address
00006280  00000000 00000000                        5619+         DS    FD                   gap
00006288  00000000 00000000                        5620+V1O105   DS    XL16              V1 output
00006290  00000000 00000000
00006298  00000000 00000000                        5621+         DS    FD                   gap
000062A0  40404040 40404040                        5622+         DC    CL8' '            was cross check skipped?
000062A8  00000000 00000000                        5623+         DS    FD                   gap
000062B0  00000000 00000000                        5624+XCO105   DS    XL16              Cross check Output
000062B8  00000000 00000000
000062C0  00000000 00000000                        5625+         DS    FD                   gap
                                                   5626+*
000062C8                                           5627+X105     DS    0F
000062C8  E310 5014 0014                00000014   5628+         LGF   R1,V2ADDR         load v2 source
000062CE  E721 0000 0006                00000000   5629+         VL    v2,0(R1)
000062D4  E310 5018 0014                00000018   5630+         LGF   R1,V3ADDR         load v3 source
000062DA  E731 0000 0006                00000000   5631+         VL    v3,0(R1)

000062E0                                           5634+         DS    0H                      E7B3 VR
000062E0  E7                                       5635+         DC    X'E7'                   - VECTOR REMAINDER
000062E1  12                                       5636+         DC    X'12'                    v1, v2
000062E2  30                                       5637+         DC    X'30'                    v3, reserved
000062E3  0840                                     5638+         DC    HL2'2112'                reserved, m5, m4, RXB
000062E5  B3                                       5639+         DC    X'B3'

000062E6  E710 5030 000E                00006288   5641+         VST   V1,V1O105         save v1 output
000062EC  07FB                                     5642+         BR    R11               return
000062F0                                           5643+RE105    DC    0F                xl16 expected result
000062F0                                           5644+         DROP  R5
000062F0  00000000 00000000                        5645          DC    XL16'0000000000000000 0000000000000000'   result
000062F8  00000000 00000000
00006300  7FFFFFFF 7FFFFFFF                        5646          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00006308  7FFFFFFF 7FFFFFFF
00006310  00000000 00000000                        5647          DC    XL16'0000000000000000 0000000000000001'   v3
00006318  00000000 00000001
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   5648
                                                   5649          VRR_C VR,X'4',X'8',N
00006320                                           5650+         DS    0FD
00006320                     00006320              5651+         USING *,R5              base for test data and test routine
00006320  00006390                                 5652+T106     DC    A(X106)           address of test routine
00006324  006A                                     5653+         DC    H'106'            test number
00006326  00                                       5654+         DC    X'00'
00006327  D5                                       5655+         DC    CL1'N'            S or Y = skip cross check
00006328  04                                       5656+         DC    X'4'               m4
00006329  08                                       5657+         DC    X'8'               m5
0000632A  E5D94040 40404040                        5658+         DC    CL8'VR'           instruction name
00006334  000063C8                                 5659+         DC    A(RE106+16)       address of v2 source
00006338  000063D8                                 5660+         DC    A(RE106+32)       address of v3 source
0000633C  00000010                                 5661+         DC    A(16)             result length
00006340  000063B8                                 5662+REA106   DC    A(RE106)          result address
00006348  00000000 00000000                        5663+         DS    FD                   gap
00006350  00000000 00000000                        5664+V1O106   DS    XL16              V1 output
00006358  00000000 00000000
00006360  00000000 00000000                        5665+         DS    FD                   gap
00006368  40404040 40404040                        5666+         DC    CL8' '            was cross check skipped?
00006370  00000000 00000000                        5667+         DS    FD                   gap
00006378  00000000 00000000                        5668+XCO106   DS    XL16              Cross check Output
00006380  00000000 00000000
00006388  00000000 00000000                        5669+         DS    FD                   gap
                                                   5670+*
00006390                                           5671+X106     DS    0F
00006390  E310 5014 0014                 00000014  5672+         LGF   R1,V2ADDR         load v2 source
00006396  E721 0000 0006                 00000000  5673+         VL    v2,0(R1)
0000639C  E310 5018 0014                 00000018  5674+         LGF   R1,V3ADDR         load v3 source
000063A2  E731 0000 0006                 00000000  5675+         VL    v3,0(R1)

000063A8                                           5678+         DS    0H                      E7B3 VR
000063A8  E7                                       5679+         DC    X'E7'                   - VECTOR REMAINDER
000063A9  12                                       5680+         DC    X'12'                    v1, v2
000063AA  30                                       5681+         DC    X'30'                    v3, reserved
000063AB  0840                                     5682+         DC    HL2'2112'                reserved, m5, m4, RXB
000063AD  B3                                       5683+         DC    X'B3'

000063AE  E710 5030 000E                 00006350  5685+         VST   V1,V1O106         save v1 output
000063B4  07FB                                     5686+         BR    R11               return
000063B8                                           5687+RE106    DC    0F                xl16 expected result
000063B8                                           5688+         DROP  R5
000063B8  00000000 00000000                        5689          DC    XL16'0000000000000000 0000000000000001'   result
000063C0  00000000 00000001
000063C8  7FFFFFFF 7FFFFFFF                        5690          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
000063D0  7FFFFFFF 7FFFFFFF
000063D8  00000000 00000000                        5691          DC    XL16'0000000000000000 0000000000000002'   v3
000063E0  00000000 00000002
                                                   5692
                                                   5693          VRR_C VR,X'4',X'8',N
000063E8                                           5694+         DS    0FD
000063E8                     000063E8              5695+         USING *,R5              base for test data and test routine
000063E8  00006458                                 5696+T107     DC    A(X107)           address of test routine
000063EC  006B                                     5697+         DC    H'107'            test number
000063EE  00                                       5698+         DC    X'00'
000063EF  D5                                       5699+         DC    CL1'N'            S or Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000063F0  04                                       5700+         DC    X'4'                m4
000063F1  08                                       5701+         DC    X'8'                m5
000063F2  E5D94040 40404040                        5702+         DC    CL8'VR'            instruction name
000063FC  00006490                                 5703+         DC    A(RE107+16)        address of v2 source
00006400  000064A0                                 5704+         DC    A(RE107+32)        address of v3 source
00006404  00000010                                 5705+         DC    A(16)              result length
00006408  00006480                                 5706+REA107   DC    A(RE107)           result address
00006410  00000000 00000000                        5707+         DS    FD                    gap
00006418  00000000 00000000                        5708+V1O107   DS    XL16               V1 output
00006420  00000000 00000000
00006428  00000000 00000000                        5709+         DS    FD                    gap
00006430  40404040 40404040                        5710+         DC    CL8' '             was cross check skipped?
00006438  00000000 00000000                        5711+         DS    FD                    gap
00006440  00000000 00000000                        5712+XCO107   DS    XL16               Cross check Output
00006448  00000000 00000000
00006450  00000000 00000000                        5713+         DS    FD                    gap
                                                   5714+*
00006458                                           5715+X107     DS    0F
00006458  E310 5014 0014                 00000014  5716+         LGF   R1,V2ADDR          load v2 source
0000645E  E721 0000 0006                 00000000  5717+         VL    v2,0(R1)
00006464  E310 5018 0014                 00000018  5718+         LGF   R1,V3ADDR          load v3 source
0000646A  E731 0000 0006                 00000000  5719+         VL    v3,0(R1)

00006470                                           5722+         DS    0H                       E7B3 VR
00006470  E7                                       5723+         DC    X'E7'                    - VECTOR REMAINDER
00006471  12                                       5724+         DC    X'12'                     v1, v2
00006472  30                                       5725+         DC    X'30'                     v3, reserved
00006473  0840                                     5726+         DC    HL2'2112'                 reserved, m5, m4, RXB
00006475  B3                                       5727+         DC    X'B3'

00006476  E710 5030 000E                 00006418  5729+         VST   V1,V1O107          save v1 output
0000647C  07FB                                     5730+         BR    R11                return
00006480                                           5731+RE107    DC    0F                 xl16 expected result
00006480                                           5732+         DROP  R5
00006480  00000000 00000000                        5733          DC    XL16'0000000000000000 0000000000000003'   result
00006488  00000000 00000003
00006490  7FFFFFFF 7FFFFFFF                        5734          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00006498  7FFFFFFF 7FFFFFFF
000064A0  00000000 00000000                        5735          DC    XL16'0000000000000000 000000000000000A'   v3
000064A8  00000000 0000000A
                                                   5736
                                                   5737          VRR_C VR,X'4',X'8',N
000064B0                                           5738+         DS    0FD
000064B0                      000064B0             5739+         USING *,R5               base for test data and test routine
000064B0  00006520                                 5740+T108     DC    A(X108)            address of test routine
000064B4  006C                                     5741+         DC    H'108'             test number
000064B6  00                                       5742+         DC    X'00'
000064B7  D5                                       5743+         DC    CL1'N'             S or Y = skip cross check
000064B8  04                                       5744+         DC    X'4'                m4
000064B9  08                                       5745+         DC    X'8'                m5
000064BA  E5D94040 40404040                        5746+         DC    CL8'VR'            instruction name
000064C4  00006558                                 5747+         DC    A(RE108+16)        address of v2 source
000064C8  00006568                                 5748+         DC    A(RE108+32)        address of v3 source
000064CC  00000010                                 5749+         DC    A(16)              result length
000064D0  00006548                                 5750+REA108   DC    A(RE108)           result address
000064D8  00000000 00000000                        5751+         DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000064E0  00000000 00000000                        5752+V1O108   DS    XL16              V1 output
000064E8  00000000 00000000
000064F0  00000000 00000000                        5753+         DS    FD                   gap
000064F8  40404040 40404040                        5754+         DC    CL8' '            was cross check skipped?
00006500  00000000 00000000                        5755+         DS    FD                   gap
00006508  00000000 00000000                        5756+XCO108   DS    XL16              Cross check Output
00006510  00000000 00000000
00006518  00000000 00000000                        5757+         DS    FD                   gap
                                                   5758+*
00006520                                           5759+X108     DS    0F
00006520  E310 5014 0014                 00000014  5760+         LGF   R1,V2ADDR         load v2 source
00006526  E721 0000 0006                 00000000  5761+         VL    v2,0(R1)
0000652C  E310 5018 0014                 00000018  5762+         LGF   R1,V3ADDR         load v3 source
00006532  E731 0000 0006                 00000000  5763+         VL    v3,0(R1)

00006538                                           5766+         DS    0H                     E7B3 VR
00006538  E7                                       5767+         DC    X'E7'                  - VECTOR REMAINDER
00006539  12                                       5768+         DC    X'12'                   v1, v2
0000653A  30                                       5769+         DC    X'30'                   v3, reserved
0000653B  0840                                     5770+         DC    HL2'2112'               reserved, m5, m4, RXB
0000653D  B3                                       5771+         DC    X'B3'

0000653E  E710 5030 000E                 000064E0  5773+         VST   V1,V1O108         save v1 output
00006544  07FB                                     5774+         BR    R11               return
00006548                                           5775+RE108    DC    0F                xl16 expected result
00006548                                           5776+         DROP  R5
00006548  00000000 00000000                        5777          DC    XL16'0000000000000000 000000000000000D'   result
00006550  00000000 0000000D
00006558  7FFFFFFF 7FFFFFFF                        5778          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00006560  7FFFFFFF 7FFFFFFF
00006568  00000000 00000000                        5779          DC    XL16'0000000000000000 000000000000000F'   v3
00006570  00000000 0000000F
                                                   5780
                                                   5781          VRR_C VR,X'4',X'8',N
00006578                                           5782+         DS    0FD
00006578                         00006578          5783+         USING *,R5              base for test data and test routine
00006578  000065E8                                 5784+T109     DC    A(X109)           address of test routine
0000657C  006D                                     5785+         DC    H'109'            test number
0000657E  00                                       5786+         DC    X'00'
0000657F  D5                                       5787+         DC    CL1'N'            S or Y = skip cross check
00006580  04                                       5788+         DC    X'4'               m4
00006581  08                                       5789+         DC    X'8'               m5
00006582  E5D94040 40404040                        5790+         DC    CL8'VR'           instruction name
0000658C  00006620                                 5791+         DC    A(RE109+16)       address of v2 source
00006590  00006630                                 5792+         DC    A(RE109+32)       address of v3 source
00006594  00000010                                 5793+         DC    A(16)             result length
00006598  00006610                                 5794+REA109   DC    A(RE109)          result address
000065A0  00000000 00000000                        5795+         DS    FD                   gap
000065A8  00000000 00000000                        5796+V1O109   DS    XL16              V1 output
000065B0  00000000 00000000
000065B8  00000000 00000000                        5797+         DS    FD                   gap
000065C0  40404040 40404040                        5798+         DC    CL8' '            was cross check skipped?
000065C8  00000000 00000000                        5799+         DS    FD                   gap
000065D0  00000000 00000000                        5800+XCO109   DS    XL16              Cross check Output
000065D8  00000000 00000000
000065E0  00000000 00000000                        5801+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   5802+*
000065E8                                           5803+X109     DS    0F
000065E8  E310 5014 0014                 00000014  5804+         LGF   R1,V2ADDR         load v2 source
000065EE  E721 0000 0006                 00000000  5805+         VL    v2,0(R1)
000065F4  E310 5018 0014                 00000018  5806+         LGF   R1,V3ADDR         load v3 source
000065FA  E731 0000 0006                 00000000  5807+         VL    v3,0(R1)

00006600                                           5810+         DS    0H                      E7B3 VR
00006600  E7                                       5811+         DC    X'E7'                   - VECTOR REMAINDER
00006601  12                                       5812+         DC    X'12'                    v1, v2
00006602  30                                       5813+         DC    X'30'                    v3, reserved
00006603  0840                                     5814+         DC    HL2'2112'                reserved, m5, m4, RXB
00006605  B3                                       5815+         DC    X'B3'

00006606  E710 5030 000E                 000065A8  5817+         VST   V1,V1O109         save v1 output
0000660C  07FB                                     5818+         BR    R11               return
00006610                                           5819+RE109    DC    0F                xl16 expected result
00006610                                           5820+         DROP  R5
00006610  00008101 02028303                        5821          DC    XL16'0000810102028303 040485050606870F'   result
00006618  04048505 0606870F
00006620  7FFFFFFF 7FFFFFFF                        5822          DC    XL16'7FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'   v2
00006628  7FFFFFFF 7FFFFFFF
00006630  01020304 05060708                        5823          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00006638  090A0B0C 0D0E0F10
                                                   5824
                                                   5825          VRR_C VR,X'4',X'8',N
00006640                                           5826+         DS    0FD
00006640                      00006640             5827+         USING *,R5              base for test data and test routine
00006640  000066B0                                 5828+T110     DC    A(X110)           address of test routine
00006644  006E                                     5829+         DC    H'110'            test number
00006646  00                                       5830+         DC    X'00'
00006647  D5                                       5831+         DC    CL1'N'            S or Y = skip cross check
00006648  04                                       5832+         DC    X'4'               m4
00006649  08                                       5833+         DC    X'8'               m5
0000664A  E5D94040 40404040                        5834+         DC    CL8'VR'           instruction name
00006654  000066E8                                 5835+         DC    A(RE110+16)       address of v2 source
00006658  000066F8                                 5836+         DC    A(RE110+32)       address of v3 source
0000665C  00000010                                 5837+         DC    A(16)             result length
00006660  000066D8                                 5838+REA110   DC    A(RE110)          result address
00006668  00000000 00000000                        5839+         DS    FD                   gap
00006670  00000000 00000000                        5840+V1O110   DS    XL16              V1 output
00006678  00000000 00000000
00006680  00000000 00000000                        5841+         DS    FD                   gap
00006688  40404040 40404040                        5842+         DC    CL8' '            was cross check skipped?
00006690  00000000 00000000                        5843+         DS    FD                   gap
00006698  00000000 00000000                        5844+XCO110   DS    XL16              Cross check Output
000066A0  00000000 00000000
000066A8  00000000 00000000                        5845+         DS    FD                   gap
                                                   5846+*
000066B0                                           5847+X110     DS    0F
000066B0  E310 5014 0014                 00000014  5848+         LGF   R1,V2ADDR         load v2 source
000066B6  E721 0000 0006                 00000000  5849+         VL    v2,0(R1)
000066BC  E310 5018 0014                 00000018  5850+         LGF   R1,V3ADDR         load v3 source
000066C2  E731 0000 0006                 00000000  5851+         VL    v3,0(R1)

000066C8                                           5854+         DS    0H                      E7B3 VR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000066C8  E7                                       5855+         DC    X'E7'                    - VECTOR REMAINDER
000066C9  12                                       5856+         DC    X'12'                     v1, v2
000066CA  30                                       5857+         DC    X'30'                     v3, reserved
000066CB  0840                                     5858+         DC    HL2'2112'                 reserved, m5, m4, RXB
000066CD  B3                                       5859+         DC    X'B3'

000066CE  E710 5030 000E                 00006670  5861+         VST   V1,V1O110         save v1 output
000066D4  07FB                                     5862+         BR    R11               return
000066D8                                           5863+RE110    DC    0F                xl16 expected result
000066D8                                           5864+         DROP  R5
000066D8  00000000 00000000                        5865          DC    XL16'0000000000000000 0000000000000000'   result
000066E0  00000000 00000000
000066E8  FFFFFFFF FFFFFFFF                        5866          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
000066F0  FFFFFFFF FFFFFF00
000066F8  00000000 00000000                        5867          DC    XL16'0000000000000000 0000000000000001'   v3
00006700  00000000 00000001
                                                   5868
                                                   5869          VRR_C VR,X'4',X'8',N
00006708                                           5870+         DS    0FD
00006708                         00006708          5871+         USING *,R5              base for test data and test routine
00006708  00006778                                 5872+T111     DC    A(X111)           address of test routine
0000670C  006F                                     5873+         DC    H'111'            test number
0000670E  00                                       5874+         DC    X'00'
0000670F  D5                                       5875+         DC    CL1'N'            S or Y = skip cross check
00006710  04                                       5876+         DC    X'4'               m4
00006711  08                                       5877+         DC    X'8'               m5
00006712  E5D94040 40404040                        5878+         DC    CL8'VR'           instruction name
0000671C  000067B0                                 5879+         DC    A(RE111+16)       address of v2 source
00006720  000067C0                                 5880+         DC    A(RE111+32)       address of v3 source
00006724  00000010                                 5881+         DC    A(16)             result length
00006728  000067A0                                 5882+REA111   DC    A(RE111)          result address
00006730  00000000 00000000                        5883+         DS    FD                   gap
00006738  00000000 00000000                        5884+V1O111   DS    XL16              V1 output
00006740  00000000 00000000
00006748  00000000 00000000                        5885+         DS    FD                   gap
00006750  40404040 40404040                        5886+         DC    CL8' '            was cross check skipped?
00006758  00000000 00000000                        5887+         DS    FD                   gap
00006760  00000000 00000000                        5888+XCO111   DS    XL16              Cross check Output
00006768  00000000 00000000
00006770  00000000 00000000                        5889+         DS    FD                   gap
                                                   5890+*
00006778                                           5891+X111     DS    0F
00006778  E310 5014 0014                 00000014  5892+         LGF   R1,V2ADDR         load v2 source
0000677E  E721 0000 0006                 00000000  5893+         VL    v2,0(R1)
00006784  E310 5018 0014                 00000018  5894+         LGF   R1,V3ADDR         load v3 source
0000678A  E731 0000 0006                 00000000  5895+         VL    v3,0(R1)

00006790                                           5898+         DS    0H                       E7B3 VR
00006790  E7                                       5899+         DC    X'E7'                    - VECTOR REMAINDER
00006791  12                                       5900+         DC    X'12'                     v1, v2
00006792  30                                       5901+         DC    X'30'                     v3, reserved
00006793  0840                                     5902+         DC    HL2'2112'                 reserved, m5, m4, RXB
00006795  B3                                       5903+         DC    X'B3'

00006796  E710 5030 000E                 00006738  5905+         VST   V1,V1O111         save v1 output
0000679C  07FB                                     5906+         BR    R11               return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000067A0                                           5907+RE111    DC    0F                xl16 expected result
000067A0                                           5908+         DROP  R5
000067A0  00000000 00000000                        5909          DC    XL16'0000000000000000 0000000000000000'   result
000067A8  00000000 00000000
000067B0  FFFFFFFF FFFFFFFF                        5910          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
000067B8  FFFFFFFF FFFFFF00
000067C0  00000000 00000000                        5911          DC    XL16'0000000000000000 0000000000000002'   v3
000067C8  00000000 00000002
                                                   5912
                                                   5913          VRR_C VR,X'4',X'8',N
000067D0                                           5914+         DS    0FD
000067D0                      000067D0             5915+         USING *,R5              base for test data and test routine
000067D0  00006840                                 5916+T112     DC    A(X112)           address of test routine
000067D4  0070                                     5917+         DC    H'112'            test number
000067D6  00                                       5918+         DC    X'00'
000067D7  D5                                       5919+         DC    CL1'N'            S or Y = skip cross check
000067D8  04                                       5920+         DC    X'4'               m4
000067D9  08                                       5921+         DC    X'8'               m5
000067DA  E5D94040 40404040                        5922+         DC    CL8'VR'           instruction name
000067E4  00006878                                 5923+         DC    A(RE112+16)       address of v2 source
000067E8  00006888                                 5924+         DC    A(RE112+32)       address of v3 source
000067EC  00000010                                 5925+         DC    A(16)             result length
000067F0  00006868                                 5926+REA112   DC    A(RE112)          result address
000067F8  00000000 00000000                        5927+         DS    FD                   gap
00006800  00000000 00000000                        5928+V1O112   DS    XL16              V1 output
00006808  00000000 00000000
00006810  00000000 00000000                        5929+         DS    FD                   gap
00006818  40404040 40404040                        5930+         DC    CL8' '            was cross check skipped?
00006820  00000000 00000000                        5931+         DS    FD                   gap
00006828  00000000 00000000                        5932+XCO112   DS    XL16              Cross check Output
00006830  00000000 00000000
00006838  00000000 00000000                        5933+         DS    FD                   gap
                                                   5934+*
00006840                                           5935+X112     DS    0F
00006840  E310 5014 0014                 00000014  5936+         LGF   R1,V2ADDR         load v2 source
00006846  E721 0000 0006                 00000000  5937+         VL    v2,0(R1)
0000684C  E310 5018 0014                 00000018  5938+         LGF   R1,V3ADDR         load v3 source
00006852  E731 0000 0006                 00000000  5939+         VL    v3,0(R1)

00006858                                           5942+         DS    0H                      E7B3 VR
00006858  E7                                       5943+         DC    X'E7'                   - VECTOR REMAINDER
00006859  12                                       5944+         DC    X'12'                    v1, v2
0000685A  30                                       5945+         DC    X'30'                    v3, reserved
0000685B  0840                                     5946+         DC    HL2'2112'                reserved, m5, m4, RXB
0000685D  B3                                       5947+         DC    X'B3'

0000685E  E710 5030 000E                 00006800  5949+         VST   V1,V1O112         save v1 output
00006864  07FB                                     5950+         BR    R11               return
00006868                                           5951+RE112    DC    0F                xl16 expected result
00006868                                           5952+         DROP  R5
00006868  FFFFFFFF FFFFFFFF                        5953          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFA'   result
00006870  FFFFFFFF FFFFFFFA
00006878  FFFFFFFF FFFFFFFF                        5954          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
00006880  FFFFFFFF FFFFFF00
00006888  00000000 00000000                        5955          DC    XL16'0000000000000000 000000000000000A'   v3
00006890  00000000 0000000A
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   5956
                                                   5957          VRR_C VR,X'4',X'8',N
00006898                                           5958+         DS    0FD
00006898                     00006898              5959+         USING *,R5              base for test data and test routine
00006898  00006908                                 5960+T113     DC    A(X113)           address of test routine
0000689C  0071                                     5961+         DC    H'113'            test number
0000689E  00                                       5962+         DC    X'00'
0000689F  D5                                       5963+         DC    CL1'N'            S or Y = skip cross check
000068A0  04                                       5964+         DC    X'4'               m4
000068A1  08                                       5965+         DC    X'8'               m5
000068A2  E5D94040 40404040                        5966+         DC    CL8'VR'           instruction name
000068AC  00006940                                 5967+         DC    A(RE113+16)       address of v2 source
000068B0  00006950                                 5968+         DC    A(RE113+32)       address of v3 source
000068B4  00000010                                 5969+         DC    A(16)             result length
000068B8  00006930                                 5970+REA113   DC    A(RE113)          result address
000068C0  00000000 00000000                        5971+         DS    FD                   gap
000068C8  00000000 00000000                        5972+V1O113   DS    XL16              V1 output
000068D0  00000000 00000000
000068D8  00000000 00000000                        5973+         DS    FD                   gap
000068E0  40404040 40404040                        5974+         DC    CL8' '            was cross check skipped?
000068E8  00000000 00000000                        5975+         DS    FD                   gap
000068F0  00000000 00000000                        5976+XCO113   DS    XL16              Cross check Output
000068F8  00000000 00000000
00006900  00000000 00000000                        5977+         DS    FD                   gap
                                                   5978+*
00006908                                           5979+X113     DS    0F
00006908  E310 5014 0014                 00000014  5980+         LGF   R1,V2ADDR         load v2 source
0000690E  E721 0000 0006                 00000000  5981+         VL    v2,0(R1)
00006914  E310 5018 0014                 00000018  5982+         LGF   R1,V3ADDR         load v3 source
0000691A  E731 0000 0006                 00000000  5983+         VL    v3,0(R1)

00006920                                           5986+         DS    0H                      E7B3 VR
00006920  E7                                       5987+         DC    X'E7'                   - VECTOR REMAINDER
00006921  12                                       5988+         DC    X'12'                    v1, v2
00006922  30                                       5989+         DC    X'30'                    v3, reserved
00006923  0840                                     5990+         DC    HL2'2112'                reserved, m5, m4, RXB
00006925  B3                                       5991+         DC    X'B3'

00006926  E710 5030 000E                 000068C8  5993+         VST   V1,V1O113         save v1 output
0000692C  07FB                                     5994+         BR    R11               return
00006930                                           5995+RE113    DC    0F                xl16 expected result
00006930                                           5996+         DROP  R5
00006930  FFFFFFFF FFFFFFFF                        5997          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   result
00006938  FFFFFFFF FFFFFFFF
00006940  FFFFFFFF FFFFFFFF                        5998          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v2
00006948  FFFFFFFF FFFFFF00
00006950  00000000 00000000                        5999          DC    XL16'0000000000000000 000000000000000F'   v3
00006958  00000000 0000000F
                                                   6000
                                                   6001          VRR_C VR,X'4',X'8',N
00006960                                           6002+         DS    0FD
00006960                     00006960              6003+         USING *,R5              base for test data and test routine
00006960  000069D0                                 6004+T114     DC    A(X114)           address of test routine
00006964  0072                                     6005+         DC    H'114'            test number
00006966  00                                       6006+         DC    X'00'
00006967  D5                                       6007+         DC    CL1'N'            S or Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006968  04                                       6008+         DC    X'4'                m4
00006969  08                                       6009+         DC    X'8'                m5
0000696A  E5D94040 40404040                        6010+         DC    CL8'VR'            instruction name
00006974  00006A08                                 6011+         DC    A(RE114+16)        address of v2 source
00006978  00006A18                                 6012+         DC    A(RE114+32)        address of v3 source
0000697C  00000010                                 6013+         DC    A(16)              result length
00006980  000069F8                                 6014+REA114   DC    A(RE114)           result address
00006988  00000000 00000000                        6015+         DS    FD                    gap
00006990  00000000 00000000                        6016+V1O114   DS    XL16               V1 output
00006998  00000000 00000000
000069A0  00000000 00000000                        6017+         DS    FD                    gap
000069A8  40404040 40404040                        6018+         DC    CL8' '             was cross check skipped?
000069B0  00000000 00000000                        6019+         DS    FD                    gap
000069B8  00000000 00000000                        6020+XCO114   DS    XL16               Cross check Output
000069C0  00000000 00000000
000069C8  00000000 00000000                        6021+         DS    FD                    gap
                                                   6022+*
000069D0                                           6023+X114     DS    0F
000069D0  E310 5014 0014                 00000014  6024+         LGF   R1,V2ADDR          load v2 source
000069D6  E721 0000 0006                 00000000  6025+         VL    v2,0(R1)
000069DC  E310 5018 0014                 00000018  6026+         LGF   R1,V3ADDR          load v3 source
000069E2  E731 0000 0006                 00000000  6027+         VL    v3,0(R1)

000069E8                                           6030+         DS    0H                       E7B3 VR
000069E8  E7                                       6031+         DC    X'E7'                    - VECTOR REMAINDER
000069E9  12                                       6032+         DC    X'12'                     v1, v2
000069EA  30                                       6033+         DC    X'30'                     v3, reserved
000069EB  0840                                     6034+         DC    HL2'2112'                 reserved, m5, m4, RXB
000069ED  B3                                       6035+         DC    X'B3'

000069EE  E710 5030 000E                 00006990  6037+         VST   V1,V1O114          save v1 output
000069F4  07FB                                     6038+         BR    R11                return
000069F8                                           6039+RE114    DC    0F                 xl16 expected result
000069F8                                           6040+         DROP  R5
000069F8  FFFF7EFE 7DFD7CFC                        6041          DC    XL16'FFFF7EFE7DFD7CFC 7BFB7AFA79F979EF'   result
00006A00  7BFB7AFA 79F979EF
00006A08  80000000 00000000                        6042          DC    XL16'8000000000000000 00000000000000FF'   v2
00006A10  00000000 000000FF
00006A18  01020304 05060708                        6043          DC    XL16'0102030405060708 090A0B0C0D0E0F10'   v3
00006A20  090A0B0C 0D0E0F10
                                                   6044
                                                   6045          VRR_C VR,X'4',X'8',N
00006A28                                           6046+         DS    0FD
00006A28                         00006A28          6047+         USING *,R5               base for test data and test routine
00006A28  00006A98                                 6048+T115     DC    A(X115)            address of test routine
00006A2C  0073                                     6049+         DC    H'115'             test number
00006A2E  00                                       6050+         DC    X'00'
00006A2F  D5                                       6051+         DC    CL1'N'             S or Y = skip cross check
00006A30  04                                       6052+         DC    X'4'                m4
00006A31  08                                       6053+         DC    X'8'                m5
00006A32  E5D94040 40404040                        6054+         DC    CL8'VR'            instruction name
00006A3C  00006AD0                                 6055+         DC    A(RE115+16)        address of v2 source
00006A40  00006AE0                                 6056+         DC    A(RE115+32)        address of v3 source
00006A44  00000010                                 6057+         DC    A(16)              result length
00006A48  00006AC0                                 6058+REA115   DC    A(RE115)           result address
00006A50  00000000 00000000                        6059+         DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006A58  00000000 00000000                        6060+V1O115   DS    XL16              V1 output
00006A60  00000000 00000000
00006A68  00000000 00000000                        6061+         DS    FD                   gap
00006A70  40404040 40404040                        6062+         DC    CL8' '            was cross check skipped?
00006A78  00000000 00000000                        6063+         DS    FD                   gap
00006A80  00000000 00000000                        6064+XCO115   DS    XL16              Cross check Output
00006A88  00000000 00000000
00006A90  00000000 00000000                        6065+         DS    FD                   gap
                                                   6066+*
00006A98                                           6067+X115     DS    0F
00006A98  E310 5014 0014                 00000014  6068+         LGF   R1,V2ADDR         load v2 source
00006A9E  E721 0000 0006                 00000000  6069+         VL    v2,0(R1)
00006AA4  E310 5018 0014                 00000018  6070+         LGF   R1,V3ADDR         load v3 source
00006AAA  E731 0000 0006                 00000000  6071+         VL    v3,0(R1)

00006AB0                                           6074+         DS    0H                     E7B3 VR
00006AB0  E7                                       6075+         DC    X'E7'                  - VECTOR REMAINDER
00006AB1  12                                       6076+         DC    X'12'                   v1, v2
00006AB2  30                                       6077+         DC    X'30'                   v3, reserved
00006AB3  0840                                     6078+         DC    HL2'2112'               reserved, m5, m4, RXB
00006AB5  B3                                       6079+         DC    X'B3'

00006AB6  E710 5030 000E                 00006A58  6081+         VST   V1,V1O115         save v1 output
00006ABC  07FB                                     6082+         BR    R11               return
00006AC0                                           6083+RE115    DC    0F                xl16 expected result
00006AC0                                           6084+         DROP  R5
00006AC0  FFFFFFFF FFFFFFFF                        6085          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   result
00006AC8  FFFFFFFF FFFFFFFF
00006AD0  80000000 00000000                        6086          DC    XL16'8000000000000000 00000000000000FF'   v2
00006AD8  00000000 000000FF
00006AE0  FFFFFFFF FFFFFFFF                        6087          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFF00'   v3
00006AE8  FFFFFFFF FFFFFF00
                                                   6088
                                                   6089          VRR_C VR,X'4',X'8',N
00006AF0                                           6090+         DS    0FD
00006AF0                         00006AF0          6091+         USING *,R5              base for test data and test routine
00006AF0  00006B60                                 6092+T116     DC    A(X116)           address of test routine
00006AF4  0074                                     6093+         DC    H'116'            test number
00006AF6  00                                       6094+         DC    X'00'
00006AF7  D5                                       6095+         DC    CL1'N'            S or Y = skip cross check
00006AF8  04                                       6096+         DC    X'4'               m4
00006AF9  08                                       6097+         DC    X'8'               m5
00006AFA  E5D94040 40404040                        6098+         DC    CL8'VR'           instruction name
00006B04  00006B98                                 6099+         DC    A(RE116+16)       address of v2 source
00006B08  00006BA8                                 6100+         DC    A(RE116+32)       address of v3 source
00006B0C  00000010                                 6101+         DC    A(16)             result length
00006B10  00006B88                                 6102+REA116   DC    A(RE116)          result address
00006B18  00000000 00000000                        6103+         DS    FD                   gap
00006B20  00000000 00000000                        6104+V1O116   DS    XL16              V1 output
00006B28  00000000 00000000
00006B30  00000000 00000000                        6105+         DS    FD                   gap
00006B38  40404040 40404040                        6106+         DC    CL8' '            was cross check skipped?
00006B40  00000000 00000000                        6107+         DS    FD                   gap
00006B48  00000000 00000000                        6108+XCO116   DS    XL16              Cross check Output
00006B50  00000000 00000000
00006B58  00000000 00000000                        6109+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   6110+*
00006B60                                           6111+X116     DS    0F
00006B60  E310 5014 0014                 00000014  6112+         LGF   R1,V2ADDR         load v2 source
00006B66  E721 0000 0006                 00000000  6113+         VL    v2,0(R1)
00006B6C  E310 5018 0014                 00000018  6114+         LGF   R1,V3ADDR         load v3 source
00006B72  E731 0000 0006                 00000000  6115+         VL    v3,0(R1)

00006B78                                           6118+         DS    0H                      E7B3 VR
00006B78  E7                                       6119+         DC    X'E7'                   - VECTOR REMAINDER
00006B79  12                                       6120+         DC    X'12'                    v1, v2
00006B7A  30                                       6121+         DC    X'30'                    v3, reserved
00006B7B  0840                                     6122+         DC    HL2'2112'                reserved, m5, m4, RXB
00006B7D  B3                                       6123+         DC    X'B3'

00006B7E  E710 5030 000E                 00006B20  6125+         VST   V1,V1O116         save v1 output
00006B84  07FB                                     6126+         BR    R11               return
00006B88                                           6127+RE116    DC    0F                xl16 expected result
00006B88                                           6128+         DROP  R5
00006B88  FEFDFCFB 7AF9F8F7                        6129          DC    XL16'FEFDFCFB7AF9F8F7 76F5F4F372F1F1EF'   result
00006B90  76F5F4F3 72F1F1EF
00006B98  80000000 00000000                        6130          DC    XL16'8000000000000000 00000000000000FF'   v2
00006BA0  00000000 000000FF
00006BA8  81020304 85060708                        6131          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   v3
00006BB0  890A0B0C 8D0E0F10
                                                   6132
                                                   6133          VRR_C VR,X'4',X'8',N
00006BB8                                           6134+         DS    0FD
00006BB8                      00006BB8             6135+         USING *,R5              base for test data and test routine
00006BB8  00006C28                                 6136+T117     DC    A(X117)           address of test routine
00006BBC  0075                                     6137+         DC    H'117'            test number
00006BBE  00                                       6138+         DC    X'00'
00006BBF  D5                                       6139+         DC    CL1'N'            S or Y = skip cross check
00006BC0  04                                       6140+         DC    X'4'               m4
00006BC1  08                                       6141+         DC    X'8'               m5
00006BC2  E5D94040 40404040                        6142+         DC    CL8'VR'           instruction name
00006BCC  00006C60                                 6143+         DC    A(RE117+16)       address of v2 source
00006BD0  00006C70                                 6144+         DC    A(RE117+32)       address of v3 source
00006BD4  00000010                                 6145+         DC    A(16)             result length
00006BD8  00006C50                                 6146+REA117   DC    A(RE117)          result address
00006BE0  00000000 00000000                        6147+         DS    FD                   gap
00006BE8  00000000 00000000                        6148+V1O117   DS    XL16              V1 output
00006BF0  00000000 00000000
00006BF8  00000000 00000000                        6149+         DS    FD                   gap
00006C00  40404040 40404040                        6150+         DC    CL8' '            was cross check skipped?
00006C08  00000000 00000000                        6151+         DS    FD                   gap
00006C10  00000000 00000000                        6152+XCO117   DS    XL16              Cross check Output
00006C18  00000000 00000000
00006C20  00000000 00000000                        6153+         DS    FD                   gap
                                                   6154+*
00006C28                                           6155+X117     DS    0F
00006C28  E310 5014 0014                 00000014  6156+         LGF   R1,V2ADDR         load v2 source
00006C2E  E721 0000 0006                 00000000  6157+         VL    v2,0(R1)
00006C34  E310 5018 0014                 00000018  6158+         LGF   R1,V3ADDR         load v3 source
00006C3A  E731 0000 0006                 00000000  6159+         VL    v3,0(R1)

00006C40                                           6162+         DS    0H                      E7B3 VR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006C40  E7                                      6163+         DC    X'E7'                    - VECTOR REMAINDER
00006C41  12                                      6164+         DC    X'12'                     v1, v2
00006C42  30                                      6165+         DC    X'30'                     v3, reserved
00006C43  0840                                    6166+         DC    HL2'2112'                 reserved, m5, m4, RXB
00006C45  B3                                      6167+         DC    X'B3'

00006C46  E710 5030 000E                00006BE8  6169+         VST   V1,V1O117         save v1 output
00006C4C  07FB                                    6170+         BR    R11               return
00006C50                                          6171+RE117    DC    0F                xl16 expected result
00006C50                                          6172+         DROP  R5
00006C50  81020304 85060708                       6173          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   result
00006C58  890A0B0C 8D0E0F10
00006C60  81020304 85060708                       6174          DC    XL16'8102030485060708 890A0B0C8D0E0F10'   v2
00006C68  890A0B0C 8D0E0F10
00006C70  80000000 00000000                       6175          DC    XL16'8000000000000000 00000000000000FF'   v3
00006C78  00000000 000000FF
                                                  6176
                                                  6177 * skip XC
                                                  6178          VRR_C VR,X'4',X'8',S
00006C80                                          6179+         DS    0FD
00006C80                        00006C80          6180+         USING *,R5              base for test data and test routine
00006C80  00006CF0                                6181+T118     DC    A(X118)           address of test routine
00006C84  0076                                    6182+         DC    H'118'            test number
00006C86  00                                      6183+         DC    X'00'
00006C87  E2                                      6184+         DC    CL1'S'            S or Y = skip cross check
00006C88  04                                      6185+         DC    X'4'               m4
00006C89  08                                      6186+         DC    X'8'               m5
00006C8A  E5D94040 40404040                       6187+         DC    CL8'VR'           instruction name
00006C94  00006D28                                6188+         DC    A(RE118+16)       address of v2 source
00006C98  00006D38                                6189+         DC    A(RE118+32)       address of v3 source
00006C9C  00000010                                6190+         DC    A(16)             result length
00006CA0  00006D18                                6191+REA118   DC    A(RE118)          result address
00006CA8  00000000 00000000                       6192+         DS    FD                   gap
00006CB0  00000000 00000000                       6193+V1O118   DS    XL16              V1 output
00006CB8  00000000 00000000
00006CC0  00000000 00000000                       6194+         DS    FD                   gap
00006CC8  40404040 40404040                       6195+         DC    CL8' '            was cross check skipped?
00006CD0  00000000 00000000                       6196+         DS    FD                   gap
00006CD8  00000000 00000000                       6197+XCO118   DS    XL16              Cross check Output
00006CE0  00000000 00000000
00006CE8  00000000 00000000                       6198+         DS    FD                   gap
                                                  6199+*
00006CF0                                          6200+X118     DS    0F
00006CF0  E310 5014 0014                00000014  6201+         LGF   R1,V2ADDR         load v2 source
00006CF6  E721 0000 0006                00000000  6202+         VL    v2,0(R1)
00006CFC  E310 5018 0014                00000018  6203+         LGF   R1,V3ADDR         load v3 source
00006D02  E731 0000 0006                00000000  6204+         VL    v3,0(R1)

00006D08                                          6207+         DS    0H                       E7B3 VR
00006D08  E7                                      6208+         DC    X'E7'                    - VECTOR REMAINDER
00006D09  12                                      6209+         DC    X'12'                     v1, v2
00006D0A  30                                      6210+         DC    X'30'                     v3, reserved
00006D0B  0840                                    6211+         DC    HL2'2112'                 reserved, m5, m4, RXB
00006D0D  B3                                      6212+         DC    X'B3'

00006D0E  E710 5030 000E                00006CB0  6214+         VST   V1,V1O118         save v1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006D14  07FB                                     6215+         BR    R11                return
00006D18                                           6216+RE118    DC    0F                 xl16 expected result
00006D18                                           6217+         DROP  R5
00006D18  00000000 00000000                        6218          DC    XL16'0000000000000000 0000000000000000'   result
00006D20  00000000 00000000
00006D28  80000000 00000000                        6219          DC    XL16'8000000000000000 0000000000000000'   v2
00006D30  00000000 00000000
00006D38  FFFFFFFF FFFFFFFF                        6220          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'   v3
00006D40  FFFFFFFF FFFFFFFF
                                                   6221
                                                   6222
                                                   6223
                                                   6224
                                                   6225
                                                   6226
00006D48  00000000                                 6227          DC    F'0'     END OF TABLE
00006D4C  00000000                                 6228          DC    F'0'
                                                   6229 *
                                                   6230 * table of pointers to individual load test
                                                   6231 *
00006D50                                           6232 E7TESTS  DS    0F
                                                   6233          PTTABLE
00006D50                                           6234+TTABLE   DS    0F
00006D50  00001118                                 6235+         DC    A(T1)
00006D54  000011E0                                 6236+         DC    A(T2)
00006D58  000012A8                                 6237+         DC    A(T3)
00006D5C  00001370                                 6238+         DC    A(T4)
00006D60  00001438                                 6239+         DC    A(T5)
00006D64  00001500                                 6240+         DC    A(T6)
00006D68  000015C8                                 6241+         DC    A(T7)
00006D6C  00001690                                 6242+         DC    A(T8)
00006D70  00001758                                 6243+         DC    A(T9)
00006D74  00001820                                 6244+         DC    A(T10)
00006D78  000018E8                                 6245+         DC    A(T11)
00006D7C  000019B0                                 6246+         DC    A(T12)
00006D80  00001A78                                 6247+         DC    A(T13)
00006D84  00001B40                                 6248+         DC    A(T14)
00006D88  00001C08                                 6249+         DC    A(T15)
00006D8C  00001CD0                                 6250+         DC    A(T16)
00006D90  00001D98                                 6251+         DC    A(T17)
00006D94  00001E60                                 6252+         DC    A(T18)
00006D98  00001F28                                 6253+         DC    A(T19)
00006D9C  00001FF0                                 6254+         DC    A(T20)
00006DA0  000020B8                                 6255+         DC    A(T21)
00006DA4  00002180                                 6256+         DC    A(T22)
00006DA8  00002248                                 6257+         DC    A(T23)
00006DAC  00002310                                 6258+         DC    A(T24)
00006DB0  000023D8                                 6259+         DC    A(T25)
00006DB4  000024A0                                 6260+         DC    A(T26)
00006DB8  00002568                                 6261+         DC    A(T27)
00006DBC  00002630                                 6262+         DC    A(T28)
00006DC0  000026F8                                 6263+         DC    A(T29)
00006DC4  000027C0                                 6264+         DC    A(T30)
00006DC8  00002888                                 6265+         DC    A(T31)
00006DCC  00002950                                 6266+         DC    A(T32)
00006DD0  00002A18                                 6267+         DC    A(T33)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006DD4  00002AE0                          6268+         DC    A(T34)
00006DD8  00002BA8                          6269+         DC    A(T35)
00006DDC  00002C70                          6270+         DC    A(T36)
00006DE0  00002D38                          6271+         DC    A(T37)
00006DE4  00002E00                          6272+         DC    A(T38)
00006DE8  00002EC8                          6273+         DC    A(T39)
00006DEC  00002F90                          6274+         DC    A(T40)
00006DF0  00003058                          6275+         DC    A(T41)
00006DF4  00003120                          6276+         DC    A(T42)
00006DF8  000031E8                          6277+         DC    A(T43)
00006DFC  000032B0                          6278+         DC    A(T44)
00006E00  00003378                          6279+         DC    A(T45)
00006E04  00003440                          6280+         DC    A(T46)
00006E08  00003508                          6281+         DC    A(T47)
00006E0C  000035D0                          6282+         DC    A(T48)
00006E10  00003698                          6283+         DC    A(T49)
00006E14  00003760                          6284+         DC    A(T50)
00006E18  00003828                          6285+         DC    A(T51)
00006E1C  000038F0                          6286+         DC    A(T52)
00006E20  000039B8                          6287+         DC    A(T53)
00006E24  00003A80                          6288+         DC    A(T54)
00006E28  00003B48                          6289+         DC    A(T55)
00006E2C  00003C10                          6290+         DC    A(T56)
00006E30  00003CD8                          6291+         DC    A(T57)
00006E34  00003DA0                          6292+         DC    A(T58)
00006E38  00003E68                          6293+         DC    A(T59)
00006E3C  00003F30                          6294+         DC    A(T60)
00006E40  00003FF8                          6295+         DC    A(T61)
00006E44  000040C0                          6296+         DC    A(T62)
00006E48  00004188                          6297+         DC    A(T63)
00006E4C  00004250                          6298+         DC    A(T64)
00006E50  00004318                          6299+         DC    A(T65)
00006E54  000043E0                          6300+         DC    A(T66)
00006E58  000044A8                          6301+         DC    A(T67)
00006E5C  00004570                          6302+         DC    A(T68)
00006E60  00004638                          6303+         DC    A(T69)
00006E64  00004700                          6304+         DC    A(T70)
00006E68  000047C8                          6305+         DC    A(T71)
00006E6C  00004890                          6306+         DC    A(T72)
00006E70  00004958                          6307+         DC    A(T73)
00006E74  00004A20                          6308+         DC    A(T74)
00006E78  00004AE8                          6309+         DC    A(T75)
00006E7C  00004BB0                          6310+         DC    A(T76)
00006E80  00004C78                          6311+         DC    A(T77)
00006E84  00004D40                          6312+         DC    A(T78)
00006E88  00004E08                          6313+         DC    A(T79)
00006E8C  00004ED0                          6314+         DC    A(T80)
00006E90  00004F98                          6315+         DC    A(T81)
00006E94  00005060                          6316+         DC    A(T82)
00006E98  00005128                          6317+         DC    A(T83)
00006E9C  000051F0                          6318+         DC    A(T84)
00006EA0  000052B8                          6319+         DC    A(T85)
00006EA4  00005380                          6320+         DC    A(T86)
00006EA8  00005448                          6321+         DC    A(T87)
00006EAC  00005510                          6322+         DC    A(T88)
00006EB0  000055D8                          6323+         DC    A(T89)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00006EB4  000056A0                                 6324+          DC    A(T90)
00006EB8  00005768                                 6325+          DC    A(T91)
00006EBC  00005830                                 6326+          DC    A(T92)
00006EC0  000058F8                                 6327+          DC    A(T93)
00006EC4  000059C0                                 6328+          DC    A(T94)
00006EC8  00005A88                                 6329+          DC    A(T95)
00006ECC  00005B50                                 6330+          DC    A(T96)
00006ED0  00005C18                                 6331+          DC    A(T97)
00006ED4  00005CE0                                 6332+          DC    A(T98)
00006ED8  00005DA8                                 6333+          DC    A(T99)
00006EDC  00005E70                                 6334+          DC    A(T100)
00006EE0  00005F38                                 6335+          DC    A(T101)
00006EE4  00006000                                 6336+          DC    A(T102)
00006EE8  000060C8                                 6337+          DC    A(T103)
00006EEC  00006190                                 6338+          DC    A(T104)
00006EF0  00006258                                 6339+          DC    A(T105)
00006EF4  00006320                                 6340+          DC    A(T106)
00006EF8  000063E8                                 6341+          DC    A(T107)
00006EFC  000064B0                                 6342+          DC    A(T108)
00006F00  00006578                                 6343+          DC    A(T109)
00006F04  00006640                                 6344+          DC    A(T110)
00006F08  00006708                                 6345+          DC    A(T111)
00006F0C  000067D0                                 6346+          DC    A(T112)
00006F10  00006898                                 6347+          DC    A(T113)
00006F14  00006960                                 6348+          DC    A(T114)
00006F18  00006A28                                 6349+          DC    A(T115)
00006F1C  00006AF0                                 6350+          DC    A(T116)
00006F20  00006BB8                                 6351+          DC    A(T117)
00006F24  00006C80                                 6352+          DC    A(T118)
                                                   6353+*
00006F28  00000000                                 6354+          DC    A(0)              END OF TABLE
00006F2C  00000000                                 6355+          DC    A(0)
                                                   6356
00006F30  00000000                                 6357           DC    F'0'     END OF TABLE
00006F34  00000000                                 6358           DC    F'0'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                         6360 ************************************************************************
                                         6361 *        Register equates
                                         6362 ************************************************************************


                             00000000  00000001  6364 R0       EQU   0
                             00000001  00000001  6365 R1       EQU   1
                             00000002  00000001  6366 R2       EQU   2
                             00000003  00000001  6367 R3       EQU   3
                             00000004  00000001  6368 R4       EQU   4
                             00000005  00000001  6369 R5       EQU   5
                             00000006  00000001  6370 R6       EQU   6
                             00000007  00000001  6371 R7       EQU   7
                             00000008  00000001  6372 R8       EQU   8
                             00000009  00000001  6373 R9       EQU   9
                             0000000A  00000001  6374 R10      EQU   10
                             0000000B  00000001  6375 R11      EQU   11
                             0000000C  00000001  6376 R12      EQU   12
                             0000000D  00000001  6377 R13      EQU   13
                             0000000E  00000001  6378 R14      EQU   14
                             0000000F  00000001  6379 R15      EQU   15


                                         6381 ************************************************************************
                                         6382 *        Register equates
                                         6383 ************************************************************************


                             00000000  00000001  6385 V0       EQU   0
                             00000001  00000001  6386 V1       EQU   1
                             00000002  00000001  6387 V2       EQU   2
                             00000003  00000001  6388 V3       EQU   3
                             00000004  00000001  6389 V4       EQU   4
                             00000005  00000001  6390 V5       EQU   5
                             00000006  00000001  6391 V6       EQU   6
                             00000007  00000001  6392 V7       EQU   7
                             00000008  00000001  6393 V8       EQU   8
                             00000009  00000001  6394 V9       EQU   9
                             0000000A  00000001  6395 V10      EQU   10
                             0000000B  00000001  6396 V11      EQU   11
                             0000000C  00000001  6397 V12      EQU   12
                             0000000D  00000001  6398 V13      EQU   13
                             0000000E  00000001  6399 V14      EQU   14
                             0000000F  00000001  6400 V15      EQU   15
                             00000010  00000001  6401 V16      EQU   16
                             00000011  00000001  6402 V17      EQU   17
                             00000012  00000001  6403 V18      EQU   18
                             00000013  00000001  6404 V19      EQU   19
                             00000014  00000001  6405 V20      EQU   20
                             00000015  00000001  6406 V21      EQU   21
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2    STMT

                              00000016  00000001  6407 V22      EQU   22
                              00000017  00000001  6408 V23      EQU   23
                              00000018  00000001  6409 V24      EQU   24
                              00000019  00000001  6410 V25      EQU   25
                              0000001A  00000001  6411 V26      EQU   26
                              0000001B  00000001  6412 V27      EQU   27
                              0000001C  00000001  6413 V28      EQU   28
                              0000001D  00000001  6414 V29      EQU   29
                              0000001E  00000001  6415 V30      EQU   30
                              0000001F  00000001  6416 V31      EQU   31
                                                  6417
                                                  6418          END
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
BEGIN	I	00000200	2	283	249	279	280	281									
CTLR0	F	0000071C	4	767	293	294	295	296									
DECNUM	C	000010D6	16	850	603	605	612	614	619	621	675	677	683	685	690	692	
E7TEST	4	00000000	112	864	342												
E7TESTS	F	00006D50	4	6232	335												
EDIT	X	000010AA	18	845	604	613	620	676	684	691							
ENDTEST	U	0000058C	1	661	340												
EOJ	I	00000700	4	757	328	664											
EOJPSW	D	000006F0	8	755	757												
FAILCONT	U	0000057C	1	651													
FAILED	F	00001000	4	806	631	653	662										
FAILMSG	U	00000578	1	645	358												
FAILPSW	D	00000708	8	759	761												
FAILTEST	I	00000718	4	761	665												
FB0001	F	00000280	8	312	316	317	319										
IMAGE	1	00000000	28472	0													
K	U	00000400	1	790	791	792	793										
K64	U	00010000	1	792													
M4	U	00000008	1	869	396	411	426	445	460	475	494	509	524	543	558	573	611
					682												
M5	U	00000009	1	870	618	689											
MB	U	00100000	1	793													
MSG	I	00000638	4	717	327	700											
MSGCMD	C	00000686	9	747	730	731											
MSGMSG	C	0000068F	95	748	724	745	722										
MSGMVC	I	00000680	6	745	728												
MSGOK	I	0000064E	2	726	723												
MSGRET	I	0000066E	4	741	734	737											
MSGSAVE	F	00000674	4	744	720	741											
NEXTE7	U	000002D4	1	337	361	656											
OPNAME	C	0000000A	8	872	378	380	382	384	608	680							
PAGE	U	00001000	1	791													
PRT3	C	000010C0	18	848	604	605	606	613	614	615	620	621	622	676	677	678	684
					685	686	691	692	693								
PRTLINE	C	00001008	13	812	822	699											
PRTLNG	U	00000051	1	822	698												
PRTM4	C	0000104F	2	817	686												
PRTM5	C	00001056	2	820	693												
PRTNAME	C	0000103E	8	815	680												
PRTNUM	C	00001015	3	813	678												
R0	U	00000000	1	6364	243	293	296	316	318	319	320	325	344	345	625	630	631
					652	653	697	698	701	717	720	722	724	726	741		
R1	U	00000001	1	6365	326	356	357	594	595	626	662	663	699	731	745	1016	1017
					1018	1019	1060	1061	1062	1063	1104	1105	1106	1107	1148	1149	1150
					1151	1192	1193	1194	1195	1238	1239	1240	1241	1282	1283	1284	1285
					1326	1327	1328	1329	1370	1371	1372	1373	1414	1415	1416	1417	1458
					1459	1460	1461	1504	1505	1506	1507	1548	1549	1550	1551	1592	1593
					1594	1595	1636	1637	1638	1639	1680	1681	1682	1683	1724	1725	1726
					1727	1768	1769	1770	1771	1812	1813	1814	1815	1856	1857	1858	1859
					1905	1906	1907	1908	1949	1950	1951	1952	1993	1994	1995	1996	2037
					2038	2039	2040	2081	2082	2083	2084	2127	2128	2129	2130	2171	2172
					2173	2174	2215	2216	2217	2218	2259	2260	2261	2262	2303	2304	2305
					2306	2347	2348	2349	2350	2393	2394	2395	2396	2437	2438	2439	2440
					2481	2482	2483	2484	2525	2526	2527	2528	2569	2570	2571	2572	2613
					2614	2615	2616	2657	2658	2659	2660	2701	2702	2703	2704	2745	2746
					2747	2748	2794	2795	2796	2797	2838	2839	2840	2841	2882	2883	2884

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
					2885	2926	2927	2928	2929	2970	2971	2972	2973	3014	3015	3016	3017
					3058	3059	3060	3061	3102	3103	3104	3105	3146	3147	3148	3149	3191
					3192	3193	3194	3237	3238	3239	3240	3281	3282	3283	3284	3325	3326
					3327	3328	3369	3370	3371	3372	3413	3414	3415	3416	3457	3458	3459
					3460	3501	3502	3503	3504	3545	3546	3547	3548	3589	3590	3591	3592
					3633	3634	3635	3636	3677	3678	3679	3680	3722	3723	3724	3725	3768
					3769	3770	3771	3812	3813	3814	3815	3856	3857	3858	3859	3900	3901
					3902	3903	3944	3945	3946	3947	3988	3989	3990	3991	4032	4033	4034
					4035	4076	4077	4078	4079	4120	4121	4122	4123	4164	4165	4166	4167
					4208	4209	4210	4211	4252	4253	4254	4255	4296	4297	4298	4299	4340
					4341	4342	4343	4384	4385	4386	4387	4428	4429	4430	4431	4473	4474
					4475	4476	4522	4523	4524	4525	4566	4567	4568	4569	4610	4611	4612
					4613	4654	4655	4656	4657	4698	4699	4700	4701	4742	4743	4744	4745
					4786	4787	4788	4789	4830	4831	4832	4833	4874	4875	4876	4877	4919
					4920	4921	4922	4965	4966	4967	4968	5009	5010	5011	5012	5053	5054
					5055	5056	5097	5098	5099	5100	5141	5142	5143	5144	5185	5186	5187
					5188	5229	5230	5231	5232	5273	5274	5275	5276	5317	5318	5319	5320
					5361	5362	5363	5364	5405	5406	5407	5408	5450	5451	5452	5453	5496
					5497	5498	5499	5540	5541	5542	5543	5584	5585	5586	5587	5628	5629
					5630	5631	5672	5673	5674	5675	5716	5717	5718	5719	5760	5761	5762
					5763	5804	5805	5806	5807	5848	5849	5850	5851	5892	5893	5894	5895
					5936	5937	5938	5939	5980	5981	5982	5983	6024	6025	6026	6027	6068
					6069	6070	6071	6112	6113	6114	6115	6156	6157	6158	6159	6201	6202
					6203	6204											
R10	U	0000000A	1	6374	281	290	291										
R11	U	0000000B	1	6375	348	349	1030	1074	1118	1162	1206	1252	1296	1340	1384	1428	1472
					1518	1562	1606	1650	1694	1738	1782	1826	1870	1919	1963	2007	2051
					2095	2141	2185	2229	2273	2317	2361	2407	2451	2495	2539	2583	2627
					2671	2715	2759	2808	2852	2896	2940	2984	3028	3072	3116	3160	3205
					3251	3295	3339	3383	3427	3471	3515	3559	3603	3647	3691	3736	3782
					3826	3870	3914	3958	4002	4046	4090	4134	4178	4222	4266	4310	4354
					4398	4442	4487	4536	4580	4624	4668	4712	4756	4800	4844	4888	4933
					4979	5023	5067	5111	5155	5199	5243	5287	5331	5375	5419	5464	5510
					5554	5598	5642	5686	5730	5774	5818	5862	5906	5950	5994	6038	6082
					6126	6170	6215										
R12	U	0000000C	1	6376	335	338	360	655									
R13	U	0000000D	1	6377													
R14	U	0000000E	1	6378													
R15	U	0000000F	1	6379	352	371	373	387	427	476	525	574	597	624	627	628	632
					646	671	704	705									
R2	U	00000002	1	6366	327	602	603	610	611	612	617	618	619	674	675	682	683
					688	689	690	697	700	701	718	720	726	727	728	730	736
					741	742											
R3	U	00000003	1	6367													
R4	U	00000004	1	6368													
R5	U	00000005	1	6369	338	339	342	672	703	995	1032	1039	1076	1083	1120	1127	1164
					1171	1208	1217	1254	1261	1298	1305	1342	1349	1386	1393	1430	1437
					1474	1483	1520	1527	1564	1571	1608	1615	1652	1659	1696	1703	1740
					1747	1784	1791	1828	1835	1872	1884	1921	1928	1965	1972	2009	2016
					2053	2060	2097	2106	2143	2150	2187	2194	2231	2238	2275	2282	2319
					2326	2363	2372	2409	2416	2453	2460	2497	2504	2541	2548	2585	2592
					2629	2636	2673	2680	2717	2724	2761	2773	2810	2817	2854	2861	2898
					2905	2942	2949	2986	2993	3030	3037	3074	3081	3118	3125	3162	3170
					3207	3216	3253	3260	3297	3304	3341	3348	3385	3392	3429	3436	3473
					3480	3517	3524	3561	3568	3605	3612	3649	3656	3693	3701	3738	3747
					3784	3791	3828	3835	3872	3879	3916	3923	3960	3967	4004	4011	4048

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
					4055	4092	4099	4136	4143	4180	4187	4224	4231	4268	4275	4312	4319
					4356	4363	4400	4407	4444	4452	4489	4501	4538	4545	4582	4589	4626
					4633	4670	4677	4714	4721	4758	4765	4802	4809	4846	4853	4890	4898
					4935	4944	4981	4988	5025	5032	5069	5076	5113	5120	5157	5164	5201
					5208	5245	5252	5289	5296	5333	5340	5377	5384	5421	5429	5466	5475
					5512	5519	5556	5563	5600	5607	5644	5651	5688	5695	5732	5739	5776
					5783	5820	5827	5864	5871	5908	5915	5952	5959	5996	6003	6040	6047
					6084	6091	6128	6135	6172	6180	6217						
R6	U	00000006	1	6370													
R7	U	00000007	1	6371													
R8	U	00000008	1	6372	279	283	284	285	287								
R9	U	00000009	1	6373	280	287	288	290									
RE1	F	000011B0	4	1031	1003	1004	1006										
RE10	F	000018B8	4	1429	1401	1402	1404										
RE100	F	00005F08	4	5420	5392	5393	5395										
RE101	F	00005FD0	4	5465	5437	5438	5440										
RE102	F	00006098	4	5511	5483	5484	5486										
RE103	F	00006160	4	5555	5527	5528	5530										
RE104	F	00006228	4	5599	5571	5572	5574										
RE105	F	000062F0	4	5643	5615	5616	5618										
RE106	F	000063B8	4	5687	5659	5660	5662										
RE107	F	00006480	4	5731	5703	5704	5706										
RE108	F	00006548	4	5775	5747	5748	5750										
RE109	F	00006610	4	5819	5791	5792	5794										
RE11	F	00001980	4	1473	1445	1446	1448										
RE110	F	000066D8	4	5863	5835	5836	5838										
RE111	F	000067A0	4	5907	5879	5880	5882										
RE112	F	00006868	4	5951	5923	5924	5926										
RE113	F	00006930	4	5995	5967	5968	5970										
RE114	F	000069F8	4	6039	6011	6012	6014										
RE115	F	00006AC0	4	6083	6055	6056	6058										
RE116	F	00006B88	4	6127	6099	6100	6102										
RE117	F	00006C50	4	6171	6143	6144	6146										
RE118	F	00006D18	4	6216	6188	6189	6191										
RE12	F	00001A48	4	1519	1491	1492	1494										
RE13	F	00001B10	4	1563	1535	1536	1538										
RE14	F	00001BD8	4	1607	1579	1580	1582										
RE15	F	00001CA0	4	1651	1623	1624	1626										
RE16	F	00001D68	4	1695	1667	1668	1670										
RE17	F	00001E30	4	1739	1711	1712	1714										
RE18	F	00001EF8	4	1783	1755	1756	1758										
RE19	F	00001FC0	4	1827	1799	1800	1802										
RE2	F	00001278	4	1075	1047	1048	1050										
RE20	F	00002088	4	1871	1843	1844	1846										
RE21	F	00002150	4	1920	1892	1893	1895										
RE22	F	00002218	4	1964	1936	1937	1939										
RE23	F	000022E0	4	2008	1980	1981	1983										
RE24	F	000023A8	4	2052	2024	2025	2027										
RE25	F	00002470	4	2096	2068	2069	2071										
RE26	F	00002538	4	2142	2114	2115	2117										
RE27	F	00002600	4	2186	2158	2159	2161										
RE28	F	000026C8	4	2230	2202	2203	2205										
RE29	F	00002790	4	2274	2246	2247	2249										
RE3	F	00001340	4	1119	1091	1092	1094										
RE30	F	00002858	4	2318	2290	2291	2293										
RE31	F	00002920	4	2362	2334	2335	2337										

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
RE32	F	000029E8	4	2408	2380	2381	2383
RE33	F	00002AB0	4	2452	2424	2425	2427
RE34	F	00002B78	4	2496	2468	2469	2471
RE35	F	00002C40	4	2540	2512	2513	2515
RE36	F	00002D08	4	2584	2556	2557	2559
RE37	F	00002DD0	4	2628	2600	2601	2603
RE38	F	00002E98	4	2672	2644	2645	2647
RE39	F	00002F60	4	2716	2688	2689	2691
RE4	F	00001408	4	1163	1135	1136	1138
RE40	F	00003028	4	2760	2732	2733	2735
RE41	F	000030F0	4	2809	2781	2782	2784
RE42	F	000031B8	4	2853	2825	2826	2828
RE43	F	00003280	4	2897	2869	2870	2872
RE44	F	00003348	4	2941	2913	2914	2916
RE45	F	00003410	4	2985	2957	2958	2960
RE46	F	000034D8	4	3029	3001	3002	3004
RE47	F	000035A0	4	3073	3045	3046	3048
RE48	F	00003668	4	3117	3089	3090	3092
RE49	F	00003730	4	3161	3133	3134	3136
RE5	F	000014D0	4	1207	1179	1180	1182
RE50	F	000037F8	4	3206	3178	3179	3181
RE51	F	000038C0	4	3252	3224	3225	3227
RE52	F	00003988	4	3296	3268	3269	3271
RE53	F	00003A50	4	3340	3312	3313	3315
RE54	F	00003B18	4	3384	3356	3357	3359
RE55	F	00003BE0	4	3428	3400	3401	3403
RE56	F	00003CA8	4	3472	3444	3445	3447
RE57	F	00003D70	4	3516	3488	3489	3491
RE58	F	00003E38	4	3560	3532	3533	3535
RE59	F	00003F00	4	3604	3576	3577	3579
RE6	F	00001598	4	1253	1225	1226	1228
RE60	F	00003FC8	4	3648	3620	3621	3623
RE61	F	00004090	4	3692	3664	3665	3667
RE62	F	00004158	4	3737	3709	3710	3712
RE63	F	00004220	4	3783	3755	3756	3758
RE64	F	000042E8	4	3827	3799	3800	3802
RE65	F	000043B0	4	3871	3843	3844	3846
RE66	F	00004478	4	3915	3887	3888	3890
RE67	F	00004540	4	3959	3931	3932	3934
RE68	F	00004608	4	4003	3975	3976	3978
RE69	F	000046D0	4	4047	4019	4020	4022
RE7	F	00001660	4	1297	1269	1270	1272
RE70	F	00004798	4	4091	4063	4064	4066
RE71	F	00004860	4	4135	4107	4108	4110
RE72	F	00004928	4	4179	4151	4152	4154
RE73	F	000049F0	4	4223	4195	4196	4198
RE74	F	00004AB8	4	4267	4239	4240	4242
RE75	F	00004B80	4	4311	4283	4284	4286
RE76	F	00004C48	4	4355	4327	4328	4330
RE77	F	00004D10	4	4399	4371	4372	4374
RE78	F	00004DD8	4	4443	4415	4416	4418
RE79	F	00004EA0	4	4488	4460	4461	4463
RE8	F	00001728	4	1341	1313	1314	1316
RE80	F	00004F68	4	4537	4509	4510	4512
RE81	F	00005030	4	4581	4553	4554	4556
RE82	F	000050F8	4	4625	4597	4598	4600

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
RE83	F	000051C0	4	4669	4641	4642	4644
RE84	F	00005288	4	4713	4685	4686	4688
RE85	F	00005350	4	4757	4729	4730	4732
RE86	F	00005418	4	4801	4773	4774	4776
RE87	F	000054E0	4	4845	4817	4818	4820
RE88	F	000055A8	4	4889	4861	4862	4864
RE89	F	00005670	4	4934	4906	4907	4909
RE9	F	000017F0	4	1385	1357	1358	1360
RE90	F	00005738	4	4980	4952	4953	4955
RE91	F	00005800	4	5024	4996	4997	4999
RE92	F	000058C8	4	5068	5040	5041	5043
RE93	F	00005990	4	5112	5084	5085	5087
RE94	F	00005A58	4	5156	5128	5129	5131
RE95	F	00005B20	4	5200	5172	5173	5175
RE96	F	00005BE8	4	5244	5216	5217	5219
RE97	F	00005CB0	4	5288	5260	5261	5263
RE98	F	00005D78	4	5332	5304	5305	5307
RE99	F	00005E40	4	5376	5348	5349	5351
REA1	A	00001138	4	1006			
REA10	A	00001840	4	1404			
REA100	A	00005E90	4	5395			
REA101	A	00005F58	4	5440			
REA102	A	00006020	4	5486			
REA103	A	000060E8	4	5530			
REA104	A	000061B0	4	5574			
REA105	A	00006278	4	5618			
REA106	A	00006340	4	5662			
REA107	A	00006408	4	5706			
REA108	A	000064D0	4	5750			
REA109	A	00006598	4	5794			
REA11	A	00001908	4	1448			
REA110	A	00006660	4	5838			
REA111	A	00006728	4	5882			
REA112	A	000067F0	4	5926			
REA113	A	000068B8	4	5970			
REA114	A	00006980	4	6014			
REA115	A	00006A48	4	6058			
REA116	A	00006B10	4	6102			
REA117	A	00006BD8	4	6146			
REA118	A	00006CA0	4	6191			
REA12	A	000019D0	4	1494			
REA13	A	00001A98	4	1538			
REA14	A	00001B60	4	1582			
REA15	A	00001C28	4	1626			
REA16	A	00001CF0	4	1670			
REA17	A	00001DB8	4	1714			
REA18	A	00001E80	4	1758			
REA19	A	00001F48	4	1802			
REA2	A	00001200	4	1050			
REA20	A	00002010	4	1846			
REA21	A	000020D8	4	1895			
REA22	A	000021A0	4	1939			
REA23	A	00002268	4	1983			
REA24	A	00002330	4	2027			
REA25	A	000023F8	4	2071			
REA26	A	000024C0	4	2117			

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
REA27	A	00002588	4	2161	
REA28	A	00002650	4	2205	
REA29	A	00002718	4	2249	
REA3	A	000012C8	4	1094	
REA30	A	000027E0	4	2293	
REA31	A	000028A8	4	2337	
REA32	A	00002970	4	2383	
REA33	A	00002A38	4	2427	
REA34	A	00002B00	4	2471	
REA35	A	00002BC8	4	2515	
REA36	A	00002C90	4	2559	
REA37	A	00002D58	4	2603	
REA38	A	00002E20	4	2647	
REA39	A	00002EE8	4	2691	
REA4	A	00001390	4	1138	
REA40	A	00002FB0	4	2735	
REA41	A	00003078	4	2784	
REA42	A	00003140	4	2828	
REA43	A	00003208	4	2872	
REA44	A	000032D0	4	2916	
REA45	A	00003398	4	2960	
REA46	A	00003460	4	3004	
REA47	A	00003528	4	3048	
REA48	A	000035F0	4	3092	
REA49	A	000036B8	4	3136	
REA5	A	00001458	4	1182	
REA50	A	00003780	4	3181	
REA51	A	00003848	4	3227	
REA52	A	00003910	4	3271	
REA53	A	000039D8	4	3315	
REA54	A	00003AA0	4	3359	
REA55	A	00003B68	4	3403	
REA56	A	00003C30	4	3447	
REA57	A	00003CF8	4	3491	
REA58	A	00003DC0	4	3535	
REA59	A	00003E88	4	3579	
REA6	A	00001520	4	1228	
REA60	A	00003F50	4	3623	
REA61	A	00004018	4	3667	
REA62	A	000040E0	4	3712	
REA63	A	000041A8	4	3758	
REA64	A	00004270	4	3802	
REA65	A	00004338	4	3846	
REA66	A	00004400	4	3890	
REA67	A	000044C8	4	3934	
REA68	A	00004590	4	3978	
REA69	A	00004658	4	4022	
REA7	A	000015E8	4	1272	
REA70	A	00004720	4	4066	
REA71	A	000047E8	4	4110	
REA72	A	000048B0	4	4154	
REA73	A	00004978	4	4198	
REA74	A	00004A40	4	4242	
REA75	A	00004B08	4	4286	
REA76	A	00004BD0	4	4330	
REA77	A	00004C98	4	4374	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
REA78	A	00004D60	4	4418		
REA79	A	00004E28	4	4463		
REA8	A	000016B0	4	1316		
REA80	A	00004EF0	4	4512		
REA81	A	00004FB8	4	4556		
REA82	A	00005080	4	4600		
REA83	A	00005148	4	4644		
REA84	A	00005210	4	4688		
REA85	A	000052D8	4	4732		
REA86	A	000053A0	4	4776		
REA87	A	00005468	4	4820		
REA88	A	00005530	4	4864		
REA89	A	000055F8	4	4909		
REA9	A	00001778	4	1360		
REA90	A	000056C0	4	4955		
REA91	A	00005788	4	4999		
REA92	A	00005850	4	5043		
REA93	A	00005918	4	5087		
REA94	A	000059E0	4	5131		
REA95	A	00005AA8	4	5175		
REA96	A	00005B70	4	5219		
REA97	A	00005C38	4	5263		
REA98	A	00005D00	4	5307		
REA99	A	00005DC8	4	5351		
READDR	A	00000020	4	876	356	
REG2LOW	U	000000DD	1	796		
REG2PATT	U	AABBCCDD	1	795		
RELEN	A	0000001C	4	875		
RPTDWSAV	D	00000628	8	710	697	701
RPTERROR	I	0000059A	4	671	627	646
RPTSAVE	F	0000061C	4	707	671	704
RPTSVR5	F	00000620	4	708	672	703
SKIPXC	C	00000048	8	880	369	377
SKL0001	U	0000004E	1	309	325	
SKT0001	C	0000022A	20	306	309	326
SVOLDPSW	U	00000140	0	245		
T1	A	00001118	4	996	6235	
T10	A	00001820	4	1394	6244	
T100	A	00005E70	4	5385	6334	
T101	A	00005F38	4	5430	6335	
T102	A	00006000	4	5476	6336	
T103	A	000060C8	4	5520	6337	
T104	A	00006190	4	5564	6338	
T105	A	00006258	4	5608	6339	
T106	A	00006320	4	5652	6340	
T107	A	000063E8	4	5696	6341	
T108	A	000064B0	4	5740	6342	
T109	A	00006578	4	5784	6343	
T11	A	000018E8	4	1438	6245	
T110	A	00006640	4	5828	6344	
T111	A	00006708	4	5872	6345	
T112	A	000067D0	4	5916	6346	
T113	A	00006898	4	5960	6347	
T114	A	00006960	4	6004	6348	
T115	A	00006A28	4	6048	6349	
T116	A	00006AF0	4	6092	6350	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T117	A	00006BB8	4	6136	6351
T118	A	00006C80	4	6181	6352
T12	A	000019B0	4	1484	6246
T13	A	00001A78	4	1528	6247
T14	A	00001B40	4	1572	6248
T15	A	00001C08	4	1616	6249
T16	A	00001CD0	4	1660	6250
T17	A	00001D98	4	1704	6251
T18	A	00001E60	4	1748	6252
T19	A	00001F28	4	1792	6253
T2	A	000011E0	4	1040	6236
T20	A	00001FF0	4	1836	6254
T21	A	000020B8	4	1885	6255
T22	A	00002180	4	1929	6256
T23	A	00002248	4	1973	6257
T24	A	00002310	4	2017	6258
T25	A	000023D8	4	2061	6259
T26	A	000024A0	4	2107	6260
T27	A	00002568	4	2151	6261
T28	A	00002630	4	2195	6262
T29	A	000026F8	4	2239	6263
T3	A	000012A8	4	1084	6237
T30	A	000027C0	4	2283	6264
T31	A	00002888	4	2327	6265
T32	A	00002950	4	2373	6266
T33	A	00002A18	4	2417	6267
T34	A	00002AE0	4	2461	6268
T35	A	00002BA8	4	2505	6269
T36	A	00002C70	4	2549	6270
T37	A	00002D38	4	2593	6271
T38	A	00002E00	4	2637	6272
T39	A	00002EC8	4	2681	6273
T4	A	00001370	4	1128	6238
T40	A	00002F90	4	2725	6274
T41	A	00003058	4	2774	6275
T42	A	00003120	4	2818	6276
T43	A	000031E8	4	2862	6277
T44	A	000032B0	4	2906	6278
T45	A	00003378	4	2950	6279
T46	A	00003440	4	2994	6280
T47	A	00003508	4	3038	6281
T48	A	000035D0	4	3082	6282
T49	A	00003698	4	3126	6283
T5	A	00001438	4	1172	6239
T50	A	00003760	4	3171	6284
T51	A	00003828	4	3217	6285
T52	A	000038F0	4	3261	6286
T53	A	000039B8	4	3305	6287
T54	A	00003A80	4	3349	6288
T55	A	00003B48	4	3393	6289
T56	A	00003C10	4	3437	6290
T57	A	00003CD8	4	3481	6291
T58	A	00003DA0	4	3525	6292
T59	A	00003E68	4	3569	6293
T6	A	00001500	4	1218	6240
T60	A	00003F30	4	3613	6294

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
T61	A	00003FF8	4	3657	6295												
T62	A	000040C0	4	3702	6296												
T63	A	00004188	4	3748	6297												
T64	A	00004250	4	3792	6298												
T65	A	00004318	4	3836	6299												
T66	A	000043E0	4	3880	6300												
T67	A	000044A8	4	3924	6301												
T68	A	00004570	4	3968	6302												
T69	A	00004638	4	4012	6303												
T7	A	000015C8	4	1262	6241												
T70	A	00004700	4	4056	6304												
T71	A	000047C8	4	4100	6305												
T72	A	00004890	4	4144	6306												
T73	A	00004958	4	4188	6307												
T74	A	00004A20	4	4232	6308												
T75	A	00004AE8	4	4276	6309												
T76	A	00004BB0	4	4320	6310												
T77	A	00004C78	4	4364	6311												
T78	A	00004D40	4	4408	6312												
T79	A	00004E08	4	4453	6313												
T8	A	00001690	4	1306	6242												
T80	A	00004ED0	4	4502	6314												
T81	A	00004F98	4	4546	6315												
T82	A	00005060	4	4590	6316												
T83	A	00005128	4	4634	6317												
T84	A	000051F0	4	4678	6318												
T85	A	000052B8	4	4722	6319												
T86	A	00005380	4	4766	6320												
T87	A	00005448	4	4810	6321												
T88	A	00005510	4	4854	6322												
T89	A	000055D8	4	4899	6323												
T9	A	00001758	4	1350	6243												
T90	A	000056A0	4	4945	6324												
T91	A	00005768	4	4989	6325												
T92	A	00005830	4	5033	6326												
T93	A	000058F8	4	5077	6327												
T94	A	000059C0	4	5121	6328												
T95	A	00005A88	4	5165	6329												
T96	A	00005B50	4	5209	6330												
T97	A	00005C18	4	5253	6331												
T98	A	00005CE0	4	5297	6332												
T99	A	00005DA8	4	5341	6333												
TESTING	F	00001004	4	807	345												
TNUM	H	00000004	2	866	344	602	674										
TSUB	A	00000000	4	865	348												
TTABLE	F	00006D50	4	6234													
V0	U	00000000	1	6385													
V1	U	00000001	1	6386	347	394	407	422	437	443	456	471	486	492	505	520	535
					541	554	569	584	592	1029	1073	1117	1161	1205	1251	1295	1339
					1383	1427	1471	1517	1561	1605	1649	1693	1737	1781	1825	1869	1918
					1962	2006	2050	2094	2140	2184	2228	2272	2316	2360	2406	2450	2494
					2538	2582	2626	2670	2714	2758	2807	2851	2895	2939	2983	3027	3071
					3115	3159	3204	3250	3294	3338	3382	3426	3470	3514	3558	3602	3646
					3690	3735	3781	3825	3869	3913	3957	4001	4045	4089	4133	4177	4221
					4265	4309	4353	4397	4441	4486	4535	4579	4623	4667	4711	4755	4799
					4843	4887	4932	4978	5022	5066	5110	5154	5198	5242	5286	5330	5374

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
					5418 5463 5509 5553 5597 5641 5685 5729 5773 5817 5861 5905 5949 5993 6037 6081 6125 6169 6214
V10	U	0000000A	1	6395	
V11	U	0000000B	1	6396	
V12	U	0000000C	1	6397	
V13	U	0000000D	1	6398	
V14	U	0000000E	1	6399	
V15	U	0000000F	1	6400	
V16	U	00000010	1	6401	
V17	U	00000011	1	6402	
V18	U	00000012	1	6403	
V19	U	00000013	1	6404	
V1FUDGE	X	000010F8	16	857	347
V1O1	X	00001148	16	1008	1029
V1O10	X	00001850	16	1406	1427
V1O100	X	00005EA0	16	5397	5418
V1O101	X	00005F68	16	5442	5463
V1O102	X	00006030	16	5488	5509
V1O103	X	000060F8	16	5532	5553
V1O104	X	000061C0	16	5576	5597
V1O105	X	00006288	16	5620	5641
V1O106	X	00006350	16	5664	5685
V1O107	X	00006418	16	5708	5729
V1O108	X	000064E0	16	5752	5773
V1O109	X	000065A8	16	5796	5817
V1O11	X	00001918	16	1450	1471
V1O110	X	00006670	16	5840	5861
V1O111	X	00006738	16	5884	5905
V1O112	X	00006800	16	5928	5949
V1O113	X	000068C8	16	5972	5993
V1O114	X	00006990	16	6016	6037
V1O115	X	00006A58	16	6060	6081
V1O116	X	00006B20	16	6104	6125
V1O117	X	00006BE8	16	6148	6169
V1O118	X	00006CB0	16	6193	6214
V1O12	X	000019E0	16	1496	1517
V1O13	X	00001AA8	16	1540	1561
V1O14	X	00001B70	16	1584	1605
V1O15	X	00001C38	16	1628	1649
V1O16	X	00001D00	16	1672	1693
V1O17	X	00001DC8	16	1716	1737
V1O18	X	00001E90	16	1760	1781
V1O19	X	00001F58	16	1804	1825
V1O2	X	00001210	16	1052	1073
V1O20	X	00002020	16	1848	1869
V1O21	X	000020E8	16	1897	1918
V1O22	X	000021B0	16	1941	1962
V1O23	X	00002278	16	1985	2006
V1O24	X	00002340	16	2029	2050
V1O25	X	00002408	16	2073	2094
V1O26	X	000024D0	16	2119	2140
V1O27	X	00002598	16	2163	2184
V1O28	X	00002660	16	2207	2228
V1O29	X	00002728	16	2251	2272
V1O3	X	000012D8	16	1096	1117
V1O30	X	000027F0	16	2295	2316

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V1031	X	000028B8	16	2339	2360
V1032	X	00002980	16	2385	2406
V1033	X	00002A48	16	2429	2450
V1034	X	00002B10	16	2473	2494
V1035	X	00002BD8	16	2517	2538
V1036	X	00002CA0	16	2561	2582
V1037	X	00002D68	16	2605	2626
V1038	X	00002E30	16	2649	2670
V1039	X	00002EF8	16	2693	2714
V104	X	000013A0	16	1140	1161
V1040	X	00002FC0	16	2737	2758
V1041	X	00003088	16	2786	2807
V1042	X	00003150	16	2830	2851
V1043	X	00003218	16	2874	2895
V1044	X	000032E0	16	2918	2939
V1045	X	000033A8	16	2962	2983
V1046	X	00003470	16	3006	3027
V1047	X	00003538	16	3050	3071
V1048	X	00003600	16	3094	3115
V1049	X	000036C8	16	3138	3159
V105	X	00001468	16	1184	1205
V1050	X	00003790	16	3183	3204
V1051	X	00003858	16	3229	3250
V1052	X	00003920	16	3273	3294
V1053	X	000039E8	16	3317	3338
V1054	X	00003AB0	16	3361	3382
V1055	X	00003B78	16	3405	3426
V1056	X	00003C40	16	3449	3470
V1057	X	00003D08	16	3493	3514
V1058	X	00003DD0	16	3537	3558
V1059	X	00003E98	16	3581	3602
V106	X	00001530	16	1230	1251
V1060	X	00003F60	16	3625	3646
V1061	X	00004028	16	3669	3690
V1062	X	000040F0	16	3714	3735
V1063	X	000041B8	16	3760	3781
V1064	X	00004280	16	3804	3825
V1065	X	00004348	16	3848	3869
V1066	X	00004410	16	3892	3913
V1067	X	000044D8	16	3936	3957
V1068	X	000045A0	16	3980	4001
V1069	X	00004668	16	4024	4045
V107	X	000015F8	16	1274	1295
V1070	X	00004730	16	4068	4089
V1071	X	000047F8	16	4112	4133
V1072	X	000048C0	16	4156	4177
V1073	X	00004988	16	4200	4221
V1074	X	00004A50	16	4244	4265
V1075	X	00004B18	16	4288	4309
V1076	X	00004BE0	16	4332	4353
V1077	X	00004CA8	16	4376	4397
V1078	X	00004D70	16	4420	4441
V1079	X	00004E38	16	4465	4486
V108	X	000016C0	16	1318	1339
V1080	X	00004F00	16	4514	4535
V1081	X	00004FC8	16	4558	4579

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
V1082	X	00005090	16	4602	4623												
V1083	X	00005158	16	4646	4667												
V1084	X	00005220	16	4690	4711												
V1085	X	000052E8	16	4734	4755												
V1086	X	000053B0	16	4778	4799												
V1087	X	00005478	16	4822	4843												
V1088	X	00005540	16	4866	4887												
V1089	X	00005608	16	4911	4932												
V109	X	00001788	16	1362	1383												
V1090	X	000056D0	16	4957	4978												
V1091	X	00005798	16	5001	5022												
V1092	X	00005860	16	5045	5066												
V1093	X	00005928	16	5089	5110												
V1094	X	000059F0	16	5133	5154												
V1095	X	00005AB8	16	5177	5198												
V1096	X	00005B80	16	5221	5242												
V1097	X	00005C48	16	5265	5286												
V1098	X	00005D10	16	5309	5330												
V1099	X	00005DD8	16	5353	5374												
V1OUTPUT	X	00000030	16	878	357												
V2	U	00000002	1	6387	1017	1061	1105	1149	1193	1239	1283	1327	1371	1415	1459	1505	1549
					1593	1637	1681	1725	1769	1813	1857	1906	1950	1994	2038	2082	2128
					2172	2216	2260	2304	2348	2394	2438	2482	2526	2570	2614	2658	2702
					2746	2795	2839	2883	2927	2971	3015	3059	3103	3147	3192	3238	3282
					3326	3370	3414	3458	3502	3546	3590	3634	3678	3723	3769	3813	3857
					3901	3945	3989	4033	4077	4121	4165	4209	4253	4297	4341	4385	4429
					4474	4523	4567	4611	4655	4699	4743	4787	4831	4875	4920	4966	5010
					5054	5098	5142	5186	5230	5274	5318	5362	5406	5451	5497	5541	5585
					5629	5673	5717	5761	5805	5849	5893	5937	5981	6025	6069	6113	6157
					6202												
V20	U	00000014	1	6405	407	422	437	456	471	486	505	520	535	554	569	584	591
V21	U	00000015	1	6406	394	407	422	437	443	456	471	486	492	505	520	535	541
					554	569	584	592									
V22	U	00000016	1	6407													
V23	U	00000017	1	6408													
V24	U	00000018	1	6409													
V25	U	00000019	1	6410													
V26	U	0000001A	1	6411													
V27	U	0000001B	1	6412													
V28	U	0000001C	1	6413													
V29	U	0000001D	1	6414													
V2ADDR	A	00000014	4	873	594	1016	1060	1104	1148	1192	1238	1282	1326	1370	1414	1458	1504
					1548	1592	1636	1680	1724	1768	1812	1856	1905	1949	1993	2037	2081
					2127	2171	2215	2259	2303	2347	2393	2437	2481	2525	2569	2613	2657
					2701	2745	2794	2838	2882	2926	2970	3014	3058	3102	3146	3191	3237
					3281	3325	3369	3413	3457	3501	3545	3589	3633	3677	3722	3768	3812
					3856	3900	3944	3988	4032	4076	4120	4164	4208	4252	4296	4340	4384
					4428	4473	4522	4566	4610	4654	4698	4742	4786	4830	4874	4919	4965
					5009	5053	5097	5141	5185	5229	5273	5317	5361	5405	5450	5496	5540
					5584	5628	5672	5716	5760	5804	5848	5892	5936	5980	6024	6068	6112
					6156	6201											
V3	U	00000003	1	6388	407	422	437	456	471	486	505	520	535	554	569	584	1019
					1063	1107	1151	1195	1241	1285	1329	1373	1417	1461	1507	1551	1595
					1639	1683	1727	1771	1815	1859	1908	1952	1996	2040	2084	2130	2174
					2218	2262	2306	2350	2396	2440	2484	2528	2572	2616	2660	2704	2748
					2797	2841	2885	2929	2973	3017	3061	3105	3149	3194	3240	3284	3328

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
					3372 3416 3460 3504 3548 3592 3636 3680 3725 3771 3815 3859 3903
					3947 3991 4035 4079 4123 4167 4211 4255 4299 4343 4387 4431 4476
					4525 4569 4613 4657 4701 4745 4789 4833 4877 4922 4968 5012 5056
					5100 5144 5188 5232 5276 5320 5364 5408 5453 5499 5543 5587 5631
					5675 5719 5763 5807 5851 5895 5939 5983 6027 6071 6115 6159 6204
V30	U	0000001E	1	6415	
V31	U	0000001F	1	6416	
V3ADDR	A	00000018	4	874	1018 1062 1106 1150 1194 1240 1284 1328 1372 1416 1460 1506 1550
					1594 1638 1682 1726 1770 1814 1858 1907 1951 1995 2039 2083 2129
					2173 2217 2261 2305 2349 2395 2439 2483 2527 2571 2615 2659 2703
					2747 2796 2840 2884 2928 2972 3016 3060 3104 3148 3193 3239 3283
					3327 3371 3415 3459 3503 3547 3591 3635 3679 3724 3770 3814 3858
					3902 3946 3990 4034 4078 4122 4166 4210 4254 4298 4342 4386 4430
					4475 4524 4568 4612 4656 4700 4744 4788 4832 4876 4921 4967 5011
					5055 5099 5143 5187 5231 5275 5319 5363 5407 5452 5498 5542 5586
					5630 5674 5718 5762 5806 5850 5894 5938 5982 6026 6070 6114 6158
					6203
V4	U	00000004	1	6389	
V5	U	00000005	1	6390	
V6	U	00000006	1	6391	
V7	U	00000007	1	6392	
V8	U	00000008	1	6393	
V9	U	00000009	1	6394	
X0001	U	000002A8	1	315	303 316
X1	F	00001188	4	1015	996
X10	F	00001890	4	1413	1394
X100	F	00005EE0	4	5404	5385
X101	F	00005FA8	4	5449	5430
X102	F	00006070	4	5495	5476
X103	F	00006138	4	5539	5520
X104	F	00006200	4	5583	5564
X105	F	000062C8	4	5627	5608
X106	F	00006390	4	5671	5652
X107	F	00006458	4	5715	5696
X108	F	00006520	4	5759	5740
X109	F	000065E8	4	5803	5784
X11	F	00001958	4	1457	1438
X110	F	000066B0	4	5847	5828
X111	F	00006778	4	5891	5872
X112	F	00006840	4	5935	5916
X113	F	00006908	4	5979	5960
X114	F	000069D0	4	6023	6004
X115	F	00006A98	4	6067	6048
X116	F	00006B60	4	6111	6092
X117	F	00006C28	4	6155	6136
X118	F	00006CF0	4	6200	6181
X12	F	00001A20	4	1503	1484
X13	F	00001AE8	4	1547	1528
X14	F	00001BB0	4	1591	1572
X15	F	00001C78	4	1635	1616
X16	F	00001D40	4	1679	1660
X17	F	00001E08	4	1723	1704
X18	F	00001ED0	4	1767	1748
X19	F	00001F98	4	1811	1792
X2	F	00001250	4	1059	1040
X20	F	00002060	4	1855	1836

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
X21	F	00002128	4	1904	1885
X22	F	000021F0	4	1948	1929
X23	F	000022B8	4	1992	1973
X24	F	00002380	4	2036	2017
X25	F	00002448	4	2080	2061
X26	F	00002510	4	2126	2107
X27	F	000025D8	4	2170	2151
X28	F	000026A0	4	2214	2195
X29	F	00002768	4	2258	2239
X3	F	00001318	4	1103	1084
X30	F	00002830	4	2302	2283
X31	F	000028F8	4	2346	2327
X32	F	000029C0	4	2392	2373
X33	F	00002A88	4	2436	2417
X34	F	00002B50	4	2480	2461
X35	F	00002C18	4	2524	2505
X36	F	00002CE0	4	2568	2549
X37	F	00002DA8	4	2612	2593
X38	F	00002E70	4	2656	2637
X39	F	00002F38	4	2700	2681
X4	F	000013E0	4	1147	1128
X40	F	00003000	4	2744	2725
X41	F	000030C8	4	2793	2774
X42	F	00003190	4	2837	2818
X43	F	00003258	4	2881	2862
X44	F	00003320	4	2925	2906
X45	F	000033E8	4	2969	2950
X46	F	000034B0	4	3013	2994
X47	F	00003578	4	3057	3038
X48	F	00003640	4	3101	3082
X49	F	00003708	4	3145	3126
X5	F	000014A8	4	1191	1172
X50	F	000037D0	4	3190	3171
X51	F	00003898	4	3236	3217
X52	F	00003960	4	3280	3261
X53	F	00003A28	4	3324	3305
X54	F	00003AF0	4	3368	3349
X55	F	00003BB8	4	3412	3393
X56	F	00003C80	4	3456	3437
X57	F	00003D48	4	3500	3481
X58	F	00003E10	4	3544	3525
X59	F	00003ED8	4	3588	3569
X6	F	00001570	4	1237	1218
X60	F	00003FA0	4	3632	3613
X61	F	00004068	4	3676	3657
X62	F	00004130	4	3721	3702
X63	F	000041F8	4	3767	3748
X64	F	000042C0	4	3811	3792
X65	F	00004388	4	3855	3836
X66	F	00004450	4	3899	3880
X67	F	00004518	4	3943	3924
X68	F	000045E0	4	3987	3968
X69	F	000046A8	4	4031	4012
X7	F	00001638	4	1281	1262
X70	F	00004770	4	4075	4056
X71	F	00004838	4	4119	4100

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES										
X72	F	00004900	4	4163	4144										
X73	F	000049C8	4	4207	4188										
X74	F	00004A90	4	4251	4232										
X75	F	00004B58	4	4295	4276										
X76	F	00004C20	4	4339	4320										
X77	F	00004CE8	4	4383	4364										
X78	F	00004DB0	4	4427	4408										
X79	F	00004E78	4	4472	4453										
X8	F	00001700	4	1325	1306										
X80	F	00004F40	4	4521	4502										
X81	F	00005008	4	4565	4546										
X82	F	000050D0	4	4609	4590										
X83	F	00005198	4	4653	4634										
X84	F	00005260	4	4697	4678										
X85	F	00005328	4	4741	4722										
X86	F	000053F0	4	4785	4766										
X87	F	000054B8	4	4829	4810										
X88	F	00005580	4	4873	4854										
X89	F	00005648	4	4918	4899										
X9	F	000017C8	4	1369	1350										
X90	F	00005710	4	4964	4945										
X91	F	000057D8	4	5008	4989										
X92	F	000058A0	4	5052	5033										
X93	F	00005968	4	5096	5077										
X94	F	00005A30	4	5140	5121										
X95	F	00005AF8	4	5184	5165										
X96	F	00005BC0	4	5228	5209										
X97	F	00005C88	4	5272	5253										
X98	F	00005D50	4	5316	5297										
X99	F	00005E18	4	5360	5341										
XC0001	U	000002D0	1	329	321										
XCCONT	H	000004A8	2	590	408	423	438	457	472	487	506	521	536	555	570	585
XCFAILMSG	H	000004C6	2	601	596										
XCHECK	U	0000030E	1	368	352										
XCO1	X	00001170	16	1012											
XCO10	X	00001878	16	1410											
XCO100	X	00005EC8	16	5401											
XCO101	X	00005F90	16	5446											
XCO102	X	00006058	16	5492											
XCO103	X	00006120	16	5536											
XCO104	X	000061E8	16	5580											
XCO105	X	000062B0	16	5624											
XCO106	X	00006378	16	5668											
XCO107	X	00006440	16	5712											
XCO108	X	00006508	16	5756											
XCO109	X	000065D0	16	5800											
XCO11	X	00001940	16	1454											
XCO110	X	00006698	16	5844											
XCO111	X	00006760	16	5888											
XCO112	X	00006828	16	5932											
XCO113	X	000068F0	16	5976											
XCO114	X	000069B8	16	6020											
XCO115	X	00006A80	16	6064											
XCO116	X	00006B48	16	6108											
XCO117	X	00006C10	16	6152											
XCO118	X	00006CD8	16	6197											

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
XCO12	X	00001A08	16	1500	
XCO13	X	00001AD0	16	1544	
XCO14	X	00001B98	16	1588	
XCO15	X	00001C60	16	1632	
XCO16	X	00001D28	16	1676	
XCO17	X	00001DF0	16	1720	
XCO18	X	00001EB8	16	1764	
XCO19	X	00001F80	16	1808	
XCO2	X	00001238	16	1056	
XCO20	X	00002048	16	1852	
XCO21	X	00002110	16	1901	
XCO22	X	000021D8	16	1945	
XCO23	X	000022A0	16	1989	
XCO24	X	00002368	16	2033	
XCO25	X	00002430	16	2077	
XCO26	X	000024F8	16	2123	
XCO27	X	000025C0	16	2167	
XCO28	X	00002688	16	2211	
XCO29	X	00002750	16	2255	
XCO3	X	00001300	16	1100	
XCO30	X	00002818	16	2299	
XCO31	X	000028E0	16	2343	
XCO32	X	000029A8	16	2389	
XCO33	X	00002A70	16	2433	
XCO34	X	00002B38	16	2477	
XCO35	X	00002C00	16	2521	
XCO36	X	00002CC8	16	2565	
XCO37	X	00002D90	16	2609	
XCO38	X	00002E58	16	2653	
XCO39	X	00002F20	16	2697	
XCO4	X	000013C8	16	1144	
XCO40	X	00002FE8	16	2741	
XCO41	X	000030B0	16	2790	
XCO42	X	00003178	16	2834	
XCO43	X	00003240	16	2878	
XCO44	X	00003308	16	2922	
XCO45	X	000033D0	16	2966	
XCO46	X	00003498	16	3010	
XCO47	X	00003560	16	3054	
XCO48	X	00003628	16	3098	
XCO49	X	000036F0	16	3142	
XCO5	X	00001490	16	1188	
XCO50	X	000037B8	16	3187	
XCO51	X	00003880	16	3233	
XCO52	X	00003948	16	3277	
XCO53	X	00003A10	16	3321	
XCO54	X	00003AD8	16	3365	
XCO55	X	00003BA0	16	3409	
XCO56	X	00003C68	16	3453	
XCO57	X	00003D30	16	3497	
XCO58	X	00003DF8	16	3541	
XCO59	X	00003EC0	16	3585	
XCO6	X	00001558	16	1234	
XCO60	X	00003F88	16	3629	
XCO61	X	00004050	16	3673	
XCO62	X	00004118	16	3718	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
XCO63	X	000041E0	16	3764		
XCO64	X	000042A8	16	3808		
XCO65	X	00004370	16	3852		
XCO66	X	00004438	16	3896		
XCO67	X	00004500	16	3940		
XCO68	X	000045C8	16	3984		
XCO69	X	00004690	16	4028		
XCO7	X	00001620	16	1278		
XCO70	X	00004758	16	4072		
XCO71	X	00004820	16	4116		
XCO72	X	000048E8	16	4160		
XCO73	X	000049B0	16	4204		
XCO74	X	00004A78	16	4248		
XCO75	X	00004B40	16	4292		
XCO76	X	00004C08	16	4336		
XCO77	X	00004CD0	16	4380		
XCO78	X	00004D98	16	4424		
XCO79	X	00004E60	16	4469		
XCO8	X	000016E8	16	1322		
XCO80	X	00004F28	16	4518		
XCO81	X	00004FF0	16	4562		
XCO82	X	000050B8	16	4606		
XCO83	X	00005180	16	4650		
XCO84	X	00005248	16	4694		
XCO85	X	00005310	16	4738		
XCO86	X	000053D8	16	4782		
XCO87	X	000054A0	16	4826		
XCO88	X	00005568	16	4870		
XCO89	X	00005630	16	4915		
XCO9	X	000017B0	16	1366		
XCO90	X	000056F8	16	4961		
XCO91	X	000057C0	16	5005		
XCO92	X	00005888	16	5049		
XCO93	X	00005950	16	5093		
XCO94	X	00005A18	16	5137		
XCO95	X	00005AE0	16	5181		
XCO96	X	00005BA8	16	5225		
XCO97	X	00005C70	16	5269		
XCO98	X	00005D38	16	5313		
XCO99	X	00005E00	16	5357		
XCOUTPUT	X	00000058	16	882	591	595
XCPLINE	C	00001059	13	830	840	626
XCPLNG	U	00000051	1	840	625	
XCPM4	C	000010A0	2	835	615	
XCPM5	C	000010A7	2	838	622	
XCPNAME	C	0000108F	8	833	608	
XCPTNUM	C	00001066	3	831	606	
XCR15	F	00000570	8	638	624	628
XCRESULT	X	00000540	16	635		
XCSKIP	C	00000007	1	868	370	372
XCV1	X	00000550	16	636		
XCV2	X	00000560	16	637		
XCVD	H	00000400	2	491	383	
XCVD3	H	00000420	2	508	495	
XCVD4	H	0000043A	2	523	510	
XCVDL	H	00000358	2	393	379	

```
     SYMBOL         TYPE   VALUE      LENGTH    DEFN  REFERENCES

XCVDL3               H    00000378        2    410   397
XCVDL4               H    00000392        2    425   412
XCVR                 H    00000454        2    540   385
XCVR3                H    00000474        2    557   544
XCVR4                H    0000048E        2    572   559
XCVRL                H    000003AC        2    442   381
XCVRL3               H    000003CC        2    459   446
XCVRL4               H    000003E6        2    474   461
ZVE7TST              J    00000000    28472    242   245   247   251   255   805   243
=A(E7TESTS)          A    0000075C        4    778   335
=AL2(L'MSGMSG)       R    00000766        2    781   722
=CL1'S'              C    00000768        1    782   370
=CL1'Y'              C    00000769        1    783   372
=CL8'        '       C    00000730        8    772   377
=CL8'SKIP XC '       C    00000728        8    771   369
=CL8'VD      '       C    00000748        8    775   382
=CL8'VDL     '       C    00000738        8    773   378
=CL8'VR      '       C    00000750        8    776   384
=CL8'VRL     '       C    00000740        8    774   380
=F'1'                F    00000760        4    779   630   652
=F'2'                F    00000758        4    777   320
=H'0'                H    00000764        2    780   717
=X'2'                X    0000076A        1    784   396   445   494   543
=X'3'                X    0000076B        1    785   411   460   509   558
=X'4'                X    0000076C        1    786   426   475   524   573
```

MACRO	DEFN	REFERENCES																
FCHECK	79	302																
PTTABLE	954	6233																
VD	195	545	560	575	2798	2842	2886	2930	2974	3018	3062	3106	3150	3195	3241	3285	3329	3373
		3417	3461	3505	3549	3593	3637	3681	3726	3772	3816	3860	3904	3948	3992	4036	4080	4124
		4168	4212	4256	4300	4344	4388	4432	4477									
VDL	139	447	462	477	1020	1064	1108	1152	1196	1242	1286	1330	1374	1418	1462	1508	1552	1596
		1640	1684	1728	1772	1816	1860											
VR	223	496	511	526	4526	4570	4614	4658	4702	4746	4790	4834	4878	4923	4969	5013	5057	5101
		5145	5189	5233	5277	5321	5365	5409	5454	5500	5544	5588	5632	5676	5720	5764	5808	5852
		5896	5940	5984	6028	6072	6116	6160	6205									
VRL	167	398	413	428	1909	1953	1997	2041	2085	2131	2175	2219	2263	2307	2351	2397	2441	2485
		2529	2573	2617	2661	2705	2749											
VRR_C	901	993	1037	1081	1125	1169	1215	1259	1303	1347	1391	1435	1481	1525	1569	1613	1657	1701
		1745	1789	1833	1882	1926	1970	2014	2058	2104	2148	2192	2236	2280	2324	2370	2414	2458
		2502	2546	2590	2634	2678	2722	2771	2815	2859	2903	2947	2991	3035	3079	3123	3168	3214
		3258	3302	3346	3390	3434	3478	3522	3566	3610	3654	3699	3745	3789	3833	3877	3921	3965
		4009	4053	4097	4141	4185	4229	4273	4317	4361	4405	4450	4499	4543	4587	4631	4675	4719
		4763	4807	4851	4896	4942	4986	5030	5074	5118	5162	5206	5250	5294	5338	5382	5427	5473
		5517	5561	5605	5649	5693	5737	5781	5825	5869	5913	5957	6001	6045	6089	6133	6178	

```
   DESC     SYMBOL   SIZE     POS        ADDR

Entry: 0

Image      IMAGE    28472  0000-6F37  0000-6F37
  Region            28472  0000-6F37  0000-6F37
    CSECT  ZVE7TST  28472  0000-6F37  0000-6F37
```

```
   STMT                        FILE NAME
1     /devstor/sharedvfp/tests/zvector-e7-30-DivideRem.asm

** NO ERRORS FOUND **
```